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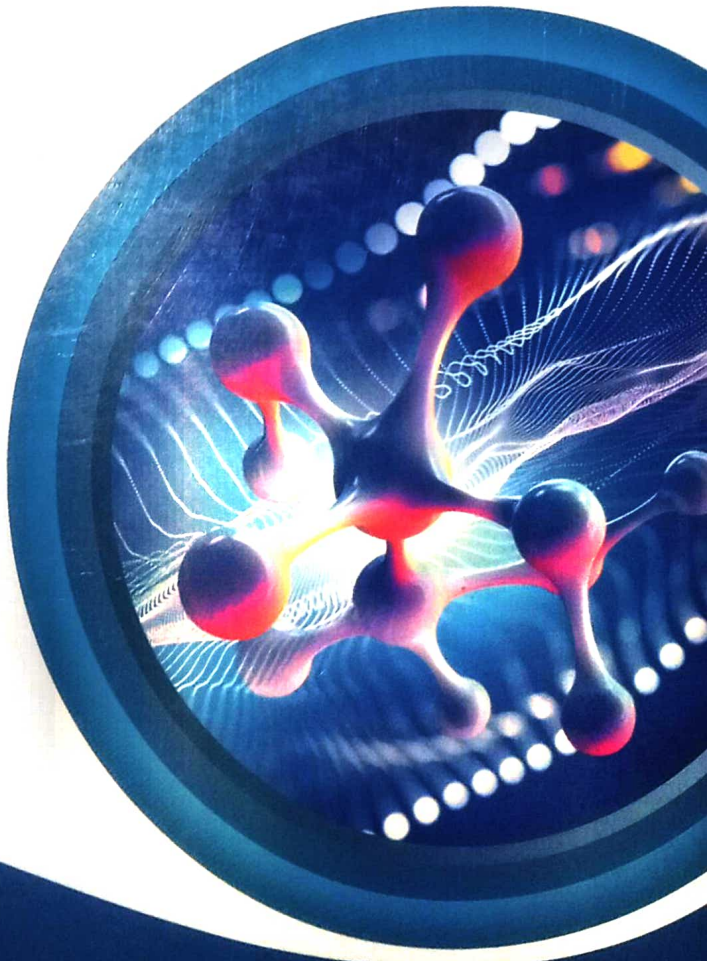
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CHEMISTRY
FULL COURSE STUDY MATERIAL

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Class XI

- Some Basic Concepts of Chemistry
- Atomic Structure
- States of Matter



Module-1

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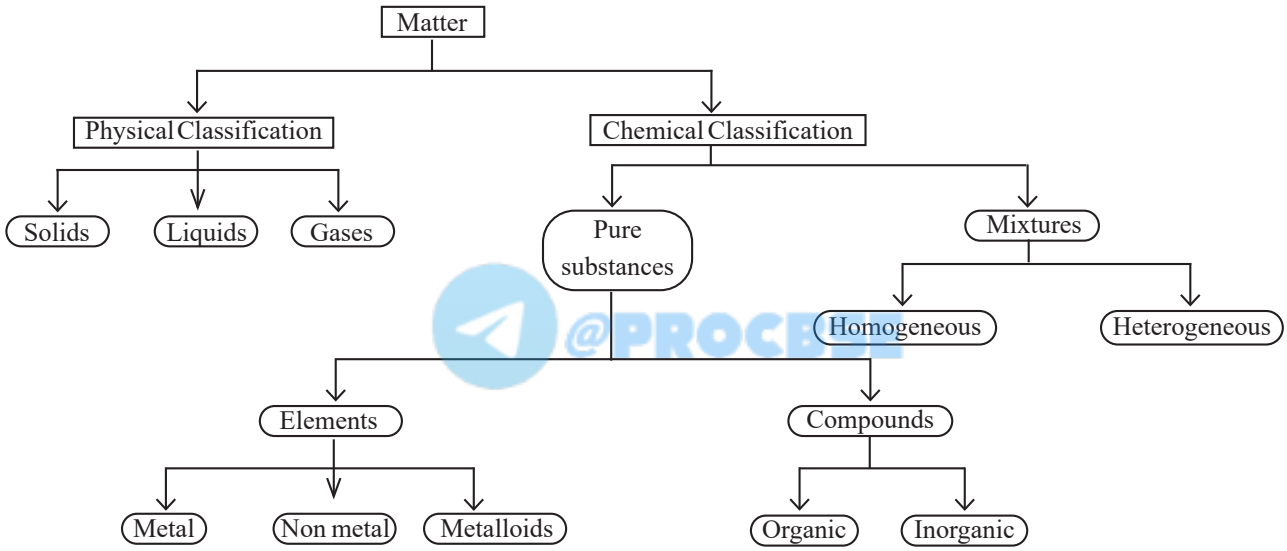
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CHAPTER

1 Some Basic Concepts of Chemistry

CLASSIFICATION OF MATTER



Law of conservation of mass [Lavoisier]: In a chemical change total mass remains conserved i.e., mass before the reaction is always equal to mass after the reaction.

$$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell)$$

1 mole $\frac{1}{2}$ mole 1 mole

Mass before the reaction = $1 \times 2 + \frac{1}{2} \times 32 = 18$ gm

Mass after the reaction = $1 \times 18 = 18$ gm

Law of constant composition [Proust]: All chemical compounds are found to have constant composition irrespective of their method of prepration or sources.

e.g. In H_2O, hydrogen & oxygen combine in 2 : 1 molar ratio, this ratio remains constant whether it is tap water, river water or seawater or produced by any chemical reaction.

Law of Multiple Proportions [Dalton]: When one element combines with the other element to form two or more different compounds, the mass of one element, which combines with a constant mass of the other bear a simple ratio to one another.

e.g. Carbon is found to form two oxides which contain 42.9% & 27.3% of carbon respectively show that these figures shows the law of multiple proportion.

	First oxide	Second oxide
Carbon	42.9 %	27.3 %
Oxygen	57.1 %	72.7%

Given

In th first oxide, 57.1 parts by mass of oxygen combine with 42.9 parts of carbon.

1 part of oxygen will combine with $\frac{42.9}{57.1}$ part of carbon = 0.751

Similarly in 2nd oxide, 1 part of oxygen will combine with $\frac{27.3}{72.7}$ part of carbon = 0.376.

The ratio of carbon that combine with the same mass of oxygen = 0.751 : 0.376 = 2 : 1.

This is a simple whole no. ratio this means above data shows the law of multiple proportion.

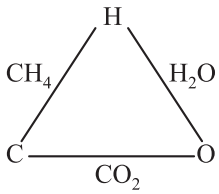
Laws of Reciprocal Proportion: The ratio of the masses of two elements A and B which combine separately with a fixed mass of the third element C is either the same or some simple multiple of the ratio of the masses in which A and B combine directly with each other.

Like CH_4, CO_2 and H_2O

$CH_4 \rightarrow C : H = 12 : 4$

$CO_2 \rightarrow C : O = 12 : 32$

$H_2O \rightarrow H : O = 4 : 32$ or $2 : 16$



Gay-Lussac's Law of Combining Volume: According to him, elements combine in a simple ratio of atoms, gases combine in a simple ratio of their volumes provided all measurements should be done at the same temperature and pressure.

$$H_2(g) + Cl_2(g) \longrightarrow 2HCl$$

1 vol. 1 vol. 2 vol.

Avogadro's hypothesis: Equal volume of all gases have equal number of molecules (not atoms) at same temperature and pressure condition.

N.T.P. (Normal Temperature and Pressure) at N.T.P. condition: Temperature = 0°C or 273 K, Pressure = 1 atm = 760 mm of Hg and volume of one mole of gas at NTP is found to be experimentally equal to 22.4 litres which is known as molar volume.

Note: Measuring the volume is equivalent to counting the number of molecules of the gas.

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Example 1: A 15.9 g sample of sodium carbonate is added to a solution of acetic acid weighing 20.0 g. The two substances react, releasing carbon dioxide gas to the atmosphere. After reaction, the contents of the reaction vessel weigh 29.3 g. What is the mass of carbon dioxide given off during the reaction?

Sol. The total mass of reactants taken = 15.9 + 20.0 = 35.9 gm. From the conservation of mass, the final mass of the contents of the vessel should also be 35.9 gm. But it is only 29.3 gm. The difference is due to the mass of released carbon dioxide gas.

Hence, the mass of carbon dioxide gas released = 35.9 – 29.3 = 6.6 gm

Example 2: The following are results of analysis of two samples of the same or two different compounds of phosphorus and chlorine. From these results, decide whether the two samples are from the same or different compounds. Also state the law, which will be obeyed by the given samples.

	Amount P	Amount Cl
Compound A	1.156 gm	3.971 gm
Compound B	1.542 gm	5.297 gm

Sol. The mass ratio of phosphorus and chlorine in compound A, $m_P : m_{Cl} = 1.156:3.971 = 0.2911:1.000$

The mass ratio of phosphorus and chlorine in compound B, $m_P : m_{Cl} = 1.542:5.297 = 0.2911:1.000$

As the mass ratio is same, both the compounds are **same** and the samples obey the **law of definite proportion**.

Example 3: 2.5 ml of a gaseous hydrocarbon exactly requires 12.5 ml oxygen for complete combustion and produces 7.5 ml carbon dioxide and 10.0 ml water vapour. All the volumes are measured at the same pressure and temperature. Show that the data illustrates Gay Lussac's law of volume combination.

Sol. $V_{hydrocarbon} : V_{oxygen} : V_{carbon\ dioxide} : V_{water\ vapour} = 2.5 : 12.5 : 7.5 : 10.0$

$= 1 : 5 : 3 : 4$ (simple ratio)

Hence, the data is according to the **law of volume combination**.

Concept Application

1. A sample of pure carbon dioxide, irrespective of its source contains 27.27% carbon and 72.73% oxygen. The data support:
   (*a*) Law of constant composition.
   (*b*) Law of conservation of mass.
   (*c*) Law of reciprocal proportions.
   (*d*) Law of multiple proportions.
2. The percentage of hydrogen in water and hydrogen peroxide is 11.1 and 5.9 respectively. These figures illustrate:
   (*a*) Law of multiple proportions.
   (*b*) Law of conservation of mass.
   (*c*) Law of constant proportions.
   (*d*) Law of combining volumes.
3. 1.0 g of an oxide of A contained 0.5 g of A. 4.0 g of another oxide of A contained 1.6 g of A. The data indicate the law of:
   (*a*) Reciprocal proportions.
   (*b*) Constant proportions.
   (*c*) Conservation of energy.
   (*d*) Multiple proportions.

BASIC DEFINITIONS

Relative Atomic Mass: One of the most important concept come out from Dalton's atomic theory was that of relative atomic mass or relative atomic weight. This is done by expressing mass of one atom with respect to a fixed standard. Dalton used hydrogen as the

standard (H = 1). Later on oxygen (O = 16) replaced hydrogen as the reference. Therefore relative atomic mass is given as:

On hydrogen scale: Relative atomic mass (R.A.M)

$$= \frac{\text{Mass of one atom of an element}}{\text{Mass of one hydrogen atom}}$$

On oxygen scale: Relative atomic mass (R.A.M)

$$= \frac{\text{Mass of one atom of an element}}{\frac{1}{16} \times \text{Mass of one oxygen atom}}$$

- The present standard unit which was adopted internationally in 1961, is based on the mass of one carbon-12 atom.

  Relative atomic mass (R.A.M)

$$= \frac{\text{Mass of one atom of an element}}{\frac{1}{12} \times \text{Mass of one C} - 12 \text{ atom}}$$

Atomic Mass Unit (or amu): The atomic mass unit (amu) is equal to $\left(\frac{1}{12}\right)^{th}$ mass of one atom of carbon-12 isotope.

$\therefore$ 1 amu = $\frac{1}{12}$ × mass of one C-12 atom

$\simeq$ mass of one nucleon in C-12 atom.

= 1.66×10^{-24} gm or 1.66×10^{-27} kg

- Today, amu has been replaced by 'u' which is known as unified mass

Atomic & Molecular Mass: It is the mass of 1 atom of a substance it is expressed in amu.

- Atomic mass = R.A.M × 1 amu

Relative molecular mass

$$= \frac{\text{mass of one molecule of the substance}}{\frac{1}{12} \times \text{mass of one} - \text{C-12 atom}}$$

- Molecular mass = Relative molecular mass × 1 amu

Note: Relative atomic mass is nothing but the number of nucleons present in the atom.

Gram Atomic Mass: The atomic mass of an element expressed in gram is called gram atomic mass of the element.

or

It is also defined as mass of 6.02×10^{23} atoms.

or

It is also defined as the mass of one mole atoms.

For example for oxygen atom: Atomic mass of 'O' atom = mass of one 'O' atom = 16 amu

gram atomic mass = mass of 6.02×10^{23} 'O' atoms = 16 amu × 6.02×10^{23}

= $16 \times 1.66 \times 10^{-24}$ g × 6.02×10^{23} = 16 g

($\because$ $1.66 \times 10^{-24} \times 6.02 \times 10^{23} \simeq 1$)

Gram Molecular Mass: The molecular mass of a substance expressed in gram is called the gram-molecular mass of the substance.

or

It is also defined as mass of 6.02×10^{23} molecules.

or

It is also defined as the mass of one mole molecules.

For example for 'O_2' molecule: Molecular mass of 'O_2' molecule

= mass of one 'O_2' molecule

= 2 × mass of one 'O' atom

= 2 × 16 amu

= 32 amu

gram molecular mass = mass of 6.02×10^{23} 'O_2' molecules

= 32 amu × 6.02×10^{23}

= $32 \times 1.66 \times 10^{-24}$ gm × 6.02×10^{23} = 32 gm

RELATIONSHIP BETWEEN GRAM AND AMU

1 amu = $\frac{1}{12}$ wt. of one C - 12 atom.

For 1 mole C = 12 gm = 6.023×10^{23} atoms

wt. of 6.023×10^{23} atoms = 12 gm

wt. of 1 atom of C = $\frac{12}{N_A}$ gm

($N_A \rightarrow$ Avogadro's number = 6.23×10^{23})

1 amu = $\frac{1}{12}$ wt. of one C - 12 atom

$$= \frac{1}{12} \times \frac{12}{N_A} \text{ gm}$$

1 amu = $\frac{1}{N_A}$ gm

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Train Your Brain

Example 4: Find the relative atomic mass of 'O' atom and its atomic mass.

Sol. The number of nucleons present in 'O' atom is 16.

$\therefore$ Relative atomic mass of 'O' atom = 16.

Atomic mass = R.A.M × 1 amu = 16 × 1 amu = 16 amu

Example 5: The weight of one atom of uranium is 235 amu. Its actual weight in g:

Sol. 3.95×10^{-22}

$235 \times 1.67 \times 10^{-24}$ g

= 3.95×10^{-22} g

Concept Application

4. The Relative molecular mass of ammonia is:

(*a*) 17 amu (*b*) 22 amu

(*c*) 28 amu (*d*) 44 amu

5. The mass of an atom of sodium is:

(*a*) 23 amu (*b*) 23gm

(*c*) 46 amu (*d*) 12 amu

6. The atomic mass & molecular mass of hydrogen is:
(*a*) 1amu & 2amu (*b*) 2 amu & 4 amu
(*c*) 3amu & 6amu (*d*) 4 amu & 8 amu

7. One 'u'stands for the mass of:
(*a*) An atom of carbon-12.
(*b*) 1/12th of carbon-12.
(*c*) 1/12th of hydrogen atom.
(*d*) One atom of any of the elements.

8. Mass of 1 amu in g:
(*a*) 1.66×10^{24} (*b*) 1.66×10^{-24}
(*c*) 1.008 (*d*) 9.1×10^{-28}

MOLE CONCEPT

Mole

Mole is a chemical counting SI unit and defined as follows:

A mole is the amount of a substance that contains as many entities (atoms, molecules or other particles) as there are atoms in exactly 0.012 kg (or 12 gm) of the carbon-12 isotope.

From mass spectrometer we found that there are 6.023×10^{23} atoms present in 12 gm of C-12 isotope.

The number of entities in 1 mol is so important that it is given a separate name and symbol known as Avogadro constant denoted by N_A.

i.e., on the whole we can say that 1 mole is the collection of 6.02×10^{23} entities. Here entities may represent atoms, ions, molecules or even pens, chair, paper etc also include in this but as this number (N_A) is very large therefore it is used only for very small things.

How Big is a Mole?

Amount of water in world's oceans (litres) | Age of earth (seconds) | Population of earth

Avogadro's number 602,200,000,000,000,000,000,000

Distance from earth to sun (centimeters)

Note: In modern practice gram-atom and gram-molecule are termed as mole.

METHODS OF CALCULATIONS OF MOLE

(*a*) If no. of some species is given, then no. of moles $= \frac{\text{Given no.}}{N_A}$

(*b*) If weight of a given species is given, then no. of moles = $\frac{\text{Given wt.}}{\text{Atomic wt.}}$ (for atoms),

or $\frac{\text{Given wt.}}{\text{Molecular wt.}}$ (for molecules)

(*c*) If volume of a gas is given along with its temperature (T) and pressure (P) use $n = \frac{PV}{RT}$

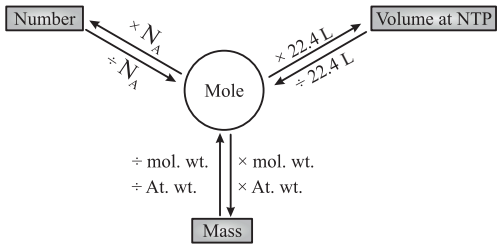
where R = 0.0821 lit-atm/mol–K (when P is in atmosphere and V is in litre.)

1 mole of any gas at STP (0°C & 1 bar) occupies 22.7 litre.

1 mole of any gas at STP (0°C & 1 atm) occupies 22.4 litre.

Atom: Atom is smallest particle which can not be divided into its constituents.

Y-map: Interconversion of mole - volume, mass and number of particles:



Note: Measuring the volume is equivalent to counting the number of molecules of the gas.

Train Your Brain

Example 6: Total number of atoms of all elements present in 1 mole of ammonium dichromate is?
(*a*) 14 (*b*) 19
(*c*) 6×10^{23} (*d*) 114×10^{23}

Sol. (*d*) $(NH_4)_2Cr_2O_7 = 19 \times 6.02 \times 10^{23} = 114 \times 10^{23}$ atoms.

Example 7: How many atoms of oxygen are their in 16 g of oxygen?

Sol. Let x atoms of oxygen are present

So, $16 \times 1.66 \times 10^{-24} \times x = 16$ g

$x = \frac{1}{1.66 \times 10^{-24}} = N_A$

Example 8: Calculate the volume in litres of 20 g hydrogen gas at NTP.

Sol. No. of moles of hydrogen gas $= \frac{\text{Mass}}{\text{Molecular mass}}$

$= \frac{20 \text{ gm}}{2 \text{ gm}} = 10$ mol

Volume of hydrogen gas at NTP = 10×22.4 L

Example 9: The number of atoms contained in 11.2 L of SO_2 at N.T.P. are:
(*a*) $3/2 \times 6.02 \times 10^{23}$ (*b*) $2 \times 6.02 \times 10^{23}$
(*c*) 6.02×10^{23} (*d*) $4 \times 6.02 \times 10^{23}$

Sol. (*a*) 22.4 litre gas has = 1 mole

1 litre gas has $= \frac{1}{22.4} \times 11.2 = \frac{1}{2}$ mole

$= S + 2O = 3$

$= \frac{3}{2} \times 6.02 \times 10^{23}$

Concept Application

9. The number of particles present in 1 mol of nitrogen atom are:
(*a*) 6.022×10^{25} (*b*) 6.022×10^{24}
(*c*) 6.022×10^{23} (*d*) 6.022×10^{22}

10. The mass of one molecule of water is approximately:
(*a*) 3×10^{-23} g (*b*) 18 g
(*c*) 1.5×10^{-23} g (*d*) 4.5×10^{-23} g

11. The molar mass of ferrous sulphate ($FeSO_4.7H_2O$) is:
(*a*) 152 gm (*b*) 278 gm
(*c*) 137 gm (*d*) None of these

ELEMENTAL ANALYSIS

For n mole of a compound ($C_3H_7O_2$);
Moles of C = 3n
Moles of H = 7n
Moles of O = 2n

Example: Find the wt. of water present in 1.61 g of $Na_2SO_4.10H_2O$.vv

Sol. Moles of $Na_2SO_4.10H_2O = \frac{\text{wt. in gram}}{\text{Molecular wt.}} = \frac{1.61}{322}$
$= 0.005$ moles
Moles of water = 10 × moles of $Na_2SO_4.10H_2O$
$= 10 \times 0.005 = 0.05$
wt. of water $= 0.05 \times 18 = 0.9$ gm

PERCENTAGE FORMULAE COMPOSITION

% of elevment in compound

$$= \frac{\text{Atomic weight of element} \times \text{Number of atom} \times 100}{\text{Total molecular weight of compound}}$$

Here we are going to find out the percentage of each element in the compound by knowing the molecular formula of compound.

We know that according to law of definite proportions any sample of a pure compound always possess constant ratio with their combining elements.

Example: Every molecule of ammonia always has formula NH_3 irrespective of method of preparation or sources. i.e. 1 mole of ammonia always contains 1 mol of N and 3 mole of H. In other words 17 gm of NH_3 always contains 14 gm of N and 3 gm of H. Now find out % of each element in the compound.

Sol. Mass % of N in NH_3 =

$$\frac{\text{Mass of N in 1 mol } NH_3}{\text{Mass of 1 mol of } NH_3} \times 100 = \frac{14 \text{ gm}}{17} \times 100 = 82.35\ \%$$

Mass % of H in NH_3 =

$$\frac{3 \times \text{Mass of H is 1 mol } NH_3}{\text{Mass of 1 mole of } NH_3} \times 1 = \frac{3}{17} \times 100 = 17.65\ \%$$

DENSITY

(*a*) Absolute density (*b*) Relative density

$$\text{Absolute density} = \frac{\text{Mass}}{\text{Volume}}$$

$$\text{Relative density} = \frac{\text{Density of substance}}{\text{Density of standard substance}}$$

$$\text{Specific gravity} = \frac{\text{Density of substance}}{\text{Density of } H_2O \text{ at } 4^\circ C}$$

Vapour density: It is defined only for gas.
It is a density of gas with respect to H_2 gas at same temperature & pressure.

$$\text{V.D.} = \frac{d_{gas}}{d_{H_2}} = \frac{PM_{gas}/RT}{PM_{H_2}/RT} = \frac{M_{gas}}{M_{H_2}} = \frac{M}{2}$$

$$\text{V.D.} = \frac{M}{2}$$

Train Your Brain

Example 10: What is the V.D. of SO_2 with respect to CH_4?

Sol. $\text{V.D.} = \frac{\text{M.W. of } SO_2}{\text{M.W. of } CH_4}$

$\text{V.D.} = \frac{64}{16} = 4$

Example 11: 7.5 litre of the particular gas at S.T.P. weighs 16 gram. What is the V.D. of gas?

Sol. Moles at S.T.P. $= \frac{7.5}{22.4} = \frac{16}{M}$

$M = 48$ gram $\text{V.D.} = \frac{48}{2} = 24.$

Example 12: Find the density of $CO_2(g)$ with respect to $N_2O(g)$.

Sol. $\text{R.D.} = \frac{\text{M.wt. of } CO_2}{\text{M.wt. of } N_2O} = \frac{44}{44} = 1$

Exampl 13: Find the vapour density of N_2O_5.

Sol. $\text{V.D.} = \frac{\text{Mol. wt. of } N_2O_5}{2} = 54.$

Concept Application

12. The vapour density of carbon dioxide is:
(*a*) 44 (*b*) 32 (*c*) 22 (*d*) 12

13. The density of air is 0.001293 g/cm^3 at STP. Identify which of the following statement is correct?
(*a*) Vapour density is 14.48.
(*b*) Molecular weight is 28.96.
(*c*) Vapour density is 0.001293 g/cm^3.
(*d*) Vapour density and molecular weight cannot be determined.

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EMPIRICAL AND MOLECULAR FORMULA

We have just seen that knowing the molecular formula of the compound we can calculate percentage composition of the elements. Conversely if we know the percentage composition of the elements initially, we can calculate the relative number of atoms of each element in the molecules of the compound. This gives us the empirical formula of the compound. Further if the molecular mass is known then the molecular formula can easily be determined.

The empirical formula of a compound is a chemical formula showing the relative number of atoms in the simplest ratio. An empirical formula represents the simplest whole number ratio of various atoms present in a compound.

The molecular formula gives the actual number of atoms of each element in a molecule. The molecular formula shows the exact number of different types of atoms present in a molecule of a compound.

The molecular formula is an integral multiple of the empirical formula.

i.e. Molecular formula = Empirical formula × n

where $n = \frac{\text{Molecular Formula Mass}}{\text{Empirical Formula Mass}}$



Train Your Brain

Example 14: Acetylene and benzene both have the empirical formula CH. The molecular masses of acetylene and benzene are 26 and 78 respectively. Deduce their molecular formulae.

Sol. ∵ Empirical Formula is CH

Step-1: The empirical formula of the compound is CH

∴ Empirical formula mass = (1 × 12) + 1 = 13.

Molecular mass = 26

Step-2: To calculate the value of 'n'

$$n = \frac{\text{Molecular mass}}{\text{Empirical formula mass}} = \frac{26}{13} = 2$$

Step-3: To calculate the molecular formula of the Compound.

Molecular formula = n × (Empirical formula of the compound)

$= 2 \times CH = C_2H_2$

Thus the molecular formula is C_2H_2

Similarly for benzene

To calculate the value of 'n'

$$n = \frac{\text{Molecular mass}}{\text{Empirical formula mass}} = \frac{78}{13} = 6.$$

Thus the molecular formula is $6 \times CH = C_6H_6$

Example 15: An organic substance containing carbon, hydrogen and oxygen gave the following percentage composition.

C = 40.684% ; H = 5.085% and O = 54.228%

The molecular weight of the compound is 118. Calculate the molecular formula of the compound.

Sol. Step-1: To calculate the empirical formula of the compound.

Element	Symbol	Percentage of element	At. mass of element	Relative no. of atoms = Percentage / At. mass	Simplest atomic ratio	Simplest whole no. atomic ratio
Carbon	C	40.687	12	$\frac{40.687}{12} = 3.390$	$\frac{3.390}{3.389} = 1$	2
Hydrogen	H	5.085	1	$\frac{5.085}{1} = 5.085$	$\frac{5.085}{3.389} = 1.5$	3
Oxygen	O	54.228	16	$\frac{54.228}{16} = 3.389$	$\frac{3.389}{3.389} = 1$	2

∴ Empirical Formula is $C_2H_3O_2$

Step-2: To calculate the empirical formula mass.

The empirical formula mass of the compound is $12 \times 2 + 3 \times 1 + 16 \times 2 = 59$.

Step-3: To calculate the value of 'n'

$$n = \frac{\text{Molecular mass}}{\text{Empirical formula mass}} = \frac{118}{59} = 2$$

Step-4: To calculate the molecular formula of the salt.

Molecular formula = n × (Empirical formula)

$= 2 \times C_2H_3O_2 = C_4H_6O_4$

Thus the molecular formula is $C_4H_6O_4$.

Example 16: Acetylene & butene have empirical formula CH & CH_2 respectively. The molecular mass of acetylene and butene are 26 & 56 respectively. Deduce their molecular formula.

Sol. C_2H_2 & C_4H_8

$$n = \frac{\text{Molecular mass}}{\text{Empirical formula mass}}$$

For Acetylene : $n = \frac{26}{13} = 2$

∴ Molecular formula = C_2H_2

For Butene: $n = \frac{56}{14} = 4$

∴ Molecular formula = C_4H_8

Example 17: An oxide of nitrogen gave the following percentage composition by mass:

N = 25.94 and O = 74.06

Calculate the empirical formula of the compound.

Sol. N_2O_5.

Element	%/Atomic mass	Simple ratio	Simple intiger ratio
N	$\frac{25.94}{14} = 1.85$	1	2
O	$\frac{74.06}{16} = 4.63$	2.5	5

So empirical formula is N_2O_5.

Concept Application

14. A compound contains 25% hydrogen and 75% carbon by mass. Determine the empirical formula of the compound.

(*a*) CH_4 (*b*) C_2H_6

(*c*) C_3H_8 (*d*) C_2H_2

15. The empirical formula of a compound of molecular mass 120 u is CH_2O. The molecular formula of the compound is:

(*a*) $C_2H_4O_2$ (*b*) $C_4H_8O_4$

(*c*) $C_3H_5C_3$ (*d*) All of these

16. Calculate the molecular formula of compound which contains 20% Ca and 80% Br (by wt.) if molecular weight of compound is 200 u. (Atomic wt. Ca = 40. Br = 80)

(*a*) $Ca_{1/2}Br$ (*b*) $CaBr_2$

(*c*) CaBr (*d*) Ca_2Br

Stoichiometry Based Concept (Problems Based on Chemical Reaction)

One of the most important aspects of a chemical equation is that when it is written in the balanced form, it gives quantitative relationships between the various reactants and products in terms of moles, masses, molecules and volumes. This is called stoichiometry (Greek word, meaning 'to measure an element'). For example, a balanced chemical equation along with the quantitative information conveyed by it is given below:

$CaCO_3$	+	$2HCl$	$\longrightarrow$	$CaCl_2$	+	H_2O	+	CO_2
1 Mole		2 Mole		1 Mole		1 Mole		1 Mole
40+12+3×16		2(1 + 35.5)		40+2×35.5		2×1+16		12+2×16
= 100 g		= 73 g		= 111 g		= 18 g		= 44 g or 22.4 L at STP

Thus,

(*i*) 1 mole of calcium carbonate reacts with 2 moles of hydrochloric acid to give 1 mole of calcium chloride, 1 mole of water and 1 mole of carbon dioxide.

(*ii*) 100 g of calcium carbonate react with 73 g hydrochloric acid to give 111 g of calcium chloride, 18 g of water and 44 g (or 22.4 litres at STP) of carbon dioxide.

Stoichiometry:

N_2 + $3H_2$ $\rightarrow$ $2NH_3$

1 mole + 3 mole $\rightarrow$ 2 mole

22.4 litre + 3 × 22.4 litre $\rightarrow$ 2 × 22.4 litre (at STP)

1 litre + 3 litre $\rightarrow$ 2 litre

1000 mL + 3000 mL $\rightarrow$ 2000 mL

1 mL + 3 mL $\rightarrow$ 2 mL

28 gm + 6 gm $\rightarrow$ 34 g

(According to the law of conservation of mass).

Note: Gram can not be represented by stoichiometry.

The quantitative information conveyed by a chemical equation helps in a number of calculations. The problems involving these calculations may be classified into the following different types:

(1) Mass - Mass Relationships i.e. mass of one of the reactants or products is given and the mass of some other reactant or product is to be calculated.

(2) Mass - Volume Relationships i.e. mass/volume of one of the reactants or products is given and the volume/mass of the other is to be calculated.

(3) Volume - Volume Relationships i.e.volume of one of the reactants or the products is given and the volume of the other is to be calculated.

The general method of calculations for all the problems of the above types consists of the following steps:

(*i*) Write down the balanced chemical equation.

(*ii*) Write the relative number of moles or the relative masses (gram atomic or molecular masses) of the reactants and the products below their formula.

(*iii*) In case of a gaseous substance, write down 22.4 litres at STP below the formula in place of 1 mole

(*iv*) Apply unitary method to make the required calculations.

Quite often one of the reactants is present in larger amount than the other as required according to the balanced equation. The amount of the product formed then depends upon the reactant which has reacted completely. This reactant is called the limiting reactant. The excess of the other is left unreacted.

Combustion reaction: For balancing the combustion reaction, first of all balance carbon atoms, then balance hydrogen atom, finally balance Oxygen atom.

For Example:

Combustion reaction of C_2H_6 : $C_2H_6 + O_2 \rightarrow CO_2 + H_2O$ (skeleton equation)

Balance carbon atoms: $C_2H_6 + O_2 \rightarrow 2CO_2 + H_2O$

Now balance hydrogen atoms: $C_2H_6 + O_2 \rightarrow 2CO_2 + 3H_2O$

Now balance oxygen atoms: $C_2H_6 + \frac{7}{2}O_2 \rightarrow 2CO_2 + 3H_2O$

Type (I) Involving Mass-Mass Relationship

Example: How much iron can be theoretically obtained in the reduction of 1 kg of Fe_2O_3?

Sol. Writing the balanced equation for the reaction.

$$n = \frac{\text{Weight}}{M_w} = \frac{1000}{160}\text{mol}$$

The equation shows that 2 mol of iron are obtained from 1 mol of ferric oxide.

Hence, the obtained no. of moles of Fe = $\frac{2\times1000}{160}$ =12.5 mol

$$= \frac{\text{Weight}}{\text{Atomic weight}} = \frac{\text{Weight}}{56}$$

Weight of iron obtained = 12.5 × 56 g = 700 g

Example: What amount of silver chloride is formed by the action of 5.850 g of sodium chloride on an excess of silver nitrate?

Sol. Writing the equation for the reaction

$$\underset{1\text{ mol}}{NaCl} + \underset{1\text{ mol}}{AgNO_3} \longrightarrow \underset{1\text{ mol}}{AgCl} + \underset{1\text{ mol}}{NaNO_3}$$

$$n = \frac{\text{Weight}}{M_w} = \frac{5.85}{58.5} = 0.1 \text{ mol}$$

1 mol of AgCl is obtained with 1 mol of NaCl.

Hence, the number of moles of AgCl obtained with 0.1 mol of NaCl = 0.1 mol

$$\because\ n = \frac{\text{Weight}}{M_W} \Rightarrow 0.1 \text{ mol} = \frac{\text{Weight}}{M_w} = \frac{\text{Weight}}{143.5}$$

$\Rightarrow$ weight = 0.1 × 143.5 g = 14.35 g.

Type (II) Mass - Volume Relationship

Example: At 100°C for complete combustion of 3g ethane the required volume of O_2 & produced volume of CO_2 at STP will be?

Sol.

$$\underset{2}{C_2H_6} + \underset{7}{7O_{2(g)}} \rightarrow \underset{4}{4CO_{2(g)}}\ \underset{6}{6H_2O_{(g)}}$$

$$n = \frac{\text{Weight}}{M_w} = \frac{3}{30} = \frac{1}{10} = 0.1 \text{ mole}$$

$\therefore$ Required moles of $O_2 = \frac{7}{2} \times 0.1 = 0.35$ mol.

Volume of O_2 at STP = 0.35 × 22.4 = 7.84 L.

And produced moles of $CO_2 = \frac{4}{2} \times 0.1 = 0.2$ mol.

Volume of CO_2 at STP = 0.2 × 22.4 = 4.48 L.

Example: In the following reaction, if 10 g of H_2, react with N_2. What will be the volume of NH_3 at STP?

$$N_2 + 3H_2 \longrightarrow 2NH_3$$

Sol. $N_2 + 3H_2 \longrightarrow 2NH_3$ 10 g

$$n = \frac{\text{Weight}}{M_w} = \frac{10}{2} = 5 \text{ mol.}$$

Produced moles of $NH_3 = \frac{2}{3} \times 5 = \frac{10}{3}$.

Volume of NH_3 at STP = $\frac{10}{3} \times 22.4$ = 74.67 L

Type (III) Volume-Volume Relationship

Example: At 100°C for complete combustion of 1.12 litre of butane (C_4H_{10}), the produced volume of $H_2O_{(g)}$ & CO_2 at STP will be.

Sol. $C_4H_{10(g)} + \frac{13}{2}O_{2(g)} \longrightarrow 4CO_{2(g)} + 5H_2O_{(g)}$

1.12 L

Volume of $H_2O_{(g)}$ at STP = 5 × 1.12 = 5.6 L.

Volume of $CO_{2(g)}$ at STP = 4 × 1.12 = 4.48 L.

Example: At 25°C for complete combustion of 5 mol propane (C_3H_8). The required volume of O_2 at STP will be?

Sol. For C_3H_8, the combustion reaction is

$C_3H_{8(g)} + 5O_{2(g)} \longrightarrow 3CO_{2(g)} + 4H_2O_{(g)}$

5 mol

Required moles of $O_2 = 5 \times 5 = 25 \text{ mol} = \frac{V}{22.4}$.

Volume of O_2 gas at STP (V) = 25 × 22.4 = 560 L.

Example: 3 litre of mixture of propane (C_3H_8) & butane (C_4H_{10}) on complete combustion gives10 litre CO_2. Find the composition of mixture.

Sol. Let the volume of propane in the mixture = x litre,

$\therefore$ The volume of butane in the mixture = (3 – x) litre

Now let us calculate the volume of CO_2 evolved with the help of chemical equation.

Step-I: Calculation of volume of CO_2 from x litre of propane.

$$\underset{x\text{ litre}}{C_3H_8} + 5O_2 \longrightarrow \underset{3x\text{ litre}}{3CO_2} + 4H_2O$$

Step-II: Calculation of volume of CO_2 from (3 – x) litre of butane. The combustion equation for butane is:

$$\underset{(3-x)\text{ litre}}{C_4H_{10}} + \frac{13}{2}O_2 \longrightarrow \underset{4(3-x)\text{ litre}}{4CO_2} + 5H_2O$$

Step-III: Calculation of composition of the mixture.

Total volume of CO_2 formed in the step (I) and step (II) = [3x + 4(3 – x)] litre

But the volume of CO_2 actually formed = 10 litre

$$3x + 4(3 - x) = 10$$

or $3x + 12 - 4x = 10$ or x = 2 litre

$\therefore$ Volume of propane = x litre = 2 litre

$\therefore$ Volume of butane = (3 – x) litre = (3 – 2) = 1 litre

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Example 18: Write a balance chemical equation for following reaction:

When ammonia (NH_3) decompose into nitrogen (N_2) gas & hydrogen (H_2) gas.

Sol. $NH_3 \rightarrow \frac{1}{2}N_2 + \frac{3}{2}H_2$ or $2NH_3 \rightarrow N_2 + 3H_2$

Example 19: When 170 g NH_3 (M =17) decomposes, how many grams of N_2 & H_2 is produced?

Sol. $NH_3 \rightarrow \frac{1}{2}N_2 + \frac{3}{2}H_2$

$$\frac{\text{moles of } NH_3}{1} = \frac{\text{moles of } N_2}{1/2} = \frac{\text{moles of } H_2}{3/2}$$

So, moles of $N_2 = \frac{1}{2} \times \frac{170}{17} = 5$

So, wt. of N_2 = 5 × 28 = 140 g

Similarly moles of $H_2 = \frac{3}{2} \times \frac{170}{17} = 15$

So, wt. of H_2 = 15 × 2 = 30 g.

Example 20: 340 g NH_3 (M = 17) when decompose, how many litres of nitrogen gas is produced at STP?

Sol. $NH_3 \rightarrow \frac{1}{2}N_2 + \frac{3}{2}H_2$

Moles of $NH_3 = \frac{340}{17} = 20$

So moles of $N_2 = \frac{1}{2} \times 20 = 10$

$\therefore$ Vol. of N_2 at STP = $10 \times 22.4 = 224$ L.

Example 21: 4 mole of $MgCO_3$ is reacted with 6 moles of HCl solution. Find the volume of CO_2 gas produced at STP, the reaction is:

$MgCO_3 + 2HCl \rightarrow MgCl_2 + CO_2 + H_2O$

Sol. Here HCl is limiting reagent. So moles of CO_2 formed = 3.

So vol. at STP = $3 \times 22.4 = 67.2$ L.

Concept Application

17. If 1.5 moles of oxygen combine with Al to form Al_2O_3, the weight of Al used in the reaction is :

(*a*) 27 g (*b*) 40.5 g

(*c*) 54 g (*d*) 81 g

18. How many liters of CO_2 at STP will be formed when 0.01 mol of H_2SO_4 reacts with excess of Na_2CO_3?

$Na_2CO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + CO_2 + H_2O$

(*a*) 22.4 L (*b*) 2.24 L

(*c*) 0.224 L (*d*) 1.12L

19. How many moles of potassium chlorate need to be heated to produce 11.2 litre oxygen at N.T.P.?

$KClO_3 \longrightarrow KCl + \frac{3}{2}O_2$

(*a*) $\frac{1}{2}$mol (*b*) $\frac{1}{3}$mol

(*c*) $\frac{1}{4}$mol (*d*) $\frac{2}{3}$mol

LIMITING REAGENT (L.R.) CONCEPT

Limiting Reagent (L.R.): The reactant which is completely consumed in a reaction is called as L.R.

Ex.

	A	+	2B	⟶	C	+	2D
Given moles	3		9				
	3 – 3		9 – 6				
	0		3		3		6

A is L.R.

Formula for checking L.R. =

$$\frac{\text{Given value (may moles, volume, or molecules)}}{\text{Stoichiometry Coefficient}}$$

Least value indicate the L.R.

Ex.

A	B
$\frac{3}{1} = 3$	$\frac{9}{2} = 4.5$

$3 < 4.5$ So A is L.R.

Identification: More than 1 initial quantites of reactants are given

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Example 22: $\underset{10\text{ mole}}{A} + \underset{10\text{ mole}}{5B} \longrightarrow C + 3D$, in this reaction which is a L.R.?

Sol.

For A	For B
$\frac{10}{1} = 10$	$\frac{10}{5} = 2$

$2 < 10$ So, B is L.R.

Example 23: $\underset{4g}{H_2(g)} + \frac{1}{2}\underset{32g}{O_2(g)} \longrightarrow H_2O(g)$; in the above reaction, what is the volume of water vapour at STP?

Sol. $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(g)$

4g 32g

For H_2	For O_2
$n = \frac{4}{2} = 2$mol	$n = \frac{32}{32} = 1$mol

For L.R. $\frac{2}{1} = 2$ $\qquad \frac{1}{\frac{1}{2}} = 2$ mol

Moles of $H_2O_{(g)}$ = 2 mol = $\frac{V}{22.4}$

Both H_2 & O_2 are L.R.

Volume of $H_2O_{(g)}$ at STP = $22.4 \times 2 = 44.8$ litre

Example 24: At NTP, In a container 100 mL N_2 and 100 mL of H_2 are mixed together. Then find out the produced volume of NH_3.

Sol. Balanced equation will be $N_2 + 3H_2 \longrightarrow 2NH_3$.

Given 100mL 100mL

For determination of Limiting reagent. Now divided the given quantities by stoichiometry coefficients

$\frac{100}{1} = 100$ $\qquad \frac{100}{3} = 33.3$ (Limiting reagent)

In this reaction H_2 is limiting reagent so reaction will proceed according to H_2.

According to stoichiometry from 3 mL of H_2 produced volume of NH_3 = 2 mL

That is from 100 mL of H_2 produced volume of $NH_3 = \frac{2}{3} \times 100 = 66.6$ mL

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Concept Application

20. 4 mole of $MgCO_3$ is reacted with 6 moles of HCl solution. Find the volume of CO_2 gas (in litres) produced at STP, the reaction is:

$MgCO_3 + 2HCl \rightarrow MgCl_2 + CO_2 + H_2O$.

(*a*) 11.2 (*b*) 22.4

(*c*) 67.2 (*d*) 44.8

21. For a reaction, $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$; identify dihydrogen ($H_2$) as a limiting reagent in the following reaction mixtures.

(*a*) 14g of N_2 + 4g of H_2 (*b*) 35g of N_2 + 8g of H_2

(*c*) 28g of N_2 + 6g of H_2 (*d*) 56g of N_2 + 10g of H_2

PERCENTAGE YIELD

The percentage yield of product

$$= \frac{\text{Actual yield}}{\text{The theoretical maximum yield}} \times 100$$

- The actual amount of any limiting reagent consumed in such incomplete reactions is given by (% yield × given moles of limiting reagent) [For reversible reactions]
- For irreversible reaction with % yield less than 100, the reactants is converted to product. (desired and waste)

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Example 25: 7.5 mL of a hydrocarbon gas was exploded with excess of oxygen. On cooling, it was found to have undergone a contraction of 15 mL. If the vapour density of the hydrocarbon is 14, determine its molecular formula. (C = 12, H = 1)

Sol. $\underset{7.5\text{ ml}}{C_xH_y} + (x + \frac{y}{4})O_2 \rightarrow X\,CO_2 + \frac{y}{2}H_2O$

On cooling the volume contraction = 15 ml

i.e. the volume of H_2O (g) = 15 ml

V.D. of hydrocarbon = 14

Molecular wt. of C_xH_y = 28

$12x + y = 28$ ...(1)

From reaction

$7.5\ \frac{y}{2} = 15 \Rightarrow y = 4$

$12x + 4 = 28$

$12x = 24$

$x = 2$

Hence hydrocarbon is C_2H_4.

Example 26: Number of moles of NH_3 produced if 140 gm of N_2 reacts with 40 gm of hydrogen. (Given % yield of reaction is 50%)

(*a*) 12 (*b*) 10

(*c*) 5 (*d*) 6

Sol. (*c*) $\underset{\substack{140\text{ gm}\\ \text{or } 5\text{ mol}}}{N_2} + \underset{\substack{40\text{ gm}\\ 20\text{ mol}}}{3H_2} \xrightarrow{50\%} 2NH_3$

$\therefore$ Number of moles of NH_3 produced = $5 \times 2 \times 0.5 \times$ = 5 mole

Concept Application

22. The percent yield for the following reaction carried out in carbon tetrachloride (CCl_4) solution is 80%.

$Br_2 + Cl_2 \longrightarrow 2BrCl$

How many moles of BrCl is formed from the reaction of 0.025 mol Br_2 and 0.025 mol Cl_2?

(*a*) 0.04 (*b*) 0.08

(*c*) 0.02 (*d*) 0.01

23. If 240 g of carbon is taken in a container to convert it completely to CO_2 but in industry it has been found that 280 g of CO was also formed along with CO_2. Find the mole percentage yield of CO_2. The reactions occuring are:

$$C + O_2 \longrightarrow CO_2; C + \frac{1}{2}O_2 \longrightarrow CO$$

(*a*) 25 (*b*) 50

(*c*) 75 (*d*) 35

CONCENTRATION OF SOLUTION

Concentration of solution can be expressed in any of the following ways.

(i) **% by wt.**: Amount of solute dissolved in 100 gm of solution.

4.9% H_2SO_4 by wt.

$\Rightarrow$ 100 gm of solution contains 4.9 gm of H_2SO_4.

(ii) **% by volume**: Volume of solute dissolved in 100 ml of solution.

x% H_2SO_4 by volume

$\Rightarrow$ 100 ml of solution contains x ml H_2SO_4.

(iii) **% wt. by volume**: wt. of solute present in 100 ml of solution.

(iv) **% volume by wt.**: Volume of solute present in 100 gm of solution.

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CONCENTRATION TERMS

Molarity (M): No. of moles of solute present in 1000 ml of solution.

$$\text{Molarity (M)} = \frac{\text{Moles of solute}}{\text{Volume of solution (L)}}$$

Molality (m): No. of moles of solute present in 1000 gm of solvent.

$$m = \frac{\text{Moles of solute}}{\text{wt. of solvent in kg}}$$

Normality (N): No. of gm equivalents of solute present in 1000 ml of solution.

$$N = \frac{\text{gm equivalents of solute}}{\text{Volume of solution (L)}}$$

Note: x ml $= \frac{x}{1000}$ L and x g $= \frac{x}{1000}$ kg

Mole fraction: The mole fraction of a particular component in a solution is defined as the number of moles of that component per mole of solution.

If a solution has n_A mole of A & n_B mole of B.

$$\text{mole fraction of A } (X_A) = \frac{n_A}{n_A + n_B}$$

$$\text{mole fraction of B } (X_B) = \frac{n_B}{n_A + n_B}$$

$X_A + X_B = 1$

Parts per million (ppm):

$$= \frac{\text{Mass of solute}}{\text{Mass of solvent}} \times 10^6 \cong \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 10^6$$

Relationship between Molarity, Molality & Density of Solution

Let the molarity of solution be 'M', molality be 'm' and the density of solution be d gm/mℓ.

Molarity implies that there are M moles of solute in 1000ml of solution, wt. of solution = density × volume

= 1000 d gm, wt. of solute = MM_1

where M_1 is the molecular wt. of solute

wt. of solvent = $(1000d - MM_1)$ gm

$(1000d - MM_1)$ gm of solvent contains M moles of solute.

1000 gm of solvent have

$$= \frac{M}{1000d - MM_1} \times 1000 \text{ mole} = \text{Molality}$$

no. of moles of solute present in 1000 gm of solvent

$$= \frac{1000 \times M}{1000d - MM_1} = \text{Molality}$$

on simplyfying $d = M\left[\frac{1}{m} + \frac{M_1}{1000}\right]$

Relationship between molality & mole fraction: Consider a binary solution consisting of two components A (Solute) and B (Solvent).

Let x_A & x_B are the mole fraction of A & B respectively.

$$x_A = \frac{n_A}{n_A + n_B}, \quad x_B = \frac{n_B}{n_A + n_B}$$

If molality of solution be m then:

$$m = \frac{n_A}{\text{mass of solvent}} \times 1000 = \frac{n_A}{n_B \times M_B} \times 1000$$

where M_B is the molecular wt. of the solvent B.

$$m = \frac{x_A}{x_B} \times \frac{1000}{M_B}$$

$$\text{molality} = \frac{\text{mole fraction of A}}{\text{mole fraction of B}} \times \frac{1000}{M_B}$$

$$m = \frac{\text{mole fraction of solute}}{\text{mole fraction of solvent}} \times \frac{1000}{\text{molecular wt. of solvent}}$$

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Example 27: 0.2 mole of HCl and 0.1 mole of barium chloride were dissolved in 0.1 L of water to produce solution. The molarity of the Cl^- ions is:

(*a*) 0.06 M (*b*) 0.09 M

(*c*) 0.12 M (*d*) 0.80 M

Sol. (*d*) $HCl \longrightarrow Cl^-$

0.2 mole

$BaCl_2 \longrightarrow 2\ Cl^-$

$2 \times 0.1 = 0.2$

Total moles of $Cl^- = 0.4$

$$M = \frac{w \times 1000}{m \times v}$$

$$\text{Molarity} = \frac{0.4 \times 1000}{500} = 0.8 \quad \because \frac{w}{m} = 0.4$$

Example 28: 149 g of potassium chloride (KCl) is dissolved in 10 L of an aqueous solution. Determine the molarity of the solution. (K = 39, Cl = 35.5)

Sol. Molecular mass of KCl = 39 + 35.5 = 74.5 g

$\therefore$ Moles of KCl $\frac{149\,g}{74.5\,g} = 2$

$\therefore$ Molarity of the solution $\frac{2}{10} = 0.2M$

Example 29: 225 g of an aqueous solution contains 5 g of urea. What is the concentration of the solution in terms of molality? (Mol. wt. of urea = 60)

Sol. Mass of urea = 5 g

Molecular mass of urea = 60

Number of moles of urea $= \frac{5}{60} = 0.083$

Mass of solvent = (255 – 5) = 250 g

$\therefore$ Molality of the solution

$$= \frac{\text{Number of moles of solute}}{\text{Mass of solvent in gram}} \times 1000$$

$$= \frac{0.083}{250} \times 1000 = 0.332 \text{ M}$$

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Example 30: 117 g NaCl is dissolved in 500 ml aqueous solution. Find the molarity of the solution.

Sol. Molarity $= \frac{117/58.5}{500/1000} = 4M.$

Example 31: Calculate the resultant molarity of following:

(*a*) 200 ml 1M HCl + 300 ml water

(*b*) 1500 ml 1M HCl + 18.25 g HCl

(*c*) 200 ml 1M HCl + 100 ml 0.5 M H_2SO_4

(*d*) 200 ml 1M HCl + 100 ml 0.5 M HCl

Sol. (*a*) 0.4M (*b*) 1.33M

(*c*) 1 M (*d*) 0.83 M

(*a*) Final molarity $= \frac{200\times1+0}{200+300} = 0.4M.$

(*b*) Final molarity $= \frac{1500\times1+\frac{18.25\times1000}{36.5}}{1500} = 133\,M$

(*c*) Final molarity of $H^+ = \frac{200\times1+100\times0.5\times2}{200+100} = 1M$

(*d*) Final molarity $= \frac{200\times1+100\times0.5}{200+100} = 0.83M.$

Example 32: 518 g of an aqueous solution contains 18 g of glucose (mol.wt. = 180). What is the molality of the solution?

Sol. wt. of solvent = 518 – 18 = 500 g.

$\Rightarrow$ So molarity $= \frac{18/180}{500/1000} = 0.2\ M$

Example 33: 0.25 g of a substance is dissolved in 6.25 g of a solvent. Calculate the percentage amount of the substance in the solution.

Sol. Wt. of solution = 0.25 + 6.25 = 6.50. 50

So % (w/w) $= \frac{0.25}{6.50}\times100 = 3.8\%$

Example 34: An aqueous solution is 1.33 molal in methanol. Determine the mole fraction of methanol & H_2O.

Sol. Molality

$$= \frac{\text{Mole fraction of solute}}{\text{Mole fraction of solvent}\times\text{mol.wt. of solvent}}\times1000$$

$$1.33 = \frac{x_A}{x_B\times M_B}\times1000,\ \frac{1.33\times18}{1000} = \frac{x_A}{x_B},\ \frac{23.94}{1000} = \frac{x_A}{x_B}$$

$\Rightarrow\ x_A = 0.02394\ x_B,\ x_A + x_B = 1 \Rightarrow 1.02394\ x_B = 1$

$x_B = \frac{1}{1.02394} = 0.98,\ x_A = 0.02$

2nd Method: Let wt. of solvent = 1000 gm,

molality = 1.33 = moles of solute

mole fraction of solute

$$= \frac{\text{moles of solute}}{\text{moles of solute} + \text{moles of solvent}},$$

$$= \frac{m}{m+\frac{1000}{18}} = \frac{1.33}{1.33+1000/18}$$

Mole fraction of solute = 0.02

Mole fraction of solvent = 1 – 0.02 = 0.98

Example 35: 225 gm of an aqueous solution contains 5 gm of urea. What is the concentration of the solution in terms of molality? (Mol. wt. of urea = 60)

Sol. Mass of urea = 5 gm

Molecular mass of urea = 60

Number of moles of urea $= \frac{5}{60} = 0.083$

Mass of solvent = (255 – 5) = 250 gm

$\therefore$ Molality of the solution =

$$\frac{\text{Number of moles of solute}}{\text{Mass of solvent in gram}}\times1000$$

$$= \frac{0.083}{250}\times1000 = 0.332\ M$$

Example 36: 0.5 g of a substance is dissolved in 25 g of a solvent. Calculate the percentage amount of the substance in the solution.

Sol. Mass of substance = 0.5 g

Mass of solvent = 25 g

$\therefore$ Percentage of the substance (%w/w)

$= \frac{0.5}{0.5+25}\times100 = 1.96$

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Concept Application

24. If 500 ml of 1 M solution of glucose is mixed with 500 ml of 1 M solution of glucose, final molarity of solution will be:

(*a*) 1 M (*b*) 0.5 M

(*c*) 2 M (*d*) 1.5 M

25. The volume of water that must be added to a mixture of 250 ml of 0.6 M HCl and 750 ml of 0.2 M HCl to obtain 0.25 M solution of HCl is:

(*a*) 750 ml (*b*) 100 ml

(*c*) 200 ml (*d*) 300 ml

26. The molarity of Cl^- in an aqueous solution which was (w/v) 2% NaCl, 4% $CaCl_2$ and 6% NH_4Cl will be:

(*a*) 0.342 (*b*) 0.721

(*c*) 1.12 (*d*) 2.18

27. 2M of 100 ml Na_2SO_4 is mixed with 3M of 100 ml NaCl solution and 1 M of 200 ml $CaCl_2$ solution. Then the ratio of the concentration of cation and anion.

(*a*) 1/2 (*b*) 2

(*c*) 1.5 (*d*) 1

28. Equal moles of H_2O and NaCl are present in a solution. Hence, molality of NaCl solution is :

(*a*) 0.55 (*b*) 55.5

(*c*) 1.00 (*d*) 0.18

29. Mole fraction of A in H_2O is 0.2. The molality of A in H_2O is:

(*a*) 13.9 (*b*) 15.5

(*c*) 14.5 (*d*) 16.8

30. What is the molarity of H_2SO_4 solution that has a density of 1.84 g/cc and contains 98% by mass of H_2SO_4? (Given atomic mass of S = 32)

(*a*) 4.18 M (*b*) 8.14M

(*c*) 18.4 M (*d*) 18 M

31. The molarity of the solution containing 2.8% (mass/volume) solution of KOH is : (Given atomic mass of K = 39) is :

(*a*) 0.1 M (*b*) 0.5 M

(*c*) 0.2 M (*d*) 1 M

Strength of Oleum

Oleum is SO_3 dissolved in 100% H_2SO_4. Sometimes, oleum is reported as more than 100% by weight, say y% (where y > 100). This means that (y - 100) grams of water, when added to 100 g of given oleum sample, will combine with all the free SO_3 in the oleum to give 100% sulphuric acid. Hence, weight % of free SO_3 in oleum = 80 (y–100)/18.

Example: If in a sample of oleum, mole fraction of SO_3 is 0.5. Label the oleum sample.

Sol. Total moles = 1

Moles of SO_3 = mole of H_2SO_4 = 0.5

Mass of $SO_3 + H_2SO_4 = 40 + 49 = 89$ gm

$SO_3 + H_2O \longrightarrow H_2SO_4$

0.5 0.5

Mass of H_2O require = $0.5 \times 18 = 9$ gm

89 gm require 9 gm H_2O

100 gm require = $\frac{9}{89} \times 100 = 10.11$ gm

% Labelling = (100 + 10.11) = 110.11%

Volume Strength of H_2O_2

Strength of H_2O_2 is represented as 10 V, 20 V, 30 V etc. 20V H_2O_2 means one litre of this sample of H_2O_2 decomposition gives 20L of O_2 gas at STP. Decomposition of H_2O_2 is given as:

$$H_2O_2 \longrightarrow H_2O + \frac{1}{2}O_2$$

1 mole $\frac{1}{2} \times 22.4$L O_2 at STP

= 34g =11.2 L O_2 at STP

$$\text{Molarity of } H_2O_2 \text{ (M)} = \frac{\text{Volume strength of } H_2O_2}{11.2}$$

Strength (in g/L): Denoted by S

Strength = Molarity × Mol. wt = Molarity × 34

Quantitative Analysis

Estimation of Nitrogen

(a) Dumas method

(b) Kjeldahl's method

Dumas Method

- In this method nitrogen present in the organic compound is converted in to N.(molecular nitrogen)
- A weighed amount of organic compound is heated with cupric oxide in an atmosphere of carbondioxide.
- Carbon and hydrogen present in the compound are oxidised to CO_2 and H_2O, while N_2 is set free.
- Some oxides of nitrogen formed are reduced to free nitrogen by passing over heated copper gauze.

$$C_xH_yN_z + \left(2x + \frac{y}{2}\right)CuO \rightarrow xCO_2 + \frac{y}{2}(H_2O) + \frac{Z}{2}N_2 + \left(2x + \frac{y}{2}\right)Cu$$

- Oxides of nitrogen + Cu → N_2 + CuO.
- The mixture of gases produced is collected over caustic potash solution (KOH solution) which absorbs CO_2.
- Nitrogen is collected in the upper part of nitro meter.

$$\%N = \frac{28}{22400} \times \frac{\text{Volume of nitrogen in ml at STP}}{\text{Weight of orgnaic compound}} \times 100$$

Kjeldahl's Method

- In this method nitrogen present in the organic compound is converted in to ammonia. (NH_3)
- A known mass of organic compound containing nitrogen is heated with concentrated sulphuric acid in presence of K_2SO_4 and $CuSO_4$ then nitrogen present in the compound is converted in to ammonium sulphate.
- Organic compound + $H_2SO_4 \rightarrow (NH_4)_2SO_4$

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- The resulting solution is distilled with excess of sodium hydroxide.

  $(NH_4)_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2NH_3 + 2H_2O$

- Ammonia evolved is absorbed in a known but excess volume of standard HCl or H_2SO_4 solution. The acid left unreacted is estimated by titration against standard solution of sodium hydroxide.

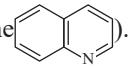
  %Nitrogen

$$= \frac{14}{1000} \times \frac{V \times N}{\text{wt. of organic compound}} \times 100$$

$$\%N = \frac{1.4 \times V \times N}{\text{Wt. of organic compound}}$$

  Where V = Volume of acid in ml neutralised by ammonia

  N = Normality of acid.

- This method is simpler and more convenient.
- It is mainly used to find percentage of nitrogen present in food stuffs, soils, fertilizersand various agricultural products.
- This method is not applicable to compounds containing nitro ($-NO_2$), Nitroso (NO), azo group ($-N=N-$), azoxy compounds ($-\overset{\overset{O}{\uparrow}}{N}=N-$) and nitrogen present in the ring (pyridin  , quinoline ). Because nitrogen present in these compounds is not quantitatively converted into ammonium sulphate.
- K_2SO_4 increases boiling point of H_2SO_4 and $CuSO_4$ acts as catalyst.

Chemical Methods Used to Find Molecular Mass

Silver Salt Method for Acids:

Organic acid form insoluble silver salts, which on heating undergoes decomposition to leave a residue of metallic silver.

$$\underset{\text{Silver salt}}{RCOOAg} \xrightarrow{\Delta} \underset{\text{Silver (residue)}}{Ag}$$

$$\frac{\text{Equivalent wt. of silver salt}}{\text{Equivalent wt. of silver}} = \frac{\text{Mass of silver salt}}{\text{Mass of silver}}$$

$$\frac{E + 108 - 1}{108} = \frac{\text{Mass of silver salt}}{\text{Mass of silver}}$$

$$E = \left[\frac{\text{Mass of silver salt}}{\text{Mass of silver}} \times 108\right] - 107$$

Molecular weight of acid = Equivalent weight of acid (E) × basicity

Example: 0.41g of the silver salt of a dibasic organic acid left a residue to 0.216g of silver on ignition.

Calculate the molecular mass of the acid.

Sol. Mass of of the silver salt taken (W) = 0.41 g , Mass of Ag formed = 0.216g

$$H_2X \longrightarrow \underset{w = 0.41\ g}{Ag_2X} \longrightarrow \underset{x = 0.216\ g}{2Ag}$$

Now molar mass acid = $n\left(\frac{108W}{x} - 107\right) gmol^{-1}$

$$= 2\left(\frac{108 \times 0.41}{0.216} - 107\right) gmol^{-1} = 196 gmol^{-1}$$

Molar mass = 196 g/mol

Platinic Chloride Method for Bases

- Organic bases combines with chloroplatinic acid (H_2PtCl_6) to insoluble platinichloride, which on ignition gives metallic platinum.
- If 'B' is mono acidic base then formula of salt will be $B_2H_2PtCl_6$.

$$B_2H_2PtCl_6 \xrightarrow{\Delta} Pt$$

$$\frac{\text{Molecular mass of platinum salt}}{\text{Atomic mass of platinum}} = \frac{\text{Mass of platinum salt}}{\text{Mass of platinum}}$$

If E in equivalent weight of base then:

$$\frac{2E + 410}{195} = \frac{\text{Mass of platinum salt}}{\text{Mass of platinum}}$$

$$E = \frac{1}{2}\left[\frac{\text{Mass of platinum salt}}{\text{Mass of platinum}} \times 195 - 410\right]$$

Molecular mass of base = Equivalent mass of base x acidity

Example: 0.30 gm chloroplatinate salt of a diacidic organic base exactly produce 0.09 gm platinum, on strong ignition. The molecular mass of organic base is (Pt = 195):

Sol. Molar mass of base is

$$= \frac{n}{2}\left(\frac{w \times 195}{x} - 410\right)$$

$$= \frac{2}{2}\left(\frac{0.3 \times 195}{0.09} - 410\right) = 240\,gm/mol$$

Determination of Molecular Mass by Victor Meyer's Method

- A known mass of the volatile substance is vapourised in victor meyer's method.
- The vapours obtained displaces an equal volume of air into graduated tube.

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❖ The volume of air displaced is calculated at STP condition by using $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

Mol. mass of volatile substance

$$= \frac{\text{Mass of substance taken}}{\text{Volume of air displaced in ml at STP}} \times 22400$$

Example: 0.15 g of substance displaced 58.9 cm^3 of air at 300 K and 746 mm pressure. Calculate the molecular mass. (Aq. tension at 300K = 26.7 mm).

Sol. Mass of substance = 0.15 g

Volume of air displaced (V) = 58.9 cm^3

Temperature (T) = 300 K

Pressure (P) = 746 – 26.7 = 719.3 mm

$$\text{Molecular mass} = \frac{719.3}{760} \times \frac{58.9}{1000} = \frac{0.15}{M} \times .0821 \times 300$$

∴ Molecular mass = 66.24 g/mol.

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Example 37: In Kjeldahl's estimation of nitrogen, the ammonia evoked from 0.5g of an organic compound neutralised 10ml of 1M H_2SO_4. Calculate the percentage of nitrogen in the compound.

Sol. 10 ml of 1M H_2SO_4 = 20 ml of M NH_3.

1000 ml of 1M ammonia contains 14g nitrogen.

20 ml of 1M ammonia contains $\frac{14 \times 20}{1000}$ g N_2.

$$\% \text{ of nitrogen} = \frac{14 \times 20 \times 100}{1000 \times 0.5} = 56.0$$

Example 38: In Kjeldahl's estimation of nitrogen, the ammonia obtained from 0.5g of an organic substance was passed into 100 cm^3 of $\frac{M}{10}H_2SO_4$. The excess of acid required 154 cm^3 of $\frac{M}{10}$NaOH for neutralisation Calculate the percentage of nitrogen in the compound.

Sol. meq. of H_2SO_4 taken = molarity × basicity volume

$$(\text{ml}) = \frac{1}{10} \times 2 \times 100 = 20$$

$$\text{m.eq. of NaOH} = \frac{1}{10} \times 1 \times 154 = 15.4$$

m.eq. of H_2SO_4 unused = 20 – 15.4 = 4.6 % of nitrogen

$$= \frac{1.4 \times \text{m.eq. of } H_2SO_4 \text{ unused}}{\text{wt. of organic compound}} = \frac{1.4 \times 4.6}{0.5} = 12.88$$

Concept Application

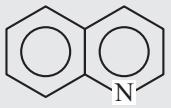
32. In Duma's method nitrogen in organic compound is estimated in the form of:

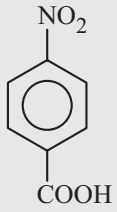
(*a*) N_2 (*b*) NO

(*c*) NH_3 (*d*) N_2O_5

33. Kjeldahl's method cannot be used for the estimation of nitrogen in:

(*a*) C_6H_5–N=N–C_6H_5

(*b*) 

(*c*) 

(*d*) N_2O_5

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Acid-Base Titration

An acid-base titration is a method to determine the strength of an acid or base by using a standard solution of an acid or base using an indicator by titrating against unknown solution*.

At equivalence point number of H^+ of an acid is equal to number of OH^- of base.

$$M_{H^+} \times V_{H^+} \times (n_f)_{acid} = M_{OH^-} \times V_{OH^-} (n_f)_{base}$$

equivalence of an acid = equivalence of base

Here: M_{H^+} = Molarity of an acid V_{H^+} = Volume of an acid

M_{OH^-} = Molarity of base V_{OH^-} = volume of base

n_f = number of H^+/OH^- ions.

NaOH → n_f = 1 HCl → n_f = 1

$Ca(OH)_2$ → n_f = 2 H_2SO_4 → n_f = 2

$Al(OH)_3$ → n_f = 3 H_3PO_4 → n_f = 3

* Unknown solution can be both an acid or base.

Indicators

Generally indicators for an acid-base titration are based on the organic moiety except $KMnO_4$ (self indicator), Indicators are change the colour of solution at a certain pH called "end point" or they help us indicating completion the reaction by showing the colour change of the medium.

The resonance of these indicators responsible for the colour change.

Most common example of indicator are listed below:

Indicator	Notation	pH range	Colour change
Phenolphthalein	HPh	8.2-10	Colourless to pink
Methyl orange	MeOH	4-6	Orange to yellow
Alizarin Yellow	DHQ	10-12	Yellow to red



Train Your Brain

Example 39: What will be the pH of the equivalence point of the reaction when 60mL of 0.25M HI is titrated with 0.5M KOH solution?

Sol. Both HI and KOH are strong acid and strong base respectively. Thus strong acid and strong base neutralize completely at equivalence point and solution becomes neutral, the pH must be 7.

Example 40: It takes 90mL of a 0.5M NaOH solution to neutralize 220mL of an HI solution. What will be the concentration of the HI solution?

Sol. The valance chemical reaction between HI and NaOH is as follow:

$HI(aq) + NaOH(aq) \rightarrow NaI(s) + H_2O(l)$

Molarity equation:

$M_A V_A (n_f)_A = M_B V_B (n_f)_B$.

M_A and V_A are the molarity and volume of an acid.

M_B and V_B are the molarity and volume of base.

Hence, $M_A = \frac{M_B V_B (n_f)_B}{V_A (n_f)_A} = \frac{0.5M \times 90mL \times 1}{220mL \times 1} = 0.20M$

The concentration of the HI solution is 0.20M.



Concept Application

34. 1.25g of solid dibasic acid is completely neutralized by 25mL of 0.25M $Ba(OH)_2$ solution, molecular mass of an acid is:

(a) 180 (b) 200

(c) 150 (d) 250

35. Find out the volume required to n eutralized of 200mL of 2M NaOH with 5M HCl.

(*a*) 70 (*b*) 80

(*c*) 90 (*d*) 100

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Short Notes

Some Useful Conversion Factors

1 Å = 10^{-10}m, 1nm = 10^{-9} m

1 pm = 10^{-12}m

1 litre = 10^{-3} m^3 = 1 dm^3

1 atm = 760 mm or torr

= 101325 Pa or Nm^{-2}

1 bar = 10^5 Nm^{-2} = 10^5 Pa

1 calorie = 4.184 J

1 electron volt(eV) = 1.6022 ×10^{-19} J

(1 J = 10^7 ergs)

(1 cal > 1 J > 1 erg > 1 eV)

Atomic Mass or Molecular Mass

Mass of one atom or molecule in a.m.u.

C → 12 amu

NH_3 → 17 amu

Actual Mass

Mass of one atom or molecule in grams:

C → 12 ×1.6 × 10^{-24} g

CH_4 → 16 ×1.6 × 10^{-24} g

Relative Atomic Mass or Relative Molecular Mass

Mass of one atom or molecule w.r.t. 1/12th or ^{12}C atom:

C → 12

CH_4 → 16

It is unitless.

Gram Atomic Mass or Gram Molecular Mass

Mass of one mole of atom or molecule:

C → 12 g

CO_2 → 44 g

It is also called molar mass.

Definition of Mole

One mole is a collection of that many entities as there are number of atoms exactly in 12 gm of C-12 isotope.

The number of atoms present in exactly 12 gm of C-12 isotope is called Avogadro's number [N_A = 6.022 × 10^{23}]

1u = 1amu = (1/12)th of mass of 1 atom of C^{12} = $\frac{1g}{N_A}$

= 1.66 ×10^{-24} g

For Elements

- 1 g atom = 1 mole of atoms = N_A atoms
- g atomic mass (GAM) = mass of N_A atoms in g
- Mole of atoms = $\frac{\text{Mass } (g)}{\text{GAM or Molar mass}}$

For Molecule

- 1g molecule = 1 mole of molecule = N_A molecule
- g molecular mass (GMM) = mass of N_A molecule in g.
- Mole of molecule = $\frac{\text{Mass } (g)}{\text{GMM or Molar mass}}$

1 Mole of Substance

- Contains 6.022 × 10^{23} particles.
- Weighs as much as molecular mass/ atomic mass/ionic mass in grams.
- If it is a gas, one mole occupies a volume of 22.4 L at 1 atm & 273 K or 22.7 L at STP.

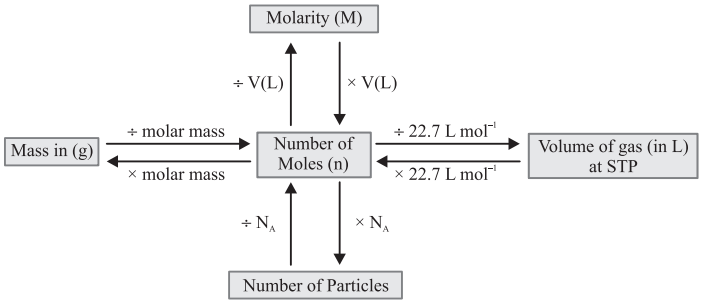
For Ionic Compounds

- 1 g formula unit = 1 mole of formula unit = N_A formula unit.
- g formula mass (GFM) = mass of N_A formula unit in g.
- Mole of formula unit = $\frac{\text{Mass } (g)}{\text{GMM or Molar mass}}$

Vapour density

Ratio of density of vapour to the density of hydrogen at similar pressure and temperature.

$$\text{Vapour density} = \frac{\text{Molar mass}}{2}$$



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Stoichiometry Based Concept

$$aA + bB \rightarrow cC + dD$$

- a,b,c,d, represents the ratios of moles, volumes [for gaseous] molecules in which the reactants react or products formed.
- a,b,c,d does not represent the ratio of masses.
- The stoichiometic amount of components may be related as:

$$\frac{\text{Moles of A reacted}}{a} = \frac{\text{Moles of B reacted}}{b} = \frac{\text{Moles of C reacted}}{c} = \frac{\text{Moles of D reacted}}{d}$$

Concept of Limiting Reagent

If data of more than one reactant is given then first convert all the data into moles then divide the moles of reactants with their respective stoichiometric coefficient. The reactant having minimum ratio will be L.R. then find the moles of product formed or excess reagent left by comparing it with L.R. through stoichiometric concept.

Percentage Purity

The percentage of a specified compound or element in an impure sample may be given as:

$$\%\text{purity} = \frac{\text{Actual mass of compound}}{\text{Total mass of sample}} \times 100$$

If impurity is unknown, it is always considered as inert (unreactive) material.

Empirical and Molecular Formula

- **Empirical formula:** Formula depicting constituent atoms in their simplest ratio.
- **Molecular formula:** Formula depicting actual number of atoms in one molecule of the compound.
- The molecular formula is generally an integral multiple of the empirical formula.

  i.e. molecular formula = empirical formula × n

  $$\text{where } n = \frac{\text{molecular formula mass}}{\text{empirical formula mass}}$$

Concentration Terms

Concentration Type	Mathematical Formula	Concept
Percentage by mass	$\%\left(\frac{w}{w}\right) = \frac{\text{Mass of solute} \times 100}{\text{Mass of solution}}$	Mass of solute (in gm) present in 100 gm of solution.
Volume percentage	$\%\left(\frac{v}{v}\right) = \frac{\text{Volume of solute} \times 100}{\text{Volume of solution}}$	Volume of solute (in cm^3) present in 100 cm^3 of solution.
Mass-volume percentage	$\%\left(\frac{w}{v}\right) = \frac{\text{Mass of solute} \times 100}{\text{Volume of solution}}$	Mass of solute (in gm) present in 100 cm^3 of solution.
Parts per million	$\text{ppm} = \frac{\text{Mass of solute} \times 10^6}{\text{Mass of solution}}$	Parts by mass of solute per million parts by mass of the solution.
Mole fraction	$X_A = \frac{\text{Mole of A}}{\text{Mole of A} + \text{Mole of B} + \text{Mole of C} + \ldots}$ $X_B = \frac{\text{Mole of B}}{\text{Mole of A} + \text{Mole of B} + \text{Mole of C} + \ldots}$	Ratio of number of moles of one component to the total number of moles.
Molarity	$M = \frac{\text{Mole of solute}}{\text{Volume of solution (in L)}}$	Moles of solute in one liter of solution.
Molality	$m = \frac{\text{Mole of solute}}{\text{Mass of solvent (Kg)}}$	Moles of solute in one kg of solvent.

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Mixing of Solutions

It is based on law of conservation of moles.

(i) Two solutions having same solute:

$$\text{Final molarity} = \frac{\text{Total moles}}{\text{Total volume}} = \frac{M_1V_1 + M_2V_2}{V_1 + V_2}$$

(ii) Dilution Effect: Final molarity, $M_2 = \frac{M_1V_1}{V_1 + V_2}$

Solved Examples

1. Find the relative atomic mass, atomic mass of the following elements.

(*i*) Na (*ii*) F (*iii*) H (*iv*) Ca (*v*) Ag

Sol. (*i*) 23, 23 amu (*ii*) 19, 19 amu

(*iii*) 1, 1.008 amu, (*iv*) 40, 40 amu,

(*v*) 108, 108 amu.

2. A sample of (C_2H_6) ethane has the same mass as 10^7 molecules of methane. How many C_2H_6 molecules does the sample contain?

Sol. Moles of $CH_4 = \frac{10^7}{N_A}$

Mass of $CH_4 = \frac{10^7}{N_A} \times 16$ = mass of C_2H_6

So Moles of $C_2H_6 = \frac{10^7 \times 16}{N_A \times 30}$

So no. of molecules of $C_2H_6 = \frac{10^7 \times 16}{N_A \times 30} \times N_A = 5.34 \times 10^6$.

3. From 160 g of SO_2 (g) sample, 1.2046×10^{24} molecules of SO_2 are removed then find out the volume of left over SO_2 (g) at NTP.

Sol. Given moles $= \frac{160}{64} = 2.5$.

Removed moles $= \frac{1.2046 \times 10^{24}}{6.023 \times 10^{23}} = 2$.

So left moles = 0.5.

Volume left at STP = $0.5 \times 22.4 = 11.2$ L.

4. 14 g of Nitrogen gas and 22 g of CO_2 gas are mixed together. Find the volume of gaseous mixture at NTP.

Sol. Moles of $N_2 = \frac{14}{28} = 0.5$.

Moles of $CO_2 = \frac{22}{44} = 0.5$.

So total moles = 0.5 + 0.5 = 1.

So vol. at STP = $1 \times 22.4 = 22.4$ L.

5. Show that in the reaction N_2 (g) + $3H_2$(g) $\rightarrow$ $2NH_3$ (g), mass is conserved.

Sol.

	N_2 (g) +	$3H_2$(g)	$\rightarrow$ $2NH_3$ (g)
moles before reaction	1	3	0
moles after reaction	0	0	2

Mass before reaction = mass of 1 mole N_2(g) + mass of 3 mole H_2(g)

= $14 \times 2 + 3 \times 2 = 34$ g

Mass after reaction = mass of 2 mole NH_3

= 2 x 17 = 34 g.

6. Calculate the mass in gm of $2N_A$ molecules of CO_2:

(*a*) 22 gm (*b*) 44 gm

(*c*) 88 gm (*d*) None of these

Sol. (*c*)

$\because$ N_A molecules of CO_2 has molecular mass = 44.

$\therefore$ $2N_A$ molecules of CO_2 has molecular mass = 44×2 = 88 gm.

7. How many carbon atoms are present in 0.35 mol of $C_6H_{12}O_6$?

(*a*) 6.023×10^{23} carbon atoms.

(*b*) 1.26×10^{23} carbon atoms.

(*c*) 1.26×10^{24} carbon atoms.

(*d*) 6.023×10^{24} carbon atoms.

Sol. (*c*)

$\because$ 1 mol of $C_6H_{12}O_6$ has 6 N_A atoms of C

$\therefore$ 0.35 mol of $C_6H_{12}O_6$ has 6×0.35 N_A atoms of C

= 2.1 N_A atoms = $2.1 \times 6.023 \times 10^{23}$ = 1.26×10^{24} carbon atoms.

8. How many molecules are in 5.23 gm of glucose ($C_6H_{12}O_6$)?

(*a*) 1.65×10^{22} (*b*) 1.75×10^{22}

(*c*) 1.75×10^{21} (*d*) None of these

Sol. (*b*)

$\because$ 180 gm glucose has = N_A molecules

$\therefore$ 5.23 gm glucose has $= \frac{5.23 \times 6.023 \times 10^{23}}{180}$

= 1.75×10^{22} molecules

9. What is the weight of 3.01×10^{23} molecules of ammonia?

(*a*) 17 gm (*b*) 8.5 gm

(*c*) 34 gm (*d*) None of these

Sol. (*b*)

$\because$ 6.023×10^{23} molecules of NH_3 has weight = 17 gm

$\therefore$ 3.01×10^{23} molecules of NH_3 has weight

$= \frac{17 \times 3.01 \times 10^{23}}{6.023 \times 10^{23}} = 8.50$ gm

10. How many molecules are present in one mℓ of water vapours at NTP?

(*a*) 1.69×10^{19} (*b*) 2.69×10^{-19}

(*c*) 1.69×10^{-19} (*d*) 2.69×10^{19}

Sol. (*d*)

$\because$ 22.4 litre water vapour at NTP has 6.023×10^{23} molecules.

$\therefore$ 1×10^{-3} litre water vapours at NTP has

$\frac{6.023 \times 10^{23}}{22.4} \times 10^{-3} = 2.69 \times 10^{19}$

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11. How many years it would take to spend Avogadro's number of rupees at the rate of 1 million rupees in one second?

(*a*) 19.098×10^{19} years

(*b*) 19.098 years

(*c*) 19.098×10^{9} years

(*d*) None of these

Sol. (*c*)

$\because$ 10^6 rupees are spent in 1sec.

$\therefore$ 6.023×10^{23} rupees are spent in

$$\frac{1\times6.023\times10^{23}}{10^6} \text{ sec}$$

or $$\frac{1\times6.023\times10^{23}}{10^6\times60\times60\times24\times365} \text{ years , } = 19.098 \times 10^9 \text{ year}$$

12. An atom of an element weighs 6.644×10^{-23} g. Calculate g atoms of element in 40 kg:

(*a*) 10 gm atom (*b*) 100 gm atom

(*c*) 1000 gm atom (*d*) 10^4 gm atom

Sol. (*c*)

$\because$ Weight of 1 atom of element

$= 6.644 \times 10^{-23}$ gm

$\therefore$ Weight of 'N' atoms of element

$= 6.644 \times 10^{-23} \times 6.023 \times 10^{23} = 40$ gm

$\because$ 40 gm of element has 1 gm atom.

$\therefore$ 40 x 10^3 gm of element has $\frac{40\times10^3}{40}$ or

$= 10^3$ gm atom.

13. The density of O_2 at NTP is 1.429g / litre. Calculate the standard molar volume of gas:

(*a*) 22.4 lit. (*b*) 11.2 lit

(*c*) 33.6 lit (*d*) 5.6 lit.

Sol. (*a*)

$\because$ 1.429 gm of O_2 gas occupies volume = 1 litre.

$\therefore$ 32 gm of O_2 gas occupies = $\frac{32}{1.429}$,

$= 22.4$ litre/mol.

14. Which of the following will weigh maximum amount?

(*a*) 40 g iron

(*b*) 1.2 g atom of N

(*c*) 1×10^{23} atoms of carbon

(*d*) 1.12 litre of O_2 at NTP

Sol. (*a*)

(*a*)Mass of iron = 40 g

(*b*) Mass of 1.2 g atom of N = $14 \times 1.2 = 16.8$ gm

(*c*) Mass of 1×10^{23} atoms of C= $\frac{12\times1\times10^{23}}{6.023\times10^{23}}$ =1.99gm.

(*d*) Mass of 1.12 litre of O_2 at NTP = $\frac{32\times1.2}{22.4}$ = 1.6 g

15. Calculate the weight of lime (CaO) obtained by heating 200 kg of 95% pure lime stone ($CaCO_3$).

(*a*) 104.4 kg (*b*) 105.4 kg

(*c*) 212.8 kg (*d*) 106.4 kg

Sol. (*d*)

$\because$ 100 kg impure sample has pure $CaCO_3 = 95$ kg

$\therefore$ 200 kg impure sample has pure $CaCO_3$

$= \frac{95\times200}{100} = 190$ kg.

$CaCO_3 \rightarrow CaO + CO_2$

$\because$ 100 kg $CaCO_3$ gives CaO = 56 kg.

$\therefore$ 190 kg $CaCO_3$ gives CaO = $\frac{56\times190}{100} = 106.4$ kg.

16. Calculate the weight of one atom of Ag. (At. wt. of Ag = 108)

(*a*) 17.93×10^{-23}gm (*b*) 16.93×10^{-23} gm

(*c*) 17.93×10^{23} gm (*d*) 36×10^{-23} gm

Sol.(*a*)

$\because$ N_A atoms of Ag weighs 108 gm

$\therefore$ 1 atom of Ag weighs = $\frac{108}{N_A}$

$= \frac{108}{6.023\times10^{23}} = 17.93\times10^{-23}$ gm

17. In 5g atom of Ag (at. wt. = 108), calculate the no. of atoms of Ag ($N = N_A$).

(*a*) 1 N (*b*) 3N

(*c*) 5 N (*d*) 7 N

Sol. (*c*)

$\because$ 1 gm atom of Ag has atoms = N

$\therefore$ 5 gm atom of Ag has atoms = 5N.

18. Calculate the number of atoms of each element present in 122.5 g of $KClO_3$.

Sol. No. of moles of $KClO_3 = \frac{122.5}{122.5} = 1$.

(mol. wt. of $KClO_3 = 122.5$)

From the formula $KClO_3$, we know that 1 mole of $KClO_3$ contains 1 mole of K atoms, 1 mole of Cl atoms and 3 moles of O atoms.

$\therefore$ No. of atoms of K = $1\times6.022\times10^{23}$

No. of atoms of Cl = $1\times6.022\times10^{23}$

No. of atoms of O = $3\times6.022\times10^{23}$

19. The vapour density of a mixture consisting of NO_2 and N_2O_4 is 38.3 at 26.7°C. Calculate the number of moles of NO_2 in 100 g of the mixture.

Sol. Wt. of NO_2 = x g.

Obs. mol. wt. (wt./mole) = $\frac{\text{wt. in g}}{\text{Total moles}}$

$= \frac{100}{\left(\frac{x}{46}+\frac{100-x}{92}\right)} = 2\times38.3.$ $\left.\right]$ (0.437 mole)

20. Calculate the following for 49 gm of H_2SO_4:

(*a*) Moles (*b*) Molecules

(*c*) Total H atoms (*d*) Total O atoms

(*e*) Total electrons

Sol. Molecular wt. of $H_2SO_4 = 98$

(*a*) Moles $= \frac{\text{wt. in gm}}{\text{molecular wt.}} = \frac{49}{98} = \frac{1}{2}\text{mole}$

(*b*) Since 1 mole $= 6.023 \times 10^{23}$ molecules.

$\frac{1}{2}\text{mole} = 6.023 \times 10^{23} \times \frac{1}{2}$ molecules $= 3.011 \times 10^{23}$ molecules.

(*c*) 1 molecule of H_2SO_4 Contains 2 H atom, so

3.011×10^{23} of H_2SO_4 contain $2 \times 3.011 \times 10^{23}$ atoms $= 6.022 \times 10^{23}$ atoms

(*d*) 1 molecules of H_2SO_4 contains 4 Oxygen atoms.

$\therefore$ 3.011×10^{23} molecular of H_2SO_4 contains $= 4 \times 3.011 \times 10^{23} = 12.044 \times 10^{23}$

(*e*) 1 molecule of H_2SO_4 contains 2H atoms + 1 S atom + 4 O atom this means 1 molecule of H_2SO_4 Contains $(2 + 16 + 4 \times 8)$ e^-

So 3.011×10^{23} molecules have $3.011 \times 10^{23} \times 50$ electrons $= 1.5055 \times 10^{25}$ e^-

21. Calculate the total ions & charge present in 4.2 gm of N^{-3}.

Sol. Mole $= \frac{\text{wt.in gm}}{\text{Ionic wt.}} = \frac{4.2}{14} = 0.3$

Total no. of ions $= 0.3 \times N_A$ ions.

Total charge $= 0.3\ N_A \times 3 \times 1.6 \times 10^{-19}$

$= 0.3 \times 6.023 \times 10^{23} \times 3 \times 1.6 \times 10^{-19} = 8.67 \times 10^4$ C

22. Find the total number of iron atom present in 224 amu iron.

Sol. Since 56 amu = 1 atom

$\therefore$ 224 amu $= \frac{1}{56} \times 224 = 4$ atom.

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Exercise-1 (Topicwise)

CALCULATION OF MOLE NO. OF ENTITY, LAW OF CHEMICAL COMBINATION, MOLAR VOLUME OF IDEAL GASES AT STP, AVERAGE MOLAR MASS

1. A sample of pure carbon dioxide, irrespective of its source contains 27.27% carbon and 72.73% oxygen. The data support:
(*a*) Law of constant composition.
(*b*) Law of conservation of mass.
(*c*) Law of reciprocal proportions.
(*d*) Law of multiple proportions.

2. The law of definite proportions is not applicable to nitrogen oxide because:
(*a*) Nitrogen atomic weight is not constant.
(*b*) Nitrogen molecular weight is variable.
(*c*) Nitrogen equivalent weight is variable.
(*d*) Oxygen atomic weight is variable.

3. 1 amu is equal to:
(*a*) $\frac{1}{12}$ of C – 12 (*b*) $\frac{1}{14}$ of O – 16
(*c*) 1 g of H_2 (*d*) 1.66×10^{-23} kg

4. In chemical scale, the relative mass of the isotopic mixture of oxygen atoms (O^{16},O^{17},O,^{18}) is assumed to be equal to:
(*a*) 16.002 (*b*) 16.00
(*c*) 17.00 (*d*) 11.00

5. 1 mol of CH_4 contains:
(*a*) 6.02×10^{23} atoms of *H*
(*b*) 4 *g*-atom of Hydrogen
(*c*) 1.81×10^{23} molecules of CH_4
(*d*) 3.0 *g* of carbon

6. 7.5 grams of a gas occupy 5.8 litres of volume at STP, the gas is:
(*a*) NO (*b*) N_2O
(*c*) CO (*d*) CO_2

7. The number of atoms in 4.25 *g* of NH_3 is approximately:
(*a*) 1×10^{23} (*b*) 2×10^{23}
(*c*) 4×10^{23} (*d*) 6×10^{23}

8. One litre of a gas at STP weights 1.16 *g*. The possible gas is:
(*a*) C_2H_2 (*b*) CO
(*c*) O_2 (*d*) CH_4

9. The mass of a molecule of water is:
(*a*) 3×10^{-26} *kg* (*b*) 3×10^{-25} *kg*
(*c*) 1.5×10^{-26} *kg* (*d*) 2.5×10^{-26} *kg*

10. If N_A is Avogadro's number, then number of valence electrons in 4.2 *g* of nitride ions (N^{3-}) is:
(*a*) 2.4 N_A (*b*) 4.2 N_A
(*c*) 1.6 N_A (*d*) 3.2 N_A

11. The number of molecules at NTP in 1 *ml* of an ideal gas will be:
(*a*) 6×10^{23} (*b*) 2.69×10^{19}
(*c*) 2.69×10^{23} (*d*) None of these

12. Volume of a gas at STP is 1.12×10^{-7} cc. The number of molecules in it are:
(*a*) 3.01×10^{20} (*b*) 3.01×10^{12}
(*c*) 3.01×10^{23} (*d*) 3.01×10^{24}

13. 4.4 g of an unknown gas occupies 2.24 L of volume at standard temperature and pressure. The gas may be:
(*a*) Carbon dioxide (*b*) Carbon monoxide
(*c*) Oxygen (*d*) Sulphur dioxide

14. The number of oxygen atoms in 4.4 *g* of CO_2 is approx.:
(*a*) 1.2×10^{23} (*b*) 6×10^{22}
(*c*) 6×10^{23} (*d*) 12×10^{23}

15. The total number of protons in 10 *g* of calcium carbonate is: ($N_A = 6.023 \times 10^{23}$)
(*a*) 1.5057×10^{24} (*b*) 2.0478×10^{24}
(*c*) 3.0115×10^{24} (*d*) 4.0956×10^{24}

16. Number of molecules in 100 *ml* each of O_2,NH_3 and CO_2 at STP are:
(*a*) In the order: $CO_2 < O_2 < NH_3$
(*b*) In the order: $NH_3 < O_2 < CO_2$
(*c*) The same in all
(*d*) In the order: $NH_3 < CO_2 < O_2$

17. The number of water molecules in 1 litre of water is:
(*a*) 18 (*b*) 18×1000
(*c*) N_A (*d*) 55.55 N_A

18. 2 g of oxygen contains number of atoms equal to that in:
(*a*) 0.5 g of hydrogen (*b*) 4 g of sulphur
(*c*) 7 g of nitrogen (*d*) 2.3 g of sodium

EMPIRICAL FORMULA, % COMPOSITION OF A GIVEN COMPOUND BY MASS, % BY MOLE, MINIMUM MOLECULAR MASS DETERMINATION.

19. Caffeine has a molecular weight of 194. If it contains 28.9% by mass of nitrogen, number of atoms of nitrogen in one molecule of caffeine is:
(*a*) 4 (*b*) 6
(*c*) 2 (*d*) 3

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20. The percentage of oxygen in NaOH is:
(*a*) 40 (*b*) 60
(*c*) 8 (*d*) 10

21. A compound (60 *g*) on analysis gave C = 24 g, H = 4 *g*, O = 32 *g*. Its empirical formula is:
(*a*) $C_2H_2O_2$ (*b*) C_2H_2O
(*c*) CH_2O_2 (*d*) CH_2O

22. What is the % of H_2O in $Fe(CNS)_3.3H_2O$?
(*a*) 45 (*b*) 30
(*c*) 19 (*d*) 25

23. A hydrocarbon contains 86% carbon, 488 ml of the hydrocarbon weigh 1.68 *g* at STP. Then, the hydrocarbon is an:
(*a*) Alkane (*b*) Alkene
(*c*) Alkyne (*d*) Arene

24. The simplest formula of a compound containing 50% of element X (atomic mass 10) and 50% of element *Y* (atomic mass 20) is:
(*a*) XY (*b*) X_2Y
(*c*) XY_3 (*d*) X_2Y_3

STOICHIOMETRY, EQUATION BASED CALCULATIONS

25. How much of NaOH is required to neutralise 1500 cm^3 of 0.1 N HCl (Na = 23)?
(*a*) 40 *g* (*b*) 4 *g*
(*c*) 6 *g* (*d*) 60 *g*

26. In the reaction;
$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$, when 1 mole of ammonia and 1 mole of O_2 are made to react to completion:
(*a*) 1.0 mole of H_2O is produced.
(*b*) 1.0 mole of NO will be produced.
(*c*) All the oxygen will be consumed.
(*d*) All the ammonia will be consumed.

27. Haemoglobin contains 0.33% of iron by weight. The molecular weight of haemoglobin is approximately 67,200. The number of iron atoms (At. wt. of *Fe* = 56) present in one molecule of haemoglobin is:
(*a*) 6 (*b*) 1
(*c*) 4 (*d*) 2

28. H_2 evolved at STP on complete reaction of 27 *g* of aluminium with excess of aqueous NaOH would be:
(*a*) 22.4 litres (*b*) 44.8 litres
(*c*) 67.2 litres (*d*) 33.6 litres

29. What is the concentration of nitrate ions if equal volumes of 0.1 M $AgNO_3$ and 0.1 M NaCl are mixed together?
(*a*) 0.1 N (*b*) 0.2 M
(*c*) 0.05 M (*d*) 0.25 M

30. 12 g of Mg (at. mass 24) will react completely with acid to give:
(*a*) One mole of H_2
(*b*) 1/2 mole of H_2
(*c*) 2/3 mole of O_2
(*d*) Both 1/2 mol of H_2 and 1/2 mol of O_2

31. 100 g $CaCO_3$ reacts with 1 litre 1 N HCl. On completion of reaction, how much weight of CO_2 will be obtained?
(*a*) 5.5 g (*b*) 11 g
(*c*) 22 g (*d*) 33 g

32. What weight of HNO_3 is needed to convert 5 g of iodine into iodic acid according to the reaction,
$I_2 + HNO_3 \rightarrow HIO_3 + NO_2 + H_2O$
(*a*) 12.4 g (*b*) 24.8 g
(*c*) 0.248 g (*d*) 49.6 g

33. How much Cl_2 at STP is liberated when 1 mole $KMnO_4$ reacts with HCl?
(*a*) 11.2 L (*b*) 22.4 L
(*c*) 44.8 L (*d*) 56 L

34. NX is produced by the following step of reactions:
$M + X_2 \rightarrow MX_2$, $3MX_2 + X_2 \rightarrow M_3X_8$, $M_3X_8 + N_2CO_3 \rightarrow NX + CO_2 + M_3O_4$
How much M (metal) is consumed to produce 206 g of NX. (Take at wt. of M = 56, N = 23, X = 80)
(*a*) 42 g (*b*) 56 g
(*c*) $\frac{14}{3}$ (*d*) $\frac{7}{4}$

35. 27 g of Al will react completely with how many grams of oxygen?
(*a*) 8 g (*b*) 16 g
(*c*) 32 g (*d*) 24 g

36. If 0.50 mole of $BaC1_2$ is mixed with 0.20 mol of Na_3PO_4, the maximum number of moles of $Ba_3(PO_4)_2$ that can be formed is:
(*a*) 0.70 (*b*) 0.50
(*c*) 0.20 (*d*) 0.10

37. How many moles of potassium chlorate need to be heated to produce 11.2 litre oxygen at N.T.P.:
(*a*) $\frac{1}{2}$mol (*b*) $\frac{1}{3}$mol
(*c*) $\frac{1}{4}$mol (*d*) $\frac{2}{3}$mol

38. 0.5 mole of H_2SO_4 is mixed with 0.2 mole of $Ca(OH)_2$. The maximum number of moles of $CaSO_4$ formed is:
(*a*) 0.2 (*b*) 0.5
(*c*) 0.4 (*d*) 1.5

CONCENTRATION TERM

39. What is the normality of a 1 *M* solution of H_3PO_4?
(*a*) 0.5 N (*b*) 1.0 N
(*c*) 2.0 N (*d*) 3.0 N

40. What is the molarity of NaOH solution if 250 mL of it contains 1 mg of NaOH?

(*a*) 10^{-1} M (*b*) 10^{-2} M

(*c*) 10^{-4} M (*d*) 10^{-3} M

41. What weight of sodium hydroxide is required to neutralise 100 ml of 0.1 N HCl?

(*a*) 4 g (*b*) 0.04 g

(*c*) 0.4 g (*d*) 40 g

42. The amount of oxalic acid (eq.wt. 63) required to prepare 500 ml of its 0.10 N solution is:

(*a*) 0.315 g (*b*) 3.150 g

(*c*) 6.300 g (*d*) 63.00 g

43. Molarity of H_2SO_4 (density 1.8 g/mL) is 18 M. The molality of this H_2SO_4 is:

(*a*) 36 (*b*) 200

(*c*) 500 (*d*) 18

44. The percentage of sodium in a breakfast cereal be labeled as 110 mg of sodium per 100 g of cereal is

(*a*) 11% (*b*) 1.10%

(*c*) 0.11% (*d*) 1.10%

45. 2 g of $CaCO_3$(s) was treated with 0.1 M HCl (500 ml). The volume of CO_2 evolved at STP after heating the solution is:

(*a*) 0.448 L (*b*) 0.224 L

(*c*) 4.48 L (*d*) 44.8 L

46. 0.16 g of dibasic acid required 25 ml of N/10 NaOH for complete neutralization. Molecular weight of acid is:

(*a*) 32 (*b*) 64

(*c*) 128 (*d*) 256

47. A molal solution is one that contains one mole of a solute in:

(*a*) 1000 g of the solvent.

(*b*) One litre of the solvent.

(*c*) One litre of the solution.

(*d*) 22.4 litres of the solution.

48. An aqueous solution of 6.3 g oxalic acid dihydrate is made up to 250 ml. The volume of 0.1 N NaOH required to completely neutralize 10 ml of this solution is:

(*a*) 40 ml (*b*) 20 ml

(*c*) 10 ml (*d*) 4 ml

49. 20 mL of x M HCl neutralizes completely 10 mL of 0.1 M $NaHCO_3$ and a further 5 mL of 0.2 M Na_2CO_3 solution to methyl orange end point. The value of x is:

(*a*) 0.167 M (*b*) 0.133 M

(*c*) 0.150 M (*d*) 0.200 M

50. A solution contains Na_2CO_3 and $NaHCO_3$. 10 mL of the solution required 2.5 mL of 0.1M H_2SO_4 for neutralisation using phenolphthalein as indicator. Methyl orange is then added when a further 2.5 mL of 0.2 M H_2SO_4 was required. The amount of Na_2CO_3 and $NaHCO_3$ in 1 litre of the solution is:

(*a*) 5.3 g and 4.2 g (*b*) 3.3 g and 6.2 g

(*c*) 4.2 g and 5.3 g (*d*) 6.2 g and 3.3 g

51. Hydrogen peroxide in aqueous solution decomposes on warming to give oxygen according to the equation,

$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$

Under conditions where 1 mole of gas occupies 24 dm^3. 100 cm^3 of X M solution of H_2O_2 produces 3 dm^3 of O2. Thus X is:

(*a*) 2.5 (*b*) 1

(*c*) 0.5 (*d*) 0.25

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Exercise-2 (Learning Plus)

1. A sample of calcium phosphate $Ca_3(PO_4)_2$ contains 8 mol of O atoms. The number of mole of Ca atoms in the sample is:

(*a*) 4 (*b*) 1.5

(*c*) 3 (*d*) 8

2. Ratio of masses of H_2SO_4 and $Al_2(SO_4)_3$ each containing 32 grams of **S** is _____.

(*a*) 0.86 (*b*) 1.72

(*c*) 0.43 (*d*) 2.15

3. Which has maximum number of atoms of oxygen?

(*a*) 10 ml $H_2O(l)$

(*b*) 0.1 mole of V_2O_5

(*c*) 12 gm O_3(g)

(*d*) 12.044×10^{22} molecules of CO_2

4. Mass of one atom of the element A is 3.9854×10^{-23}. How many atoms are contained in 1g of the element A?

(*a*) 2.509×10^{22}

(*b*) 6.022×10^{23}

(*c*) 12.044×10^{23}

(*d*) None of these

5. The number of atoms present in 0.5 g-atoms of nitrogen is same as the atoms in:

(*a*) 12 g of C (*b*) 32 g of S

(*c*) 8 g of oxygen (*d*) 24 g of Mg

6. How many moles of magnesium phosphate $Mg_3(PO_4)_2$ will contain 0.25 mole of oxygen atoms?

(*a*) 0.02 (*b*) 3.125×10^{-2}

(*c*) 1.25×10^{-2} (*d*) 2.5×10^{-2}

7. 64 g of an organic compound has 24 g carbon and 8 g hydrogen and the rest is oxygen. The empirical formula of the compound is:

(*a*) CH_4O (*b*) CH_2O

(*c*) C_2H_4O (*d*) None of these

8. Two elements X (atomic mass = 75) and Y (atomic mass = 16) combine to give a compound having 75.8% of X. The formula of the compound is:

(*a*) X_2Y_3 (*b*) X_2Y

(*c*) X_2Y_2 (*d*) XY

9. A definite amount of gaseous hydrocarbon was burnt with just sufficient amount of O_2. The volume of all reactants was 600 ml, after the explosion the volume of the products [CO_2(g) and H_2O(g)] was found to be 700 ml under the similar conditions. The molecular formula of the compound is:

(*a*) C_3H_8 (*b*) C_3H_6

(*c*) C_3H_4 (*d*) C_4H_{10}

10. Mole fraction of ethyl alcohol in aqueous ethyl alcohol (C_2H_5OH) solution is 0.25. Hence, percentage of ethyl alcohol by weight is:

(*a*) 54% (*b*) 25%

(*c*) 75% (*d*) 46%

11. 74 gm of sample on complete combustion gives 132 gm CO_2 and 54 gm of H_2O. The molecular formula of the compound may be:

(*a*) C_5H_{12} (*b*) $C_4H_{10}O$

(*c*) $C_3H_6O_2$ (*d*) $C_3H_7O_2$

12. Weight of oxygen in Fe_2O_3 and FeO is in the simple ratio for the same amount of iron, is:

(*a*) 3 : 2 (*b*) 1 : 2

(*c*) 2 : 1 (*d*) 3 : 1

13. A person needs on average of 2.0 mg of riboflavin (vitamin B_2) per day. How many gm of butter should be taken by the person per day if it is the only source of riboflavin? Butter contains 5.5 microgram riboflavin per gm.

(*a*) 363.6 gm (*b*) 2.75 mg

(*c*) 11 gm (*d*) 19.8 gm

14. The oxide of a metal contains 30% oxygen by weight. If the atomic ratio of metal and oxygen is 2 : 3, determine the atomic weight of metal.

(*a*) 12 (*b*) 56

(*c*) 27 (*d*) 52

15. When a mixture of 10 mole of SO_2 , 15 mole of O_2 was passed over catalyst , 8 mole of SO_3 was formed. How many mole of $SO_2$and O_2 did not enter into combination?

(*a*) 2 moles of SO_2, 11 moles of O_2

(*b*) 3 moles of SO_2, 11.5 moles of O_2

(*c*) 2 moles of SO_2, 4 moles of O_2

(*d*) 8 moles of SO_2, 4 moles of O_2

16. $C_6H_5OH(g) + O_2(g) \rightarrow CO_2(g) + H_2O(l)$

Magnitude of volume change if 30 ml of C_6H_5OH (g) is burnt with excess amount of oxygen, is:

(*a*) 30 ml (*b*) 60 ml

(*c*) 20 ml (*d*) 10 ml

17. Mass of sucrose $C_{12}H_{22}O_{11}$ produced by mixing 84 gm of carbon, 12 gm of hydrogen and 56 lit. O_2 at 1 atm & 273 K according to given reaction, is:

$C(s) + H_2(g) + O_2(g) \rightarrow C_{12}H_{22}O_{11}(s)$

(*a*) 138.5 (*b*) 155.5

(*c*) 172.5 (*d*) 199.5

18. What volume (in ml) of 0.2 M H_2SO_4 solution should be mixed with the 40 ml of 0.1 M NaOH solution such that the resulting solution has the concentration of H_2SO_4 as $\frac{6}{55}$ M.

(*a*) 70 (*b*) 45

(*c*) 30 (*d*) 58

19. For the reaction; $2x + 3y + 4z \rightarrow 5w$, initially if 1 mol of x, 3 mol of y and 4 mol of z is taken. If 1.25 mol of w is obtained then % yield of this reaction is:

(*a*) 50% (*b*) 60%

(*c*) 70% (*d*) 40%

20. If 10 g of Ag reacts with 1 g of sulphur , the amount of Ag_2S formed will be:

(*a*) 7.75 g (*b*) 0.775 g

(*c*) 11 g (*d*) 10 g

21. A solution of A (MM = 20) and B (MM = 10), [Mole fraction X_B = 0.6] having density 0.7 gm/ml then molarity and molality of B in this solution will be _____ and _____ respectively.

(*a*) 30 M, 75 m (*b*) 40 M, 75 m

(*c*) 30 M, 65 m (*d*) 50 M, 55 m

22. 125 ml of 8% w/w NaOH solution sp. gravity = 1 is added to 125 ml of 10 % w/v HCl solution. The nature of resultant solution would be _____.

(*a*) Acidic (*b*) Basic

(*c*) Neutral (*d*) None of these

23. 36.5 % (w/w) HCl has density equal to 1.20 g mL^{-1}. The molarity (M) and molality (m), respectively, are:

(*a*) 15.7, 15.7 (*b*) 12, 12

(*c*) 15.7, 12 (*d*) 12, 15.7

24. An aqueous solution of ethanol has density 1.025 g/mL and it is 2 M. What is the molality of this solution?

(*a*) 1.79 (*b*) 2.143

(*c*) 1.951 (*d*) None of these.

25. 500 mL of a glucose solution contains 6.02×10^{22} molecules of glucose. The concentration of the solution is:

(*a*) 0.1 M (*b*) 1.0 M

(*c*) 0.2 M (*d*) 2.0 M

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26. Equal moles of H_2O and NaCl are present in a solution. Hence, molality of NaCl solution is:

(*a*) 0.55 (*b*) 55.5

(*c*) 1.00 (*d*) 0.18

27. Decreasing order of mass of pure NaOH in each of the aqueous solution.

I. 50 g of 40% (W/W) NaOH

II. 50 ml of 50% (W/V) NaOH (d_{sol} = 1.2 g/ml).

III. 50 g of 15 M NaOH (d_{sol} = 1 g/ml).

(*a*) I, II, III (*b*) III, II, I

(*c*) II, III, I (*d*) III = II = I

28. Mole fraction of A in H_2O is 0.2. The molality of A in H_2O is:

(*a*) 13.9 (*b*) 15.5

(*c*) 14.5 (*d*) 16.8

29. What is the molarity of H_2SO_4 solution that has a density of 1.84 g/cc and contains 98% by mass of H_2SO_4? (Given: atomic mass of S = 32)

(*a*) 4.18 M (*b*) 8.14 M

(*c*) 18.4 M (*d*) 18 M

30. The molarity of the solution containing 2.8% (W/V) solution of KOH is: (Given: atomic mass of K = 39)

(*a*) 0.1 M (*b*) 0.5 M

(*c*) 0.2 M (*d*) 1 M

31. A solution of $FeCl_3$ is $\frac{M}{30}$, its molarity for Cl^- ion will be:

(*a*) $\frac{M}{90}$ (*b*) $\frac{M}{30}$

(*c*) $\frac{M}{10}$ (*d*) $\frac{M}{5}$

32. If 500 ml of 1 M solution of glucose is mixed with 500 ml of 1 M solution of glucose. Final molarity of solution will be:

(*a*) 1 M (*b*) 0.5 M

(*c*) 2 M (*d*) 1.5 M

33. What volume of a 0.8 M solution contains 100 milli moles of the solute?

(*a*) 100 mL (*b*) 125 mL

(*c*) 500 mL (*d*) 62.5 mL

34. The molarity of Cl^- in an aqueous solution which was (w/V) 2% NaCl, 4% $CaCl_2$ and 6% NH_4Cl, will be:

(*a*) 0.342 (*b*) 0.721

(*c*) 1.12 (*d*) 2.18

35. 2 M of 100 mL Na_2SO_4 is mixed with 3 M of 100 mL NaCl solution and 1 M of 200 mL $CaCl_2$ solution. Then, the ratio of the concentration of cation and anion is:

(*a*) 1/2 (*b*) 2

(*c*) 1.5 (*d*) 1

36. 4.4 g of CO_2 and 2.24 litre of H_2 at STP are mixed in a container. The total number of molecules present in the container will be:

(*a*) 6.023×10^{23} (*b*) 1.2046×10^{23}

(*c*) 6.023×10^{22} (*d*) 6.023×10^{24}

37. Two elements A (atomic mass = 75) and B (atomic mass = 16) combine to yield a compound. The % by weight of A in the compound was found to be 75.08. The formula of the compound is:

(*a*) A_2B (*b*) A_2B_3

(*c*) AB (*d*) AB_2

38. 10 g of $CaCO_3$ on heating gives 5 g of the residue (as CaO). The percent yield of the reaction is approximately:

(*a*) 50% (*b*) 72%

(*c*) 89% (*d*) 100%

39. 33.6 g of an impure sample of sodium bicarbonate when heated strongly gave 4.4 g of CO_2. The percentage purity of $NaHCO_3$ would be:

(*a*) 25% (*b*) 50%

(*c*) 75% (*d*) 100%

40. A compound was found to contain nitrogen and oxygen in the ratio 28 gm and 80 gm respectively. The formula of compound is:

(*a*) NO (*b*) N_2O_3

(*c*) N_2O_5 (*d*) N_2O_4

41. The total number of electrons in one molecule of carbon dioxide is:

(*a*) 22 (*b*) 44

(*c*) 66 (*d*) 88

42. A gaseous mixture contains oxygen and nitrogen in the ratio of 1 : 4 by weight. Therefore the ratio of their number of molecules is:

(*a*) 1 : 4 (*b*) 1 : 8

(*c*) 7 : 32 (*d*) 3 : 16

43. 2.76 g of silver carbonate on being strongly heated yields a residue weighing:

(*a*) 2.16 g (*b*) 2.48 g

(*c*) 2.32 g (*d*) 2.64 g

44. 12 g of alkaline earth metal gives 14.8 g of its nitride. Atomic weight of metal is:

(*a*) 12 (/) 20

(*c*) 40 (*d*) 14.8

45. How many liters of CO_2 at STP will be formed when 0.01 mol of H_2SO_4 reacts with excess of Na_2CO_3.

$Na_2CO_3 + H_2SO_4 \rightarrow Na_2SO_4 + CO_2 + H_2O$

(*a*) 22.4 L (*b*) 2.24 L

(*c*) 0.224 L (*d*) 1.12 L

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46. Equal weight of 'X' (At. wt. = 36) and 'Y' (At. wt. = 24) are reacted to form the compound X_2Y_3. Then

(*a*) X is the limiting reagent

(*b*) Y is the limiting reagent

(*c*) No reactant is left over and mass of X_2Y_3 formed is double the mass of 'X' taken

(*d*) None of these

47. A person adds 1.71 gram of sugar ($C_{12}H_{22}O_{11}$) in order to sweeten his tea. The number of carbon atoms added are (mol. mass of sugar = 342)

(*a*) 3.6×10^{22} (*b*) 7.2×10^{21}

(*c*) 0.05×10^{20} (*d*) 6.6×10^{22}

48. A compound contains 38.8 % C, 16.0 % H and 45.2 % N. The formula of the compound would be

(*a*) CH_3NH_2 (*b*) CH_3CN

(*c*) C_2H_5CN (*d*) $CH_2(NH)_2$

49. When 100 g of ethylene polymerises entirely to polyethene, the weight of polyethene formed as per the equation $n(C_2H_4) \longrightarrow (-CH_2-CH_2-)_n$ is:

(*a*) (n/2)g (*b*) 100g

(*c*) (100/n)g (*d*) 100ng

50. What is the molarity of H_2SO_4 solution that has a density of 1.84 g/cc and contains 98% by mass of H_2SO_4? (Given atomic mass of S = 32)

(*a*) 4.18 M (*b*) 8.14 M

(*c*) 18.4 M (*d*) 18 M

Exercise-3 (JEE Advanced Level)

MULTIPLE CORRECT TYPE QUESTIONS

1. Which is/are correct statements about 1.7 g of NH_3?

(*a*) It contain 0.3 mol H – atoms.

(*b*) It contain 2.408×10^{23} atoms.

(*c*) Mass % of hydrogen is 17.65%.

(*d*) It contains 0.3 mol N-atoms.

2. If 27 g of carbon is mixed with 88 g of oxygen and is allowed to burn to produce CO_2, then:

(*a*) Oxygen is the limiting reagent.

(*b*) Volume of CO_2 gas produced at NTP is 50.4 L.

(*c*) C and O combine in mass ratio of 3.8.

(*d*) Volume of unreacted O_2 at STP is 11.2 L.

3. $\left(CH(C_2H_5)-COOH\right)_n + AgNO_3$ (Excess) $\longrightarrow$ Silver salt $\longrightarrow$ Ag (metal)

If 0.5 mole of silver salt is taken and weight of residue obtained is 216 g. (Ag = 108 g/mol), then which of the following is correct?

(*a*) n = 4

(*b*) n = 2

(*c*) M.wt. of silver salt is 718 g/mol

(*d*) M.wt. of silver salt is 388 g/mol

4. For the following reaction;

$N_2 + 3H_2 \longrightarrow 2NH_3$

Identify the compositions which will produce same amount of NH_3.

(*a*) 140 g N_2 & 35 g H_2

(*b*) 18 g H_2 & 52 g N_2

(*c*) Total 20 moles of mixture having N_2 & H_2 present in stoichiometric ratio (No limiting reagent).

(*d*) 136 gm of mixture having mass fraction of $H_2 = \frac{6}{34}$.

5. For the following reaction : $Na_2CO_3 + 2HCl \longrightarrow 2NaCl + \mathbf{CO_2} + H_2O$

106.0 g of Na_2CO_3 reacts with 109.5 g of HCl.

Which of the following is/are correct?

(*a*) The HCl is in excess amount.

(*b*) 117.0 g of NaCl is formed.

(*c*) The volume of CO_2 produced at NTP is 22.4 L.

(*d*) None of these

6. A sample of a mixture of $CaCl_2$ and NaCl weighing 4.44 g was treated to precipitate all the Ca as $CaCO_3$, which was then heated and quantitatively converted to 1.12 g of CaO. (At . wt. Ca = 40, Na = 23, Cl = 35.5)

(*a*) Mixture contains 50% NaCl

(*b*) Mixture contains 60% $CaCl_2$

(*c*) Mass of $CaCl_2$ is 2.22 g

(*d*) Mass of $CaCl_2$ is 1.11 g

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7. $A + B \rightarrow A_3B_2$ (unbalanced)
$A_3B_2 + C \rightarrow A_3B_2C_2$ (unbalanced)
Above two reactions are carried out by taking 3 moles each of A and B and one mole of C. Then, which option is/are correct?
(*a*) 1 mole of $A_3B_2C_2$ is formed.
(*b*) 1/2 mole of $A_3B_2C_2$ is formed.
(*c*) 1/2 mole of A_3B_2 is formed.
(*d*) 1/2 mole of A_3B_2 is left finally.

8. 21.2 g sample of impure Na_2CO_3 is dissolved and reacted with a solution of $CaCl_2$, the weight of precipitate of $CaCO_3$ is 10.0 g. Which of the following statements is/are correct?
(*a*) The % purity of Na_2CO_3 is 50%.
(*b*) The percentage purity of Na_2CO_3 is 60%.
(*c*) $n_{Na_2CO_3} = n_{CaCO_3} = 0.1$ mol.
(*d*) The number of moles of NaCl formed is 0.1 mol.

9. A mixture of C_3H_8 and O_2 having total volume of 100 ml in an eudiometry tube is sparked & it is observed that a contraction of 45 mL is observed. What can be the composition of the reacting mixture?
(*a*) 15 mL C_3H_8 & 85 mL O_2
(*b*) 25 mL C_3H_8 & 75 mL O_2
(*c*) 45 mL C_3H_8 & 55 mL O_2
(*d*) 55 mL C_3H_8 & 45 mL O_2

10. The **incorrect** statement(s) regarding 2 M $MgCl_2$ aqueous solution is/are: ($d_{solution}$ = 1.09 gm/ml)
(*a*) Molality of Cl^- is 4.44 m.
(*b*) Mole fraction of $MgCl_2$ is exactly 0.035.
(*c*) The conc. of $MgCl_2$ is 19% w/v.
(*d*) The conc. of $MgCl_2$ is 19×10^4 ppm.

11. Solution containing 23 g HCOOH is/are:
(*a*) 46 g of 70% $\left(\frac{w}{v}\right)$ HCOOH ($d_{solution}$ = 1.40 g/mL)
(*b*) 50 g of 10 M HCOOH ($d_{solution}$ = 1 g/mL)
(*c*) 50 g of 25% $\left(\frac{w}{w}\right)$ HCOOH
(*d*) 46 g of 5 M HCOOH ($d_{solution}$ = 1 g/mL)

12. Which of the following solution contains same molar concentration?
(*a*) 166 g KI per liter solution
(*b*) 33.0 g $(NH_4)_2SO_4$ in 200 mL solution
(*c*) 25.0 g $CuSO_4.5H_2O$ in 100 mL solution
(*d*) 27.0 mg Al^{3+} per mL solution

13. If 100 ml of 1M H_2SO_4 solution is mixed with 100 ml of 9.8%(w/w) H_2SO_4 solution (d = 1 g/mL), then:
(*a*) Concentration of solution remains same.
(*b*) Volume of solution become 200 mL.
(*c*) Mass of H_2SO_4 in the solution is 98 g.
(*d*) Mass of H_2SO_4 in the solution is 19.6 g.

COMPREHENSION BASED QUESTIONS

Comprehension (Q. 14 to 16): According to the Avogadro's law, equal number of moles of gases occupy the same volume at identical condition of temperature and pressure. Even if we have a mixture of non-reacting gases, then Avogadro's law is still obeyed by assuming mixture as a new gas.

Now let us assume air to consist of 80% by volume of nitrogen (N_2) and 20% by volume of oxygen (O_2). If air is taken at STP, then its 1 mol would occupy 22.4 L. 1 mol of air would contain 0.8 mol of N_2 and 0.2 mol of O_2, hence the mole fractions of N_2 and O_2 are given by $X_{N_2} = 0.8$ and $X_{O_2} = 0.2$.

14. Volume occupied by air at NTP containing exactly 11.2 g of nitrogen, is:
(*a*) 22.4 L (*b*) 8.96 L
(*c*) 11.2 L (*d*) 2.24 L

15. If air is treated as a solution of O_2 and N_2, then % w/w of oxygen is:
(*a*) $\frac{10}{9}$ (*b*) $\frac{200}{9}$
(*c*) $\frac{700}{9}$ (*d*) $\frac{350}{9}$

16. Density of air at NTP is:
(*a*) 1 g/L (*b*) $\frac{9}{7}$ g/L
(*c*) $\frac{2}{7}$ g/L (*d*) Can't be determined

Comprehension (Q. 17 to 19): A chemist decided to determine the molecular formula of an unknown compound. He collects following informations:

I. Compound contain 2 : 1 'H' to 'O' atoms (number of atoms).
II. Compound has 40% C by mass.
III. Approximate molecular mass of the compound is 178 g.
IV. Compound contains C, H and O only.

17. What is the % by mass of oxygen in the compound?
(*a*) 53.33% (*b*) 88.88%
(*c*) 33.33% (*d*) None of these

18. What is the empirical formula of the compound?
(*a*) CH_3O (*b*) CH_2O
(*c*) C_2H_2O (*d*) CH_3O_2

19. Which of the following could be molecular formula of compound?
(*a*) $C_6H_6O_6$
(*b*) $C_6H_{12}O_6$
(*c*) $C_6H_{14}O_{12}$
(*d*) $C_6H_{14}O_6$

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MATCH THE COLUMN TYPE QUESTIONS

20. One type of artificial diamond (commonly called YAG for yttrium aluminium garnet) can be represented by the formula $Y_3Al_5O_{12}$.[Y = 89, Al =27]

Column-I		Column-II	
Element		Weight percentage	
A.	Y	p.	22.73%
B.	Al	q.	32.32%
C.	O	r.	44.95%

(*a*) A → r; B → p; C → q

(*b*) A → q; B → p; C → r

(*c*) A → r; B → q; C → p

(*d*) A → p; B → r; C → q

21.

Column-I		Column-II	
A.	100 mL of 0.2 M $AlCl_3$ solution + 400 ml of 0.1 M HCl solution	p.	Total concentration of cation(s) = 0.12 M
B.	50 mL of 0.4 M KCl + 50 ml H_2O	q.	$[SO_4^{2-}] = 0.06$ M
C.	30 mL of 0.2 M K_2SO_4 + 70 ml H_2O	r.	$[SO_4^{2-}] = 2.5$ M
D.	200 mL 24.5% (w/v) H_2SO_4	s.	$[Cl^-] = 0.2$ M

(*a*) A → r,s; B → s; C → p,q; D → r

(*b*) A → p,s; B → s; C → p,q; D → r

(*c*) A → p,s; B → s; C → s,q; D → r

(*d*) A → p,s; B → p,s; C → p,q; D → r

22.

Column-I		Column-II	
A.	A gaseous organic compound containing C = 52.17%, H = 13.04% & O = 34.78% (by weight) having molar mass 46 g/mol.	p.	One mole of compound contains $4N_A$ atoms of hydrogen.
B.	0.3 g of an organic compound containing C, H and O on combustion yields 0.44 g of CO_2 and 0.18 g of H_2O, with two O	q.	The empirical formula of the compound is same as its molecular formula.
C.	A hydrocarbon containing C = 42.857% and H = 57.143% (by mole) containing 3C atoms.	r.	Combustion products of one mole of compound contains larger number of moles of CO_2 than that of H_2O.
D.	A hydrocarbon containing 10.5 g carbon per gram of hydrogen having vapour density 46.	s.	CO_2 gas produced by the combustion of 0.25 mole of compound occupies a volume of 11.2 L at NTP.

(*a*) A → p,s; B → p,s; C → p,q,r; D → q,r

(*b*) A → q,s; B → q,s; C → p,q,r; D → q,r

(*c*) A → q,s; B → p,s; C → r,s; D → q,r

(*d*) A → q,s; B → p,s; C → p,q,r; D → q,r

NUMERICAL TYPE QUESTIONS

23. Find the oxidation state of osmium (Os) in OsO_4.

24. Find the oxidation number of nitrogen in $(NH_4)_2SO_4$.

25. Molarity of H_2SO_4 is 18 M. Its density is 1.8 g/cm^3, calculate its molality.

26. Find the percentage of oxygen in NaOH.

27. Calculate the mass of NaOH that is required to neutralise 1500 cm^3 of 0.1 N HCl (Atomic mass: Na = 23).

28. One mole of potassium chlorate ($KClO_3$) is thermally decomposed and excess of aluminium is burnt in the gaseous product. How many mol of aluminium oxide (Al_2O_3) are formed?

29. 100 g $CaCO_3$ reacts with 1litre 1 N HCl. On completion of reaction, how much weight of CO_2 will be obtained?

30. 2 M of 100 ml Na_2SO_4 is mixed with 3 M of 100 ml NaCl solution and 1 M of 200 ml $CaCl_2$ solution. Then, the ratio of the concentration of cation and anion, is___.

31. What is the quantity of water that should be added to 16 g. methanol to make the mole fraction of methanol as 0.25.

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Exercise-4 (Past Year Questions)

JEE MAIN

1. 1 gram of a carbonate (M_2CO_3) on treatment with excess HCl produces 0.01186 mole of CO_2, the molar mass of M_2CO_3 in g mol^{-1} is: **(2017)**

(*a*) 1186 (*b*) 84.3

(*c*) 1186 (*d*) 11.86

2. The most abundant elements by mass in the body of a healthy human adult are:
Oxygen (61.4%) : Carbon (22.9%), Hydrogen (10.0%), and Nitrogen (2.6%). The weight which a 75 kg person would gain if all ^1H atoms are replaced by ^2H atoms, is: **(2017)**

(*a*) 15 kg (*b*) 37.5 kg

(*c*) 7.5 kg (*d*) 10 kg

3. The ratio of mass percent of C and H of an organic compound ($C_XH_YO_Z$) is 6 : 1. If one molecule of the above compound ($C_XH_YO_Z$) contains half as much oxygen as required to burn one molecule of compound C_XH_Y completely to CO_2 and H_2O. The empirical formula of compound $C_XH_YO_Z$ is: **(2018)**

(*a*) C_2H_4O (*b*) $C_3H_4O_2$

(*c*) $C_2H_4O_3$ (*d*) $C_3H_6O_3$

4. A 10 mg effervescent tablet containing sodium bicarbonate and oxalic acid releases 0.25 ml of CO_2 at T = 298.15 K and p = 1 bar. If molar volume of CO_2 is 25.0 L under such condition, what is the percentage of sodium bicarbonate in each tablet? **(2019)**
[Molar mass of $NaHCO_3$ = 84 g mol^{-1}]

(*a*) 0.84 (*b*) 33.6

(*c*) 16.8 (*d*) 8.4

5. Water filled in two glasses A and B have BOD values of 10 and 20 respectively. The correct statement regarding them is: **(2019)**

(*a*) B is more polluted than A.

(*b*) A is suitable for drinking, whereas B is not.

(*c*) Both A and B are suitable for drinking.

(*d*) A is more polluted than B.

6. For the following reaction, the mass of water produced from 445 g of $C_{57}H_{110}O_6$ is: **(2019)**

$2C_{57}H_{110}O_6\,(s) + 163\,O_2\,(g) \rightarrow 114\,CO_2\,(g) + 110\,H_2O\,(l)$

(*a*) 490 g (*b*) 445 g

(*c*) 495 g (*d*) 4 g

7. 50 mL of 0.5 M oxalic acid is needed to neutralize 25 ml of sodium hydroxide solution. The amount of NaOH in 50 mL of the given sodium hydroxide solution, is: **(2019)**

(*a*) 4 g (*b*) 10 g

(*c*) 20 g (*d*) 80 g

8. A solution of sodium sulphate contains 92 g of Na^+ ions per kilogram of water. The molality of Na^+ ions in that solution in mol kg^{-1} is: **(2019)**

(*a*) 12 (*b*) 4

(*c*) 8 (*d*) 16

9. The hardness of water sample (in terms of equivalents of $CaCO_3$) containing 10^{-3} M $CaSO_4$ is:
(molar mass of $CaSO_4$ = 136 g mol^{-1}) **(2019)**

(*a*) 10 ppm (*b*) 50 ppm

(*c*) 90 ppm (*d*) 100 ppm

10. The amount of sugar ($C_{12}H_{22}O_{11}$) required to prepare 2 L of its 0.1 M aqueous solution is: **(2019)**

(*a*) 136.8 g (*b*) 17.1 g

(*c*) 68.4g (*d*) 34.2 g

11. 8 g of NaOH is dissolved in 18 g of H_2O. Mole fraction of NaOH in solution and molality (in mol kg^{-1}) of the solution respectively, are: **(2019)**

(*a*) 0.2, 22.20 (*b*) 0.2, 11.11

(*c*) 0.167, 11.11 (*b*) 0.167, 22.20

12. The volume strength of 1 M H_2O_2 is (Molar mass of H_2O_2 = 34 g mol^{-1}) **(2019)**

(*a*) 5.6 (*b*) 16.8

(*c*) 11.35 (*d*) 22.4

13. 100 mL of a water sample contains 0.81 g of calcium bicarbonate and 0.73 g of magnesium bicarbonate. The hardness of this water sample expressed in terms of equivalents of $CaCO_3$ is: (molar mass of calcium bicarbonate is 162 g mol^{-1} and magnesium bicarbonate is 146 g mol^{-1}) **(2019)**

(*a*) 1,000 ppm (*b*) 10,000 ppm

(*c*) 100 ppm (*d*) 5,000 ppm

14. For a reaction,

$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$;

identify dihydrogen (H_2) as a limiting reagent in the following reaction mixtures. **(2019)**

(*a*) 14 g of N_2 + 4 g of H_2

(*b*) 28 g of N_2 + 6 g of H_2

(*c*) 56 g of N_2 + 10 g of H_2

(*d*) 35 g of N_2 + 8 g of H_2

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15. The percentage composition of carbon by mole in methane is: **(2019)**

(*a*) 80% (*b*) 25%

(*c*) 75% (*d*) 20%

16. The maximum prescribed concentration of copper in drinking water is: **(2019)**

(*a*) 5 ppm (*b*) 0.5 ppm

(*c*) 0.05 ppm (*d*) 3 ppm

17. The percentage strength of 11.2 volumes solution of H_2O_2 is : [Given that molar mass of H = 1 g mol^{-1} and O = 16 g mol^{-1}] **(2019)**

(*a*) 13.6% (*b*) 3.4%

(*c*) 34% (*d*) 1.7%

18. 25 g of an unknown hydrocarbon upon burning produces 88 g of CO_2 and 9 of H_2O. This unknown hydrocarbon contains. **(2019)**

(*a*) 20 g of carbon and 5 g of hydrogen.

(*b*) 24 g of carbon and 1 g of hydrogen.

(*c*) 18 g of carbon and 7 g of hydrogen.

(*d*) 22 g of carbon and 3 g of hydrogen.

19. What would be the molality of 20% (mass/mass) aqueous solution of KI? **(2019)**

(molar mass of KI = 166 g mol^{-1})

(*a*) 1.08 (*b*) 1.48

(*c*) 1.51 (*d*) 1.35

20. 5 moles of AB_2 weigh 125 × 10^{-3} kg and 10 moles of A_2B_2 weigh 300 × 10^{-3} kg. The molar mass of $A_{(M_A)}$ and molar mass of $B_{(M_B)}$ in kg mol^{-1} are: **(2019)**

(*a*) $M_A = 50 \times 10^{-3}$ and $M_B = 25 \times 10^{-3}$

(*b*) $M_A = 25 \times 10^{-3}$ and $M_B = 50 \times 10^{-3}$

(*c*) $M_A = 5 \times 10^{-3}$ and $M_B = 10 \times 10^{-3}$

(*d*) $M_A = 10 \times 10^{-3}$ and $M_B = 5 \times 10^{-3}$

21. The mole fraction of a solvent in aqueous solution of a solute is 0.8. The molality (in mol kg^{-1}) of the aqueous solution is: **(2019)**

(*a*) 13.88×10^{-1} (*b*) 13.88×10^{-2}

(*c*) 13.88 (*d*) 13.88×10^{-3}

22. At 300 K and 1 atmospheric pressure, 10 mL of a hydrocarbon required 55 mL of O_2 for complete combustion and 40 mL of CO_2 is formed. The formula of the hydrocarbon is: **(2019)**

(*a*) C_4H_8 (*b*) C_4H_7Cl

(*c*) C_4H_{10} (*d*) C_4H_6

23. The minimum amount of O_2(g) consumed per gram of reactant is for the reaction: **(2019)**

(Given atomic mass: Fe = 56, O = 16, Mg = 24, P = 31, C = 12, H = 1)

(*a*) $C_3H_8(g) + 5\ O_2(g) \rightarrow 3\ CO_2(g) + 4\ H_2O(l)$

(*b*) $P_4(s) + 5\ O_2(g) \rightarrow P_4O_{10}(s)$

(*c*) $4\ Fe(s) + 3\ O_2(g) \rightarrow 2Fe_2O_3(s)$

(*d*) $2\ Mg(s) + O_2(g) \rightarrow 2\ MgO(s)$

24. Amongst the following statements, that which was not proposed by Dalton was: **(2020)**

(*a*) When gases combine or reproduced in a chemical reaction, they do so in a simple ratio by volume provided all gases are at the same T & P.

(*b*) Matter consists of indivisible atoms.

(*c*) Chemical reactions involve reorganization of atoms. These are neither created nor destroyed in a chemical reaction.

(*d*) All the atoms of a given element have identical properties including identical mass. Atoms of different elements differ in mass.

25. The volume (in mL) of 0.125 M $AgNO_3$ required to quantitatively precipitate chloride ions in 0.3 g of $[Co(NH_3)_6]Cl_3$ is _____. **(2020)**

$M_{[Co(NH_3)_6]Cl_3}$=267.46 g/mol

M_{AgNO_3} = 169.87 g/mol

26. $NaClO_3$ is used, even in spacecrafts, to produces O_2. The daily consumption of pure O_2 by a person is 492 L at 1 atm, 300 K. How much amount of $NaClO_3$, in grams, is required to produce O_2 for the daily consumption of a person at 1 atm, 300 K? _____. **(2020)**

$NaClO_3(s) + Fe(s) \rightarrow O_2(g) + NaCl(s) + FeO(s)$

R = 0.082 L atm mol^{-1} K^{-1}

27. The hardness of a water sample containing 10^{-3} M $MgSO_4$ expressed as $CaCO_3$ equivalents (in ppm) is _____. (molar mass of $MgSO_4$ is 120.37 g/mol) **(2020)**

28. 6.023×10^{22} molecules are present in 10 g of a substance 'x'. The molarity of a solution containing 5 g of substance 'x' in 2 L solution is _____ × 10^{-3}. **(2020)**

29. The strengths of 5.6 volumes hydrogen peroxide (of density 1 g/mL) in terms of mass percentage and molarity (M), respectively, are **(2020)**

(Take molar mass of hydrogen peroxide as 34 g/ mol)

(*a*) 1.7 and 0.5 (*b*) 0.85 and 0.5

(*c*) 1.7 and 0.25 (*d*) 0.85 and 0.25

30. The mole fraction of glucose ($C_6H_{12}O_6$) in an aqueous binary solution is 0.1. The mass percentage of water in it, to the nearest integer, is _____. **(2020)**

31. The volume strength of 8.9 M H_2O_2 solution calculated at 273 K and 1 atm is _____. **(2020)**

(R = 0.0821 L atm K^{-1} mol^{-1})

(Rounded off to the nearest integer)

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32. The mass of ammonia in grams produced when 2.8 kg of dinitrogen quantitatively reacts with 1 kg of dihydrogen is _____. **(2020)**

33. The minimum number of moles of O_2 required for complete combustion of 1 mole of propane and 2 moles of butane is _____. **(2020)**

34. The average molar mass of chlorine is 35.5 g mol^{-1}. The ratio of ^{35}Cl to ^{37}Cl in naturally occuring chlorine is close to **(2020)**

(*a*) 1 : 1 (*b*) 2 : 1

(*c*) 3 : 1 (*d*) 4 : 1

35. A solution of two components containing n_1 moles of the 1st component and n_2 moles of the 2nd component is prepared. M_1 and M_2 are the molecular weights of component 1 and 2 respectively. If d is the density of the solution in g mL^{-1}, C_2 is the molarity and x_2 is the mole fraction of the 2nd component, then C_2 can be expressed as **(2020)**

(*a*) $C_2 = \frac{1000x_2}{M_1 + x_2(M_2 - M_1)}$

(*b*) $C_2 = \frac{1000\,d\,x_2}{M_1 + x_2(M_2 - M_1)}$

(*c*) $C_2 = \frac{d\,x_2}{M_2 + x_2(M_2 - M_1)}$

(*d*) $C_2 = \frac{dx_1}{M_2 + x_2(M_2 - M_1)}$

36. If 80 g of copper sulphate $CuSO_4 \cdot 5H_2O$ is dissolved in deionised water to make 5L of solution. The concentration of the copper sulphate solution is $x \times 10^{-3}$ mol L^{-1}. The value of x is [Atomic masses Cu: 63.54 U, 8 : 32 U, 0 : 16 U, H: 1 U] **(2021)**

37. The number of atoms in 8g of sodium is $x \times 10^{23}$. The value of x is (Nearest integer)

[Given: $N_A = 6.02 \times 10^{23}$ mol^{-1}, Atomic mass of Na = 23.0u] **(2021)**

38. Sodium oxide reacts with water to produce sodium hydroxide. 20.0 g of sodium oxide is dissolved in 500 mL of water. Neglecting the change in volume, the concentration of the resulting NaOH solution is $y \times 10^{-1}$ M. The value of y is (Nearest integer). **(2021)**

[Atomic mass Na = 23.0, O = 16.0, H = 1.0]

39. The molarity of the solution prepared by dissolving 6.3 g of oxalic acid ($H_2C_2O_4.2H_2O$) in 250 mL of water in mol L^{-1} is $x \times 10^{-2}$. The value of x is (Nearest integer) [Atomic mass: H: 1.0, C: 12.0, O: 16.0] **(2021)**

40. If a rocket runs on a fuel ($C_{15}H_{30}$) and liquid oxygen, the weight of oxygen required and CO_2 released for every litre of fuel respectively are:

(Given: density of the fuel is 0.756 g/mL) **(2022)**

41. 56.0 L of nitrogen gas is mixed with excess of hydrogen gas and it is found that 20 L of ammonia gas is produced. The volume of unused nitrogen gas is found to be____L. **(2022)**

42. Chlorophyll extracted from the crushed green leaves was dissolved in water to make 2 L solution of Mg of concenttration 48 ppm. The number of atoms of Mg in this solution is $x \times 10^{20}$ atoms. The value of x is_____.

(Given : Atomic mass of Mg is 24 g mol^{-1}, $N_A = 6.02 \times 10^{23}$ mol^{-1}) **(2022)**

43. When 800 mL of 0.5 M nitric acid is heated in a beaker, its volume is reduced to half and 11.5 g of nitric acid is evaporated. The molarity of the remaining nitric acid solution is $x \times 10^{-2}$ M.

(Molar mass of nitric acid is 63 g mol^{-1}) **(2022)**

JEE ADVANCED

44. Reaction of Br_2 with Na_2CO_3 in aqueous solution gives sodium bromide and sodium bromate with evolution of CO_2 gas. The number of sodium bromide molecules involved in the balanced chemical equation is _____. **(2012)**

45. Dissolving 120 g of urea (mol. wt. 60) in 1000 g of water gave a solution of density 1.15 g/mL. The molarity of the solution is: **(2012)**

(*a*) 1.78 M (*b*) 2.00 M

(*c*) 2.05 M (*d*) 2.22 M

46. 29.2% (w/w) HCl stock solution has a density of 1.25 g mL^{-1}. The molecular weight of HCl is 36.5 g mol^{-1}. The volume (mL) of stock solution required to prepare a 200 mL solution of 0.4 M HCl is _____. **(2012)**

47. A compound H_2X with molar weight of 80 g is dissolved in a solvent having density of 0.4 g ml^{-1}. Assuming no change in volume upon dissolution, the **molality** of a 3.2 molar solution is _____. **(2014)**

48. The mole fraction of a solute in a solution is 0.1. At 298 K, molarity of this solution is the same as its molality. Density of this solution at 298 K is 2.0 g cm^{-3}. The ratio of molecular weight of the solute and solvent, $\left(\frac{MW_{solute}}{MW_{solvent}}\right)$, is _____. **(2016)**

49. In neutral or faintly alkaline solution, 8 moles of permanganate anion quantitatively oxidize thiosulphate anion to produce X moles of a sulphur containing product. The magnitude of X is **(2016)**

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50. The ammonia prepared by treating ammonium sulphate with calcium hydroxide is completely used by $NiCl_2.6H_2O$ to form a stable coordination compound. Assume that both the reactions are 100% complete. If 1584 g of ammonium sulphate and 952 g of $NiCl_2.6H_2O$ are used in the preparation, the combined weight (in grams) of gypsum and the nickel-ammonia coordination compound thus produced is_____.

(Atomic weights in g mol^{-1} : H = 1, N = 14, O = 16, S = 32, Cl = 35.5, Ca = 40, Ni = 59) **(2018)**

51. The mole fraction of urea in an aqueous urea solution containing 900 g of water is 0.05. If the density of the solution is 1.2 g cm^{-3}, the molarity of urea solution is ___. **(2019)**

(Given data: Molar masses of urea and water are 60 g mol^{-1} and 18 g mol, respectively)

52. Aluminium reacts with sulfuric acid to form aluminium sulfate and hydrogen. What is the volume of hydrogen gas in liters (L) produced at 300 K and 1.0 atm pressure, when 5.4 g of aluminium and 50.0 mL of 5.0 M sulfuric acid are combined for the reaction?

(Use molar mass of aluminium as 27.0 g mol^{-1}, R = 0.082 atm L mol^{-1} K^{-1}) **(2020)**

53. To check the principle of multiple proportions, a series of pure binary compounds (P_mQ_n) were analyzed and their composition is tabulated below. The correct option(s) is(are) **(2022)**

Compound	Weight % of P	Weight % of Q
1	50	50
2	44.4	55.6
3	40	60

(*a*) If empirical formula of compound 3 is P_3Q_4. then the empirical formula of compound 2 is P_2A_5-.

(*b*) If empirical formula of compound 3 is P_2Q_5 and atomic weight of clement P is 20, then the atomic weight of Q is 45.

(*c*) If empirical formula of compound 2 is PQ, then the empirical formula of the compound 1 is P_5Q_4.

(*d*) If atomic weight of P and Q are 70 and 35, respectively, then the empirical formula of compound 1 is P_2Q.

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ANSWER KEY

CONCEPT APPLICATION

1. (*a*)	**2.** (*a*)	**3.** (*d*)	**4.** (*a*)	**5.** (*a*)	**6.** (*a*)	**7.** (*b*)	**8.** (*b*)	**9.** (*c*)	**10.** (*a*)
11. (*b*)	**12.** (*c*)	**13.** (*a,b*)	**14.** (*a*)	**15.** (*b*)	**16.** (*b*)	**17.** (*c*)	**18.** (*c*)	**19.** (*b*)	**20.** (*c*)
21. (*d*)	**22.** (*a*)	**23.** (*b*)	**24.** (*a*)	**25.** (*c*)	**26.** (*d*)	**27.** (*d*)	**28.** (*b*)	**29.** (*a*)	**30.** (*c*)
31. (*b*)	**32.** (*a*)	**33.** (*d*)	**34.** (*b*)	**35.** (*b*)					

EXERCISE-1 (TOPICWISE)

1. (*a*)	**2.** (*c*)	**3.** (*a*)	**4.** (*b*)	**5.** (*b*)	**6.** (*a*)	**7.** (*d*)	**8.** (*a*)	**9.** (*a*)	**10.** (*a*)
11. (*b*)	**12.** (*b*)	**13.** (*a*)	**14.** (*a*)	**15.** (*c*)	**16.** (*c*)	**17.** (*d*)	**18.** (*b*)	**19.** (*a*)	**20.** (*a*)
21. (*d*)	**22.** (*c*)	**23.** (*b*)	**24.** (*b*)	**25.** (*c*)	**26.** (*c*)	**27.** (*c*)	**28.** (*d*)	**29.** (*c*)	**30.** (*b*)
31. (*c*)	**32.** (*a*)	**33.** (*d*)	**34.** (*a*)	**35.** (*d*)	**36.** (*d*)	**37.** (*b*)	**38.** (*a*)	**39.** (*d*)	**40.** (*c*)
41. (*c*)	**42.** (*b*)	**43.** (*c*)	**44.** (*c*)	**45.** (*a*)	**46.** (*c*)	**47.** (*a*)	**48.** (*a*)	**49.** (*c*)	**50.** (*a*)
51. (*a*)									

EXERCISE-2 (LEARNING PLUS)

1. (*c*)	**2.** (*a*)	**3.** (*c*)	**4.** (*a*)	**5.** (*c*)	**6.** (*b*)	**7.** (*a*)	**8.** (*a*)	**9.** (*a*)	**10.** (*d*)
11. (*c*)	**12.** (*a*)	**13.** (*a*)	**14.** (*b*)	**15.** (*a*)	**16.** (*b*)	**17.** (*b*)	**18.** (*a*)	**19.** (*a*)	**20.** (*a*)
21. (*a*)	**22.** (*a*)	**23.** (*d*)	**24.** (*b*)	**25.** (*c*)	**26.** (*b*)	**27.** (*b*)	**28.** (*a*)	**29.** (*c*)	**30.** (*b*)
31. (*c*)	**32.** (*a*)	**33.** (*b*)	**34.** (*d*)	**35.** (*d*)	**36.** (*b*)	**37.** (*b*)	**38.** (*c*)	**39.** (*b*)	**40.** (*c*)
41. (*a*)	**42.** (*c*)	**43.** (*a*)	**44.** (*c*)	**45.** (*c*)	**46.** (*c*)	**47.** (*a*)	**48.** (*a*)	**49.** (*b*)	**50.** (*c*)

EXERCISE-3 (JEE ADVANCED LEVEL)

1. (*a,b,c*)	**2.** (*b,c,d*)	**3.** (*a,c*)	**4.** (*a,c*)	**5.** (*a,b,c*)	**6.** (*a,c*)	**7.** (*b,d*)	**8.** (*a,c*)	**9.** (*a,b*)	**10.** (*b,d*)
11. (*a,b*)	**12.** (*a,c,d*)	**13.** (*a,b,d*)	**14.** (*c*)	**15.** (*b*)	**16.** (*b*)	**17.** (*a*)	**18.** (*b*)	**19.** (*b*)	**20.** (*a*)
21. (*b*)	**22.** (*d*)	**23.** [+8]	**24.** [–3]	**25.** [500]	**26.** [40]	**27.** [6g]	**28.** [1]	**29.** [22g]	**30.** [1]
31. [27g]									

EXERCISE-4 (PAST YEAR QUESTIONS)

JEE Main

1. (*b*)	**2.** (*c*)	**3.** (*c*)	**4.** (*d*)	**5.** (*a*)	**6.** (*c*)	**7.** (*a*)	**8.** (*b*)	**9.** (*d*)	**10.** (*c*)
11. (*c*)	**12.** (*c*)	**13.** (*b*)	**14.** (*c*)	**15.** (*d*)	**16.** (*d*)	**17.** (*b*)	**18.** (*b*)	**19.** (*c*)	**20.** (*c*)
21. (*c*)	**22.** (*d*)	**23.** (*c*)	**24.** (*a*)	**25.** [26.80 to 27.00]		**26.** [2120.00 to 2140.00]		**27.** [100.00 to 100.00]	
28. [25.00]	**29.** (*a*)	**30.** [47]	**31.** [100]	**32.** [3400]	**33.** [18]	**34.** (*c*)	**35.** (*b*)	**36.** [64]	**37.** [2]
38. [13]	**39.** [20]	**40.** [2592 g and 2376 g]		**41.** [46]	**42.** [24]	**43.** [54]			

JEE Advanced

44. [5]	**45.** (*c*)	**46.** (8 mL)	**47.** [8]	**48.** [9]	**49.** [6]	**50.** [29.92]	**51.** [2.98 or 2.99]	**52.** [6.15]
53. (*b,c*)								

CPROBE

CHAPTER

2 Atomic Structure

DALTON'S ATOMIC THEORY

Dalton proposed the atomic theory on the basis of the law of conservation of mass and law of definite proportions. He also proposed the law of multiple proportion as a logical consequence of this theory. The salient features of this theory are:

(*a*) Each element is composed by extremely small particles called atoms.

(*b*) Atoms of a particular element are all like but differ with the atoms of other elements.

(*c*) Atom of each element is an ultimate particle, and has a characteristic mass but is structureless.

(*d*) Atom is indestructible i.e. it can neither be created nor destroyed by simple chemical reactions.

(*e*) Atom of an element takes part in chemical reaction to form molecule.

(*f*) In a given compound, the relative number and kind of atom are same.

(*g*) Atoms of different elements combine in fixed ratio of small whole numbers to form compound atoms (now called molecules).

Merits and Demerits of Dalton's Theory

(i) Merits:

(*a*) Dalton's theory explains the law of conservation of mass and some other laws of chemical combination.

(*b*) Atoms of elements take part in chemical reaction is true till today.

(ii) Demerits:

(*a*) There is no mention of atomic weights of elements.

(*b*) He could not explain that why do atoms of same element combined with each other.

(*c*) The law of definite proportion fails if different isotopes are used.

FUNDAMENTAL PARTICLES

(a) Properties of Electron:

(*a*) Electron was discovered by Sir J.J. Thomson.

(*b*) The charge on the electron is -1.6×10^{-19} coulomb (Millikan)

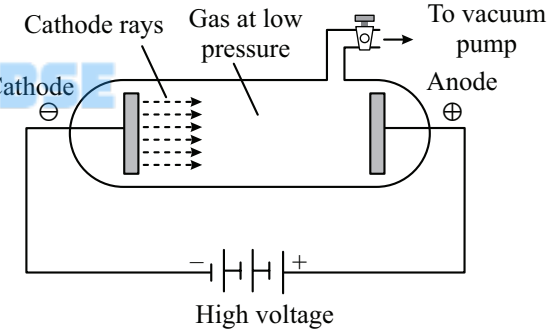
(*c*) The molar mass of electron is 5.48×10^{-4} gm/mole.

(*d*) The value of e/m has been found to be -1.7588×10^{11} C/kg.

(*e*) The first precise measurement of the charge on the electron was made by Robert A. Millikan. in 1909 by oil drop experiment. Its value was found to be -1.6022×10^{-19} coulomb.

(*f*) The mass of electron can be calculated from the value of e/m and the value of which is 9.1096×10^{-31} Kg.

Cathode Rays



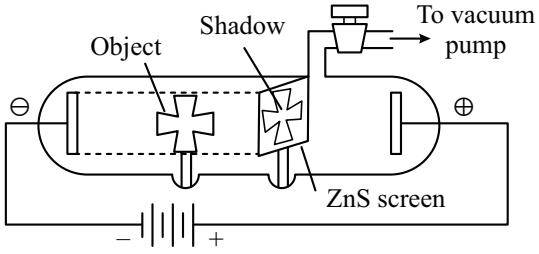
(*a*) The electron was discovered as a result of the studies of the passage of electricity through gases at extremely low pressures known as discharge tube experiments.

(*b*) When a high voltage of the order of 10,000 volts or more was impressed across the electrodes, some sort of invisible rays moved from the negative electrode to the positive electrodes these rays are called as cathode rays.

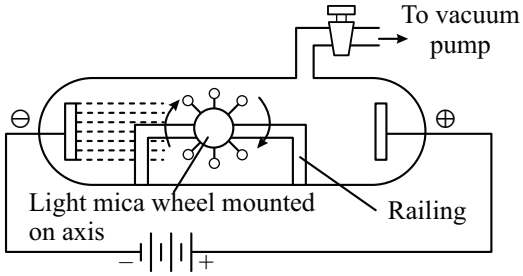
(*c*) Cathode rays have the following properties.

(*i*) Path of travelling is straight from the cathode with a very high velocity.

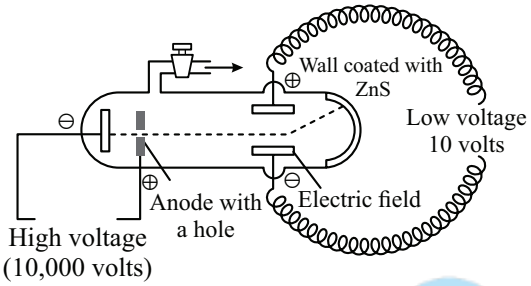
As it produces shadow of an object placed in its path.



(*ii*) Cathode rays produce mechanical effects. If a small pedal wheel is placed between the electrodes, it rotates. This indicates that the cathode rays consist of material part rays possess heating effect.



(*iii*) When electric and magnetic fields are applied to the cathode rays in the discharge tube, the rays are deflected thus establishing that they consist of charged particles.



(*iv*) Cathode rays produce X-rays when they strike against hard metals like tungsten, copper etc.

(*v*) When the cathode rays are allowed to strike a thin metal foil, it gets heated up. Thus the cathode rays possess heating effect.

(*vi*) They produce a green glow when strike the glass wall beyond the anode. Light is emitted when they strike the zinc sulphide screen.

(*vii*) They produce a green glow when strike the glass wall beyond the anode. Light is emitted when they strike the zinc sulphide screen.

(*viii*) Cathode rays penetrate through thin sheets of aluminium and other metals.

(*ix*) They affect the photographic plates.

(*x*) The ratio of charge to mass i.e. charge/mass is same for all the cathode rays irrespective of the gas used in the tube.

(b) Properties of Proton

(*a*) Proton was discovered by Goldestein.

(*b*) Proton carries a charge of $+1.602 \times 10^{-19}$ coulomb, i.e., one unit positive charge.

(*c*) Mass of proton is 1.672×10^{-27} kg or 1.0072 amu.

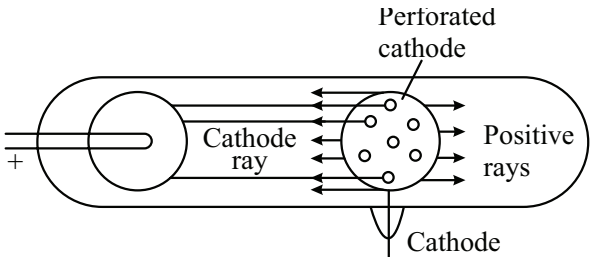
(*d*) A proton is defined as a sub-atomic particle which has a mass nearly 1 amu and a charge of +1 unit.

Positive Rays-Discovery of Proton

(*a*) The existence of positively charged particles in an atom was shown by E. Goldstein in 1886.

(*b*) He repeated the same discharge tube experiments by using a perforated cathode.

(*c*) It was observed that when a high potential difference was applied between the electrodes, not only cathode rays were produced but also a new type of rays were produced simultaneously from anode moving towards cathode and passed through the holes or canal of the cathode. These termed as canal ray or cathode ray.



(*d*) Characteristics of Anode Rays are as follows.

(*i*) These rays travel in straight lines and cast shadow of the object placed in their path.

(*ii*) The anode rays are deflected by the magnetic and electric fields like cathode rays but direction is different that mean these rays are positively charged.

(*iii*) These rays have kinetic energy and produces heating effect also.

(*iv*) The e/m ratio of for these rays is smaller than that of electrons.

(*v*) Unlike cathode rays, their e/m value is dependent upon the nature of the gas taken in the tube.

(*vi*) These rays produce flashes of light on ZnS screen.

(*vii*) These rays can pass through thin metal foils.

(*viii*) They are capable to produce ionisation in gases.

(*ix*) They can produce physical and chemical changes.

(c) Properties of Neutron

(*i*) This was discovered 20 years after the structure of atom was elucidated by Rutherford.

(*ii*) It has been found that for all atoms except hydrogen atomic mass is more than the atomic number. Thus Rutherford (1920) suggested that in an atom, there must be present at least a third type of fundamental particle.

(*iii*) It should be electrically neutral and posses mass nearly equal to that of proton. He proposed its name as neutron.

(*iv*) Chadwick (1932), bombarded beryllium with a stream of α-particles and observed electrically and magnetically neutral radiations.

(*v*) There were neutral particles which was called neutron. Nuclear reaction is as follows:

$$_4Be^9 + {}_2He^4 \longrightarrow {}_6C^{12} + {}_0n^1$$

(*vi*) A neutron is a subatomic particle which has a mass 1.675×10^{-24}g, approximately 1 amu, or nearly equal to the mass of proton on hydrogen atom and carrying no electrical charge.

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Train Your Brain

Example 1: Which has highest e/m ratio:

(*a*) He^{2+} (*b*) H^+

(*c*) He^{1+} (*d*) H

Sol. (*b*) Mass of H^+ is minimum

Example 2: Arrange the following particles in increasing order of values of e/m ratio: Electron (*e*), proton (p), neutron (n) and α-particle (α):

(*a*) n, p, e, a (*b*) n, a, p, e

(*c*) n, p, a, e (*d*) e, p, n, a

Sol. (*b*)

	Electron	Proton	Neutron	α-particle
e	1 unit	1 unit	zero	2 units
m	1/1837 unit	1 unit	1 unit	4-units
e/m	1837	1	zero	1/2

Example 3: Mass of neutron is ..... times the mass of electron:

(*a*) 1840 (*b*) 1480

(*c*) 2000 (*d*) None of these

Sol. (*a*) Mass of neutron = 1.675×10^{-27} kg, mass of electron = 9.108×10^{-31} kg.

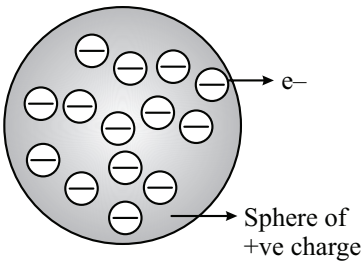
Concept Application

1. The mass of cathode ray particle is:
   (*a*) Same for different gases
   (*b*) Different for different gases
   (*c*) Minimum for H_2 gas
   (*d*) Different for same gases
2. The ratio of the "e/m" (specific charge) values of a electron and an α-particle is -
   (*a*) 2 : 1 (*b*) 1 : 1
   (*c*) 1 : 2 (*d*) None of these

THOMSON'S MODEL OF ATOM (1904)

Thomson was the first to propose a detailed model of the atom.

Thomson proposed that an atom consists of a uniform sphere of positive charge in which the electrons are distributed more or less uniformly.



This model of atom is known as "Plum-Pudding model" or "Raisin Pudding Model" or "Water Melon Model".

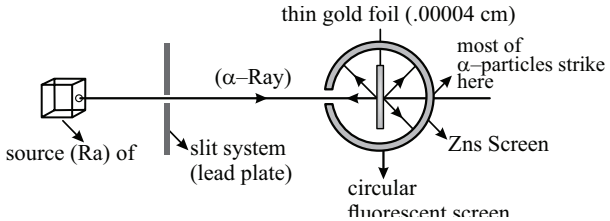
Drawbacks: An important drawback of this model is that the mass of the atoms is considered to be evenly spread over that atom.

It is a static model. It does not reflect the movement of electron.

It couldn't explain the stability of an atom.

RUTHERFORD'S SCATTERING EXPERIMENT

α-scattering experiment



Rutherford Observed that

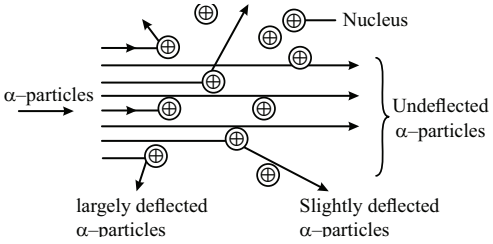
(*i*) Most of the α-particles (nearly 99.9%) went straight without suffering any deflection.

(*ii*) A few of them got deflected through small angles.

(*iii*) A very few (about one in 20,000) did not pass through the foil at all but suffered large deflections (more than 90°) or even came back in the direction from which they have come i.e. a deflection of 180°.

Following conclusions were drawn from the above observations:

(1) Since most of the α-particle went straight through the metal foil undeflected, it means that there must be very large empty space within the atom.



(2) Since few of the α-particles were deflected from their original paths through moderate angles; it was concluded that whole of the +ve charge is concentrated and the space occupied by this positive charge is very small in the atom.

When α-particles come closer to this point, they suffer a force of repulsion and deviate from their paths.

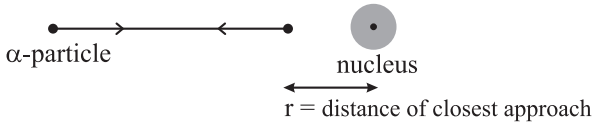
The positively charged heavy mass which occupies only a small volume in an atom is called **nucleus**. It is supposed to be present at the centre of the atom.

A very few of the α-particles suffered strong deflections on even returned on their path indicating that the nucleus is rigid and α-particles recoil due to direct collision with the heavy positively charged mass.

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DISTANCE OF CLOSEST APPROACH

Rutherford estimated the size of nucleus by calculating the distance of closest approach.



The initial kinetic energy of α-particle must be equal to potential energy at distance of closest approach.

$$\text{K.E.} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q_1.q_2}{r} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{(2e).(Ze)}{r}$$

$$\boxed{r = \frac{2Ze^2}{(4\pi\varepsilon_0)(\text{K.E.})}}$$

where Z = atomic number of element used in foil.

(iv) Almost all mass of the atom is densely concentrated in extremely small region (nucleus).

(v) The nucleus is surrounded by electrons that move around the nucleus with a very high speed in circular paths called orbits. Electrons and the nucleus are held together by electrostatic forces of attraction.

Train Your Brain

Example 4: An α-particles of kinetic energy of 10 MeV is projected towards gold nucleus. Calculate the distance of closet approach. (Atomic number of gold = 79, 1 eV = 1.6 × 10^{-19} J)

Sol. $\text{K.E.} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q_1q_2}{r}$

or $10 \times 10^6 \times 1.6 \times 10^{-19} = 9 \times 10^9 \times \frac{(2\times1.6\times10^{-19})\times(79\times1.6\times10^{-19})}{r}$

$$\frac{9\times10^9\times2\times1.6\times10^{-19}\times79}{10\times10^6} = 2.88\times10^{-16}\,\text{m}$$

Example 5: An a-particles is projected towards the following nucleus with same kinetic energy in different experiement the distance of closet approch is maximum for

(*a*) Na (Z = 11) (*b*) Ca (Z = 20)
(*c*) Ag (Z = 47) (*d*) Au (Z = 79)

Sol. (*a*) $K.E = \frac{kq_1q_2}{r}$

q_1 = α-particle

q_1 = charge on given element

therefor $r = \frac{Kq_1q_2}{K.E.}$

if K.E. = constant & $q_1 = 2e^-$

then $r \propto q_2$ total charge on other element.

Example 6: An a-particle, a proton, a deutron and a neutron are projected towards the same nucleus with the same kinetic energy in different experiement. The distance of closet approch is minimum for

(*a*) a (*b*) P (*c*) d (*d*) n

Sol. (*d*) $K.E. = \frac{kq_1q_2}{r}$

or $r = \frac{kq_1q_2}{K.E.}$

K.E. = constant

q_1 = charge on bombarded particle

q_2 (constant) = charge on target particle

lower the charge on projected particle, minumum will be closest approach.

Example 7: An a-particle having K.E. = 7.7 MeV is scattered by gold (Z = 79) nucleus through 180°. Find distance of closest approach.

Sol. K.E. = 7.7 M eV

$= 7.7 \times 10^6 \times 1.6 \times 10^{-19}$ J

$K.E. = \frac{kq_1q_2}{r}$ or $r_0 = \frac{Kq_1q_2}{K.E.}$

$$r_0 = \frac{9\times10^9\times2\times(1.6\times10^{-19})^2\times79}{7.7\times10^6\times1.6\times10^{-19}}m$$

$r_0 = 2.95 \times 10^{-14}m$ or $r_0 \approx 3.0 \times 10^{-14}\,m$.

From the above example it is clear that nuclear dimension cannot be greater than 3×10^{-14}m.

Concept Application

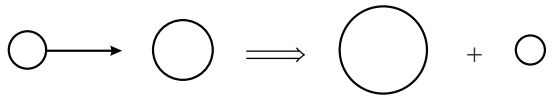
3. The approximate radius of a H-atom is 0.05 nm, and that of proton is 1.5×10^{-15} m. Assuming both the hydrogen atom and the proton to be spherical, calculate fraction of the space in an atom of hydrogen that is occupied by the nucleus.

4. Rutherford's alpha particle scattering experiment eventually led to the conclusion that

(*a*) mass and energy are related
(*b*) electrons occupy space around the nucleus
(*c*) neutrons are buried deep in the nucleus
(*d*) the point of impact with matter can be precisely determined

NEUTRON

In 1932, **chadwick** bombarded beryllium with a stream of α-particles. He observed that penetrating radiations were produced which were not affected by electric & magnetic fields. These radiations consists of neutral particles, which were called **neutrons**. The nuclear reaction can be shown as.



α–Particle	Be-atom	C-atom	Neutron
Charge = +2	Atomic No. = 4	Atomic No. = 6	Charge = 0
Mass = 4 amu	Mass = 9 amu	Mass = 12 amu	Mass = 1 amu

$[^4_2He + [^9_4Be \longrightarrow [^{12}_6C + [^1_0n$

Thus a neutron is a sub atomic particle which has a mass 1.675×10^{-24} g approximately 1amu, or nearly equal to the mass of proton or hydrogen atom and carrying no electrical charge.

The specific charge (e/m value) of a **neutron** is zero.

APPLICATIONS OF RUTHERFORD MODEL

On the basis of scattering experiments, Rutherford proposed model of the atom, which is known as nuclear atomic model. According to this model-

$$\frac{D_A}{D_N} = \frac{\text{Diameter of the atom}}{\text{Diameter of the nucleus}} = \frac{10^{-8}}{10^{-13}} = 10^5,$$

$R \propto A^{1/3} \Rightarrow R = R_0 A^{1/3}$ cm

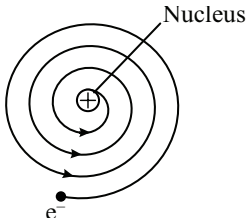
Where $R_0 = 1.33 \times 10^{-13}$ (a constant) and A = mass number (p + n) and R = radius of the nucleus.

$$\boxed{R = 1.33 \times 10^{-13}\ A^{1/3}\ \text{cm}}$$

$$\frac{\text{volume of the atom}}{\text{volume of the nucleus}} = \frac{(10^{-8})^3}{(10^{-13})^3} = 10^{15}$$

Drawbacks of Rutherford Model

(1) This theory could **not** explain stability of atom. According to Maxwell, electron loses it energy continuously in the form of electromagnetic radiations. As a result of this, the e^- should loss energy at every turn and move closer and closer to the nucleus following a spiral path. The ultimate result will be that it will fall into the nucleus, thereby making the atom unstable.



(2) If the electrons loss energy continuously, the observed spectrum should be continuous but the actual observed spectrum consists of well defined lines of definite frequencies. Hence, the loss of energy by electron is **not** continuous in an atom.

ATOMIC NUMBER (Z) AND MASS NUMBER (A)

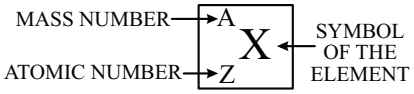
Atomic number (Z) of an element

= Total number of protons present in the nucleus.

= Total number of electrons present in the neutral atom.

- Since the electrons have negligible mass, the entire mass of the atom is mainly due to protons and neutrons only. Since these particles are present in the nucleus, therefore they are collectively called **nucleons**.
- As each of these particles has one unit mass on the atomic mass scale, therefore the sum of the number of protons and neutrons will be nearly equal to the mass of the atom.

  Mass number of an element = No. of protons (Z) + No. of neutrons (n).
- The mass number of an element is nearly equal to the atomic mass of that element.
- The atomic number (Z) and mass number (A) of an element 'X' are usually represented along with the symbol of the element as



(*i*) Isotopes: Such atoms of the same element having same atomic number but different mass numbers are called isotopes.

^1_1H, ^2_1D and ^3_1T and named as protium, deuterium (D) and tritium (T) respectively. Ordinary hydrogen is protium.

(*ii*) Isobars: Such atoms of different elements which have same mass numbers (and of course different atomic numbers) are called isobars.

e.g. $^{40}_{18}Ar$, $^{40}_{19}K$, $^{40}_{20}Ca$.

(*iii*) Isotones: Such atoms of different elements which contain the same number of neutrons are called isotones.

e.g. $^{14}_6C$, $^{15}_7N$, $^{16}_8O$.

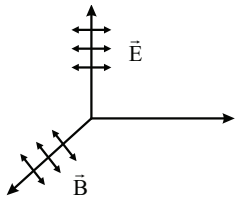
(*iv*) Isoelectronic: The species (atoms or ions) containing the same number of electrons are called isoelectronic.

For example, O^{2-}, F^-, Na^+, Mg^{2+}, Al^{3+}, Ne all contain 10 electrons each and hence they are isoelectronic.

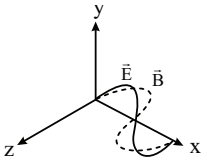
WAVES AND IT'S CHARACTERISTICS

Electromagnetic Wave Radiation

The oscillating electrical/magnetic field are electromagnetic radiations. Experimentally, the direction of oscillations of electrical and magnetic field are perpendicular to each other.

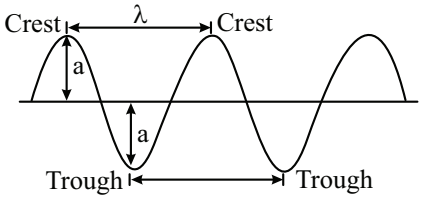


$\vec{E}$ = Electric field, $\vec{B}$ = Magnetic field



Direction of wave propogation.

Some important characteristics of a wave:



(i) Wavelength of a wave is defined as the distance between any two consecutive crests or troughs. It is represented by λ (Lambda) and is expressed in Å or m or cm or nm (nanometer) or pm (picometer).
1 Å = 10^{-8} cm = 10^{-10} m
1 nm = 10^{-9} m, 1 pm = 10^{-12} m

(ii) Frequency of a wave is defined as the number of waves passing through a point in one second. It is represented by ν (nu) and is expressed in Hertz (Hz) or cycles/sec or simply sec^{-1} or s^{-1}.
1 Hz = 1 cycle/sec

(iii) Velocity of a wave is defined as the linear distance travelled by the wave in one second. It is represented by v and is expressed in cm/sec or m/sec (ms^{-1}).

(iv) Amplitude of a wave is the height of the crest or the depth of the trough. It is represented by 'a' and is expressed in the units of length.

(v) Wave number is defined as the number of waves present in 1 cm length. Evidently, it will be equal to the reciprocal of the wavelength. It is represented by $\bar{\nu}$ (read as nu bar).

$$\bar{\nu} = \frac{1}{\lambda}$$

If λ (Lambda) is expressed in cm, $\bar{\nu}$ will have the units cm^{-1}.

Relationship between velocity, wavelength and frequency of a wave. As frequency is the number of waves passing through a point per second and λ is the length of each wave, hence their product will give the velocity of the wave.

Order of wavelength in Electromagnetic spectrum:

Cosmic rays < γ – rays < X-rays < Ultraviolet rays < Visible < Infrared < Micro waves < Radio waves.

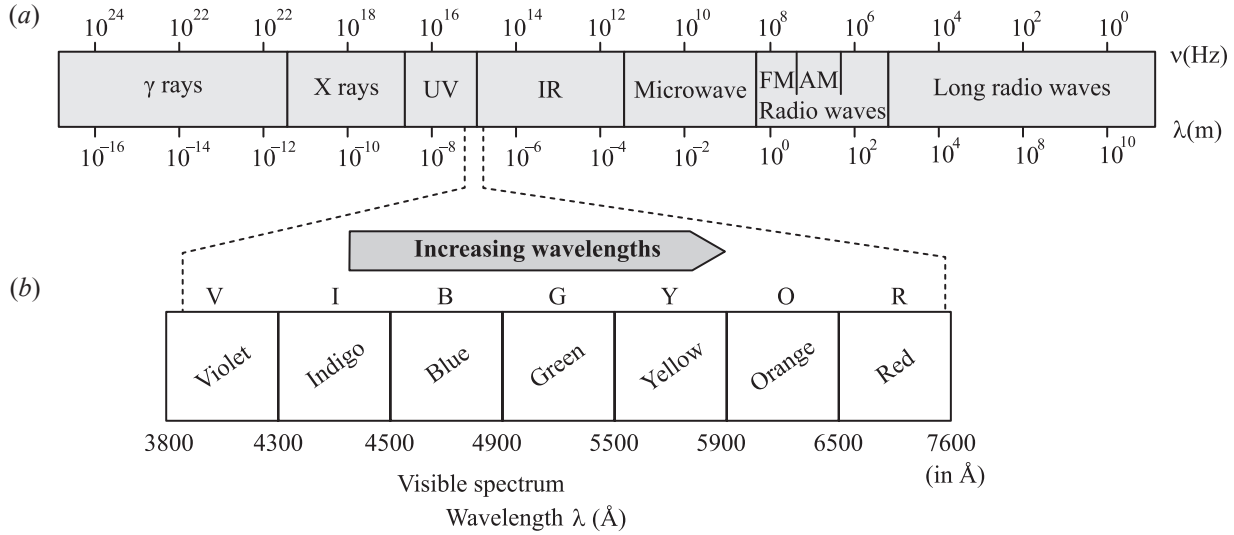
NATURE OF LIGHT

- **Maxwell electromagnetic wave theory (wave nature of light):** An accelerated electrically charged particle produces and transmits electrical and magnetic field. These are transmitted in the form of waves known as electromagnetic waves or electromagnetic radiations.
  He stated that light also possess electrical and magnetic field and it is also known as electromagnetic radiations.
- **Characteristics of Electromagnetic Radiations:**
  (*i*) In these electromagnetic radiation electrical and magnetic field oscillates perpendicular to each other and it also propagates perpendicular to both field.
  (*ii*) All these electromagnetic radiation do not require any medium and can travel in vacuum.
  (*iii*) Velocity of all electromagnetic radiation is 3×10^8 m/s in vacuum.
  (*iv*) Energy of an electromagnetic wave is directly proportional to intensity and it is independent of frequency.
  (*v*) They also show diffraction and interference and therefore, Maxwell concluded light to be wave nature.
  But Maxwell theory couldn't explain the results of photoelectric effect and black body radiations.

ELECTRO MAGNETIC SPECTRUM

The arrangement of various types of electromagnetic radiations in the order of their increasing or decreasing wavelength or frequencies is known as electromagnetic spectrum. The wavelength decreases in the following order.

Radio waves > microwaves > Infrared > Television waves > Visible > Ultraviolet > X-rays > γ-rays > cosmic rays



(a) The spectrum of electromagnetic radiation.
(b) Visible spectrum.
The visible region is only a small part of the entire spectrum.

PLANCK'S QUANTUM THEORY

Particle nature of light: He stated that a body radiates energy in the form of discontinuous energy packets or bundles. Each bundle of energy is known as quantum and quantum of light is known as photons.

Energy of each quantum is directly proportional to frequency of radiation.

$E \propto \nu$

$E = h\nu$ $\quad h = 6.62 \times 10^{-34}$ J-s

↓

Planck's constant

Total energy absorbed or emitted by a body will be whole no. integral multiple of energy of quantum.

$$E = \frac{nhc}{\lambda}$$

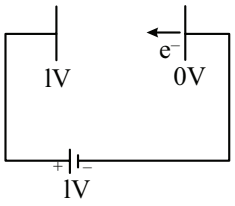
One electron volt (e.v.): Energy gained by an electron when it is accelerated from rest through a potential difference of 1 volt.

Note: Positive charge always moves from high potential to low potential and –ve charge always moves from low potential to high potential if set free.

If a charge 'q' is accelerated through a potential difference of 'V' volt then its kinetic energy will be increased by q × V.

$1eV = 1.6 \times 10^{-19}$ C × 1 volt

$\therefore$ $1eV = 1.6 \times 10^{-19}$J



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Example 8: Calculate the no. of photons emitted by 60 watt bulb in 10 hrs. When light of wavelength 6000 Å is emitted by it.

Sol. $P = \frac{E}{t} = \frac{nh\upsilon}{t}$ or $P = \frac{nhc}{\lambda t}$ or $n = \frac{P \times \lambda \times t}{h \times c}$

$$or \quad n = \frac{60 \times 6000 \times 10^{-10} \times 10 \times 60 \times 60}{6.626 \times 10^{-34} \times 3 \times 10^{8}}$$

or $n = 6.52 \times 10^{24}$

Example 9: Certain sun glasses having small of AgCl incorporated in the lenses, on expousure to light of appropriate wavelength turns to gray colour to reduce the glare following the reactions:

$$AgCl \xrightarrow{h\nu} Ag(Gray) + Cl$$

If the heat of reaction for the decomposition of AgCl is 248 kJ mol^{-1}, what maximum wavelength is needed to induce the desired process?

Sol. Energy needed to change = 248×10^3 J/mol

If photon is used for this purpose, then according to Einstein law one molecule absorbs one photon.

Therefore,

$$\therefore \quad N_A \cdot \frac{hc}{\lambda} = 248 \times 10^3$$

$$\lambda = \frac{6.626 \times 10^{-34} \times 3.0 \times 10^{8} \times 6.023 \times 10^{23}}{248 \times 10^{3}}$$

$= 4.83 \times 10^{-7}$ m

Example 10: A bulb is rated as 110 watt. If it emits 25% of absorbed energy as red light (λ = 6626Å), how many photons are emitted out by the bulb per second?

Sol. $P \times \eta = \frac{n \times h \times c}{\lambda \times t}$

$$or \quad 110 \times \frac{25}{100} = n \times \frac{6.626 \times 10^{-34} \times 3 \times 10^{8}}{6626 \times 10^{-10}}$$

or $n = 9.2 \times 10^{19}$

Example 11: Wavelength of micro wave radiation is 0.08 m. How many moles of photons is needed to increase the temperature of 400 gm water from 25° to 45°C, assuming 25% efficiency?

Specific heat capacity = 4.2 J/K.gm

Sol. $\left(\frac{nhc}{\lambda}\right) \times \frac{25}{100} = ms\Delta t$

$$n \times \frac{6.626510^{-34} \times 3 \times 10^{8}}{0.08} \times \frac{25}{100} = 400 \times 4.2 \times 20$$

$\therefore$ Number of moles of photon = $\frac{n}{N_A}$

$$or \quad \text{Number of moles of photon} = \frac{540.86 \times 10^{26}}{6.023 \times 10^{23}}$$

$= 9.0 \times 10^{14}$ mol.

Example 12: A dye absorbs the radiation of 4000 Å and fluoresces the radiation of 5000 Å. If only 40% of the absorbed energy is emitted out, calculate the ratio of number of quanta emitted out and the number of quanta absorbed.

$$E_a \times \frac{40}{100} = E_e$$

$$n_a \times \frac{hc}{4000Å} \times \frac{40}{100} = n_e \times \frac{hc}{5000Å}$$

$$\frac{n_e}{n_a} = \frac{40}{100} \times \frac{5000}{4000} \times \frac{1}{2}$$

$$\frac{n_e}{n_a} = \frac{1}{4}$$

Concept Application

5. Photon of which light has maximum energy:

(*a*) Red (*b*) Blue

(*c*) Violet (*d*) Green

6. The Vividh Bharati Station of All India Radio, Delhi broadcasts on a frequency of 1368 kHz (kilo hertz). Calculate the wavelength and wave number of the electromagnetic radiation emitted by the transmitter.

(*a*) (219.3 m, 4.56×10^{-3} m^{-1})

(*b*) (219.2 m, 4.46×10^{-3} m^{-1})

(*c*) (218.3 m, 4.56×10^{-3} m^{-1})

(*d*) (219.5 m, 4.58×10^{-3} m^{-1})

7. Two bulbs 'A' and 'B' emit red light and yellow light at 8000 Å and 4000Å respectively. The number of photons emitted by both the bulbs per second is the same. If the red bulb is labelled as 100 watts, find the wattage of the yellow bulb.

(*a*) 100 watt (*b*) 200 watt.

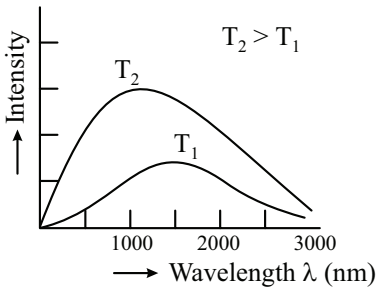
(*c*) 300 watt (*d*) 400 watt.

APPLICATION OF PLANCK'S QUANTUM THEORY

Black Body Radiation:

When solids are heated they emit radiation over a wide range of wavelengths.

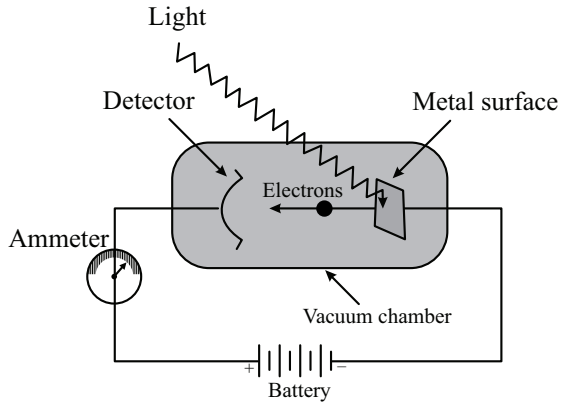
The ideal body, which emits and absorbs all frequencies, is called a black body and the radiation emitted by such a body is called black body radiation. The exact frequency distribution of the emitted radiation (i.e., intensity versus frequency curve of the radiation) from a black body depends only on its temperature.



The above experimental results cannot be explained satisfactorily on the basis of the wave theory of light. Planck suggested that atoms and molecules could emit (or absorb) energy only in discrete quantities.

Photoelectric Effect:

When certain metals (for example Potassium, Rubidium, Caesium etc.) were exposed to a beam of light electrons were ejected as shown in Fig.



The phenomenon is called **Photoelectric effect**. The results observed in this experiment were:

- The electrons are ejected from the metal surface as soon as the beam of light strikes the surface, i.e., there is no time lag between the striking of light beam and the ejection of electrons from the metal surface.
- The number of electrons ejected is proportional to the intensity or brightness of light.

TERMS USED IN PHOTO ELECTRIC EFFECT

1. **Work function (w):** It is the minimum amount of energy required to cause a photo emission from the metal surface. It is also known as threshold energy or Binding energy. Work function depends upon ionisation energy and therefore w is minimum for alkali metals).
2. **Threshold frequency (ν_0):** The minimum value of frequency that can cause photo emissions. If $\nu < \nu_0$, then there is no photo emission.

$$w = h\nu_0 \Rightarrow w = \frac{hc}{\lambda_0}$$

3. **Threshold wavelength:** (λ_0) The maximum value of wavelength that can cause photo emission.

   If $\lambda > \lambda_0$, then photo emission is not possible.
4. **Intensity (I):** Energy falling on metal surface of unit area of unit time.

$$I = \frac{E}{At} = \frac{nh\nu}{At}$$

5. **Photo intensity (I_p):** It is number of photons falling per unit area per unit time.

$$I_P = \frac{n}{At}$$

Relation between I and I_p:

$I = I_p h\nu$

Photo intensity is independent of frequency while intensity depends on frequency.

Power: Total energy radiated per unit time.

$$P = \frac{E}{t} = \frac{nh\nu}{t}$$

@PROGREE

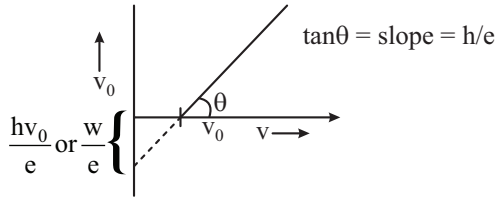
SATURATION CURRENT & STOPPING POTENTIAL

It is the minimum potential require to stop the fastest moving electrons completely or it is the minimum potential at which photo current becomes zero.

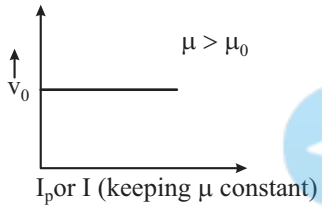
$eV_0 = h\nu - w$

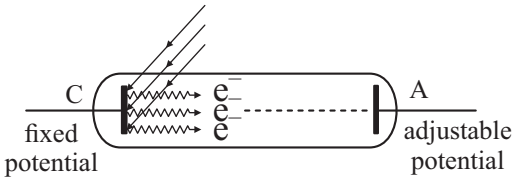
$eV_0 = h\nu - h\nu_0$

$$\Rightarrow V_0 = \frac{h\nu}{e} - \frac{h\nu_0}{e}$$



It can be commented that stopping potential increase with increase in frequency. How ever if photo intensity is changed there is no effect on stopping potential.





Case-I: $V_C = V_A$

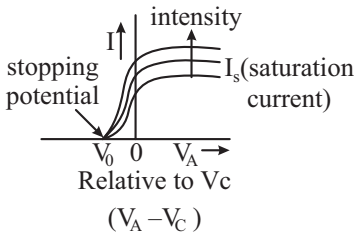
Some of the ejected electrons reach at electrode A resulting photocurrent.

Case-II: $V_C < V_A$

As electrode (A) is at high potential it attracts the electron & even a slower electron will reach at electrode A. It will result in increase in photocurrent. Further increase in the potential difference, a situation may result when the slowest photocurent electron reach at electrode. It results maximum called saturation current. Further increase in potential will not increase photocurrent.

Case-III: $V_C > V_A$

As electrode (A) is at low potential. It will repel electron resulting decrease in photocurrent. Further decrease in potential at electrode (A) may result a situation when the fastest electron just fails to reach at (A) and the photocurrent drops to zero. The potential of (A) relative to C to just stop photocurrent is called stopping potential.

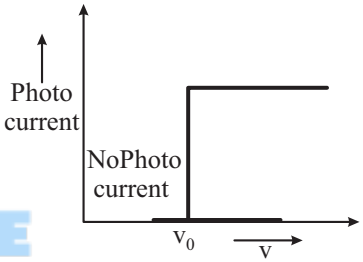


On increasing the intensity of light the stopping potential does not change because the maximum K.E. of photoelectron does not change. But the photocurrent increases, because the number of photon falling on the surface increases.

EFFECT OF VARIATION OF FREQUENCY

(a) Effect of Photo Emission: $I = I_p h\nu$

If frequency of subjected photon increases (intensity increases keeping photo intensity constant) then there is no change in no. of ejected photo electrons as well as no change in photo current.



(b) Effect on kinetic Energy: Average K. E. as well as K. E. max increases with increases in frequency.

$h\nu - W = KE_{max}$.

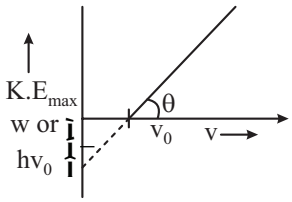
$KE_{max} = h\nu - h\nu_0$

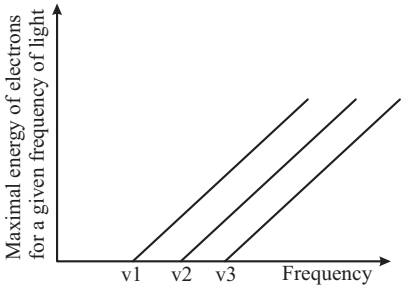
$y = mx + c$

$$KE_{max} = h\nu - \frac{hc}{\lambda_0}$$

$KE_{max} = h\nu - w$

tanθ = slope = h



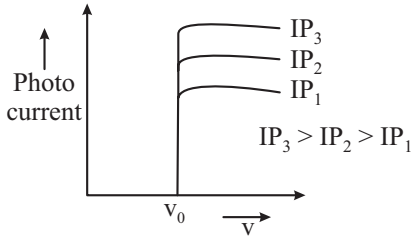


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The diagram shows the interdependence between light frequency and the maximal energy of electrons emitted from metal. It shows the interdependence for three different metals. See that it clearly shows the limiting frequencies different for different metals.

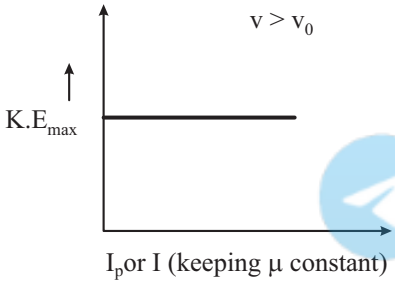
Effect of Variation of Photo Intensity

On increasing intensity, keeping frequency constant (i.e. increasing photo intensity) no. of ejected photo electrons increases as well as photo current increases.



Effect on Kinetic Energy

Average K.E. and $K.E_{max}$ remains constant with change in photo intensity



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Example 13: In ultraviolet light of wavelength 200 nm is used in an experiment of photoelectric effect with lithium cathode (w = 2.5 eV). Then calculate

(*i*) $K.E_{max}$ (*ii*) Stopping potential

Sol. $K.E._{max} = \frac{hc}{\lambda} - w$

$$K.E._{max} = \frac{6.626\times10^{-34}\times3\times10^{8}}{200\times10^{-7}\times1.6\times10^{-19}} - 2.5\ eV$$

$\Rightarrow 6.2 - 2.5$

$\Rightarrow 3.7$ eV

Stopping Potential = 3.7 V

Example 14: A photon of wavelength 3000 Å strikes a metal surface, the work function of the metal being 2.20 eV. Calculate (i) the energy of the photon in eV (ii) the kinetic energy of the emitted photo electron and (iii) the velocity of the photo electron.

Sol. (*i*) Energy of the photon

$$E = h\nu = \frac{hc}{\lambda} = \frac{\left(6.6\times10^{-34}\,Js\right)\left(3\times10^{8}\,ms^{-1}\right)}{3\times10^{-7}\,m} = 6.6\times10^{-19}\,J$$

$1\ ev = 1.6\times10^{-19}\,J$

$$\text{Therefore } E = \frac{6.6x10^{-19}\,J}{1.6\times10^{-19}\,J/eV} = 4.125\ eV$$

(*ii*) Kinetic energy of the emitted photo electron

Work function = 2.20 eV

Therefore, KE = 4.125 – 2.20

= 1.925 eV = 3.08×10^{-19} J

(*iii*) Velocity of the photo electron

$$KE = \frac{1}{2}mv^2 = 3.08\times10^{-19}\,J$$

Therefore, velocity (v) =

$$\sqrt{\frac{2\times3.08\times10^{-19}}{9.1\times10^{-31}}} = 8.22\times10^{5}\,ms^{-1}$$

Example 15: The threshold frequency ν_0 for a metal is 6×10^{14} s^{-1}. Calculate the kinetic energy of an electron emitted when radiation of frequency $\nu = 1.1 \times 10^{15}$ s^{-1} hits the metal.

Sol. K.E. = $\frac{1}{2}m_eV^2 = h(\nu - \nu_0)$

$\therefore$ K.E. = $(6.626 \times 10^{-34})(1.1 \times 10^{15} - 6 \times 10^{14})$

= 3.3×10^{-19} J

Concept Application

8. A photon of energy hν is absorbed by a free electron of a metal having work function w < hν. Then :

(*a*) The electron is sure to come out

(*b*) The electron is sure to come out with a kinetic energy (hν – w)

(*c*) Either the electron does not come out or it comes with a kinetic energy (hν – w)

(*d*) It may come out with a kinetic energy less than (hν – w)

9. A beam of white light is dispersed into its wavelength components by a Quartz prism and falls on a thin sheet of potassium metal. What is the correct decreasing order of kinetic energy of the electron emitted by the different light component.

(*a*) blue > green > orange > yellow

(*b*) violet > blue > orange > red

(*c*) yellow > green > blue > violet

(*d*) orange > yellow > blue > violet

10. A laser producing monochromatic lights of different wavelength is used to eject electron from the sheet of gold having threshold frequency 6.15 x 10^{14} s^{-1} which of the following incident radiation will be suitable for the ejection of electrons:

(*a*) 1.5 moles of photons having frequency 3.05×10^{14} s^{-1}

(*b*) 0.5 moles of photon of frequency 12.3×10^{12} s^{-1}

(*c*) One photon with frequency 5.16×10^{15} s^{-1}

(*d*) All of the above

11. The number of photoelectrons emitted depends upon:

(*a*) The intensity of the incident radiation

(*b*) The frequency of the incident radiation

(*c*) The product of intensity and frequency of incident radiation

(*d*) None of these

BOHR'S MODEL

It was the first model based on Planck's quantum theory and the model explained stability of atom and line spectrum of hydrogen.

Postulates of Bohr's Model

(*i*) An atom consist of centrally located small, dense, positively charged nucleus and electrons are revolving around the nucleus in circular paths known as circular orbits and coulombic force of attraction between nucleus and electron is balanced by centrifugal force of the revolving electron.

(*ii*) Out of the infinite circular orbits only those circular orbits are possible in which angular momentum of electron is integral multiple of h/2π i.e. angular momentum of an electron can have fixed values like $\frac{h}{2\pi}, \frac{2h}{2\pi}, \frac{3h}{2\pi}$ etc. i.e., angular momentum of electron is quantised.

$$mvr = \frac{nh}{2\pi}$$

m and v are mass and velocities of electron respectively and r is radius of orbit and n is integer which is later related with orbit number and shell number and h is Plank's constant.

(*iii*) The energy of these circular orbits have fixed values and hence electron in an atom can have only certain values of energy. It is characteristic of an orbit and it cannot have any orbit value of its own. And therefore energy of an electron is also quantised.

(*iv*) As long as electrons remains in these fixed orbits, it dosen't lose energy i.e. energy of an electron is stationary (not changing with time) and therefore these orbits are known as allowed energy levels or stationary states and this explains the stability of atom.

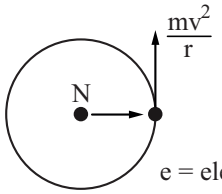
(*v*) The energy levels are designated as K, L, M, N and numbered as 1, 2, 3, 4 etc from nucleus outwards and as the distance of the shells or energy level from the nucleus increases the energy of the energy level also increases i.e., $E_N > E_M > E_L > E_k$

(*vi*) The emission or absorption of energy in the form of radiations can only occur when an electron jump from one stationary orbit to other.

$\Delta E = E_{higher} - E_{lower}$

$\Delta E = h\nu$ where $h\nu$ is the energy of absorbed photon or emitted photon which corresponds to the difference in energy levels. Energy is absorbed when electron jumps from lower energy level (normal state) to higher energy level (exited, unstable state) and energy is emitted when electrons jumps from higher energy level to lower energy level.

- Bohr's model is applicable for one electron species only, like H, He^+, Li^{+2}, Be^{+3} etc.
- Derivation of Radius of different orbits in one electron species (using Bohr's model):



e = electron
N = Nucleus

$$mvr = \frac{nh}{2\pi} \quad ...(i)$$

$q_1 = e,\ q_2 = Ze$

$$\frac{mv^2}{r} = \frac{kq_1q_2}{r^2} = \frac{kze^2}{r^2} \quad ...(ii)$$

$$\Rightarrow \frac{mn^2h^2}{4\pi^2m^2r^2.r} = \frac{kZe^2}{r^2}$$

$$\Rightarrow r = \frac{mn^2h^2}{4\pi^2m^2kZe^2} \Rightarrow r = \frac{n^2h^2}{4\pi^2mkZe^2}$$

$$\Rightarrow \frac{h^2}{4\pi^2mke^2}\times\frac{n^2}{Z}$$

$$= \frac{(6.625\times10^{-34})^2}{4\pi^2\times9.1\times10^{-31}\times9\times10^9\times(1.6\times10^{-19})^2}\times\frac{n^2}{Z}$$

$$r = 0.529 \times \frac{n^2}{z}A^\circ \quad \Rightarrow \quad r \propto \frac{n^2}{z}$$

For a particular atom $r \propto n^2$

Radius of 1st orbit of H atom

r = 0.529 Å

Example 14: Calculate ratio of radius of 2nd orbit of H atom to 3rd orbit of Li^{+2} ion:

Sol. $= \frac{\text{Radius of } 2^{nd} \text{ orbit of H atom}}{\text{Radius of } 3^{rd} \text{ orbit of } Li^{+2} \text{ atom}}$

$$= \frac{n^2}{z}\times\frac{z_1}{n_1^2} = \frac{4}{3}$$

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Derivation of Velocity of Electron in Bohr's Orbit

$v = \frac{nh}{2\pi mr}$, putting value of r.

$v = \frac{nh \times 4\pi^2 mkZe^2}{2\pi mn^2h^2}$

$v = \frac{2\pi kZe^2}{h.n.} = \frac{2\pi ke^2}{h} \times \frac{Z}{n}$

$v = 2.18 \times 10^6 \frac{Z}{n}$ m/s

$v \propto \frac{Z}{n}$

Derivation of Total Energy of Electron/System

T.E. of system = K.E. of e^- + P.E. of system (nucleus and e^-)

kinetic energy of electron $= \frac{1}{2}mv^2 = \frac{1}{2}\frac{kze^2}{r}$

$PE = -\frac{kze^2}{r}$

$T.E = \frac{-KZe^2}{2r} \qquad \left[r = \frac{n^2h^2}{4\pi^2m^2kze^2}\right]$

$T.E. = -13.6 \times \frac{z^2}{n^2}$ eV/atom ...(i)

$T.E. = -2.18 \times 10^{-18} \frac{z^2}{n^2}$ J/atom

As shell no. or distance increases, the value of T.E. and P.E. increases (however magnitude decreases) and becomes maximum at infinity i.e., zero.

Negative sign indicates that electron is under the influence of attractive forces of nucleus.

$K.E. = -\frac{P.E.}{2}$

$T.E. = \frac{P.E.}{2}$

T.E = – K.E.

Calculation of Energy of Energy Level in H atom

(*i*) When n = 1 (ground level)

K.E. = 13.6 eV (atom)

P.E. = –27.2 eV / atom

T.E. = –13.6 eV/atom

(*ii*) When n = 2/2nd energy level / 1st excited state

$K.E. = \frac{13.6}{4}$ eV/atom = 3.4 eV/ atom

P.E. = – 6.8 eV/atom

T.E. = – 3.4 eV/atom

$E_2 - E_1$ = –3.4 + 13.6 eV/atom = 10.2 eV/atom

(*iii*) When n = 3 / 3rd energy level/2nd excited state.

$K.E. = \frac{13.6}{9}$ eV/atom = 1.51 eV/atom

P.E. = –3.02 eV/atom

T.E. = –1.51 eV/atom

$E_3 - E_2$ = –1.51 eV/atom + 3.4 eV/atom

= 1.89 eV atom.

(*iv*) When n = 4 / 3rd excited state

$K.E. = \frac{13.6}{4^2} \Rightarrow \frac{13.6}{16 \times 10} = 0.85$ eV/atom

P.E. = –1.70 eV/atom

T.E. = –0.85 eV/atom

$E_4 - E_3 = 0.66$

As distance increases (n increases) energy of the energy level increases but energy difference between consecutive energy level keeps on decreasing i.e., maximum energy difference between 2 to 1 (consecutive).

Conclusion from Equation of Energy

(*a*) The negative sign of energy indicates that there is attraction between the negatively charged electron and positively charged nucleus.

(*b*) All the quantities on R.H.S. in the energy equation are constant for an element having atomic number Z except 'n' which is an integer such as 1,2,3, etc. i.e. the energy of an electron is constant as long as the value of 'n' is kept constant.

(*c*) The energy of an electron is inversely proportional to the square of 'n' with negative sign.

Failures/Limitations of Bohr's theory

(*a*) He could not explain the line spectra of atoms containing more than one electron.

(*b*) He also could not explain the presence of multiple spectral lines.

(*c*) He was unable to explain the splitting of spectral lines in magnetic field (Zeeman effect) and in electric field (Stark effect).

(*d*) No conclusion was given for the principle of quantisation of angular momentum.

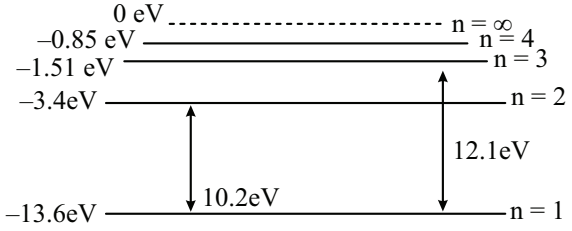
(*e*) He was unable to explain the de-Broglie's concept of dual nature of matter.

(*f*) He could not explain Heisenberg's uncertainty principle.

ENERGY LEVEL DIAGRAM

(*i*) Orbit of lowest energy is placed at the bottom, and all other orbits are placed above this.

(*ii*) The gap between two orbits is proportional to the energy difference of the orbits.



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Definition Valid for Single Electron System

(*i*) Ground state: Lowest energy state of any atom or ion is called ground state of the atom, it is n = 1.

Ground state energy of H–atom = – 13.6 eV

Ground state energy of He^+ Ion = – 54.4 eV

(*ii*) Excited State: States of atom other than the ground state are called excited states:

n = 2	First excited state
n = 3	Second excited state
n = 4	Third excited state
n = (n + 1)	n^{th} excited state

(*iii*) Ionisation energy (IE): Minimum energy required to move an electron from ground state to n = ∞ is called ionisation energy of the atom or ion.

Ionisation energy of H–atom = 13.6 ev

Ionisation energy of He^+ ion = 54.4 ev

Ionisation energy of Li^{+2} ion = 122.4 ev

(*iv*) Ionisation Potential (I.P.): Potential difference through which a free electron must be accelerated from rest, such that its kinetic energy becomes equal to ionisation energy of the atom is called ionisation potential of the atom.

I.P. of H atom = 13.6 V, I.P. of He^+ Ion= 54.4 V

(*v*) Excitation Energy: Energy required to move an electron from ground state of the atom to any other state of the atom is called excitation energy of that state.

Excitation energy of 2nd state = Excitation energy of 1st excited state = 1st excitation energy = 10.2 ev.

(*vi*) Excitation Potential: Potential difference through which an electron must be accelerated from rest to higher energy level. So that its kinetic energy become equal to excitation energy of any state is called excitation potential of that state.

Excitation potential of third state = excitation potential of second exicited state = second excitation potential = 12.09 V.

(*vii*) Binding Energy 'or' Separation Energy: Energy required to move an electron from any state to n = ∞ is called binding energy of that state.

Binding energy of ground state = I.E. of atom or ion.



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Example 16: A single electron system has ionization energy 11180 kJ mol^{-1} . Find the number of protons in the nucleus of the system.

Sol. I.E. = $\frac{Z^2}{n^2} \times 21.69 \times 10^{-19}$ J/atom

$$\frac{11180\times10^3}{6.023\times10^{23}} = \frac{Z^2}{1^2} \times 21.69 \times 10^{-19}$$

or z = 3

Example 17: What are the frequency and wavelength of a photon emitted during a transition from n = 5 state to the n = 2 state in the hydrogen atom?

Sol. Since $n_i = 5$ and $n_f = 2$, this transition gives rise to a spectral line in the visible region of the Balmer series.

$$\Delta E = 2.18 \times 10^{-18} \text{ J} \left[\frac{1}{5^2} - \frac{1}{2^2}\right] = -4.58 \times 10^{-19} \text{ J}$$

It is an emission energy

The frequency of the photon (taking energy in terms of magnitude) is given by

$$\nu = \frac{\Delta E}{h} = \frac{4.58\times10^{-19}\text{ J}}{6.626\times10^{-34}\text{ Js}} = 6.91 \times 10^{14} \text{ Hz and}$$

$$c = \nu\lambda \text{ or } \lambda = \frac{3\times10^8}{6.91\times10^{14}} \text{ or } \lambda = 434\ nm$$

Example 18: The ionization energy of He^+ is 19.6×10^{-18} J $atom^{-1}$. The energy of the first stationary state of Li^{+2} will be

(*a*) 21.2×10^{-18} J/atom (*b*) 44.10×10^{-18} J/atom

(*c*) 63.2×10^{-18} J/atom (*d*) 84.2×10^{-18} J/atom

Sol. (*b*) E_1 for $Li^{+2} = E_1$ for $H \times Z^2_{Li} = E_1$ for $H \times 9$

E_1 for $He^+ = E_1$ for $H \times Z^2_{He} = E_1$ for $H \times 4$

or E_1 for $Li^{+2} = \frac{9}{4}$ E_1 for He^+

$= 19.6 \times 10^{-18} \times \frac{9}{4}$

$= 44.10 \times 10^{-18}$ J/atom



Concept Application

12. Consider Bohr's theory for hydrogen atom. The magnitude of angular momentum, orbit radius and velocity of the electron in nth energy state in a hydrogen atom are *l*, r & v respectively. Find out the value of 'x', if product of v, r and *l* (vr*l*) is directly proportional to n^x.

(*a*) (x = 1) (*b*) (x = 2)

(*c*) (x = 3) (*d*) (x = 4)

13. A single electron ion has nuclear charge + Ze where Z is atomic number and e is electronic charge. It requires 16.52 eV to excite the electron from the second Bohr orbit to third Bohr orbit. Then atonic number, kinetic enrgy of electron in first Bohr orbit respectively:

(*a*) Z = 3, K.E. = 122.4 eV

(*b*) Z = 3, K.E. = 122.2 eV

(*c*) Z = 2, K.E. = 122.4 eV

(*d*) Z = 3, K.E. = 121.4 eV

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14. Correct order of radius of the Ist orbit of H, He^+, Li^{2+}, Be^{3+} is:

(*a*) $H > He^+ > Li^{2+} > Be^{3+}$

(*b*) $Be^{3+} > Li^{2+} > He^+ > H$

(*c*) $He^+ > Be^{3+} > Li^{2+} > H$

(*d*) $He^+ > H > Li^{2+} > Be^{3+}$

15. Match the following

A. Energy of ground state of He^+	(i) + 6.04 eV
B. Potential energy of I orbit of H-atom	(ii) –27.2 eV
C. Kinetic energy of II excited state of He^+	(iii) 54.4 V
D. Ionisation potential of He^+	(iv) – 54.4 eV

(*a*) A – (i), B – (ii), C – (iii), D – (iv)

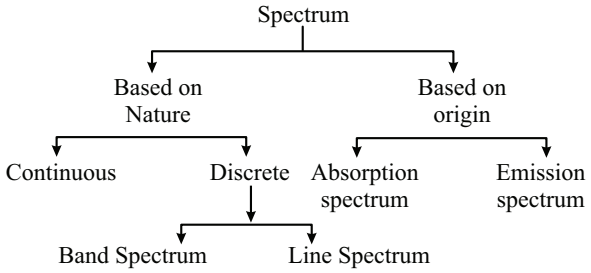
(*b*) A – (iv), B – (iii), C – (ii), D – (i)

(*c*) A – (iv), B – (ii), C – (i), D – (iii)

(*d*) A – (ii), B – (iii), C – (i), D – (iv)

SPECTRUM

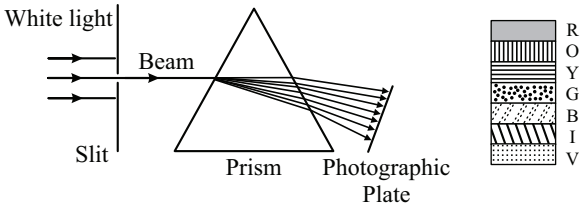
Study of Emission and Absorption Spectra: An instrument used to separate the radiation of different wavelengths (or frequencies) is called spectroscope or a spectrograph. Photograph (or the pattern) of the emergent radiation recorded on the film is called a spectrogram or simply a spectrum of the given radiation The branch or science dealing with the study of spectra is called **spectroscopy**.



(*a*) Emission spectra: When the radiation emitted from some source e.g. from the sun or by passing electric discharge through a gas at low pressure or by heating some substance to high temperature etc, is passed directly through the prism and then received on the photographic plate, the spectrum obtained is called 'Emission spectrum'.

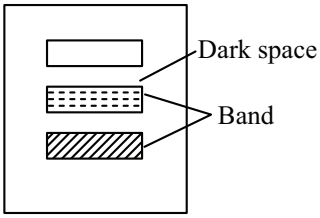
Depending upon the source of radiation, the emission spectra are mainly of two type:

Continuous spectra: When white light from any source such as sun, a bulb or any hot glowing body is analysed by passing through a prism it is observed that it splits up into seven different wide band of colours from violet to red. These colours are so continuous that each of them merges into the next. Hence the spectrum is called continuous spectrum.



Discrete spectra : It is of two type:

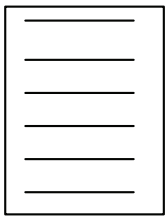
(*i*) Band spectrum



Band spectrum contains colourful continuous bands sepearted by some dark space.

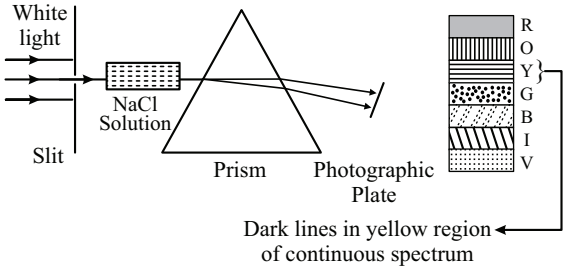
Generally molecular spectrum are band spectrum.

(*ii*) Line Spectrum:



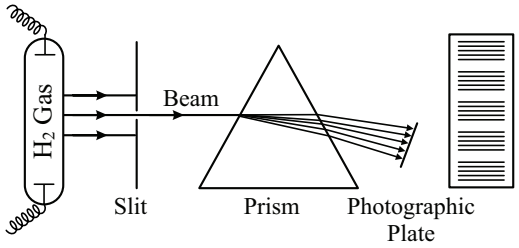
This is the ordered arrangement of lines of particular wavelength seperated by dark space e.g. hydrogen spectrum. Line spectrum can be obtained from atoms.

(*b*) Absorption spectra: When white light from any source is first passed through the solution or vapours of a chemical substance and then analysed by the spectroscope, it is observed that some dark lines are obtained in the continuous spectrum. These dark lines are supposed to result from the fact that when white light (containing radiations of many wavelengths) is passed through the chemical substance, radiations of certain wavelengths are absorbed, depending upon the nature of the element.



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Emission Spectrum of Hydrogen



When hydrogen gas at low pressure is taken in the discharge tube and the light emitted on passing electric discharge is examined with a spectroscope, the spectrum obtained is called the emission spectrum of hydrogen.

Hydrogen Spectrum (Before Bohr's Model)

The spectrum of H atom is observed as discontinue line spectra. The line spectra of different in 3 region UV, visible and IR. The different lines observed H spectrum were classified in to different series and named after their discoverers.

Rydberg gave an empirical formula to calculate wavelength, which is applicable to all series.

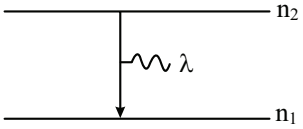
$$\frac{1}{\lambda} = R_H\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$R_H = 109677\ cm^{-1}$ Rydberg's constant.

n_1 and n_2 are integers.

Explanation of Hydrogen Spectrum using Bohr's Model

When electron in an excited atom comes back from higher energy level (n_2) to lower ever level (n_1) then it emits a photon, having energy equal to difference in energy levels.



$$h\nu = \Delta E = E_{n_2} - E_{n_1}$$

$$h\nu = \frac{-2\pi^2 mk^2e^4z^2}{n_2^2h^2} - \left(-\frac{2\pi^2 mk^2e^4z^2}{n_1^2h^2}\right)$$

$$h\nu = \frac{2\pi mk^2e^4z^2}{h^2}\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$$\frac{1}{\lambda} = \frac{2\pi mk^2e^4z^2}{h^3C}\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$$\frac{1}{\lambda} = R_H\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

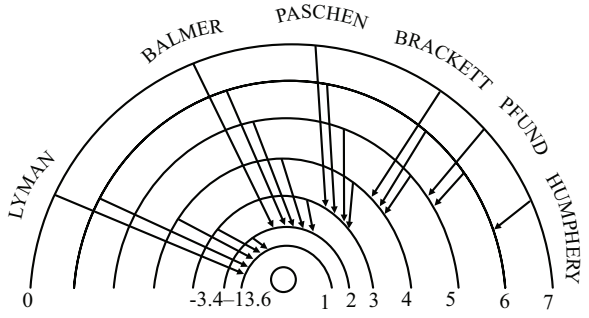
The theoretical value is very closed to observed value.

∴ Bohr's model provides theoretical explanation of H-spectrum.

Wavelength or wave no. of any line of any one electron species can be calculated as:

$$\frac{1}{\lambda} = R_H Z^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right], \qquad \frac{hc}{\lambda} = \Delta E$$

Different Series in Hydrogen Spectrum



Series	Discovered by	Regions	n_2	n_1
Lyman	Lyman	U.V. region	$n_2 = 2,3,4...$	$n_1=1$
Balmer	Balmer	Visible region	$n_2 = 3,4,5...$	$n_1=2$
Paschen	Paschen	Infra red (I.R.)	$n_2 = 4,5,6...$	$n_1=3$
Brackett	Brackett	I.R. region	$n_2 = 5,6,7...$	$n_1=4$
Pfund	Pfund	I.R. region	$n_2 = 6,7,8...$	$n_1=5$
Humphery	Humphery	Far I.R. region	$n_2 = 7,8,9...$	$n_1=6$

Key Points

First line / Starting line / Initial line (λ_{max} and ν_{min})

Last line / limiting line / Series limit (λ_{min} and ν_{max})

First line of any series = α line

Second line of any series = β line

Third line of any series = γ line

Total no. of emission lines between n_2 & n_1

$$\frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

For transition from any orbit 'n' to n = 1 , total no. of emission lines = $\frac{n(n-1)}{2}$

NO. OF PHOTONS EMITTED BY A SAMPLE OF H-ATOM

If an electron is in any higher state n and makes a transition to ground state, then total no. of different photons emitted is equal to $\frac{n \times (n-1)}{2}$.

If an electron is in any higher state $n = n_2$ and makes a transition to another excited state $n = n_1$, then total no. of different photons emitted is equal to $\frac{\Delta n(\Delta n + 1)}{2}$, where $\Delta n = n_2 - n_1$.

Note : In case of single isolated atom if electron make transition from n^{th} state to the ground state then max. number of spectral lines observed = (n – 1).

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SPECTRA LINES OF HYDROGEN ATOM

Lyman Series

- It is first spectral series of H.
- It was found out in ultraviolet region in 1898 by Lyman.
- It's value of $n_1 = 1$ and $n_2 = 2,3,4$ where 'n_1' is ground state and 'n_2' is called excited state of electron present in a H - atom.
- $\frac{1}{\lambda} = R_H\left[\frac{1}{1^2} - \frac{1}{n_2^2}\right]$ where $n_2 > 1$ always.
- The wavelength of marginal line = $\frac{n_1^2}{R_H}$ for all series. So for Lyman series $\lambda = \frac{1}{R_H}$.
- I[st] line of lyman series = $2 \rightarrow 1$
- II[nd] line of lyman series = $3 \rightarrow 1$
- Last line of lyman series = $\infty \rightarrow 1$
- First line of any spectral series is the longest (λ_{max}) line.
- Last line of any spectral series is the shortest (λ_{min}) line.

Series limit: It is the last line of any spectral series.

Wave no of I[st] line of Lyman series

$$= \frac{1}{\lambda} = \bar{\nu} = R \times 1^2\left(\frac{1}{1^2} - \frac{1}{2^2}\right)$$

$$\bar{\nu} = R \times 1^2\left(\frac{4-1}{4}\right)$$

$$\bar{\nu} = \frac{R \times 3}{4} = \frac{3R}{4}$$

$$\therefore \left[\lambda = \frac{4}{3R}\right]$$

Wave no of last line of Lyman series

$$\bar{\nu} = R \times 1^2\left(\frac{1}{1^2} - \frac{1}{\infty^2}\right) \quad \bar{\nu} = R$$

Balmer Series

- It is the second series of H-spectrum.
- It was found out in 1892 in visible region by Balmer.
- It's value of $n_1 = 2$ and $n_2 = 3,4,5,$.....
- The wavelength of marginal line of Balmer series = $\frac{n_1^2}{R_H}$ $= \frac{2^2}{R_H} = \frac{4}{R_H}$
- $\frac{1}{\lambda} = R_H\left(\frac{1}{2^2} - \frac{1}{n_2^2}\right)$ where $n_2 > 2$ always.

1[st] line of balmer series = $3 \rightarrow 2$

last line of balmer series = $\infty \rightarrow 2$

($\bar{\nu}$) 1[st] line = $R \times 1\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = \frac{5R}{36}$

($\bar{\nu}$) last line = $R\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right) = \frac{R}{4}$

Paschen Series

(*a*) It is the third series of H - spectrum.

(*b*) It was found out in infrared region by Paschen.

(*c*) It's value of $n_1 = 3$ and $n_2 = 4,5,6$ .....

(*d*) The wavelength of marginal line of Paschen series $\frac{n_1^2}{R_H} = \frac{3^2}{R_H} = \frac{9}{R_H}$.

(*e*) $\frac{1}{\lambda} = R_H\left[\frac{1}{3^2} - \frac{1}{n_2^2}\right]$ where $n_2 > 3$ always.

Brackett Series

(*a*) It is fourth series of H - spectrum.

(*b*) It was found out in infrared region by Brackett.

(*c*) It's value of $n_1 = 4$ and $n_2 = 5,6,7$ .....

(*d*) The wavelength of marginal line of brackett series

$$= \frac{n_1^2}{R_H} = \frac{4^2}{R_H} = \frac{16}{R_H}$$

(*e*) $\frac{1}{\lambda} = R_H\left[\frac{1}{4^2} - \frac{1}{n_2^2}\right]$ where $n_2 > 4$ always.

Pfund Series

(*a*) It is fifth series of H- spectrum.

(*b*) It was found out in infrared region by Pfund.

(*c*) It's value of $n_1 = 5$ and $n_2 = 6,7,8$ ..... where n_1 is ground state and n_2 is excited state.

(*d*) The wavelength of marginal line of Pfund series

$$= \frac{n_1^2}{R_H} = \frac{5^2}{R_H} = \frac{25}{R_H}$$

(*e*) $\frac{1}{\lambda} = R_H\left[\frac{1}{5^2} - \frac{1}{n_2^2}\right]$ where $n_2 > 5$ always.

Humphry Series

(*a*) It is the sixth series of H - spectrum.

(*b*) It was found out in infrared region by Humphry.

(*c*) It's value of $n_1 = 6$ and $n_2 = 7, 8, 9$ .....

(*d*) The wavelength of marginal line of Humphry series

$$= \frac{n_1^2}{R_H} = \frac{6^2}{R_H} = \frac{36}{R_H}$$

(*e*) $\frac{1}{\lambda} = R_H\left[\frac{1}{6^2} - \frac{1}{n_2^2}\right]$ where $n_2 > 6$.

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Train Your Brain

Example 19: Calculate wavelength for 2nd line of Balmer series of He^+ ion.

Sol. $\frac{1}{\lambda} = R(2)^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

$n_1 = 2 \quad n_2 = 4$

$\frac{1}{\lambda} = R(2^2)\left[\frac{1}{2^2} - \frac{1}{4^2}\right]$

$\frac{1}{\lambda} = \frac{3R}{4} \qquad \lambda = \frac{4}{3R}$

Example 20: The ratio of $(E_2 - E_1)$ to $(E_4 - E_3)$ for He^+ ion is approximately equal to (where E_n is the energy of nth orbit).

(*a*) 10 (*b*) 15

(*c*) 17 (*d*) 12

Sol. (*b*) $\frac{\Delta E_{2\to1}}{\Delta E_{4\to3}} = \frac{13.6(2)^2\left[\frac{1}{(1)^2} - \frac{1}{(2)^2}\right]}{13.6(2)^2\left[\frac{1}{(3)^2} - \frac{1}{(4)^2}\right]} = 15$

Example 21: If the binding energy of 2nd excited state of a hydrogen like sample is 24 eV approximately, then the ionisation energy of the sample is approximately.

(*a*) 54.4 eV (*b*) 24 eV

(*c*) 122.4 eV (*d*) 216 eV

Sol. (*d*) $\frac{13.6(Z)^2}{(3)^2} = 24$

I.E. $= 13.6(Z)^2 = (24 \times 9) = 216$ eV

Example 22: If electron make transition from 7th excited state to 2nd state in H atom sample find the max. number of spectral lines observed.

Sol. $\Delta n = 8 - 2 = 6$

spectral lines $= 6\left(\frac{6+1}{2}\right) = 6 \times \frac{7}{2} = 21$

Example 23: Which transition of the Hydrogen spectrum would have the same lenght as the Balmer transition, n = 4 to n = 2 of He^+ spectrum?

(*a*) $n_2 = 2$ to $n_1 = 1$ (*b*) $n_2 = 3$ to $n_1 = 1$

(*c*) $n_2 = 4$ to $n_1 = 2$ (*d*) $n_2 = 5$ to $n_1 = 3$

Sol. (*a*) For He^+ ion, we have

$\frac{1}{\lambda} = R_H Z^2\left[\frac{1}{n^2} - \frac{1}{n_2^2}\right] = [2]^2\left[\frac{1}{2^2} - \frac{1}{4^2}\right] \times R_H$

$= \frac{3}{4}R_H$ ...(i)

Now for H atom

$\frac{1}{\lambda} = R_H\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$ ...(ii)

Equating equs (*a*) and (*b*) we have

$\frac{1}{n_1^2} - \frac{1}{n_2^2} = \frac{3}{4}$

Obviously $n_1 = 1$ and $n_2 = 2$. Hence the transition n = 2 to n = 1 in hydrogen atom will have the same length as the transition n = 4 to n = 2 in He^+ species.

Example 24: The shortest wave length in H-spectrum of Lyman series when $R_H = 109678\ cm^{-1}$ is?

(*a*) 1215.67 Å (*b*) 911.7 Å

(*c*) 1002.7 Å (*d*) 1127.30 Å

Sol. (*b*) For Lyman series $n_1 = 1$

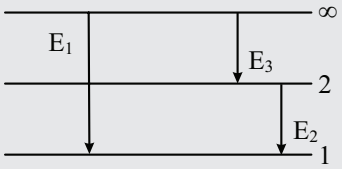
For shortest 'λ' of Lyman sereis the energy differnece in two levels showing transition should be maximum (i.e. $n_2 = \infty$).

$\frac{1}{\lambda} = R_H\left[\frac{1}{1^2} - \frac{1}{\infty^2}\right] = 109678\ cm^{-1}$

$\therefore \lambda = 9.117 \times 10^{-8} = 911.7$ Å

Example 25: Let ν_1 be the frequency of series limit of lyman series, ν_2 be the frequency of 1st line of lyman series, ν_3 be the frequency of series limit of Balmer series. Then find relation between ν_1, ν_2 and ν_3.

Sol.



$E_1 = E_2 + E_3$

$\Rightarrow h\nu_1 = h\nu_2 + h\nu_3$

$\Rightarrow \nu_1 = \nu_2 + \nu_3$

Concept Application

16. What electron transition in the He^+ spectrum would have the same wavelength as the first Lyman transition of hydrogen.

(*a*) (n = 4 to n = 1) (*b*) (n = 4 to n = 2)

(*c*) (n = 3 to n = 2) (*d*) (n = 2 to n = 2)

17. At what atomic number would a transition from n = 2 to n = 1 energy level result in emission of photon of $\lambda = 3 \times 10{-}8$ m.

(*a*) (z = 1) (*b*) (z = 2)

(*c*) (z = 3) (*d*) (z = 4)

18. The energy of hydrogen atom in its ground state is –13.6 eV. The energy of the level corresponding to n = 5 is:
(*a*) –0.54 eV (*b*) –5.40 eV
(*c*) –0.85 eV (*d*) –2.72 eV

19. The wavelength of a spectral line for an electronic transition is inversely proportional to:
(*a*) Number of electrons undergoing transition.
(*b*) The nuclear charge of the atom.
(*c*) The velocity of an electron undergoing transition.
(*d*) The difference in the energy involved in the transition.

DUAL NATURE OF ELECTRON (De-Broglie HYPOTHESIS)

(*a*) Einstein had suggested that light can behave as a wave as well as like a particle i.e. it has dual character.

(*b*) In 1924, de-Broglie proposed that an electron behaves both as a material particle and as a wave.

(*c*) This proposed a new theory wave mechanical theory of matter. According to this theory, the electrons protons and even atom when in motion possess wave properties.

(*d*) According to de-Broglie, the wavelength associated with a particle of mass m, moving with velocity v is given by the relation, $\lambda = \frac{h}{mv}$ where h is Planck's constant.

Derivation of de-Broglie Relationship

For a photon,

$$E = \frac{hc}{\lambda} \quad ...(i)$$

$$E = mc^2 \quad ...(ii)$$

$$mc^2 = \frac{hc}{\lambda}$$

$$\lambda = \frac{h}{mc} = \frac{h}{p}$$

He concluded that as electromagnetic radiation have some associated mass or associated momentum, in the same way every moving particle of mass 'm' and velocity 'v' is associated with waves and these associated waves are known as matter waves or de-Broglie's waves.

$\lambda = h / mv = h / p$

Derivation of Bohr's postulate of quantisation of angular momentum from de-Broglie's Equations: According to de-Broglie moving electron is associated with waves must be completely in phase and therefore only those orbits are possible where circumference of the orbit is integral multiple of λ i.e. $2\pi r = n\lambda$, where n is the no. of waves.

$$2\pi r = \frac{nh}{mv} \Rightarrow mvr = \frac{nh}{2\pi}$$

From the above expression it can be commented that no of waves in a shell waves = shell no.

❖ **Calculation of de-Broglie wavelength if K.E. of the particle is E:**

$$E = \frac{1}{2}mv^2$$

$$2E = mv^2$$

$$2mE = m^2v^2 = p^2$$

$$p = \sqrt{2mE}$$

$$\frac{h}{\lambda_{dB}} = \sqrt{2mE}$$

$$\lambda_{dB} = \frac{h}{\sqrt{2mE}}$$

If a charge particle at rest (having charge 'q') is accelerated by potential difference 'V' volt then:

$$\lambda_{dB} = \frac{h}{\sqrt{2mqv}},$$

$$\lambda_{dB} = \sqrt{\frac{150}{V}}\,\text{Å} \quad \text{(Only for electron).}$$

Train Your Brain

Example 26: Calculate λ_{dB} of electron have K.E. 3eV.

Sol. $\lambda_{dB} = \sqrt{\frac{150}{K.E.}} = \sqrt{\frac{150}{3}} = \sqrt{50}\,\text{Å} = 7.07\text{Å}$

Example 27: Calculate the wavelength of a moving electron having 4.55×10^{-25} J of kinetic energy.

Sol. Kinetic energy = $(\frac{1}{2}mv^2) = 4.55 \times 10^{-25}$ J

$$\therefore \quad \lambda = \frac{h}{\sqrt{2 \times m \times K.E.}}$$

$$\text{or}\lambda = \frac{6.626 \times 10^{-34}}{\sqrt{2 \times 9.11 \times 10^{-31} \times 4.55 \times 10^{-25}}}$$

$$= 7.27 \times 10^{-7} \text{ meter}$$

Concept Application

20. An electron, practically at rest, is initially accelerated through a potential difference of 100 volts. It then has a de Broglie wavelength = λ_1 Å. It then get retarded through 19 volts and then has a wavelength λ_2 Å. A further retardation through 32 volts changes the wavelength to λ_3Å, What is $\frac{\lambda_3 - \lambda_2}{\lambda_1}$?

(*a*) $\frac{20}{63}$ (*b*) $\frac{19}{63}$
(*c*) $\frac{20}{64}$ (*d*) $\frac{21}{63}$

21. An α-particle is accelerated through a potential difference of V volts from rest. The de-Broglie's wavelength associated with it is:

(*a*) $\sqrt{\frac{150}{V}}$ Å (*b*) $\frac{0.286}{\sqrt{V}}$ Å

(*c*) $\frac{0.101}{\sqrt{V}}$ Å (*d*) $\frac{0.983}{\sqrt{V}}$ Å

22. What possibly can be the ratio of the de Broglie wavelengths for two electrons each having zero initial energy and accelerated through 50 volts and 200 volts?

(*a*) 3 : 10 (*b*) 10 : 3

(*c*) 1 : 2 (*d*) 2 : 1

HEISENBERG'S UNCERTAINTY PRINCIPLE

The exact position and momentum of a fast moving particle cannot be calculated precisely at the same moment of time. If Δx is the error in the measurement of position of the particle and if Δp is the error in the measurement of momentum of the particle, then:

$\Delta x \,.\, \Delta p \geq \frac{h}{4\pi}$ or $\Delta x \,.\, (m\Delta v) \geq \frac{h}{4\pi}$

where, Δx = uncertainty in position
Δp = uncertainty in momentum
h = Plank's constant
m = mass of the particle
Δv = uncertainty in velocity

If the position of a particle is measured precisely, i.e. $\Delta x \to 0$ then $\Delta p \to \infty$.

If the momentum of the particle is measured precisely. i.e. $\Delta p \to 0$ then $\Delta x \to \infty$.

This is because of a principle of optics that if a light of wavelength 'λ' is used to locate the position of a particle then minimum error in the position measurement will be ± λ amount.

i.e. $\Delta x = \pm \lambda$

If $\Delta x \to 0$; $\lambda \to 0$

But, $p = \frac{h}{\lambda} \Rightarrow p \to \infty$

So, to make $\Delta x \to 0$, $\lambda \to 0$ a photon of very high energy is used to locate it.

∴ When this photon will collide with the electron then momentum of electron will get changed by a large amount.

$\Delta p.\Delta x \geq \frac{h}{4\pi}$ (Multiplied & divided by Δt)

$\frac{\Delta p}{\Delta t}\Delta t\,.\,\Delta x \geq \frac{h}{4\pi}$

($\frac{\Delta p}{\Delta t}$ = rate of change in momentum = F)

$F.\Delta x.\Delta t \geq \frac{h}{4\pi}$

$\Delta E \,.\, \Delta t \geq \frac{h}{4\pi}$

ΔE ⟶ uncertainty in energy

Δt ⟶ uncertainty in time

- In terms of uncertainty in energy ΔE, and uncertainty in time Δt, this principle is written as,

  $\Delta E.\Delta t \geq \frac{h}{4\pi}$
- Heisenberg replaced the concept of definite orbits by the concept of probability.

Train Your Brain

Example 28: A golf ball has a mass of 40 g, and a speed of 45 m/s. If the speed can be measured within accuracy of 2%, calculate the uncertainty in the position.

Sol. The uncertainty in the speed is 2%,

i.e., $45 \times \frac{2}{100} = 0.9 \text{ m s}^{-1}$. Using the equation

$\Delta x = \frac{h}{4\pi m \Delta v} = \frac{6.626\times10^{-34}}{4\times3.14\times40\times10^{-3}\,(0.9\,\text{m s}^{-1})}$

$= 1.46 \times 10^{-33}$ m

This is nearly ~10^{18} times smaller than the diameter of a typical atomic nucleus. As mentioned earlier for large particles, the uncertainty principle sets no meaningful limit to the precision of measurements.

Example 29: The uncertainty in measuring speed of a particle is zero. Uncertainty in measuring its position will be

(*a*) Zero (*b*) $\frac{h}{4\pi}$

(*c*) $\frac{h}{4\pi m}$ (*d*) Infinite

Sol. $\Delta x.\Delta p = \frac{h}{4\pi}$

or $\Delta x \times m\Delta v = \frac{h}{4\pi}$

or $\Delta x = \frac{h}{4\pi \times m \times \Delta v}$

or $\Delta x = \infty$ as $\Delta v = 0$

Example 30: The mass of a particle is 10^{-10} g and its diameter is 10^{-4} cm. If its speed is 10^{-6} cm/s with 0.0001% uncertainty in measurement, the minimum uncertainty in its position is

(*a*) 5.28×10^{-8} m (*b*) 5.28×10^{-7}

(*c*) 5.28×10^{-6} m (*d*) 5.28×10^{-9} m

Sol. m = 10^{-10} g or m = 10^{-13} kg

$\Delta v = 10^{-6} \times 10^{-2} \times 0.0001 \times 10^{-2}$ m/s = 10^{-14} m/s

According to Heisenberg's Uncertainty Principle

$\Delta x \times m\Delta v = \frac{h}{4\pi}$ or

$\Delta x = \frac{6.626\times10^{-34}}{4\times3.14\times10^{-13}\times10^{-14}} m$ or $\Delta x = 5.28\times10^{-8} m$

@PROCBSE

Concept Application

23. The uncertainty in the momentum of an electron is 1.0 × 10^{-5} kg ms^{-1} . The uncertainty in its position will be: (h = 6.626 × 10^{-34} Js)

(*a*) 1.05 × 10^{-28} m (*b*) 1.05 × 10^{-26} m

(*c*) 5.27 × 10^{-30} m (*d*) 5.25 × 10^{-28} m

24. If uncertainty in position and momentum of a particle is numerically equal, then the minimum uncertainty in speed of the particle is

(*a*) $\sqrt{\frac{h}{2\pi}}$ (*b*) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$

(*c*) $\sqrt{\frac{h}{\pi}}$ (*d*) $\frac{1}{m}\sqrt{\frac{h}{\pi}}$

25. Uncertainty in the position of an electron (mass = 9.1 × 10^{-31} kg) moving with a velocity 300 m/s, accurate up to 0.001 %, will be (h = 6.3 × 10^{-34} Js)

(*a*) 5.76 × 10^{-2} m (*b*) 1.92 × 10^{-2} m

(*c*) 3.84 × 10^{-2} m (*d*) 19.2 × 10^{-2} m

QUANTUM NUMBER

As we know to search a particular person in this world 4 things are needed:

1. Country to which the person belongs.
2. The city in that country where the person is residing.
3. The area in that city.
4. House number.

Similarly to locate the position of an electron in the atom 4 identification number are required and these identification number are called as quantum mumber.

1. Principal quantum number (n) → Shell (Orbit)
2. Azimuthal quantum number (l) → Sub shell
3. Magnetic quantum number (m) → Orbital
4. Spin quantum number (s) → Spin of e^-

1. Principal quantum number: Given by Bohr's.

It represents the name, size and energy of the shell to which e^- belongs.

The value of n lies between 1 to ∞.

i.e n = 1,2,3,4– – – – – ∞ corresponding name of shells are K, L, M, N, O, – – – – –.

Greater the value of n, greater is the distance from the nucleus.

$r = 0.529 \times \frac{n^2}{z}$ Å

$r_1 < r_2 < r_3 < r_4 < r_5$ – – – – – – – –.

Greater the value of n, greater is the energy of shell.

$E = -13.6 \times \frac{z^2}{n^2}$ eV/atom

$E_1 < E_2 < E_3 < E_4$ – – – – – – – –

The angular momentum of a revolving electron is mvr $= \frac{nh}{2\pi}$

Where n = Principal quantum number.

The number of electrons in a particular shell is equal to $2n^2$

2. Azimuthal quantum number / Angular quantum number/ Secondary quantum number /Subsidiary quantum number: Represented by '*l*' (Given by Sommerfeld).

It represents the shape of the subshell and orbital and orbital angular momentum.

Value of ℓ between 0 to (n – 1).

i.e. ℓ = 0,1,2 – – – – – – – – – – (n–1).

ℓ = 0(s Subshell)

ℓ = 1(p Subshell)

ℓ = 2(d Subshell)

ℓ = 3(f Subshell)

If n = 1 then $\ell = 0 \Rightarrow$ 1s i.e. in n =1 shell, only one subshell 's' is present.

If n = 2 then $\ell = 0,1 \Rightarrow$ 2s, 2p i.e. in n =2 shell, two subshell 's' & 'p' are present.

If n = 3 then $\ell = 0,1,2 \Rightarrow$3s, 3p, 3d i.e. in n =3 shell, three subshell 's' , 'p' & 'd' are present.

If n = 4 then $\ell = 0,1,2,3 \Rightarrow$ 4s,4p, 4d, 4f i.e. in n =4 shell, four subshell 's' , 'p' , 'd' & 'f' are present.

If the value of n is same then the order of energy of the various subshell will be:

s < p < d < f

4s < 4p < 4d < 4 f, 3s < 3p< 3d, 2s < 2p

If Value of ℓ is same but value of n is different then the order of energy will be:

1s < 2s < 3s < 4s < 5s < 6s

3d < 4d < 5d < 6d

4p < 5p <6p

The orbital angular momentum = $\sqrt{\ell(\ell+1)}\ \frac{h}{2\pi}$ or $\sqrt{\ell(\ell+1)}\hbar$ $\left\{\because \hbar = \frac{h}{2\pi}\right\}$ { $\hbar$ is called as 'hash'}

Orbital angular momentum : For s subshell = 0

For p Subshell = $\sqrt{2}\frac{h}{2\pi}$ or $\sqrt{2}\hbar$.

The number of electron in a particular subshell is equal to $2(2\ell + 1)$.

for s subshell number of electrons = 2 e^-

for p subshell number of electrons = 6 e^-

for d subshell number of electrons = 10 e^-

for f subshell number of electrons = 14 e^-

Shape of the subshell:

s → spherical

p → dumb bell shape

d → double dumb bell shape

f → complex shape

@PROGRESS

3. **Magnetic quantum number /Orientation quantum number (m):** given by linde. It represents the shape of different orbitals and the orientation of electron cloud (orbital).

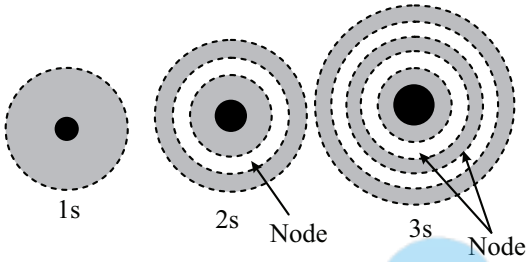
Under the influence of magnetic field each subshell is further subdivided into orbitals (The electron cloud is knwon as orbital). Magnetic quantum number describe these different distribution of electron cloud.

Value of m = all integral value from $-\ell$ to $+\ell$ including zero.

i.e. Value of m $= -\ell$ ..... 0 ..... $+\ell$

Case-I: If $\ell = 0$ then m = 0 it implies that s subshall has only one orbital called as s orbital.

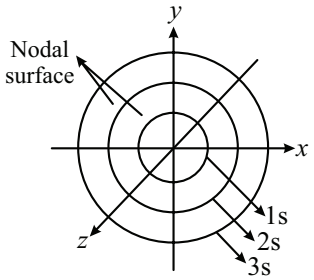
Shapes of s-orbitals: The s-orbitals are spherically symmetrical about the nucleus, i.e., the probability of finding ns electron is same in all directions from the nucleus. The size of the orbital depends on the value of principal quantum number, there is one spherically symmetrical orbital. The 1s orbital is smaller than 2s-orbital and 2s-orbital is smaller than 3s, but all are spherical in shape as shown in figure.



Although the s-orbitals belonging to different shells are spherically symmetrical, yet they differ in certain respects as explained below:

(*i*) The probability of 1s electron is found to be maximum near the nucleus and decreases as the distance from the nucleus increases. In case of 2s electrons, the probability is again maximum near the nucleus and then decreases to zero and increases again and then decreases as the distance from the nucleus from the nucleus increases. The intermediate region (a spherical shell) where the probability is zero is called a nodal surface of simply node. Thus, 2s-orbital differs from 1s-orbital in having one node within it. Similarly, 3s has two nodes. In general, any ns orbital has (n -1) nodes.

(*ii*) The size and energy of the s-orbital increases as the principal quantum number inreases, i.e., the size and energy of s-orbital increases in the order 1s < 2s < 3s ....



The s orbital of higher energy levels are also symmetrically spherical and can be represented as follows:

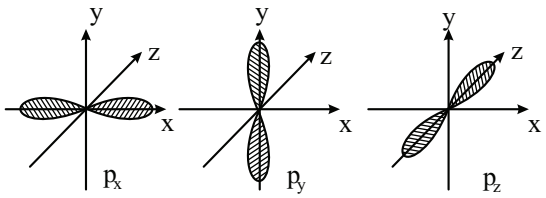
Case-II: If $\ell = 1$ (p - subshell) then m = –1, 0, 1

It implies that, p subshell have three orbitals called as p_x, p_y and p_z.

Shape of p-orbitals: There are three p-orbitals, commonly referred to as p_x, p_y and p_z. These three p-orbitals, posses equivalent energy and therefore, have same relation with the nucleus. They, however, differ in their direction & distribution of the charge.

These three p-orbitals are situated at right angle to another and are directed along x, y and z axis (figure).

Each p orbital has dumb bell shape (2 lobes which are separated from each other by a point of zero probability called nodal point or node or nucleus).

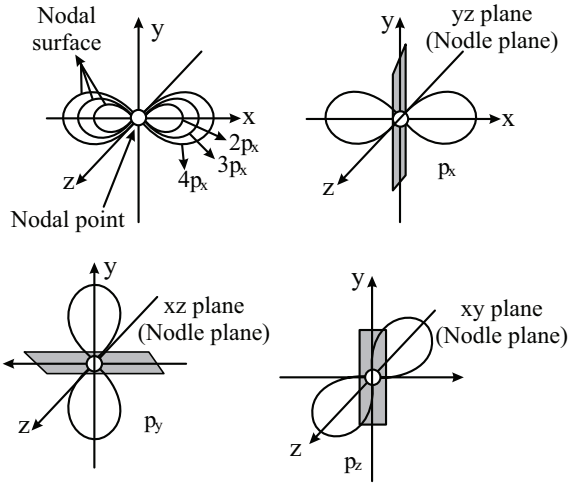


The two lobes of each orbital are separated by a plane of zero electron density called nodal plane.

Each p orbital of higher energy level are also dumb bell shape but they have nodal surface.

Nodal surface:	Orbital	Nodal	Surface	
	$2p_x$	0	$3p_x$	1
	$4p_x$	2	np_x	(n – 2)

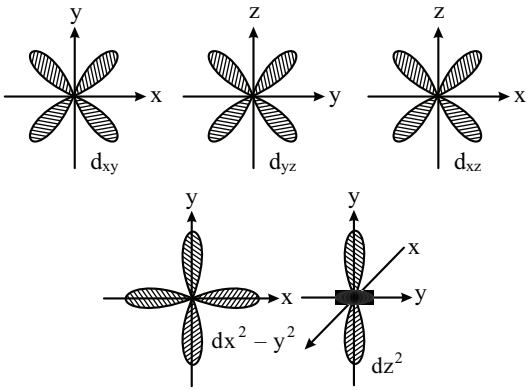
Nodal Plane:	Orbital	Nodal plane
	p_x	yz plane
	p_y	xz plane
	p_z	xy plane



Case-III: When λ = 2, 'm' has five values – 2, –1, 0, +1, +2. It implies that d subshell of any energy shell has five orbitals. All the five orbitals are not identical in shape. Four of the d orbitals d_{xy}, d_{yz}, d_{xz}, $d_{x^2-y^2}$ contain four lobes while fifth orbital d_{z^2} consists of only two lobes. The lobes d_{xy} orbital lie between x and y axis. Similar is the case for d_{yz} and d_{xz}. Four lobes of $d_{x^2-y^2}$ orbital are lying along x and y axis while the two lobes of d_{z^2} orbital are lying along z axis and contain a ring of negative charge surrounding the nucleus in xy plane. Geometry of d orbital is Double Dumb bell.

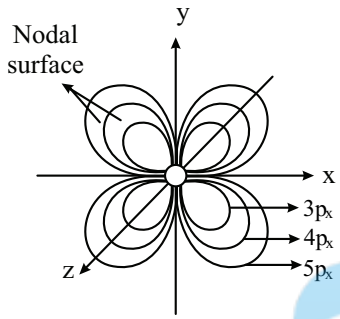
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Shape of d–orbitals: It implies that d subshell has 5 orbitals i.e. five electron cloud and can be represented as follows.



Each d-orbital of higher energy level are also double dumbled shape but they have nodal surface.

In d orbital:



(*i*) Nodal Point → 1

(*ii*) Nodal Surface → $3d_{xy}$ → 0 Nodal surface
$4d_{xy}$ → 1 Nodal surface
$5d_{xy}$ → 2 Nodal surface
nd_{xy} → (n – 3)
Number of nodal surface = n–ℓ –1

(*iii*) Nodal plane : d_{xy} → xz & yz nodal plane.
d_{xz} → xy & zy nodal plane.
d_{zy} → dzx & yx nodal plane.
$d_{x^2-y^2}$ → 0, nodal plane.
d_{z^2} → 0, nodal plane.

Note: Orbitals of d subshell are Equivalent in energy.

f subshell : When ℓ = 3 (f subshell)

Then the structure of f-orbital is very complex.

The number of values of f-orbitals in f subshell show that f subshell has 7 orbitals which are equivalent in energy.

Representation of the orbitals :

s subshell → | s |

p subshell → | p_x | p_y | p_z |

d subshell → | d_{xy} | d_{yz} | d_{z^2} | d_{xz} | $d_{x^2y^2}$ |

f subshell → | f_{x^3} | f_{y^3} | f_{z^3} | f_{xyz} | $f_{x(y^2z^2)}$ | $f_{y(z^2x^2)}$ | $f_{z(x^2y^2)}$ |

4. **Spin Quantam number (s):** Given by Gold schmidt.

It represents the direction of electron spin around its own axis.

For clock wise spin/spin up (↑) electron → $\pm ½$.

For anticlock wise spin/spin down (↓) electron → $\mp ½$.

Spin angular momentum of an e^- = $\sqrt{s(s+1)}.\frac{h}{2\pi}$.

or $\sqrt{s(s+1)}\,\hbar$

Each orbital can accomodate 2 electrons with opposite spin or spin paired.

Correct ↑↓ spin paired

Wrong ↑↑ spin paired

Train Your Brain

Example 31: For $5dz^2$, calculate the value of *n*, *l*, m and *s*.

Sol. n = 5, ℓ = 2, m = 0, s = $+\frac{1}{2}$ or $-\frac{1}{2}$

Example 32: Which of the following set of Quantum number is not possible?

(*a*) n = 3, ℓ = +2, m = 0, s = $+\frac{1}{2}$

(*b*) n = 3, ℓ = 0, m = 0, s = $-\frac{1}{2}$

(*c*) n = 3, ℓ = 0, m = 1, s = $+\frac{1}{2}$

(*d*) n = 3, ℓ = 1, m = 0, s = $-\frac{1}{2}$

Sol. (*c*) For s-subshell *l* = 0 then should be m = 0

Example 33: The orbital angular momentum of an electron is 2s orbital is

(*a*) $+\frac{1}{2}.\frac{h}{2\pi}$ (*b*) 0

(*c*) $\frac{h}{2\pi}$ (*d*) $\sqrt{2}\frac{h}{4\pi}$

Sol. The orbital angular momentum = $\sqrt{l(l+1)}\,\frac{h}{2\pi}$.

For 2s orbital, *l* = 0

$\therefore$ Orbital angular momentum = $\sqrt{0(0+1)}\,\frac{h}{2\pi} = 0$

Example 34: The number of orbitals of g-type

(*a*) 5 (*b*) 7

(*c*) 9 (*d*) 11

Sol. For g - subshell, *l* = 4.
Therefor no. of orbital = –*l* to + *l*.
i.e. –4 to +4 (including zero).
Hence in g-subshell, 9 orbitals are present.

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Concept Application

26. Number of orbitals represented by $n = 3$, $l = 2$ and $m = +2$ is

(*a*) 1 (*b*) 2

(*c*) 3 (*d*) 4

27. An electron that has the quantum numbers $n = 3$ and $m = 2$

(*a*) Must have spin quantum number value, + 1/2.

(*b*) Must have $l = 2$, 3 or 4

(*c*) Must have $l = 0$, 1 or 2

(*d*) Must have $l = 2$

28. If an electron has spin quantum number of + ½ and magnetic quantum number of –1, it cannot be present in

(*a*) s-orbital (*b*) p-orbital

(*c*) d-orbital (*d*) f-orbital

29. In which of the following orbital, electon will be closer to the nucleus?

(*a*) 6s (*b*) 4f

(*c*) 5d (*d*) 6p

30. Possible set of quantum numbers for which: $n = 4$, $l = 3$ and $s = +\frac{1}{2}$ is

(*a*) 14 (*b*) 7

(*c*) 5 (*d*) 10

Rules for Filling Subshell

1. Aufbau Principle.

2. (n + *l*) rule.

3. Hund's maximum multiplicity principle.

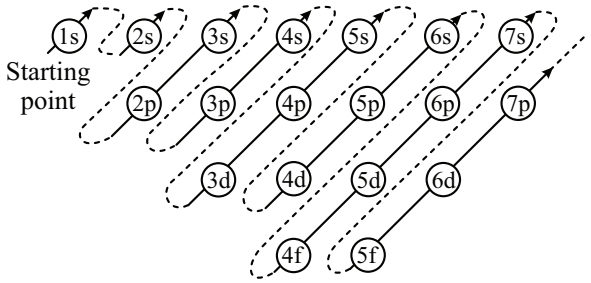
4. Pauli's exclusion principle.

1. Aufbau Principle: Aufbau is a German word and its meaning ' Building up'.

Aufbau principle gives a sequence in which various subshell are filled up depending on the relative order of the energies of various subshell.

Principle: The subshell with minimum energy is filled up first when this subshell obtained maximum quota of electrons then the next subshell of higher energy starts filling.

The sequence in which various subshell are filled are as follows.



$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6, 5s^2, 4d^{10}, 5p^6, 6s^2, 4f^{14}, 5d^{10}, 6p^6, 7s^2, 5f^{14}, 6d^{10}$, ...

For Ex.

$_1H \rightarrow 1s^1$

$_2He \rightarrow 1s^2$

$_3Li \rightarrow 1s^2, 2s^1$

$_4Be \rightarrow 1s^2, 2s^2$

$_5B \rightarrow 1s^2, 2s^2, 2p^1$

$_6C \rightarrow 1s^2, 2s^2, 2p^2$

$_7N \rightarrow 1s^2, 2s^2, 2p^3$

$_8O \rightarrow 1s^2, 2s^2, 2p^4$

$_9F \rightarrow 1s^2, 2s^2, 2p^5$

$_{10}Ne \rightarrow 1s^2, 2s^2, 2p^6$

$_{11}Na \rightarrow 1s^2, 2s^2, 2p^6, 3s^1$

$_{12}Mg \rightarrow 1s^2, 2s^2, 2p^6, 3s^2$

$_{13}A\ell \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^1$

$_{14}Si \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^2$

$_{15}p \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^3$

$_{16}S \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^4$

$_{17}Cl \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^5$

$_{18}Ar \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^6$

$_{19}K \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1$

$_{20}Ca \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2$

$_{21}Sc \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^1$

$_{22}Ti \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^2$

$_{23}V \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^3$

$_{24}Cr \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1, 3d^5$ [Exception]

$_{25}Mn \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^5$

$_{26}Fe \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^6$

$_{27}Co \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^7$

$_{28}Ni \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^8$

$_{29}Cu \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1, 3d^{10}$ [Exception]

$_{30}Zn \rightarrow 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}$

Electronic configuration can be written by following different methods:

$_{26}Fe \rightarrow$ (1) $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^6$

(2) $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^6, 4s^2$

(3) $1s^2, 2s^2p^6, 3s^2p^6d^6, 4s^2$

(4) [Ar] $4s^2$ $3d^6$

$_{26}Fe \rightarrow \underbrace{1s^2}_{}\ \underbrace{2s^2 2p^6}_{(n-2)}\ \underbrace{3s^2 3p^6 3d^6}_{(n-1)}\ \underbrace{4s^2}_{n}$

$n \rightarrow$ Outer most shell or ultimate shell or valence shell.

In this shell e^- are called as valance electron or this is called core charge.

$(n-1) \rightarrow$ Penultimate shell or core or pre valence Shell.

$(n-2) \rightarrow$ Pre penultimate shell.

If we remove the last n shell (ultimate shell) then the remaining shell collectivly be called as Kernal.

Ex. $_{26}Fe \rightarrow \underbrace{1s^2\ 2s^2 2p^6\ 3s^2 3d^6 3p^6}_{\text{Kernal}}\ 4s^2$

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2. (*n* + *l*) rule: According to it the sequence in which various subshell are filled up can also be determined with the help of (n +l) value for a given subshell.

Principle of (n+l) rule: The subshell with lowest (n+ℓ) value is filled up first, When two or more subshell have same (n+ℓ) value then the subshell with lowest value of n is filled up first.

Sub shell	n	ℓ	n + ℓ	
1s	1	0	1	
2s	2	0	2	
2p	2	1	3	(1)
3s	3	0	3	(2)
3p	3	1	4	(1)
4s	4	0	4	(2)
3d	3	2	5	(1)
4p	4	1	5	(2)
5s	5	0	5	(3)
4d	4	2	6	(1)
5p	5	1	6	(2)
6s	6	0	6	(3)

Order : $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^6$, $4s^2$, $3d^{10}$, $4p^6$, $5s^2$, $4d^{10}$, $5p^6$, $6s^2$, $4f^{14}$, $5d^{10}$, $6p^6$,$7s^2$, $5f^{14}$, $6d^{10}$, ...

3. Hund's Maximum Multiplicity Rule: (Multiplicity: Many of the same kind).

According to Hund's rule electrons are distributed among the orbitals of subshell in such a way as to give maximum number of unpaired electron with parallel spin.

Thus the orbital available in the subshell are first filled singly with parallel spin electron before they begin to pair this means that pairing of electrons occurs with the introduction of second electron in 's' subshell, fourth electron in 'p' subshell, 6th electron in 'd' Subshell & 8th e^- in 'f' subshell.

Ex. $_5B \rightarrow$ $1s^2$ ↑↓ $2s^2$ ↑↓ $2p^1$ ↑ | |

$_6C \rightarrow$ $1s^2$ ↑↓ $2s^2$ ↑↓ $2p^2$ ↑ ↑ |

$_7N \rightarrow$ $1s^2$ ↑↓ $2s^2$ ↑↓ $2p^3$ ↑ ↑ ↑

$_8O \rightarrow$ $1s^2$ ↑↓ $2s^2$ ↑↓ $2p^4$ ↑↓ ↑ ↑

$_9F \rightarrow$ $1s^2$ ↑↓ $2s^2$ ↑↓ $2p^5$ ↑↓ ↑↓ ↑

$_{10}Ne \rightarrow$ $1s^2$ ↑↓ $2s^2$ ↑↓ $2p^6$ ↑↓ ↑↓ ↑↓

4. Pauli's Exclusion principle: In 1925 Pauli stated that no two electron in an atom can have same values of all four quantum numbers. i.e. An orbital can accomodates maximum 2 electrons with opposite spin.

Ex.1. $_6C^{12} \rightarrow$ $1s^2$ ↑↓ $2s^2$ ↑↓ $2p^2$ ↑ ↑ |

p_x p_z p_y

n = 1, 2, 2

ℓ = 0, 0, 1

m = 0, 0, +1, –1, 0

s = $+\frac{1}{2}, -\frac{1}{2}$ $+\frac{1}{2}, -\frac{1}{2}$ $+\frac{1}{2}, +\frac{1}{2}$,

Ex.2 $_{17}Cl \rightarrow$ $1s^2$ ↑↓ $2s^2$ ↑↓ $2p^6$ ↑↓ ↑↓ ↑↓ $3s^2$ ↑↓ $3p^5$ ↑↓ ↑↓ ↑

n = 1, 2, 2, 3, 3

ℓ = 0, 0, 1, 0, 1

m = 0, 0, +1, –1, 0, 0, +1, –1, 0

s = $+\frac{1}{2}, -\frac{1}{2}$ $+\frac{1}{2}, -\frac{1}{2}$ $+\frac{1}{2}, -\frac{1}{2}$ $+\frac{1}{2}, -\frac{1}{2}$ $+\frac{1}{2}, -\frac{1}{2}$

$+\frac{1}{2}, -\frac{1}{2}$ $+\frac{1}{2}, -\frac{1}{2}$ $+\frac{1}{2}, -\frac{1}{2}$ $+\frac{1}{2}$

Exception of Aufbau principle: In some cases it is seen that the electronic configuration is slightly different from the arrangement given by Aufbau principle. A simple region behind this is that half filled & full filled subshell have got extra stability.

Ex.1 $_{24}Cr \rightarrow$ $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $4s^2$ $3d^4$
(wrong configuration)

↑↓ ↑ ↑ ↑ ↑ ↑

$_{24}Cr \rightarrow$ $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $4s^1$ $3d^5$
(right configuration)

↑ ↑ ↑ ↑ ↑ ↑

Ex.2 $_{29}Cu \rightarrow$ $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $4s^2$ $3d^9$
(wrong configuration)

↑↓ ↑↓ ↑↓ ↑↓ ↑↓ ↑

$_{29}Cu \rightarrow$ $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $4s^1$
$3d^{10}$ (right configuration)

↑ ↑↓ ↑↓ ↑↓ ↑↓ ↑↓

Symmetrical Distribution of Electrons

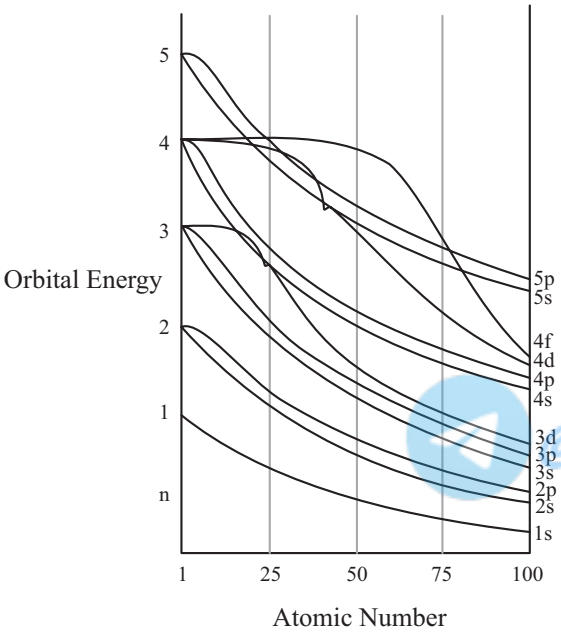
- It is well known that symmetry leads to stability.
- The completely filled or half-filled sub-shells have symmetrical distribution of electrons in them and are therefore more stable.
- Electrons in the same sub-shell (here 3d) have equal energy but different spatial distribution.
- Consequently, their shielding of one another is relatively strongly attracted by the nucleus.

Exchange Energy

- The stabilizing effect arises whenever two or more electrons with the same spin are present in the degenerate orbitals of a sub-shell.
- These electrons tend to exchange their positions and the energy released due to this exchange is called exchange energy.

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- The number of exchanges that can take place is maximum when the sub-shell is either half-filled or completely filled.
- As a result the exchange energy is maximum and so is the stability.
- You may notice that the exchange energy is at the basis of Hund's rule that electrons which enter orbitals of equal energy have parallel spins as far as possible.
- In other words, the extra stability of half-filled and completely filled sub-shell is due to:
- Relatively small shielding
- Smallercoulombic repulsion energy
- Larger exchange energy
- Symmetrical distribution of electrons



Variation of orbital energy with atomic number

Variation of orbital energy with atomic number

Train Your Brain

Example 35: Calculate the number of unpaired e^- in Cr.

Sol. $_{24}Cr \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$

In $_{24}Cr$, $6e^-$ are unpaired.

Example 36: The number of unpaired e^- in Cr^{+3}:

Sol. $Cr^{+3} \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^3$

In Cr^{+3}, $3e^-$ are unpaired.

Example 37: The number of unpaired e^- in 3d subshell of Cr^{+3}

Sol. $Cr^{+3} \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^3$

Therefor number of unpaired electron in 3d subshell is 3.

Example 38: The number of unpaired e^- in Fe^{+2} & Fe^{+3}

Sol. $Fe^{+2} \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^6$ = 4 unpaired e^-

$Fe^{+3} \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^5$ = 5 unpaired e^-

Concept Application

31. How many unpaired electrons are there in Ni^{+2} ion if the atomic number of Ni is 28?

(*a*) 2 (*b*) 3

(*c*) 4 (*d*) 1

32. Which of the following sets of quantum numbers are impossible for electrons ?

Set	n	ℓ	m	s
(i)	1	0	1	$+\frac{1}{2}$
(ii)	3	0	0	$-\frac{1}{2}$
(iii)	1	2	2	$+\frac{1}{2}$
(iv)	4	3	–3	$+\frac{1}{2}$
(v)	5	2	1	$-\frac{1}{2}$
(vi)	3	2	1	0

(*a*) (i), (iii) (*b*) (i), (ii), and (vi)

(*c*) (i), (iii), and (vi) (*d*) (i), (iii), and (iv)

33. The total spin resulting from a d^7 configuration is :

(*a*) 1 (*b*) 2

(*c*) 5/2 (*d*) 3/2

34. Given is the electronic configuration of element X:

K	L	M	N
2	8	11	2

The number of electrons present with $\ell = 2$ in an atom of element X is :

(*a*) 3 (*b*) 6

(*c*) 5 (*d*) 4

THE QUANTUM/WAVE MECHANICAL MODEL

- Quantum mechanics is a theoretical science that deals with the study of the motions of the microscopic objects that have both observable wave-like and particle-like properties.
- It specifies the laws of motion that these objects obey.
- When quantum mechanics is applied to macroscopic objects (for which wave like properties are insignificant) the results are the same as those from the classical mechanics.
- Quantum mechanical model of atom is the picture of the structure of the atom, which emerges from the application of the Schrödinger equation to atoms.

- For a particle whose energy does not vary with time, the Schrodinger equation may be written as

  $\hat{H}\Psi = E\Psi$

- where $\hat{H}$ = Hamiltonian or Total energy operator, ψ = wave function

$$\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}+\frac{8\pi^2\ m(E-V)}{h^2}\psi=0$$

  where x,y and z are Cartesian Co-ordinates, m = mass of the electron,

  E = Total energy of the electron, V = PE of the electron,

  ψ = Wave function of the electron

- For a given system for which V and m are known, we can use the Schrödinger equation to obtain value of E (the allowed energies of the particle) & ψ (the wave function).
- For H-atom, PE $(V)=-\frac{Ke^2}{r}$
- Schrodinger's wave equation becomes

$$\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}+\frac{8\pi^2\ m\left(E+\frac{Ke^2}{r}\right)}{h^2}\psi=0$$

- The wave function itself has no physical meaning, but ψ^2 is a probability density and for this to be the case, ψ must have certain properties:
  - It must be finite for all values of x;
  - It can only have one value for any value of x;
  - And $\frac{d\psi}{dx}$ must vary continuously as x varies.
- ψ must be continuous, have a continuous slope, be single-valued, & be square integrable.
- The results (solutions) of wave equation are
  - The wave function is a solution of the Schrodinger equation and describes the behavior of an electron in region of space called the atomic orbital.
  - We can find energy values that are associated with particular wave functions.
  - The quantization of energy levels arises naturally from the Schrodinger equation.
  - A wave function wis a mathematical function that contains detailed information about the behavior of an electron.
  - An atomic wave function consists of a radial component,R (r), and an angular component, A (β, ϕ).
  - The region of space defined by a wave function is called an atomic orbital.

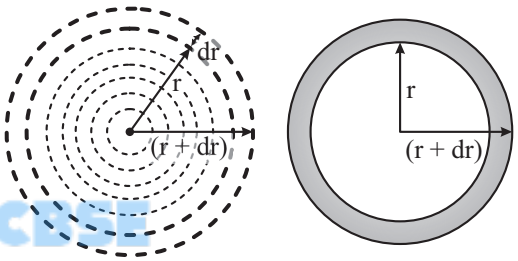
Born Interpretation of Wave Equation

- A principal tenet of quantum mechanics is that the wave function contains all the dynamical information about the system it describes.
- **In the words of Born:** "If the wave function of a particle has the value ψ at some point x, then probability of finding the particle between x and (x + dx) is proportional to $|\psi|^2$ dx.
- "For a three dimensional wave function of a particle, if the wave function of a particle has the value ψ at some point, then probability of finding a particle in a small volume (dV) around that point is $|\psi|^2$ dV".
- The wave function ψ is a probability amplitude in the sense that its square modulus ($|\psi|^2$) is a probability density.
- The physical significance of ψ is difficult to understand, but $|\psi|^2$ has a clear physical significance of being the probability density (probability per unit volume) of the location of particle.

Radial Probability Function

- According to Max Born, probability of finding an electron in a small volume dV is $|\psi|^2$ dV.
- Radial probability distribution function = R^2dV.

  (where R = radial wave function)
- In order to find the value of dV, the entire space of the electron orbital around the nucleus is supposed to be divided into small imaginary concentric shells of very small thickness dr, each.
- Let us find out the volume of the space in between two such shells.



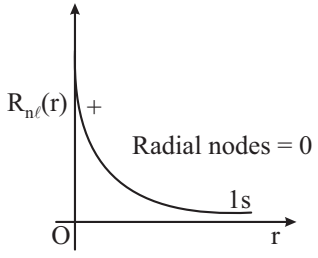
RADIAL WAVE FUNCTION FOR H-LIKE ATOMS

- $R_{n,\ell}$ = Constant × polynomial × Exponential.

(A) $R_{n,\ell}$ (r) v/s r plots:

(i) 1s-orbital:

$$R_{1,0}=2\left(\frac{Z}{a_0}\right)^{3/2}\times 1\times e^{-Zr/a_0}$$



- 1s - orbital has a finite value of $R_{n,\ell}$ (r) at nucleus.

(ii) 2s-orbital:

$$R_{2,0}=\left(\frac{Z}{a_0}\right)^{3/2}\frac{1}{2\sqrt{2}}\times\left(2-\frac{Zr}{a_0}\right)\times e^{-Zr/2a_0}$$

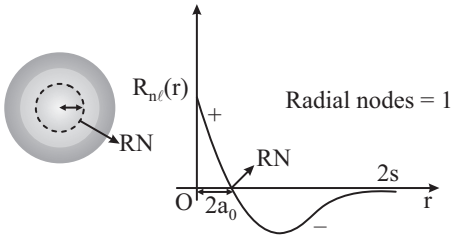
(*a*) If $2>\frac{Zr}{a_0}$, then $R_{2.0}$ is + ve.

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(*b*) If $\frac{Zr}{a_0} > 2$, then $R_{2.0}$ is – ve.

(*c*) If $\frac{Zr}{a_0} = 2$, then $R_{2.0}$ is 0 .

At Radial node, $r = \frac{2a_0}{Z}$

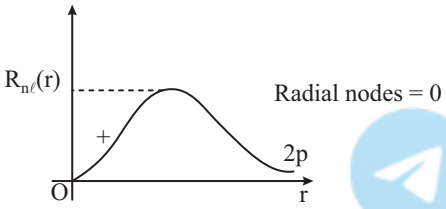


2s - orbital has a finite value of $R_{n,\ell}$ (r) at nucleus.

2s - orbital also decays exponentially but its decay slower than 1s - orbital.

(iii) 2p– subshell:

$$R_{2,1} = \left(\frac{Z}{a_o}\right)^{3/2} \frac{1}{2\sqrt{6}} \times \left(\frac{Zr}{a_o}\right) \times e^{-Zr/2a_0}$$

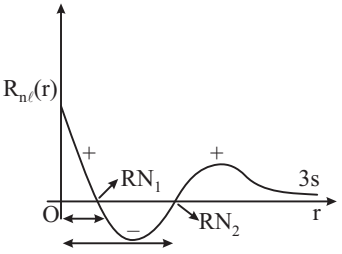


2p - orbital has zero value of $R_{n,\ell}$ (r) at nucleus.

2p - orbital has a maxima in the $R_{n,\ell}$(r) v/s r pbt.

(iv) 3s -orbital:

$$R_{3,0} = \left(\frac{Z}{a_0}\right)^{3/2} \frac{2}{81\sqrt{3}} \times \left(27 - 18\frac{Zr}{a_0} + \frac{2Z^2r^2}{a_o^2}\right) \times e^{-Zr/3a_0}$$

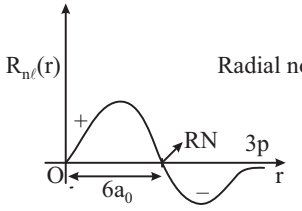


Radial nodes = 2

3s - orbital has a finits value of $R_{n,\ell}$ (r) at nucleus.

(v) 3p-subshell:

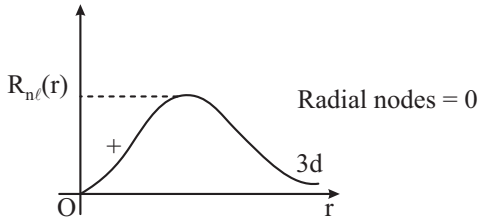
$$R_{3,1} = \left(\frac{Z}{a_0}\right)^{3/2} \frac{4}{81\sqrt{6}} \times \left(6\frac{Zr}{a_0} - \frac{Z^2r^2}{a_0^2}\right) \times e^{-Zr/3a_0}$$



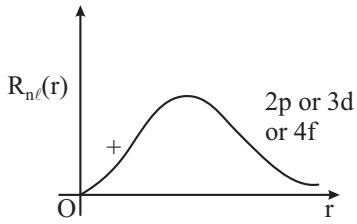
3p - orbital has zero value of $R_{n,\ell}$ (r) at nucleus.

(vi) 3d-subshell:

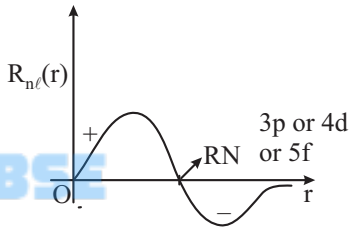
$$R_{3,2} = \left(\frac{Z}{a_0}\right)^{3/2} \frac{4}{81\sqrt{30}} \times \frac{Z^2r^2}{a_0^2} \times e^{-Zr/3a_0}$$



3d - orbital has zero value of $R_{n,\ell}$ (r) at the nucleus.



peak height: 2p > 3d ? 4f

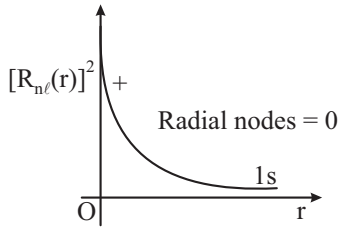


- For all orbitals other than s-orbital, $R_{n,\ell}$ (r) = 0 at the nucleus.
- For all 1s - orbitals, $R_{n,\ell}$ (r) is always positive.
- For the first orbital of other types (i.e. 2p, 3d, 4f etc), $R_{n,\ell}$ (r) is +ve everywhere except at the origin.
- For the second orbital of a given type (i.e. 2s, 3p, 4d, 5f etc.), $R_{n,\ell}$ (r) may be + ve or – ve but the radial wave function has only 1 sign change, i.e. they have 1 radial node.
- For the third orbital of a given type (i.e. 3s, 4p, 5d, 6f etc.), $R_{n,\ell}$ (r) may have 2 sign changes i.e. they have 2 radial nodes.

(B) $[R_{n,\ell}(r)]^2$ v/s r plots:

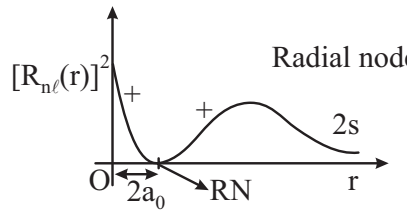
$[R_{n,\ell}(r)]^2$ would always be +ve, irrespective of the sign of $R_{n,\ell}$ (r) for a given value of r. R^2 represents the probability density of finding electron at a particular point in space.

(i) 1s – orbital:

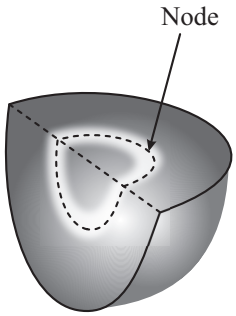


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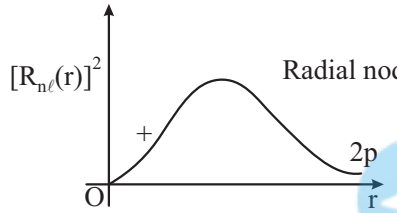
(ii) 2s – orbital:



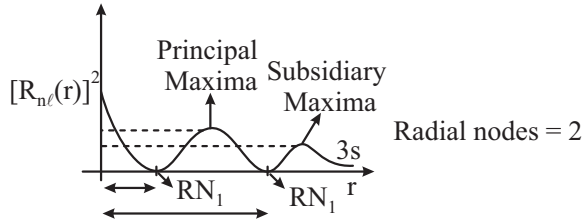
2s -orbital

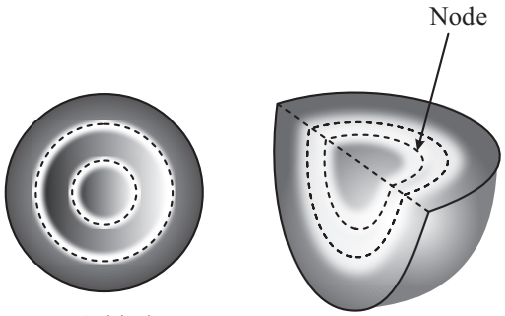


(iii) 2p – subshell:



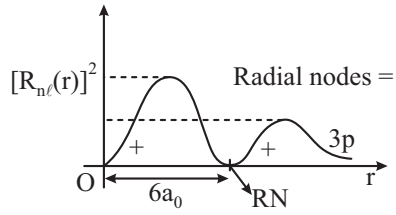
(iv) 3s -orbital:

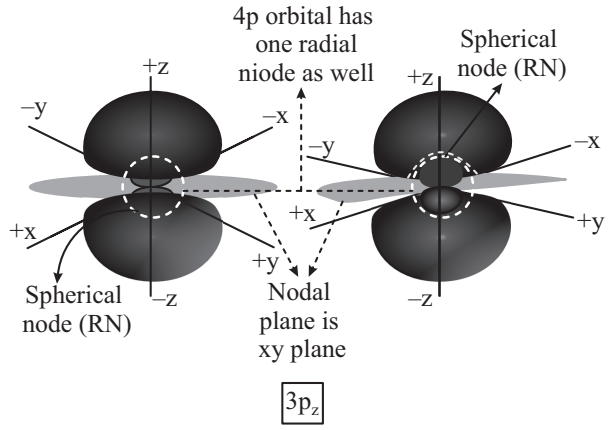




3s- Orbital

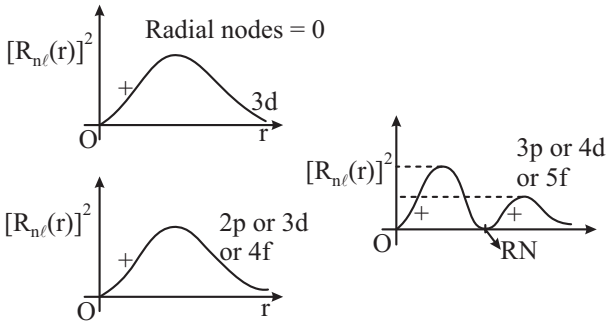
(v) 3p-subshell:





$3p_z$

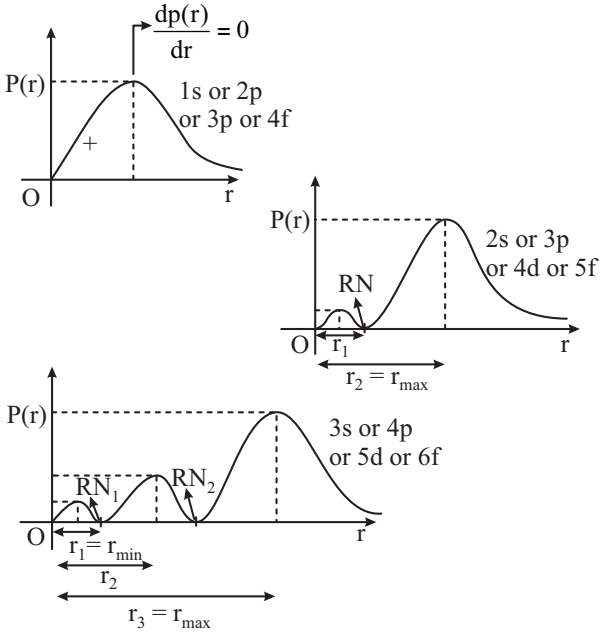
(vi) 3d-subshell:



- **Note:** first maxima is higher than second 4 second is higher than third.
- $R_{n,1}$ (r) v/s r plot, R^2 v/s r plot & $4\pi r^2 R^2$ v/s r plot are all same for '2p' orbital, '3d' orbital, '4f' orbital etc.

(C) $4\pi r2\ [R_{n,1}\ (r)]^2$ v/s r plots:

- Radial distribution function or radial probability distribution function (RPDF)
  $= p(r)\ dr = 4\pi r^2\ [R_{n,1}\ (r)]^2\ dr$
  $p(r) = 4\pi^2\ [R_{n,1}\ (r)]^2$
  The radial distribution is the probability function of choice to determine the most likely radius to find the electron for a given orbital (regardless of its direction).



Train Your Brain

Example 39: The Schrodinger wave equation for hydrogen atom is

$$\Psi_{2s} = \frac{1}{4\sqrt{2\pi}}\left(\frac{1}{a_o}\right)^{3/2}\left(2-\frac{r_o}{a_o}\right)e^{-r_o/a_o}$$

Where a_0 is Bohr's radius. If the radial node in 2s be at r_0, then find r_0 in terms of a_0.

Sol.

Ψ_{2s}^2 = Probability of finding electron within 2s sphere

$\Psi_{2s}^2 = 0$ at node

$$\therefore \frac{1}{32\pi}\left(\frac{1}{a_o}\right)^3\left(2-\frac{r_o}{a_o}\right)^2 e^{-\frac{2r_o}{a_o}} = 0$$

$$\left(2-\frac{r_o}{a_o}\right)^2 = 0; \left(2-\frac{r_o}{a_o}\right) = 0$$

$$2 = \frac{r_o}{a_o}$$

$\therefore r_0 = 2a_0$

Example 40: The Schrodinger wave equation for hydrogen atom is

$$\Psi(\text{ radial}) = \frac{1}{16\sqrt{4}}\left(\frac{Z}{a_o}\right)^{3/2}(\sigma-1)\left(\sigma^2-8\sigma+12\right)e^{-\sigma/2}$$

❖ Where a_0 and z are the constant in which answer can be expressed and $\sigma = \frac{2Zr}{a_o}$. Minimum and maximum position of radial nodes from nucleus are ... respectively.

(a) $\frac{a_o}{Z}, \frac{3a_o}{Z}$ (b) $\frac{a_o}{2Z}, \frac{a_o}{Z}$

(c) $\frac{a_o}{2Z}, \frac{3a_o}{Z}$ (d) $\frac{a_o}{2Z}, \frac{4a_o}{Z}$

Sol. At the node, Since $\psi^2 = 0$, thus $\psi = 0$.

(i) $(\sigma - 1) = 0, \sigma = 1$ $\quad\therefore\quad r_0 = \frac{a_0}{2Z}$

(ii) $(\sigma^2 - 8\sigma + 12) = 0$ $(\sigma^2 - 6\sigma - 2\sigma + 12) = 0$

$(\sigma - 6)(\sigma - 2) = 0; \sigma = 6 = \frac{2Zr_0}{a_0};$

$\therefore \quad r_0 = \frac{3a_0}{Z}$

Radial nodes are at, $\frac{a_0}{2Z}, \frac{a_0}{Z}$ *and* $\frac{3a_0}{Z}$.

Concept Application

35. For an orbital in B^{+4} radial function is:

$$R(r) = \frac{1}{9\sqrt{6}}\left(\frac{z}{a_0}\right)^{\frac{3}{4}}(4-\sigma)\sigma\ e^{-\sigma/2}$$

where $\sigma = \frac{Zr}{a_0}$ and $a_0 = 0.529$ Å ; Z = atomic number, r = radial distance from nucleus.

The radial node of orbital is at distance from nucleus.

36. The correct time independent Schrödinger's wave equation for an electron with E as total energy and V as potential energy is :

(a) $\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}+\frac{8\pi^2}{mh^2}(E-V)\psi = 0$

(b) $\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}+\frac{8\pi m}{h^2}(E-V)\psi = 0$

(c) $\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}+\frac{8\pi^2 m}{h^2}(E-V)\psi = 0$

(d) $\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}+\frac{8\pi m^2}{h}(E-V)\psi = 0$

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Short Notes

Rutherford's Scattering Experiment

Q = ne (charge is quantized)

P.E. = $\frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r}$

Mass Number and Atomic Number

Mass number of an element = No. of protons (Z) + No. of neutrons (n).

Distance of Closest Approach

Closest distance (r) = $\frac{4KZe^2}{m_\alpha v_\alpha^2}$

Wave and its Characteristics

$\bar{\nu} = \frac{1}{\lambda}$

$v = \nu \times \lambda$

$E_0 = h\nu$ (ν - Frequency of light)

$E_0 = \frac{hc}{\lambda}$ (c - speed of light)

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Photoelectric Effect

$h\nu = h\nu_0 + \frac{1}{2} m_e v^2$

Bohr's Model of Atom

- $\frac{mv^2}{r} = \frac{Ke^2Z}{r^2}$
- $mvr = \frac{nh}{2\pi}$
- $\frac{hc}{\lambda} = \Delta E$
- $\upsilon = \frac{\Delta E}{h}$
- $r = \frac{n^2h^2}{4\pi^2 mKZe^2}$
- $r_n = 0.529 \times \frac{n^2}{Z}$ Å
- $v = \frac{2\pi Ze^2K}{nh}$
- $v_n = 2.18 \times 10^6 \times \frac{Z}{n}$ m/sec
- $T = \frac{2\pi r}{v}$
- $f = \frac{v}{2\pi r}$
- T.E. = $E_n = -\frac{2\pi^2 me^4 k^2}{h^2}\left(\frac{z^2}{n^2}\right)$
- $E_n = -13.6 \frac{Z^2}{n^2}$ eV / atom
- $E_n = -2.18 \times 10^{-18} \frac{Z^2}{n^2}$ J/atom
- T.E. = $\frac{1}{2}$ P.E.
- T.E. = – K.E.

Emission Spectrum of Hydrogen

- $\frac{1}{\lambda} = \bar{\nu} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

R = Rydberg constant = 1.09678×10^7 m^{-1}

- Number of different line produce = $\frac{\Delta n(\Delta n+1)}{2}$, where $\Delta n = n_2 - n_1$.

  n_2 = higher energy orbit, n_1 = lower energy orbit.
- For single isolated atom max. number of spectral lines observed = (n – 1).

De-Broglie's Hypothesis

- $\lambda = \frac{h}{mv}$
- $\lambda = \frac{h}{\sqrt{2emV}}$
- $\lambda = \frac{12.3}{\sqrt{V}}$ Å

Heisenberg's Uncertainty

- $\Delta x \,.\, \Delta p \geq \frac{h}{4\pi}$ or $\Delta x \,.\, (m\Delta v) \geq \frac{h}{4\pi}$
- $\Delta E.\Delta t \geq \frac{h}{4\pi}$

Schrodingers Wave Equation for Hydrogenic Atom

- $\frac{\delta^2\psi}{\delta x^2} + \frac{\delta^2\psi}{\delta y^2} + \frac{\delta^2\psi}{\delta z^2} + \frac{8\pi^2 m}{h^2}(E-V)\psi = 0$

$\Psi(r,\theta,\phi) = R(r).\, Y(\theta,\phi)$

$R(r) = R(n,\ell) \propto \left(\frac{2zr}{na_0}\right)^{\ell} .e^{-zr/na_0}$

$Y(\theta,\phi) = Y(\ell, m) \propto \sin^{|m|}\theta .\, \cos^{\ell-|m|}\theta .\, e^{im\phi}$

if $\psi = e^{-\alpha x^2}$, Probabilty density $= |\psi|^2 = e^{-2\alpha x^2}$

for most probable $\frac{d|\psi|^2}{dx} = 0$

- Radial nodes = $n - \ell - 1$,
- Angular nodes = ℓ,
- Total nodes = n – 1

Quantum Numbers

- Number of subshell present in n^{th} shell = n.
- Number of orbitals present in n^{th} shell = n^2.
- The maximum number of electrons in a principal energy shell = $2n^2$.
- Angular momentum of any orbit = $\frac{nh}{2\pi}$.
- Number of orbitals in a subshell = $2\ell + 1$
- Maximum number of electrons in particular subshell = $2 \times (2\ell + 1)$.
- $L = \frac{h}{2\pi}\sqrt{\ell(\ell+1)} = \hbar\sqrt{\ell(\ell+1)}\left[\hbar = \frac{h}{2\pi}\right]$.
- Orbitals present in a main energy level is 'n^2'.
- $\mu = \sqrt{n(n+2)}$ B.M. .
- Spin angular momentum $= \frac{h}{2\pi}\sqrt{s(s+1)}$.
- Maximum spin of atom = $\frac{1}{2}\times$ No. of unpaired electron.

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Solved Examples

1. The bond dissociation energy of Cl–Cl bond in chlorine gas is 240 kJ/mol. Calculate the longest wavelength of EMR needed to dissociate bond. Assume one photon may dissociate only one bond.

Sol. $E = \frac{240\times10^3}{6\times10^{23}} = \frac{1\times6.626\times10^{-34}\times3\times10^8}{\lambda}$

2. A near ultra violet photon of wavelength 300 nm is absorbed by a gas and then emitted as two photons. One photon is of red light with wavelength 760 nm. What would be the wave length of the second photon ?

Sol. It may noted that energy of photon which adsorbed is emitted as sum of the energy of two photons.

Energy absorbed $hv = \frac{hc}{\lambda}$

3. The ionization energy of hydrogen atom is 13.6 eV. What will be the ionization energy of He^+ –?

Sol. He^+ is a hydrogen like species i.e. the electron is ionised from first orbit.

$\therefore$ Ionization energy of $He^+ = \frac{Z^2E_H}{n^2}$

$= \frac{4\times13.6}{1^2} = 54.4$ eV

4. The ionization energy of H-aotm is 13.6 eV. The ionization energy of Li^{+2} ion will be:

(*a*) 13.6 eV (*b*) 27.2 eV

(*c*) 54.4 eV (*d*) 122.4 eV

Sol. (*d*) E_1 for $Li^{+2} = E_1$ for $H \times Z^2$ [for Li, Z = 3]

$= 13.6 \times 9$

$= 122.4$ eV

5. Consider ψ (wave function) of 2s atomic orbital of H-atom is

$$\Psi_{2s} = \frac{1}{4\sqrt{2\pi a_0^{3/2}}}\left[2-\frac{r}{a_0}\right]e^{\frac{r}{2a_0}}$$

Find distance of radial node from nucleus in terms of a_0.

Sol. R(r) = 0

$$\left[2-\frac{r}{a_0}\right]e^{\frac{r}{2a_0}} = 0 \Rightarrow 2-\frac{r}{a_0} = 0 \Rightarrow r = 2a_0$$

6. (*a*) The Schrodinger wave equation for hydrogen atom is

$$\psi_{2s} = \frac{1}{4(2\pi)^{1/2}}\left(\frac{1}{2_0}\right)^{3/2}\left(2-\frac{r_0}{a_0}\right)e^{-r/a}$$

Where a_0 is Bohr's radius. Let the radial node in 2s be at r_0. Then find r_0 in terms of a_0.

(b) A base ball having mass 10 g and projected with speed of 100 m/s. Find out the value of wavelength of base ball.

Sol. (*a*) $r_0 = 2a_0$, (b) 6.626×10^{-25} Å

$$\psi_{2s} = \frac{1}{4(2\pi)^{1/2}}\left(\frac{1}{a_0}\right)^{3/2}\left(2-\frac{r_0}{a_0}\right)e^{-r/a}$$

(a) $2-\frac{r_0}{a_0} = 0$

$\frac{r_0}{a_0} = 2$

$r_0 = 2a_0$

(b) $\lambda = \frac{h}{mv}$

$= \frac{6.626\times10^{-34}}{(0.1)(100)}$

$= 6.626\times10^{-25}$ Å

7. Calculation of different types of maximum possible radiations which can be observed in (from higher energy level to lower energy level).

Sol. $n_2 = n$, $n_1 = 1$

Total no. of radiation obtained having difference wave lengths = Transition ending at 1st level + Transition ending at 2nd level +.....+ Transition ending at (n – 1) energy levels.

= Total no. of radiation in lyman series + total no. of radiation in Balmer series .....+ Total no. of radiation in (n – 1) series.

= (n – 1) + (n – 2) + (n – 3) ..... + 2 + 1

$= \frac{(n-1)(n-1+1)}{2} \Rightarrow \frac{n(n-1)}{2}$

= $(n_1–n_1) + n_2 – (n_1 + 1)$..... +2 +1

Total no. of different radiation

$= \frac{(n_2-n_1)(n_2-n_1+1)}{2} = \frac{N(N+1)}{2}$

N = difference in higher energy level and lower energy level.

8. The energy level of an atom for 1st, 2nd and third levels are E, $4E_{1\to3}$, 2E respectively. If photon of wavelengthl is ....λ emitted for a transition 3 to 1. Calculate the wavelength for transition 2 to 1 in terms of λ.

Sol. Energy of photon released = $E_3 – E_1$ ...(i)

As we know that energy of photon is given by = $\frac{hc}{\lambda}$

where λ wavelength of photon

from (1)

$$\frac{hc}{\lambda} = E_3 - E_1 = E \Rightarrow \lambda = \frac{hc}{E} \quad \text{... (ii)}$$

On transition from 2 → 1 energy level.

Energy of photon released = $E_2 – E_1$ ... (iii)

Let, wavelength of photon released be λ

$\Rightarrow$Energy of photon released $= \frac{hc}{\lambda}$

From (iii)

$$E_2 - E_1 = \frac{hc}{\lambda,}$$

$$\Rightarrow 3E = \frac{hc}{\lambda,}$$

$$\Rightarrow \lambda' = \frac{hc}{3E} = \frac{1}{3}\left(\frac{hc}{E}\right)$$

$$\Rightarrow \lambda' = \frac{1}{3}(\lambda)(\text{from (ii)})$$

$$\Rightarrow \lambda' = \frac{1}{3}$$

Hence, wavelength of photon emitted for transition 2 → 1 in $\frac{\lambda}{3}$.

9. Two electromagnetic radiations have wave numbers in the ratio 3:3. Their energies per quanta will be in the ratio

(*a*) 3:2 (*b*) 9:4

(*c*) 4:9 (*d*) 2:3

Sol. $\frac{\bar{\nu}_1}{\bar{\nu}_2} = \frac{2}{3}$ (given)

We know–

$E = \frac{nhc}{\lambda}$ or $E = nhc\,\bar{\nu}$

Where $\bar{\nu}$ = wake number

$$\therefore \frac{\frac{E_1}{n_1}}{\frac{E_2}{n_2}} = \frac{hc\bar{\nu}_1}{hc\bar{\nu}_2} \text{ or } \frac{E_1'}{E_2'} = \frac{\bar{\nu}_1}{\bar{\nu}_2} = \frac{2}{3}$$

Where, E′ = Energy per quanta

10. Hydrogen like species is observed to emit 6 wavelengths originating from all possible transitions between a group of levels these energy levels have energy between – 0.85 eV and –0.544 eV (including both these energy levels) calculate:

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(i) The quantum number of levels between which transition is taking place.

(ii) Find the atomic number of species.

Sol. (i) To bind the atomic number of the atom we have

$$n\frac{(n-1)}{2}=6$$

$n(n-1)=12$

$n^2-n-12=0$

$n^2-4n-2n-12=0$

$n(n-4)-3(n-4)=0$

$(n-4)(n-3)=0$

$n=4, 3$

$n=4$

i.e transition is taking place from

$m \rightarrow m+3$

$Em = 0.85$ ev

$Em+3 = -0.544$ ev

$$-0.85=-13.6\times\frac{z^2}{m^2} \text{ or, } z/m=0.25 \quad \text{... (i)}$$

$$-0.544=-13.6\times\frac{z^2}{(m+3)^2}$$

on solving, $\frac{z}{m+3}=0.2$ ... (ii)

on solving (i) & (ii)

$z=3, m=12$

(ii) $\Delta E_{max} = -0.544-(-0.85)=0.306$ eV

$$\lambda \text{ min} = \frac{hc}{\Delta E_{max}}=\frac{1241}{0.306}=4052.3nm$$

11. Arrange the particle in their increasing order of specific charge ratio.

$Na^+, Li^+, F^-, Mg^{2+}, Al^{3+}$

Sol. $Na^+=\frac{1}{23}; Li^+=\frac{1}{7}; F^-=\frac{1}{19}; Mg^{2+}=\frac{2}{24}=\frac{1}{12}; Al^{3+}=\frac{3}{27}=\frac{1}{9}$

$Na^+ < F^- < Mg^{2+} < Al^{3+} < Li^+$

12. Which of the following pairs have same specific charge $\left(\frac{e}{m}\right)$?

(*a*) electron & proton

(*b*) electron & positron

(*c*) proton & deutron

(*d*) α-particle & deutron

(*e*) proton & positron

Sol. (*b,d*) $M_e = M_{positron}$

$M_\alpha = m_{deutron}$

13. Through what potential difference an α-particle should be accelerated to have speed 5×10^6 m/s.

Sol. $qV=\frac{1}{2}mv^2$

$$\left(2\times1.6\times10^{-19}\times V\right)=\frac{1}{2}\times4\times1.66\times10^{-27}\times\left(5\times10^6\right)^2$$

V = 0.26 MV

14. The distance of closest approach of an α-particle fired towards a nucleus with momentum 'P' is *r*. What will be the distance of closest approach when the momentum of the α-particle is 2P?

(*a*) 1*r* (*b*) 4*r*

(*c*) *r*/2 (*d*) *r*/4

Sol. We know–

$$K.E.=\frac{p^2}{2m} \quad \text{...(i)}$$

and K. E. $=\frac{Kq_1,q_2}{r}$ ...(ii)

from (i) and (ii)

$$\frac{p^2}{2m}=\frac{kq_1q_2}{r}$$

As m, k, q_1 and q_2 are constant

$$p^2\propto\frac{1}{r} \text{ or } r\propto\frac{1}{p^2}$$

Now $\frac{r_2}{r_1}=\frac{P_1^2}{P_2^2}$

given $r_1 = r$, $P_1 = P$ $P_2 = 2P$

$$\therefore \frac{r_2}{r}=\frac{P^2}{4P^2} \text{ or } r_2=\frac{r}{4}$$

15. Wavelength of photon which have energy equal to average of energy of photons with $\lambda_1 = 4000$ Å and $\lambda_2 = 6000$ Å will be:

(*a*) 5000 Å (*b*) 4800 Å

(*c*) 9600 Å (*d*) 2400 Å

Sol. According to question $E=\frac{E_1+E_2}{2}$

$$\text{or } \frac{hc}{\lambda}=\frac{\frac{hc}{\lambda_1}+\frac{hc}{\lambda_2}}{2}$$

$$\text{or } \frac{1}{\lambda}=\left(\frac{\frac{1}{4000}+\frac{1}{6000}}{2}\right)$$

or $\lambda = 4800$ Å

16. The force of attraction on electron by the nucleus is directly proportional to

(*a*) $\frac{n^3}{Z^4}$ (*b*) $\frac{Z^3}{n^4}$

(*c*) $\frac{n^4}{Z^2}$ (*d*) $\frac{Z^2}{n^4}$

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Sol. (*d*) $F = \frac{q_1 q_2}{r^2}$ ... (i)

$q_1 = e^-$, $q_2 = Ze^-$

and $r = 0.529\ \frac{n^2}{z}$. Å

Putting all these volues in equation ... (i)

$$F = \frac{Ke^2 z}{\left(0.529\frac{n^2}{z}\text{Å}\right)^2}$$

or $F \propto \frac{z^3}{n^4}$

17. Suppose the mass of electron decreased by 25%. How will it affect the Rydberg constant?

(*a*) It remains unchanged.

(*b*) It becomes one-fourth.

(*c*) It reduces to 75% of its original value.

(*d*) It is doubled.

Sol. $R_H = \frac{e^4 me}{8 \in_0^2 h^3 c}$ *or* $R_H \propto m_e$ ... (i)

Let m'_e is the new mass after 25% reduction of orginal m_e

i.e $m'_e = m_e - \frac{m_e}{4}$ or $m'_e = \frac{3m_e}{4}$

New $R'_H = \frac{e^4 m'_e}{8 \in_0^2 h^3 c}$ or $R'_H \propto \frac{3m'_e}{4}$ ... (ii)

From (i) and (ii)

$R'_H = \frac{3}{4} R_H$ or $R'_H = 75\%\ R_H$

18. Calculate the energy per quanta of an EMR of frequency 400 MHz.

Sol. $E = nh\nu$ or $\frac{E}{n} = 6.626 \times 10^{-34} \times 4 \times 10^6$ J/quanta

or $\frac{E}{n}$ 2.65 × 10^{-25} J/quanta

19. Calculate the energy per quanta of an EMR of wavelength 662.6 nm.

Sol. $\frac{E}{n} = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{662.6 \times 10^{-9}\ m} = 3 \times 10^{-15}$ J/quanta

20. Calculate the wavelength (in Å) of an EMR of energy 3.1eV/ quanta.

Sol. $\frac{E}{n} = h\frac{c}{\lambda} \Rightarrow 3.1 \times 1.602 \times 10^{-19} = \frac{1 \times 6.62 \times 10^{-34} \times 3 \times 10^8}{\lambda}$

or λ = 4007.7Å

21. In order to see an object, 10^{-19}J must be recived by our eyes. How many photons of green light must be recieved by our eyes for its visibility. $\left(\lambda = 550\ \text{nm}, h = 6.6 \times 10^{-34}\ \text{J}\right)$

Sol. : $E = n\frac{hc}{\lambda}$

$$10^{-19} = n \times \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{550 \times 10^{-9}}$$

$n \approx 1$

22. Rydberg given the equation for all visible radiation in the hydrogen spectrum as $\lambda = \frac{kn^2}{n^2 - 4}$. The value of *k* in terms of Rydberg constant is:

(*a*) 4*R* (*b*) $\frac{R}{4}$

(*c*) $\frac{4}{R}$ (*d*) *R*

Sol. According to question

$z = 1$, $n_1 = 2$, $n_2 = n$

$$\frac{1}{\lambda} = R_H\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right](z)^2$$

or $$\frac{1}{\lambda} = R_H\left[\frac{1}{2^2} - \frac{1}{n^2}\right](1)^2$$

or $$\frac{1}{\lambda} = R_H\left[\frac{n^2 - 4}{4n^2}\right]$$

or $$\lambda = \frac{4n^2}{R_H(n^2 - 4)}$$ ... (i)

And also $\lambda = \frac{kn^2}{(n^2 - 4)}$ (given) ... (ii)

From (i) and (ii)

$k = \frac{4}{R_H}$

23. If the P.E. of an electron is –6.8 eV in hydrogen atom then find out K.E., E of orbit where electron exist & radius of orbit.

Sol. (i) P.E. = –2K.E.

–6.8 = –2K.E.

$\frac{6.8}{2}$ = K.E. K.E. = 3.4eV

(ii) E. = –K.E.

= –3.4eV

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(iii) Orbit =2nd

$$\because\ E = -13.6\times\frac{Z^2}{n^2}$$

$$\therefore\ 3.4 = -13.6\times\frac{1^2}{n^2}$$

$$\Rightarrow\ n^2 = \frac{-13.6}{-3.4} = 4$$

i.e. $n = 2$

(iv) $r = 0.529\times\frac{n^2}{Z}\text{Å}$

$$r = 0.529\times\frac{(2)^2}{1}\text{Å}$$

$$= 0.529\times 4\text{Å} = 2.16\text{Å}$$

24. The ionization energy for the hydrogen atom is 13.6eV then calculate the required energy in eV to excite it from the ground state to 1st excited state.

Sol. Ionization energy $= 13.6\text{eV}$
i.e. 1st energy state $= -13.6\text{eV}$
Energy of 1st excited state
i.e. 2nd orbit $= -3.4\text{eV}$
so, $E_2 - E_1 = -3.4 + 13.6 = 10.2\text{eV}$

25. In a hydrogen spectrum if electron moves from 6th to 2nd by transition in muli steps then find out the number of lines in spectrum

Sol. Total number of line $= 4+3+2+1+0 = 10$

$$\text{Total number of lines} = \frac{(n_2 - n_1)\left[(n_2 - n_1)+1\right]}{2}$$

$$= \frac{(6-2)(4+1)}{2} = 10$$

26. In the spectrum of H^+ ion the wavelength of α line of Balmer series is xÅ. What is the wavelength of α line of Paschen series.

Sol. $$\frac{1}{\lambda_1} = Rz^2\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$$

$$\frac{1}{\lambda_2} = Rz^2\left(\frac{1}{3^2} - \frac{1}{5^2}\right)$$

$$\frac{\lambda_2}{\lambda_1} = \frac{\frac{1}{2^2} - \frac{1}{3^2}}{\frac{1}{3^2} - \frac{1}{5^2}} \Rightarrow \frac{\lambda_2}{x} = \frac{5}{16}\times\frac{25}{4}$$

$$\lambda_2 = \frac{125}{64}x\,\text{Å}$$

27. A sample of He^+ ions in ground state absorbs the radiation of xÅ subsequently, the sample emit radiation of 6 different wavelength. Calculate the value of x.

Sol. $\frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2} = 6$ given, $n_1 = 1$

$$\frac{(n-1)(n-1+1)}{2} = 6$$

On solving
n = 4 or n = 3
6 different wavelength will be obtained only from n = 4

$$\frac{1}{x} = R_H \times z^2\left(\frac{1}{1^2} - \frac{1}{4^2}\right)$$

$$\frac{1}{x} = R_H \times z^2\left(\frac{1}{11} - \frac{1}{16}\right)$$

$$\frac{1}{x} = R_H \times \frac{\cancel{4}\times 15}{\cancel{16}\ 4}$$

or $n = \frac{4}{15R_H}$ *or* $n = \frac{4}{15\times 1.09678\times 10^7\, m^1}$

or n = 243.Å

28. The orbital angular momentum of an electron is $\sqrt{3}\frac{h}{\pi}$.

Which of the following may be the permissible value of angular momentum of this electron revolving in unknown Bohr orbit?

(*a*) $\frac{h}{\pi}$ (*b*) $\frac{h}{2\pi}$

(*c*) $\frac{3h}{2\pi}$ (*d*) $\frac{2h}{\pi}$

Sol. Orbital angular momentum $= \sqrt{l(l+1)}\frac{h}{2\pi}$

Orbital angular momentum $= \sqrt{3}\frac{h}{\pi}$

$$\therefore\ \sqrt{l(l+1)}\times\frac{h}{2\pi} = \sqrt{3}\frac{h}{\pi}$$

On Solving above equation, we get $l = -4$ and 3 (-4 not allowed as $l < 0$)
For $l = 3$, n = 4, 5, 6 (Possible)
Therefor possible angular momentum

$$= \frac{4h}{2\pi}, \frac{5h}{2\pi}, \frac{6h}{2\pi}, \frac{7h}{2\pi}, \ldots.$$

Exercise-1 (Topicwise)

DISCOVERY OF SUB ATOMIC PARTICLES, ATOMIC MODELS, NUCLEUS

1. The fraction of volume occupied by the nucleus with respect to the total volume of an atom is:
(*a*) 10^{-15} (*b*) 10^{-5}
(*c*) 10^{-30} (*d*) 10^{-10}

2. The approximate size of the nucleus of $^{64}_{28}Ni$ is:
(*a*) 3 fm (*b*) 4 fm
(*c*) 5 fm (*d*) 2 fm

3. The ratio of e/m for a cathode ray:
(*a*) Varies with a gas in a discharge tube.
(*b*) Is fixed.
(*c*) Varies with different electrodes.
(*d*) Is maximum if hydrogen is taken.

4. Cathode rays have:
(*a*) Mass only
(*b*) Charge only
(*c*) No mass and no charge
(*d*) Mass and charge both

5. Which is the correct statement about proton?
(*a*) It is a nucleus of deuterium.
(*b*) It is an ionised hydrogen molecule.
(*c*) It is an ionised hydrogen atom.
(*d*) It is an α-particle.

6. Rutherford's experiment on scattering of α-particles showed for the first time that the atom has:
(*a*) Electrons (*b*) Protons
(*c*) Neutrons (*d*) Nucleus

7. When alpha particles are sent through a thin metal foil, most of them go straight through the foil because:
(*a*) Alpha particles are much heavier than electrons.
(*b*) Alpha particles are positively charged.
(*c*) Most part of the atom is empty space.
(*d*) Alpha particles move with very high velocity.

8. Increasing order (lowest first) for the values of e/m (charge/mass) for electron (e), proton (p), neutron (n) and α-particle (α) is:
(*a*) e, p, n, α (*b*) n, p, e, α
(*c*) n, p, α, e (*d*) n, α, p, e

9. Nucleons are:
(*a*) Protons and neutrons.
(*b*) Neutrons and electrons.
(*c*) Protons and electrons.
(*d*) Protons, neutrons and electrons.

10. The discovery of neutron came very late because:
(*a*) It is present in nucleus.
(*b*) It is a fundamental particle.
(*c*) It does not move.
(*d*) It does not carry any charge.

11. Atomic number of an element represents:
(*a*) Number of neutrons in the nucleus.
(*b*) Atomic mass of an element.
(*c*) Valency of an element.
(*d*) Number of protons in the nucleus.

12. The highest value of e/m of anode rays has been observed when the discharge tube is filled with:
(*a*) Nitrogen (*b*) Oxygen
(*c*) Hydrogen (*d*) Helium

QUANTUM THEORY OF LIGHT & PHOTOELECTRIC EFFECT

13. The MRI (magentic resonance imaging) body scanners used in hospitals operate with 400 MHz radio frequency. The wavelength corresponding to this radio frequency is:
(*a*) 0.75 m (*b*) 0.75 cm
(*c*) 1.5 m (*d*) 2 cm

14. Electromagnetic radiations of wavelength 242 nm is just sufficient to ionise Sodium atom. Then the ionisation energy of Sodium in kJ $mole^{-1}$ is:
(*a*) 494.65 (*b*) 400
(*c*) 247 (*d*) 600

15. A bulb of 40 W is producing a light of wavelength 620 nm with 80% of efficiency then the number of photons emitted by the bulb in 20 seconds are (1eV = 1.6×10^{-19} J, hc = 12400 eV Å):
(*a*) 2×10^{18} (*b*) 10^{18}
(*c*) 10^{21} (*d*) 2×10^{21}

16. Light of wavelength λ falls on metal having work function hc/λ_0. Photoelectric effect will take place only if :
(*a*) $\lambda \geq \lambda_0$ (*b*) $\lambda \geq 2\lambda_0$
(*c*) $\lambda \leq \lambda_0$ (*d*) $\lambda \leq \lambda_0/2$

17. A light whose frequency is equal to 6×10^{14} Hz is incident on a metal whose work function is 2 eV ($h = 6.63 \times 10^{-34}$ Js, 1 eV = 1.6×10^{-19} J). The maximum energy of electrons emitted will be:
(*a*) 2.49 eV (*b*) 4.49 eV
(*c*) 0.49 eV (*d*) 5.49 eV

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18. A 600 W mercury lamp emits monochromatic radiation of wavelength 331.3 nm. How many photons are emitted from the lamp per second?
($h = 6.626 \times 10^{-34}$ Js, velocity of light = 3×10^8 ms^{-1})
(*a*) 1×10^{19} (*b*) 1×10^{20}
(*c*) 1×10^{21} (*d*) 1×10^{23}

19. Photoelectric effect can be explained by assuming that light:
(*a*) Is a form of transverse wave.
(*b*) Is a form of longitudinal wave.
(*c*) Can be polarised.
(*d*) Consists of quanta.

20. When the frequency of light incident on a metallic plate is doubled above threshold frequency, the KE of the emitted photoelectrons will be:
(*a*) Doubled
(*b*) Halved
(*c*) Increased but more than double of the previous KE
(*d*) Unchanged

21. A photon of 300 nm is absorbed by a gas and then emits two photons. One photon has a wavelength 496 nm then the wavelength of second photon is:
(*a*) 757 (*b*) 857
(*c*) 957 (*d*) 657

BOHR MODEL

22. Which is the correct relationship:
(*a*) E_1 of H = 1/2 E_2 of He^+ = 1/3 E_3 of Li^{2+} = 1/4 E_4 of Be^{3+}
(*b*) $E_1(H) = E_2(He^+) = E_3(Li^{2+}) = E_4(Be^{3+})$
(*c*) $E_1(H) = 2E_2(He^+) = 3E_3(Li^{2+}) = 4E_4(Be^{3+})$
(*d*) No relation

23. If velocity of an electron in 1st orbit of H atom is V, what will be the velocity of electron in 3rd orbit of Li^{+2}?
(*a*) V (*b*) V/3
(*c*) 3 V (*d*) 9 V

24. In a certain electronic transition in the hydrogen atoms from an initial state (1) to a final state (2), the difference in the orbital radius ($r_1 - r_2$) is 24 times the first Bohr radius. Identify the transition:
(*a*) 5 → 1 (*b*) 25 → 1
(*c*) 8 → 3 (*d*) 6 → 5

25. **S_1** : Potential energy of the two opposite charge system increases with the decrease in distance.
S_2 : When an electron make transition from higher orbit to lower orbit it's kinetic energy increases.
S_3 : When an electron make transtition from lower energy to higher energy state its potential energy increases.
S_4 : 11eV photon can free an electron from the 1st excited state of He^+ -ion.
(*a*) T T T T (*b*) F T T F
(*c*) T F F T (*d*) F F F F

26. Energy of Bohr orbit:
(*a*) Increases as we move away from the nucleus.
(*b*) Decreases as we move away from the nucleus.
(*c*) Remains the same as we move away from the nucleus.
(*d*) None of the above.

27. The radius of the first orbit of H-atom is r. Then, the radius of the first orbit of Li^{2+} will be:
(*a*) r/9 (*b*) r/3
(*c*) 3r (*d*) 9r

28. According to Bohr theory, the angular momentum for an electron of 5 th orbit is:
(*a*) $5\ h/\pi$ (*b*) $2.5\ h/\pi$
(*c*) $5\ \pi/h$ (*d*) $25\ h/\pi$

29. Which of the following electron transitions in hydrogen atom will require largest amount of energy?
(*a*) from n = 1 to n = 2 (*b*) from n = 2 to n = 3
(*c*) from n = ∞ to n = 1 (*d*) from n = 3 to n = 5

30. $E_n = -313.6/n^2$ kcal/mol. If the value of E = –34.84 kcal/mol, to which value does 'n' correspond?
(*a*) 4 (*b*) 3
(*c*) 2 (*d*) 1

31. If the velocity of an electron in the first Bohr orbit of a hydrogen atom is V, then its velocity in the third Bohr orbit will be:
(*a*) V/9 (*b*) V/3
(*c*) 9 V (*d*) 3 V

32. The ratio of the radius of Bohr first orbit for the electron orbiting the hydrogen nucleus to that of the electron orbiting the deuterium nucleus (mass nearly twice that of H nucleus) is approximately:
(*a*) 1 : 1 (*b*) 1 : 2
(*c*) 2 : 1 (*d*) 1 : 4

33. An electron in H-atom in its ground state absorbs 1.50 times as much as energy as the minimum required for its escape (13.6 eV) from the atom. Thus KE given to emitted electron is:
(*a*) 13.6 eV (*b*) 20.4 eV
(*c*) 34.0 eV (*d*) 6.8 eV

34. Bohr's theory is not valid for the species:
(*a*) H atom (*b*) He^+ ion
(*c*) Li^+ ion (*d*) Li^{2+} ion

35. If r_1 is the radius of the first orbit of hydrogen atom, then the radii of second, third and fourth orbitals in terms of r_1 are:
(*a*) r_1^2, r_1^3, r_1^4 (*b*) $8r_1, 27r_1, 64r_1$
(*c*) $4r_1, 9r_1, 16r_1$ (*d*) $2r_1, 6r_1, 8r_1$

SPECTRUM

36. In a sample of H-atom electrons make transition from 5th excited state upto ground state, producing all possible types of photons, then number of lines in infrared region are:
(*a*) 4 (*b*) 5
(*c*) 6 (*d*) 3

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37. Total no. of lines in Lyman series of H spectrum will be (where n = no. of orbits):

(*a*) n (*b*) n – 1

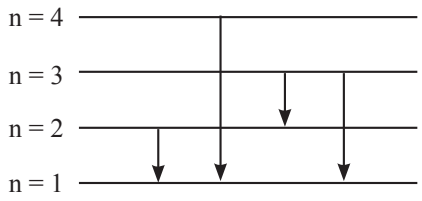
(*c*) n – 2 (*d*) n (n + 1)

38. No. of visible lines when an electron returns from 5th orbit upto ground state in H spectrum:

(*a*) 5 (*b*) 4

(*c*) 3 (*d*) 10

39. Suppose that a hypothetical atom gives a red, green, blue and violet line spectrum. Which jump according to figure would give off the red spectral line?



(*a*) 3 → 1 (*b*) 2 → 1

(*c*) 4 → 1 (*d*) 3 → 2

40. The difference between the wave number of 1st line of Balmer series and last line of paschen series for Li^{2+} ion is:

(*a*) $\frac{R}{36}$ (*b*) $\frac{5R}{36}$

(*c*) 4R (*d*) $\frac{R}{4}$

41. The spectral lines corresponding to the radiation emitted by an electron jumping from higher orbits to first orbit belong to:

(*a*) Paschen series (*b*) Balmer series

(*c*) Lyman series (*d*) None of these

42. For the Paschen series, the value of n_1 and n_2 in the expression $\Delta E = R_M \times c\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$ is:

(*a*) $n_1 = 1, n_2 = 2, 3, 4\ldots$ (*b*) $n_1 = 2, n_2 = 3, 4, 5\ldots$

(*c*) $n_1 = 3, n_2 = 4, 5, 6\ldots$ (*d*) $n_1 = 4, n_2 = 5, 6, 7\ldots$

43. If the series limit of wavelength of the Lyman series for the hydrogen atoms in 912 Å, then the series limit of wavelength for the Balmer series of the hydrogen atom is:

(*a*) 912 Å (*b*) 912 × 2 Å

(*c*) 912 × 4 Å (*d*) 912/2 Å

44. The shortest λ for the Lyman series of hydrogen atom is … (Given $R_H = 109678\ cm^{-1}$)

(*a*) 911.7 Å (*b*) 700 Å

(*c*) 600 Å (*d*) 811 Å

45. The frequency of first line of Balmer series in hydrogen atom is v_o. The frequency of corresponding line emitted by singly ionised helium atom is:

(*a*) $2v_o$ (*b*) $4v_o$

(*c*) $v_o/2$ (*d*) $v_o/4$

46. Which of the following statement is not true?

(*a*) Lyman spectral series of hydrogen atom lies in the ultraviolet region of electromagnetic radiation.

(*b*) Balmer spectral series of hydrogen atom lies in the visible region of electromagnetic radiation.

(*c*) Paschen spectral series of hydrogen atom lies in the visible region of electromagnetic radiation.

(*d*) Brackett spectral series of hydrogen atom lies in the infrared region of electromagnetic radiation.

47. The first emission line of Balmer series in H spectrum has the wave number equal to:

(*a*) $\frac{9R_H}{400}\ cm^{-1}$ (*b*) $\frac{7R_H}{144}\ cm^{-1}$

(*c*) $\frac{3R_H}{4}\ cm^{-1}$ (*d*) $\frac{5R_H}{36}\ cm^{-1}$

48. When an electron de-excites from higher orbit in H-atom, two radiations are emitted out in Paschen and Lyman series. The wavelength of radiation emitted out in Lyman series is:

(*a*) $\frac{8R}{9}$ (*b*) $\frac{3R}{4}$

(*c*) $\frac{4}{3R}$ (*d*) $\frac{9}{8R}$

De-BROGLIE WAVELENGTH AND HEISENBERG UNCERTAINITY PRINCIPLE

49. In H-atom, if 'x' is the radius of the first Bohr orbit, de Broglie wavelength of an electron in 3rd orbit is:

(*a*) 3πx (*b*) 6πx

(*c*) $\frac{9x}{2}$ (*d*) $\frac{x}{2}$

50. de-Broglie wavelength of electron in second orbit of Li^{2+} ion will be equal to de-Broglie of wavelength of electron in:

(*a*) n = 3 of H-atom (*b*) n = 4 of C^{5+} ion

(*c*) n = 6 of Be^{3+} ion (*d*) n = 3 of He^{+} ion

51. de-Broglie equation describes the relationship of wavelength associated with the motion of an electron and its:

(*a*) Mass (*b*) Energy

(*c*) Momentum (*d*) Charge

52. A body of mass 'x' kg is moving with velocity of 100 m sec^{-1}. It's de-Broglie wavelength is 6.62×10^{-35} m. Hence 'x' is (h = 6.62×10^{-34} J sec):

(*a*) 0.25 kg (*b*) 0.15 kg

(*c*) 0.2 kg (*d*) 0.1 kg

53. The wavelengths of electron waves in two orbits is 3:5. The ratio of kinetic energy of electrons will be:

(*a*) 25 : 9 (*b*) 5 : 3

(*c*) 9 : 25 (*d*) 3 : 5

54. A ball of mass 200 g moving with a velocity of 10 m sec^{-1}. If the error in measurement of velocity is 0.1%, the uncertainty in its position is:

(*a*) 3.3×10^{-31} m (*b*) 3.3×10^{-27} m

(*c*) 5.3×10^{-25} m (*d*) 2.64×10^{-32} m

@PROCBEE

55. If the radius of first Bohr orbit is x, then de Broglie wavelength of electron in 3rd orbit is nearly:

(*a*) $2\pi x$ (*b*) $6\pi x$

(*c*) 9x (*d*) x/3

56. Which of the following particles moving with same velocity would be associated with smallest de-Broglie wavelength?

(*a*) Hydrogen molecule (*b*) Oxygen molecule

(*c*) Helium molecule (*d*) Nitrogen molecule

QUANTUM NUMBERS & ELECTRONIC CONFIGURATION

57. Spin magnetic moment of X^{n+} (Z = 26) is $\sqrt{24}$ B.M. Hence number of unpaired electrons and value of n respectively are:

(*a*) 4, 2 (*b*) 2, 4

(*c*) 3, 1 (*d*) 0, 2

58. Which of the following ions has the maximum number of unpaired d-electrons?

(*a*) Zn^{2+} (*b*) Fe^{2+}

(*c*) Ni^{3+} (*d*) Cu^{+}

59. Consider the ground state of Cr atom (Z = 24). The numbers of electrons with the azimuthal quantum numbers, $\ell = 1$ and 2 are, respectively:

(*a*) 16 and 5 (*b*) 12 and 5

(*c*) 16 and 4 (*d*) 12 and 4

60. The orbital angular momentum of an electron in 2s-orbital is:

(*a*) $+\frac{1}{2}\frac{h}{2\pi}$ (*b*) Zero

(*c*) $\frac{h}{2\pi}$ (*d*) $\sqrt{2}\frac{h}{2\pi}$

61. The possible value of ℓ and m for the last electron in the Cl^- ion are:

(*a*) 1 and 2 (*b*) 2 and + 1

(*c*) 3 and –1 (*d*) 1 and –1

62. For an electron, with n = 3 has only one radial node. The orbital angular momentum of the electron will be:

(*a*) 0 (*b*) $\sqrt{6}\frac{h}{2\pi}$

(*c*) $\sqrt{2}\frac{h}{2\pi}$ (*d*) $3\left(\frac{h}{2\pi}\right)$

QUANTUM MECHANICAL MODEL OF ATOM, SHRODINGER WAVE EQUATION AND ORBITAL CONCEPT

63. The maximum radial probability in 1s-orbital occurs at a distance when: [r_0 = Bohr radius]

(*a*) $r = r_0$ (*b*) $r = 2r_0$

(*c*) $r = \frac{r_0}{2}$ (*d*) $2r = \frac{r_0}{2}$

64. The maximum probability of finding electron in the d_{xy} orbital is :

(*a*) Along the x-axis.

(*b*) Along the y-axis.

(*c*) At an angle of 45° from the x and y axis.

(*d*) At an angle of 90° from the x and y axis.

65. $3p_y$ orbital has.....nodal plane :

(*a*) XY (*b*) YZ

(*c*) ZX (*d*) All of these

66. A 3p-orbital has:

(*a*) Two non-spherical nodes.

(*b*) Two spherical nodes.

(*c*) One spherical and one non spherical nodes.

(*d*) One spherical and two non spherical nodes.

67. According to Schrodinger model nature of electron in an atom is as:

(*a*) Particle only.

(*b*) Wave only.

(*c*) Particle and wave nature simultaneous.

(*d*) Sometimes waves and sometimes particle.

68. Consider the following statements:

(A) Electron density in the XY plane in $3d_{x^2-y^2}$ orbital is zero.

(B) Electron density in the XY plane in $3d_{z^2}$ orbital is zero.

(C) 2s orbital has one nodal surface.

(D) For $2p_z$ orbital, XY is the nodal plane.

Which of these are incorrect statements?

(*a*) A & C (*b*) D & C

(*c*) Only B (*d*) A, B

69. Which of the following d-orbitals has dough-nut shape ?

(*a*) d_{xy} (*b*) d_{yz}

(*c*) $d_{x^2-y^2}$ (*d*) d_{z^2}

70. The permissible solution to the Schrodinger wave equation gave an ideal of ..... quantum number:

(*a*) 4 (*b*) 3

(*c*) 2 (*d*) 1

@PROCBSE

Exercise-2 (Learning Plus)

1. A light source of wavelength λ illuminates a metal and ejects photo-electrons with $(K.E.)_{max} = 1$ eV.

Another light source of wavelength λ/3, ejects photo-electrons from same metal with $(K.E.)_{max} = 5$eV.

Find the value of work function?

(*a*) 1 eV (*b*) 2 eV

(*c*) 0.5 eV (*d*) None of these

2. In an atom, two electrons move round the nucleus in circular orbits of radii R and 4R. The ratio of the time taken by them to complete one revolution is : (Consider Bohr model to be valid)

(*a*) 1 : 4 (*b*) 4 : 1

(*c*) 1 : 8 (*d*) 8 : 1

3. The potential energy of the electron present in the ground state of Be^{3+} ion is represented by:

(*a*) $+\frac{e^2}{\pi \in_0 r}$ (*b*) $-\frac{e}{\pi \in_0 r}$

(*c*) $-\frac{e^2}{\pi \in_0 r^2}$ (*d*) $-\frac{e^2}{\pi \in_0 r}$

4. Which transition in Li^{2+} would have the same wavelength as the $2 \rightarrow 4$ transition in He^+ ion?

(*a*) $4 \rightarrow 2$ (*b*) $2 \rightarrow 4$

(*c*) $3 \rightarrow 6$ (*d*) $6 \rightarrow 2$

5. Let υ_1 be the frequency of the series limit of the Lyman series, υ_2 be the frequency of the first line of the Lyman series, and υ_3 be the frequency of the series limit of the Balmer series :

(*a*) $\upsilon_1 - \upsilon_2 = \upsilon_3$ (*b*) $\upsilon_2 - \upsilon_1 = \upsilon_3$

(*c*) $\upsilon_3 = 1/2\ (\upsilon_1 - \upsilon_3)$ (*d*) $\upsilon_1 + \upsilon_2 = \upsilon_3$

6. If the shortest wave length of Lyman series of H atom is x, then the wave length of the first line of Balmer series of H atom will be:

(*a*) 9x/5 (*b*) 36x/5

(*c*) 5x/9 (*d*) 5x/36

7. In a sample of H-atoms, electrons de-excite from a level 'n' to 1. The total number of lines belonging to Balmer series are two. If the electrons are ionised from level 'n' by photons of energy 13 eV. Then the kinetic energy of the ejected photoelectrons will be :

(*a*) 12.15 eV (*b*) 11.49 eV

(*c*) 12.46 eV (*d*) 12.63 eV

8. A particle X moving with a certain velocity has a de-Broglie wave length of 1Å. If particle Y has a mass of 25% that of X and velocity 75% that of X, de-Broglies wave length of Y will be:

(*a*) 3 Å (*b*) 5.33 Å

(*c*) 6.88 Å (*d*) 48 Å

9. The ratio of the de-broglie wavelength of a proton and α-particles will be 1 : 2 if their:

(*a*) Velocity are in the ratio 1 : 8.

(*b*) Velocity are in the ratio 8 : 1.

(*c*) Kinetic energy are in the ratio 1 : 64.

(*d*) Kinetic energy are in the ratio 1 : 256.

10. Consider an electron in the nth orbit of a hydrogen atom in the Bohr model. The circumference of the orbit can be expressed in terms of the de Broglie wavelength λ of the electron as :

(*a*) (0.529) nλ (*b*) $\sqrt{n}\lambda$

(*c*) (13.6) λ (*d*) nλ

11. In case of $d_{x^2-y^2}$ orbital:

(*a*) Probability of finding the electron along x-axis is zero.

(*b*) Probability of finding the electron along y-axis is zero.

(*c*) Probability of finding the electron is maximum along x and y-axis.

(*d*) Probability of finding the electron is zero in x-y plane.

12. The value of the spin magnetic moment of a particular ion is 2.83 Bohr magneton. The ion is:

(*a*) Fe^{2+} (*b*) Ni^{2+}

(*c*) Mn^{2+} (*d*) Co^{3+}

13. If n and ℓ are respectively the principal and azimuthal quantum numbers, then the expression for calculating the total number of electrons in any orbit is:

(*a*) $\sum_{\ell=1}^{\ell=n} 2(2\ell+1)$ (*b*) $\sum_{\ell=1}^{\ell=n-1} 2(2\ell+1)$

(*c*) $\sum_{\ell=0}^{\ell=n+1} 2(2\ell+1)$ (*d*) $\sum_{\ell=0}^{\ell=n-1} 2(2\ell+1)$

14. The quantum numbers + 1/2 and – 1/2 for the electron spin represent :

(*a*) Rotation of the electron in clockwise and anticlockwise direction respectively.

(*b*) Rotation of the electron in anticlockwise and clockwise direction respectively.

(*c*) Magnetic moment of the electron pointing up and down respectively,

(*d*) Two quantum mechanical spin states which have no classical analogue.

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15. An electron, a proton and an alpha particle have kinetic energies of 16E, 4E and E respectively. What is the qualitative order of their de-Broglie wavelengths?

(*a*) $\lambda_e > \lambda_p = \lambda_\alpha$ (*b*) $\lambda_p = \lambda_\alpha > \lambda_e$

(*c*) $\lambda_p > \lambda_e > \lambda_\alpha$ (*d*) $\lambda_\alpha > \lambda_e >> \lambda_p$

16. In an atom, two electrons move around the nucleus in circular orbits of radii R and 4R. The ratio of the time taken by them to complete one revolution is:

(*a*) 1 : 4 (*b*) 4 : 1

(*c*) 1 : 8 (*d*) 8 : 7

17. A 3p-orbital has:

(*a*) Two non-spherical nodes.

(*b*) Two spherical nodes.

(*c*) One spherical and one non-spherical node.

(*d*) One spherical and two non-spherical nodes.

18. The difference in angular momentum associated with the electron in two successive orbits of hydrogen atom is:

(*a*) h/p (*b*) h/2p

(*c*) h/2 (*d*) (n -1) h/2p

19. An α–particle is accelerated from rest through a potential difference of 6.0 V. It's de Broglie wavelength is:

(*a*) 5 Å (*b*) 4.15 pm

(*c*) 414.6 Å (*d*) 5 nm

20. A photon of 2.55 eV is emitted out by an electronic transition in hydrogen atom. The change in de Broglie wavelength of the electron is:

(*a*) 3.32 Å (*b*) 4.98 Å

(*c*) 6.64 Å (*d*) 9.96 Å

21. An electron that has the quantum numbers n = 3 and m = 2:

(*a*) Must have spin quantum number value +1/2

(*b*) Must have l = 2, 3 or 4

(*c*) Must have l = 0, 1 or 2

(*d*) Must have l = 2

22. Electronic configuration of an element is $1s^2$, $2s^1$, $2p^2$. It is:

(*a*) Ground state configuration of B (Z = 5).

(*b*) Excited state configuration of B.

(*c*) Ground state configuration of C^+ ion (Z = 6).

(*d*) Impossible configuration.

23. What is the most probable distance of a 1s-electron in a He^+ ion. The wave function for 1s-orbital is given by $\psi = \sqrt{\left(\frac{Z^3}{\pi a_0^3}\right)} e^{-Zr/a_0}$, where a_0 = radius of first Bohr's orbit in H-atom = 52.9 pm.

(*a*) 52.9 pm (*b*) 13.25 pm

(*c*) 6.61 pm (*d*) 26.45 pm

24. The dissociation energy of H_2 is 429.0 kJ/mol. If H_2 is dissociated by illumination with radiation of wavelength 270.0 nm, then what percentage of radiant energy will be converted into kinetic energy? ($h = 6.6 \times 10^{-34}$ Js, $N_A = 6 \times 10^{23}$)

(*a*) 1.25% (*b*) 3.26%

(*c*) 5.0% (*d*) 7.5%

25. The magnitude of the orbital angular momentum of an electron is given by $L = \sqrt{5}\ h/\pi$. How many orbitals of this kind are possible, belonging to an orbit?

(*a*) 4 (*b*) 5

(*c*) 11 (*d*) 9

26. Which of the ions is not having the configuration of Ne?

(*a*) Cl^- (*b*) F^-

(*c*) Na^+ (*d*) Mg^{2+}

27. In a given atom, no two electrons can have the same values for all the four quantum numbers. This is called:

(*a*) Hund's rule (*b*) Pauli's exclusion principle

(*c*) Uncertainty principle (*d*) Aufbau principle

28. How many electrons can fit into the orbitals that comprise the 3rd quantum shell, n = 3?

(*a*) 2 (*b*) 8

(*c*) 18 (*d*) 32

29. Which of the following sets of quantum numbers is correct for an electron in 4f-orbital?

(*a*) n = 4, *l* = 3, m = +4; s = +1/2

(*b*) n = 4, *l* = 4, m = –4, s = –1/2

(*c*) n = 4, *l* = 3, m = +1, s = +1/2

(*d*) n = 3, *l* = 2, m = –2, s = +1/2

30. In any subshell, the maximum number of electrons having same values of spin quantum number is:

(*a*) $\sqrt{l(\ell+1)}$ (*b*) $l + 2$

(*c*) $2l + 1$ (*d*) $4l + 2$

31. Total number of 'm' values for n = 4 is:

(*a*) 8 (*b*) 16

(*c*) 12 (*d*) 20

32. The electronic configuration of an atom/ion can be defined by which of the following?

(*a*) Aufbau principle.

(*b*) Pauli's exclusion principle.

(*c*) Hund's rule of maximum multiplicity.

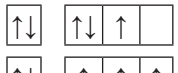
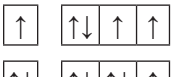
(*d*) All of these.

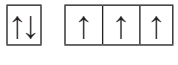
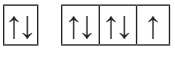
33. According to Aufbau principle, the correct order of energy of 3d, 4s and 4p-orbitals is:

(*a*) 4p < 3d < 4s (*b*) 4s < 4p < 3d

(*c*) 4s < 3d < 4p (*d*) 3d < 4s < 4p

34. The orbital diagram in which 'Aufbau principle' is violated, is:

(*a*)  (*b*) 

(*c*)  (*d*) 

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35. An ion which has 18 electrons in the outermost shell is:
(*a*) K^+ (*b*) Cu^+
(*c*) Cs^+ (*d*) Th^{4+}

36. A strong argument for the particle nature of cathode rays is:
(*a*) They can rotate paddle wheel.
(*b*) They can propagate in vaccum.
(*c*) They cast shadows.
(*d*) They are deflected by electric and magnetic fields.

37. Three isotopes of an element have mass numbers, M, (M + 1) and (M + 2). If the mean mass number is (M + 0.5), then which of the following ratios may be accepted for M, (M + 1), (M + 2) in that order?
(*a*) 1 : 1 : 1 (*b*) 4 : 1 : 1
(*c*) 3 : 2 : 1 (*d*) 2 : 1 : 1

38. Which of the following statements is not correct?
(*a*) The shape of an atomic orbital depends on the azimuthal quantum number.
(*b*) The orientation of an atomic orbital depends on the magnetic quantum number.
(*c*) The energy of an electron in an atomic orbital of multi-electron atom depends on the principal quantum number.
(*d*) The number of degenerate atomic orbitals of one type depends on the values of azimuthal and magnetic quantum numbers.

39. Gases begin to conduct electricity in discharge tube at low pressure because:
(*a*) At low pressures, gases turn to plasma.
(*b*) Colliding electrons can acquire higher kinetic energy due to increased mean free path leading to ionisation of atoms.
(*c*) Atoms break up into electrons and protons.
(*d*) The electrons in atoms can move freely at low pressure.

40. The uncertainty in momentum of an electron is 1×10^{-5} kg ms^{-1}. The uncertainty in its position will be ($h = 6.62 \times 10^{-34}$ kg m^2s):
(*a*) 1.05×10^{-28} m (*b*) 1.05×10^{-26} m
(*c*) 5.27×10^{-30} m (*d*) 5.25×10^{-28} m

Exercise-3 (JEE Advanced Level)

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MULTIPLE CORRECT TYPE QUESTIONS

1. The radius of the following orbits are double as that of the first Bohr's orbit of hydrogen atom.
(*a*) He^+ (n = 2) (*b*) Be^{+3} (n = 2)
(*c*) O^{+7} (n = 4) (*d*) C^{+5} (n = 3)

2. Which are correct statements?
(*a*) Hund's rule deals with degenerate orbitals.
(*b*) Pauli exclusion principal stated as only two electrons may exist in the same orbital and these electrons must have opposite spin.
(*c*) In ground state of atom the orbitals are filled on the basis of increasing (n + l) value.
(*d*) The maximum number of electrons in the subshell with principal quantum number 'n' is equal to $2n^2$?

3. Which statements are correct?
(*a*) ψ (orbital wave function) a mathematical function of co–ordinates of the electron.
(*b*) ψ^2 represents probability density of electron at a point.
(*c*) Schrodinger wave equation play same role in quantum mechanism as Newton's Law of motion do in classical mechanism.
(*d*) Radial probability density [R^2 (r)] is maximum close to nucleus but total probability is least.

4. Which of the following statements are correct about the cathode ray experiment?
(*a*) The cathode rays starts from anode and move towards cathode.
(*b*) Television picture tubes are cathode ray tubes and pictures result due to fluorescence on the television screen coated with certain fluorescent materials.
(*c*) In the presence of magnetic or electric field, the behaviour of cathode rays are similar to negatively charged particles.
(*d*) The characteristics of cathode rays does not depend upon the material of electrodes but nature of gas present in the cathode ray tube.

5. Choose the correct statements from among the following:
(*a*) The total number of spherical nodes in an orbital are $n - l - 1$.
(*b*) The total number of angular nodes in an orbital are l.
(*c*) The number of maxima in an orbital is $(n - l)$.
(*d*) A node is a point in space where the wave function ψ has zero amplitude.

6. The energy of an electron in the first Bohr's orbit of H-atom is –13.6 eV. The possible energy value(s) of the excited state(s) for electrons in Bohr's orbits to hydrogen is:
(*a*) –3.4 eV (*b*) –10.2 eV
(*c*) –1.51 eV (*d*) –0.85 eV

7. Which of the following statement(s) is/are correct?

(*a*) The ratio of the radii of the first three Bohr orbits of hydrogen atom is 1:8:27.

(*b*) The ratio of magnitude of total energy: kinetic energy: potential energy for electron in any orbit of hydrogen atom is 1:1:2.

(*c*) The frequency of a green light is 6×10^{14} Hz, then its wavelength is 500 nm.

(*d*) The ratio of de Broglie wavelength of a H-atom, He-atom and CH_4-molecule moving with equal kinetic energy is 4:2:1.

8. When photons of energy 4.25 eV strikes the surface of a metal 'A', the ejected photoelectrons have maximum kinetic energy T_A (in eV) and de Broglie wavelength λ_A. The maximum kinetic energy of photoelectrons liberated from another metal 'B' by photons of energy 4.20 eV is T_B (= T_A – 1.50 eV). If the de Broglie wave length of these photoelectrons is λ_B (= $2\lambda_A$), then

(*a*) the work function of 'A' is 2.25 eV

(*b*) the work function of 'B' is 3.70 eV

(*c*) T_A = 2.00 eV

(*d*) T_B = 2.75 eV

9. Which is/are correct for sodium atom in the ground state?

(*a*) There is only one unpaired electron.

(*b*) There are five pairs of electrons.

(*c*) 6 electrons are of one spin and other 5 of opposite spin.

(*d*) There are ten electrons in the same spin.

COMPREHENSION BASED QUESTIONS

Comprehension (Q. 10 to 13): The wave function for an atomic orbital of single electron atom or ion is

$$\Psi(r,\theta,\phi) = \frac{2}{3}\left(\frac{Z}{3a_0}\right)^{3/2}(1-\sigma)(12-8\sigma+\sigma^2).\sigma.e^{-\sigma/2}.\cos\theta$$

where $\sigma = \left(\frac{2Zr}{na_0}\right)$ and a_0 = 0.529 Å. All other parameters have their usual meaning.

10. The number of radial and angular nodes for the orbital is, respectively,

(*a*) 3, 1 (*b*) 2, 1

(*c*) 3, 2 (*d*) 2, 2

11. The atomic orbital should be:

(*a*) 4p (*b*) 5p

(*c*) 5d (*d*) 5f

12. If θ is the angle measured from Z-axis, then the orbital should be:

(*a*) p_x (*b*) p_y

(*c*) p_z (*d*) d_{z^2}

13. The maximum distance of radial node from the nucleus is:

(*a*) $\frac{a_0}{Z}$ (*b*) $\frac{3a_0}{Z}$

(*c*) $\frac{6a_0}{Z}$ (*d*) $\frac{15a_0}{Z}$

MATCH THE COLUMN TYPE QUESTIONS

14. Frequency = f, Time period = T, Energy of nth orbit = E_n, radius of nth orbit = r_n, Atomic number = Z, Orbit number = n

Column-I		Column-II	
A.	f	p.	n^3
B.	T	q.	Z^2
C.	E_n	r.	$\frac{1}{n^2}$
D.	$\frac{1}{r_n}$	s.	Z

(*a*) A → (q); B → (p); C → (q,r); D → (r,s)

(*b*) A → (r); B → (p); C → (q,r); D → (r,s)

(*c*) A → (q); B → (p); C → (q,r); D → (q, s)

(*d*) A → (q); B → (p); C → (s,r); D → (r,s)

15. Match List-I with List-II and select the correct answer using the codes given below in the lists (n, ℓ and m are respectively the principal, azimuthal and magnetic quantum no.)

List-I		List-II	
A.	Number of value of l for an energy level (n)	p.	0, 1, 2, ... (n – 1)
B.	Values of ℓ for a particular type of orbit	q.	+ℓ to –ℓ through zero
C.	Number of value of m for ℓ = 2	r.	5
D.	Values of 'm' for a particular type of orbital	s.	n

(*a*) A → (r); B → (p); C → (s); D → (q)

(*b*) A → (s); B → (q); C → (r); D → (p)

(*c*) A → (s); B → (p); C → (r); D → (q)

(*d*) A → (r); B → (p); C → (s); D → (q)

16. Match the column & choose the correct option among the options given.

(Note: Each statement in column-I has one or more matches in column-II.)

Column-I (For mono electronic species)		Column-II	
A.	Radius of nth orbit	p.	inversely proportional to Z
B.	Energy of electron in n^{th} orbit	q.	inversely proportional to n^2
C.	Velocity of electron in nth orbit	r.	inversely proportional to n
D.	Angular momentum of electron	s.	proportional to n

(*a*) A → (p); B → (q); C → (r); D → (s)
(*b*) A→ (p,r); B → (q); C → (r); D → (s)
(*c*) A → (p,q); B → (p,r); C → (r); D → (s)
(*d*) A → (p,q); B → (q); C → (s); D → (r)

17. Match the column & choose the correct option among the options given.

(Note: Each statement in column-I has one or more matches in column-II.)

Column-I		Column-II	
A.	5 → 1 transition in H atom	p.	spectral line in IR region
B.	7 → 3 transition in H atom	q.	spectral line in visible region
C.	Last line of Balmer series in H atom	r.	spectral line ultra violet region
D.	4 → 2 transition in He_+ ion	s.	spectral line will be observed one

(*a*) A → (r,s); B → (q,s); C → (r,s); D → (p,s)
(*b*) A → (r,s); B → (p,s); C → (q,s); D → (r,s)
(*c*) A → (r,s); B → (r,s); C → (p,s); D → (r,s)
(*d*) A → (r,s); B → (r); C → (p,s); D → (q,s)

18. Match the column & choose the correct option among the options given.

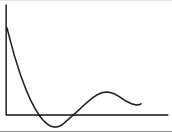
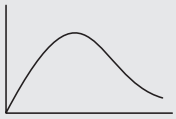
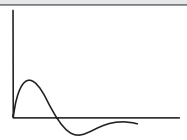
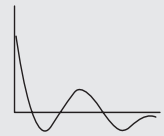
(Note: Each statement in column-I has only one match in column-II.)

Column-I		Column-II	
A.	Orbit angular momentum	p.	$\sqrt{n(n+2)}$
B.	Orbital angular momentum	q.	$\frac{nh}{2\pi}$
C.	Spin angular momentum	r.	$\sqrt{s(s+1)h}$
D.	Magnetic moment	s.	$\sqrt{l(l+1)h}$
		t.	$\sqrt{n(n+1)h}$

(*a*) A → (p); B → (r); C → (s); D → (q)
(*b*) A → (p); B → (s); C → (r); D → (q)
(*c*) A → (q); B → (s); C → (r); D → (p)
(*d*) A → (q); B → (r); C → (s); D → (p)

19. Match the columns& choose the correct option among the options given.

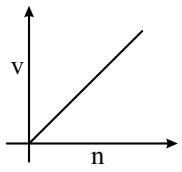
(Note: Each statement in column-I has only one match in column-II.)

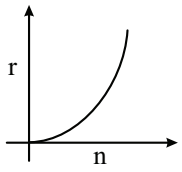
Column-I (Orbital)		Column-II (R vs. r Graph)	
A.	3s	p.	
B.	4s	q.	
C.	2p	r.	
D.	3p	s.	

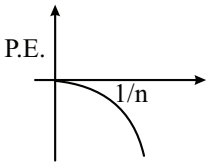
(*a*) A → (q); B → (s); C → (q); D → (r)
(*b*) A → (p); B → (q); C → (s); D → (r)
(*c*) A → (r); B → (s); C → (q); D → (q)
(*d*) A → (p); B → (s); C → (q); D → (r)

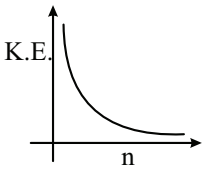
20. Select the correct statement(s):
(*a*) All electromagnetic radiation travel with speed of light in vaccum.
(*b*) Energy of photon of UV light is lower than that of yellow light.
(*c*) He^+ and H have similar spectrum.
(*d*) The total energy of an electron in unielectronic specie is greater than zero.

21. Select the correct curve(s):
If v = velocity of electron in Bohr's orbit.
r = Radius of electron in Bohr's orbit.
P.E. = Potential energy of electron in Bohr's orbit.
K.E. = Kinetic energy of electron in Bohr's orbit.

(*a*) 

(*b*) 

(*c*) 

(*d*) 

22. Which is/are **correct** statement:
(*a*) The difference in angular momentum associated with the electron present in consecutive orbits of H- atom is $(n-1)\frac{h}{2\pi}$.
(*b*) Energy difference between energy levels will be changed if, P. E. at infinity assigned value other than zero.
(*c*) Frequency of spectral line in a H-atom is in the order of $(2 \to 1) < (3 \to 1) < (4 \to 1)$.
(*d*) On moving away from the nucleus, kinetic energy of electron decreases.

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23. If there are only two H-atoms, each is in 3rd excited state then:

(*a*) Maximum number of different photons emitted is 4.

(*b*) Maximum number of different photos emitted is 3.

(*c*) Minimum number of different photons emitted is 1.

(*d*) Minimum number of different photons emitted is 2.

24. The qualitative order of de-Broglie wavelength for electron, proton and α particle is $\lambda_e > \lambda_p > \lambda_\alpha$ if:

(*a*) If kinetic energy is same for all particles.

(*b*) If the accelerating potential difference 'V' is same for all the particles (from rest).

(*c*) If velocities are same for all particles.

(*d*) None of the above.

25. In the experiment on photoelectric effect using light having frequency greater than the threshold frequency, the photocurrent will certainly increase when:

(*a*) Anode voltage is increased.

(*b*) Area of cathode surface is increased.

(*c*) Intensity of incident light is increased.

(*d*) Distance between anode and cathode is increased.

26. An electron in hydrogen atom first jumps from second excited state to first excited state and then, from first excited state to ground state. Let the ratio of wavelength, momentum and energy of photons in the two cases by x, y and z, then select the wrong answers:

(*a*) z = 1/x (*b*) x = 9/4

(*c*) y = 5/27 (*d*) z = 5/27

27. Choose the correct statement among the following:

(*a*) Radial distribution function ($\psi^2.4\pi r^2 dr$) give probability at a particular distance along any chosen direction.

(*b*) $\psi^2(r)$ give probability density at a particular distance over a spherical surface.

(*c*) For 's' orbitals $\psi(r)\psi(\theta)\psi(\phi) = \psi(x, y, z)$ is independent of θ and ϕ.

(*d*) '2p' orbital with quantum numbers. n = 2, l = 1, m = 0, also shows angular dependence.

28. Which of the following statements is/are correct for an electron of quantum numbers n = 4 and m = 2?

(*a*) The value of ℓ may be 2.

(*b*) The value of ℓ may be 3.

(*c*) The value of s may be +1/2.

(*d*) The value of ℓ may be 0, 1, 2, 3.

29. Which of the following statement(s) is (are) correct?

(*a*) The electronic configuration of Cr is [Ar] $(3d)^5$ $(4s)^1$. (Atomic number of Cr = 24)

(*b*) The magnetic quantum number may have negative values.

(*c*) In silver atom, 23 electrons have a spin of one type and 24 of the opposite type. (Atomic number of Ag = 47)

(*d*) None of these

NUMERICAL TYPE QUESTIONS

30. What is likely to be orbit number for a circular orbit of diameter 20 nm of the hydrogen atom if we assume Bohr orbit to be the same as that represented by the principal quantum number?

31. Number of possible spectral lines which may be emitted in Bracket series in H atom if electrons present in 9th excited level returns to ground level, are _____ .

32. In a sample of H-atom electrons make transition from 5th excited state to ground state, producing all possible types of photons, then number of lines in infrared region are _____.

33. The maximum number of 3d-electrons having spin quantum number s = +1/2 are _____ .

34. Magnetic moment of X^{3+} ion of 3d series is $\sqrt{35}$ BM. What is atomic number of X^{3+}?

35. The work function (ϕ) of some metals is listed below. The number of metals which will show photoelectric effect when light of 300 nm wavelength falls on the metal is

Metal	Li	Na	K	Mg	Cu	Ag	Fe	Pt	W
ϕ(eV)	2.4	2.3	2.2	3.7	4.8	4.3	4.7	6.3	4.75

36. The quantum number n corresponding to the excited state of He^+ ion if on transition to the ground state that ion emits two photons in succession with wavelengths 108.5 and 30.4 nm is

37. 'a particle' of 3.6 MeV are fired towards nucleus ${}^{A}_{Z}X$, at point of closest separation distance between 'a particle' and 'X' is 1.6×10^{-14}m. Calculate atomic number of'X' [Given: $1/4\pi e_0 = 9 \times 10^9$ in S.I. units]

38. The atomic number of hydrogen-like ion has the wavelength difference between the first line of Balmer and Lyman series equal to 59.3 nm.

39. Photons of same energy were allowed to strike on two different samples of hydrogen atoms, one having each atom in ground state and other in a particular excited state of orbit number n. The photonic beams ionize the hydrogen atoms in both the samples. If the difference in maximum kinetic energy of emitted photoelectron from both the samples is 12.75 eV, then the value of n is.

40. The angular momentum of an electron in a Bohr's orbit of H-atom is 3.1652×10^{-34} kg-m^2/sec. Calculate the wavenumber in terms of Rydberg constant (R) of the spectral line emitted when an electron falls from this level to the ground state.[Use $h = 6.626 \times 10^{-34}$ Js]

41. A photon having λ = 960 Å causes the ionization of a nitrogen atom. Give the I.E. per mole of nitrogen in KJ.

(Use : hc = 1240 eV × nm and $1\frac{eV}{atom} = 96.0$ kJ/mol.

42. The work function for a metal is 40 eV. To emit photo electrons of zero velocity from the surface of the metal the wavelength of incident light should be x nm.

43. If first ionization potential of a hypothetical atom is 16 V, then the first excitation potential will be :

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44. Electrons in the H-atoms jump from some higher level upto 3rd energy level. If three spectral lines are possible for the transition, find the initial position of electron.

45. An electron in Li^{2+} ion makes a transition from higher state n_2 to lower state $n_1 = 6$. The emitted photons is used to ionize an electron in H–atom from 2nd excited state. The electron on leaving the H–atom has a de–Broglie wavelength $\lambda = 12.016$ Å. Find the value of n_2.

Note: Use $(12.016)^2 = \frac{150\times144}{13.6\times11}$, $\lambda_{Å} = \sqrt{\frac{150}{KE_{eV}}}$

46. In a hypothetical H-atom the mass of electron & its charge is double of what we consider then calculate the total energy (in eV) of electron in the Ist orbit of such a hypothetical H-atom?

[Assuming all others concepts and parameter to be same as we considered in Bohr's model.]

47. In H-spectrum, longest wavelength of Lyman is 120 nm and shortest wavelength of Balmer is 360 nm. From this data, find longest wavelength of photon, that can ionize this H-atom.

48. A proton and an electron, both at rest initially, combine to form a H-atom in ground state. A single photon is emitted in this process. Find the wavelength (in nm) of this photon. (use : hc = 1237.6 cv × nm)

49. The ionisation energy of the hydrogen atom is given to be 13.6 eV. A photon falls on a hydrogen atom which is initially in the ground state and excites it to the (n = 4)state. Calculate the wavelength (in Å) of the photon.

50. The diameter of a dust particle of mass 10^{-3} gm is 2Å. If uncertainity in speed of this particle is $\frac{3.31}{\pi}\times10^{-3}$ m/s, then find minimum uncertaintity in its position.

51. The vapours of Hg absorb some electrons accelerated by a potential difference of 4.5 eV, as a result of which light is emitted. If the full energy of single incident e^- is supposed to be converted into light emitted by single Hg atom, find the wave no. (in m^{-1}) of the light.

52. Mr. Alakh Pandey has to decode a number "ABCDEF" where each alphabet is represented by a single digit. Suppose an orbital whose radial wave function is represented as;

$\psi_{(r)} = k_1 \cdot e^{-r/k_2}\left(r^2 - 5k_3 r + 6k_3^2\right)$.

From the following information given about each alphabet then write down the answers in the form of "ABCDEF", for above orbital.

Info A = Value of n where "n" is principal quantum number.

Info B = No. of angular nodes.

Info C = Azimuthal quantum number of subshell to orbital belongs.

Info D = No. of subshells having energy between (n + 5)s to (n + 5)p where n is principal quantum number.

Info E = Orbital angular momentum of given orbital.

Info F = Radial distance of the spherical node which is farthest from the nucleus. (Assuming $k_3 = 1$)

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Exercise-4 (Past Year Questions)

JEE MAIN

1. A stream of electrons from a heated filament was passed between two charged plates kept at a potential difference *V* esu. If *e* and *m* are charge and mass of an electron, respectively, then the value of h/λ (where λ is wavelength associated with electron wave) is given by: **(2016)**

(*a*) meV (*b*) 2meV

(*c*) $\sqrt{meV}$ (*d*) $\sqrt{2meV}$

2. The radius of the second Bohr orbit for hydrogen atom is:
(Plank's constant h = 6.6262 × 10^{-34} Js; mass of electron = 9.1091 × 10^{-31} kg; Charge of electron e = 1.60210 × 10^{-19} C; Permittivity of vaccum ε_0 = 8.854185 × 10^{-12} kg^{-1} m^{-3} A^2) **(2017)**

(*a*) 1.65 Å (*b*) 4.76 Å

(*c*) 0.529 Å (*d*) 2.12 Å

3. Among the following the energy of 2s orbital is lowest in: **(2019)**

(*a*) Na (*b*) Li

(*c*) H (*d*) K

4. Which of the following combination of statements is true regarding the interpretation of the atomic orbitals?

A. An electron in an orbital of high angular momentum stays away from the nucleus than an electron in the orbital of lower angular momentum.

B. For a given value of the principal quantum number, the size of the orbit is inversely proportional to the azimuthal quantum number.

C. According to wave mechanics, the ground state angular momentum is equal to $\frac{h}{2\pi}$.

D. The plot of (ψ) Vs (r) for various azimuthal quantum numbers, shows peak shifting towards higher r value: **(2019)**

(*a*) A, D (*b*) A, B

(*c*) A, C (*d*) B, C

5. The de-Broglie wavelength (λ) associated with a photoelectron varies with the frequency (v) of the incident radiation as [v_0 is threshold frequency]: **(2019)**

(*a*) $\lambda \propto \frac{1}{(v - v_0)}$ (*b*) $\lambda \propto \frac{1}{(v - v_0)^{\frac{1}{4}}}$

(*c*) $\lambda \propto \frac{1}{(v - v_0)^{\frac{3}{2}}}$ (*d*) $\lambda \propto \frac{1}{(v - v_0)^{\frac{1}{2}}}$

6. Heat treatment of muscular pain involves radiation of wavelength of about 900 nm.
Which spectral line of H atom is suitable for this purpose?
[$R_H = 1 \times 10^5$ cm^{-1}, h = 6.6×10^{-34} Js = 3×10^8 ms^{-1}] **(2019)**

(*a*) Paschen, $\infty \rightarrow 3$ (*b*) Paschen, $5 \rightarrow 3$
(*c*) Balmer, $\infty \rightarrow 2$ (*d*) Lyman, $\infty \rightarrow 1$

7. For emission line of atomic hydrogen from $n_1 = 8$ to $n_f = n$, the plot of wave number $(\bar{\nu})$ against $\left(\frac{1}{n^2}\right)$ will be (The Rydberg constant, R_H is in wave number unit): **(2019)**

(*a*) Linear with intercept - R_H.
(*b*) Non linear.
(*c*) Linear with slope R_H.
(*d*) Linear with slope - R_H.

8. The isotopes of hydrogen are: **(2019)**

(*a*) Tritium and protium only.
(*b*) Protium and deuterium only.
(*c*) Protium, deuterium and tritium.
(*d*) Deuterium and tritium only.

9. What is the work function of the metal if the light of wavelength 4000 Å generates photoelectrons of velocity 6×10^5 ms^{-1} from it? **(2019)**
(Mass of electron = 9×10^{-31} kg
Velocity of light = 3×10^8 ms^{-1}
Planck's constant = 6.626×10^{-34} Js
Charge of electron = 1.6×10^{-19} JeV^{-1})

(*a*) 0.9 eV (*b*) 3.1 eV
(*c*) 2.1 eV (*d*) 4.0 eV

10. The 71st electron of an element X with an atomic number of 71 enters in to the orbital: **(2019)**

(*a*) 6 p (*b*) 4f
(*c*) 5d (*d*) 6s

11. The ground state energy of hydrogen atom is –13.6 eV. The energy of second excited state of He^+ ion eV is: **(2019)**

(*a*) –54.4 (*b*) –3.4
(*c*) –6. 04 (*d*) –27.2

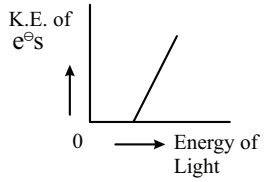
12. If the de-Brogile wavelength of the electron in nth Bohr orbit in a hydrogenic atom is equal to $1.5\pi a_0$ (a_0 is Bohr radius), then the value of n/z is: **(2019)**

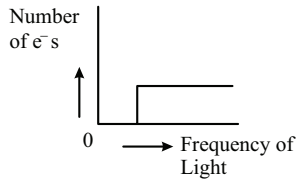
(*a*) 0.40 (*b*) 1.50
(*c*) 1.0 (*d*) 0.75

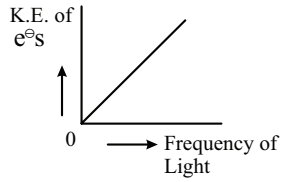
13. The total number of isotopes of hydrogen and number of radioactive isotopes among them, respectively are: **(2019)**

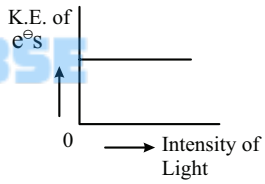
(*a*) 3 and 1 (*b*) 3 and 2
(*c*) 2 and 1 (*d*) 2 and 0

14. Which of the graphs shown below does not represent the relationship between incident light and the electron ejected from metal surface? **(2019)**

(*a*)


(*b*)


(*c*)


(*d*)


15. The quantum number of four electrons are given below; **(2019)**

(I) $n = 4, l = 2, m_l = -2, m_s = -\frac{1}{2}$

(II) $n = 3, l = 2, m_l = 1, m_s = +\frac{1}{2}$

(III) $n = 4, l = 1, m_l = 0, m_s = +\frac{1}{2}$

(IV) $n = 3, l = 1, m_l = 1, m_s = -\frac{1}{2}$

The Correct order of their increasing energies will be:

(*a*) IV < III < II < I (*b*) IV < II < III < I
(*c*) I < II < III < IV (*d*) I < III < II < IV

16. The size of the isoelectronic species $C1^-$, Ar and Ca^{2+} is affected by: **(2019)**

(*a*) Principal quantum number of valence shell.
(*b*) Nuclear charge.
(*c*) Azimuthal qunatum number of valence shell.
(*d*) Electron–electron interaction in the outer orbitals.

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17. The element having greatest difference between its first and second ionization energies, is: **(2019)**

(*a*) Ca (*b*) K (*c*) Ba (*d*) Sc

18. For any given series of spectral lines of atomic hydrogen, let $\Delta\bar{v} = \bar{v}_{max} - \bar{v}_{min}$ be the difference in maximum and minimum frequencies in cm^{-1}. The ratio $\Delta\bar{v}_{Lyman} / \Delta\bar{v}_{Balmer}$ is: **(2019)**

(*a*) 27 : 5 (*b*) 4 : 1 (*c*) 5 : 4 (*d*) 9 : 4

19. If P is the momentum of the fastest electron ejected from a metal surface after the irradiation of light having wavelenght λ, then for 1.5 P momentum of the photoelectron, the wavelength of the light should be:

(Assume kinetic energy of ejected photoelectron to be very high in comparison to work function). **(2019)**

(*a*) $\frac{1}{2}\lambda$ (*b*) $\frac{3}{4}\lambda$

(*c*) $\frac{2}{3}\lambda$ (*d*) $\frac{4}{9}\lambda$

20. Which one of the following about an electron occupying the is orbital in a hydrogen atom is incorrect? (The Bohr radius is represented by a_0) **(2019)**

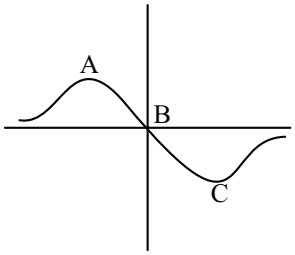
(*a*) The electron can be found at a distance $2a_0$ from the nucleus.

(*b*) The probability density of finding the electron is maximum at the nucleus.

(*c*) The magnitude of potential energy is double that of its kinetic energy on an average.

(*d*) The total energy of the electron is maximum when it is at a distance a_0 from the nucleus.

21. The electrons are more likely to be found: **(2019)**



(*a*) In the region (A) and (B) (*b*) In the region (A) and (C)

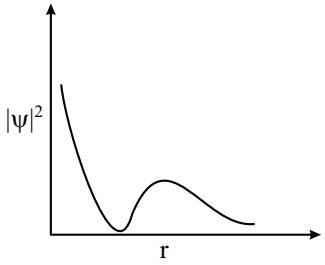
(*c*) Only in the region (C) (*d*) Only in the region (A)

22. The isoelectronic set of ions is **(2019)**

(*a*) N^{3-}, Li^+, Mg^{2+} and O^{2-} (*b*) Li^+, Na^+, O^{2-} and F^-

(*c*) F^-, Li^+, Na^+ and Mg^{2+} (*d*) N^{3-}, O^{2-}, F^- and Na^+

23. The graph between $|\psi|^2$ and r (radial distance) is shown below. This represents: **(2019)**



(*a*) 3s orbital (*b*) 1s orbital

(*c*) 2p orbital (*d*) 2s orbital

24. The ratio of the shortest wavelength of two spectral series of hydrogen spectrum is found to be about 9. The spectral series are: **(2019)**

(*a*) Paschen and Pfund (*b*) Lyman and Paschen

(*c*) Brackett and Pfund (*d*) Balmer and Brackett

25. The region in the electromagnetic spectrum where the Balmer series lines appear is: **(2020)**

(*a*) Infrared (*b*) Microwave

(*c*) Ultraviolet (*d*) Visible

26. The number of orbitals associated with quantum number n = 5, $m_s = +\frac{1}{2}$ is: **(2020)**

(*a*) 11 (*b*) 25

(*c*) 50 (*d*) 15

27. For the Balmer series in the spectrum of H atom, $\bar{v} = R_H\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$, the correct statement among (I) to (IV) are : **(2020)**

(I) As wavelength decreases, the lines in the series converge.

(II) The integer n_1 is equal to 2.

(III) The lines of longer wavelength corresponds to $n_2 = 3$.

(IV) The ionization energy of hydrogen can be calculated from wave number of these lines.

(*a*) (I), (III), (IV) (*b*) (I), (II), (IV)

(*c*) (II), (III), (IV) (*d*) (I), (II), (III)

28. Hydrogen has three isotopes (A), (B) and (C). If the number of neutrons in (A), (B), (C) respectively, are (x), (y) and (z) the sum of (x), (y) and (z) is: **(2020)**

(*a*) 4 (*b*) 2

(*c*) 3 (*d*) 1

29. The radius of the second Bohr orbit, in terms of the Bohr radius, a_0, in Li^{2+} is: **(2020)**

(*a*) $\frac{2a_0}{3}$ (*b*) $\frac{4a_0}{9}$

(*c*) $\frac{4a_0}{3}$ (*d*) $\frac{2a_0}{9}$

30. The de-Broglie wavelength of an electron in the 4th Bohr orbit is: **(2020)**

(*a*) $4\pi a_0$ (*b*) $2\pi a_0$

(*c*) $6\pi a_0$ (*d*) $8\pi a_0$

31. The work function of sodium metal is 4.41×10^{-19} J. If photons of wavelength 300 nm are incident on the metal, the kinetic energy of the ejected electrons will be: **(2020)**

($h = 6.63 \times 10^{-34}$ J s ; $c = 3 \times 10^8$ m/s) _____ $\times 10^{-21}$J.

32. The metal mainly used in devising photoelectric cells is: **(2020)**

(*a*) Na (*b*) Rb

(*c*) Li (*d*) Cs

33. The number of subshells associated with n = 4 and m = –2 quantum numbers is: **(2020)**

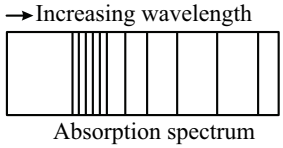
(*a*) 2 (*b*) 8

(*c*) 4 (*d*) 16

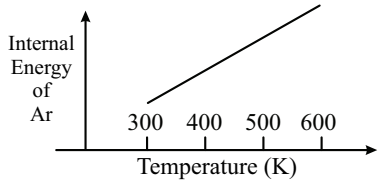
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34. The figure that is not a direct manifestation of the quantum nature of atoms is: **(2020)**

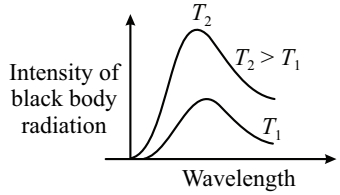
(*a*)



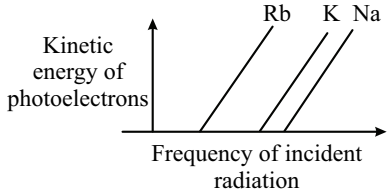
(*b*)



(*c*)



(*d*)



35. Consider the hypothetical situation where the azimuthal quantum number, *l*, takes value 0, 1, 2, ... n + 1, where n is the principal quantum number. Then, the element with atomic number: **(2020)**

(*a*) 9 is the first alkali metal.

(*b*) 6 has a 2p-valence subshell.

(*c*) 8 is the first noble gas.

(*d*) 13 has a half-filled valence subshell.

36. The photoelectric current from Na (work function, w_0 = 2.3 eV) is stopped by the output voltage of the cell Pt(s)|H_2 (g, 1 bar)|HCl(aq., pH = 1)|AgCl(s)|Ag(s).

The pH of aq. HCl required to stop the photoelectric current from K(w_0 = 2.25 eV), all other conditions remaining the same, is ____$\times 10^{-2}$. (to the nearest integer) **(2020)**

Given, $2.303\ \frac{RT}{F} = 0.06$ V; $E^0_{AgCl|Ag|Cl^-} = 0.22$ V

37. The shortest wavelength of H atom in the Lyman series is λ_1 . The longest wavelength in the Balmar series of He^+ is: **(2020)**

(*a*) $\frac{5\lambda_1}{9}$ (*b*) $\frac{36\lambda_1}{5}$

(*c*) $\frac{27\lambda_1}{5}$ (*d*) $\frac{9\lambda_1}{5}$

38. The correct statement about probability density (except at infinite distance from nucleus) is: **(2020)**

(*a*) It can never be zero for 2s orbital.

(*b*) It can be zero for 3p orbital.

(*c*) It can be zero for 1s orbital.

(*d*) It can be negative for 2p orbital.

39. The difference between radii of 3rd and 4th orbits of Li^{2+} is ΔR_1. The difference between the radii of 3rd and 4th orbits of He^+ is ΔR_2. Ratio $\Delta R_1 : \Delta R_2$ is: **(2020)**

(*a*) 3 : 2 (*b*) 8 : 3

(*c*) 2 : 3 (*d*) 3 : 8

40. The correct electronic configuration and spin only magnetic moment (BM) of Gd^{3+} (Z = 64), respectively, are: **(2020)**

(*a*) [Xe] $5f^7$ and 8.9 (*b*) [Xe] $4f^7$ and 7.9

(*c*) [Xe] $5f^7$ and 7.9 (*d*) [Xe] $4f^7$ and 8.9

41. In the sixth period, the orbitals that are filled are: **(2020)**

(*a*) 6s, 4f, 5d, 6p (*b*) 6s, 5d, 5f, 6p

(*c*) 6s, 6p, 6d, 6f (*d*) 6s, 5f, 6d, 6p

42. A 50 watt bulb emits monochromatic red light of wavelength of 795nm. The number of photons emitted per second by the bulb is $x \times 10^{20}$. The value of x is __. **(2021)**

[Given: $h = 6.63\times10^{-34}$Js and $c = 3.0\times10^{8}$ms^{-1}]

43. The value of magnetic quantum number of the outermost electron of Zn^+ ion is __. **(2021)**

44. Ge (Z = 32) in its ground state electronic configuration has completely filled orbitals with $m_1 = 0$. The value of x is __. **(2021)**

45. The number of photons emitted by a monochromatic (single frequency) infrared range finder of power 1mW and wavelength of 1000nm, in 0.1 second is $x\times10^{13}$. The value of x is __. (Nearest integer)

($h = 6.63\times10^{-34}$Js, $c = 3.00\times10^{8}$ms^{-1}) **(2021)**

46. A proton and a Li^{3+} nucleus are accelerated by the same potential. If λ_{Li} and λ_p denote the de Broglie wavelengths of Li^{3+} and proton respectively, then the value of $\frac{\lambda_{Li}}{\lambda_P}$ is $x \times 10^{-1}$.

The value of x is _____ (Rounded off to the nearest integer)

[Mass of Li^{3+} = 8.3 mass of proton] **(2021)**

47. According to Bohr's atomic theory:

(I) Kinetic enertgy of electron is $\propto \frac{Z^2}{n^2}$.

(II) The product of velocity (v) of electron and principal quantum number (n), 'vn' $\propto Z^2$.

(III) Frequency of revolution of electron in an orbit is $\propto \frac{Z^3}{n^3}$.

(IV) Coulombic force of attraction on the electron is $\propto \frac{Z^3}{n^4}$.

Choose the most appropriate answer the from the options given below: **(2021)**

(*a*) I, III and IV only (*b*) I and IV only

(*c*) I only (*d*) III only

48. Electromagnetic radiation of wavelength 663 nm is just sufficient to ionise the atom of metal A. The ionization energy of metal A in kJ mol is_____. (Rounded-off to the nearest integer)
[$h = 6.63 \times 10^{-34}$ Js, $c = 3.00 \times 10^8$ ms^{-1}, $N_A = 6.02 \times 10^{23}$ mol^{-1}] **(2021)**

49. The orbital having two radial as well as two angular nodes is: **(2021)**
(*a*) 4d (*b*) 4f (*c*) 3p (*d*) 5d

50. Which of the following forms of hydrogen emits low energy β^- particles? **(2021)**
(*a*) Deuterium ^2_1H (*b*) Protium ^1_1H
(*c*) Proton H^+ (*d*) Tritium ^3_1H

51. A ball weighing 10 g is moving with a velocity of 90 ms^{-1}. If the uncertainty in its velocity is 5% then the uncertainty in its position is....$\times 10^{-33}$ m. **(2021)**
(Given $h = 6.63 \times 10^{-34}$ Js)

52. When light of wavelength 248 nm falls on a metal of threshold energy 3.0 eV, the de-Broglie wavelength of emitted electrons is... Å.
[Use: $\sqrt{3} = 1.73$, $h = 6.63 \times 10^{-34}$ Js, $m_e = 9.1 \times 10^{-31}$ kg; $c = 3.0 \times 10^8$ ms^{-1}; $1eV = 1.6 \times 10^{-19}$ J] **(2021)**

53. The number of orbitals with $n = 5$, $m_l = +2$ is____ **(2021)**

54. A certain orbital has $n = 4$ and $m_l = -3$. The number or radial nodes in this orbital is_____. **(2021)**

55. A certain orbital has no angular nodes and two radial nodes, the orbital is: **(2021)**
(*a*) 2s (*b*) 3p (*c*) 3s (*d*) 2p

56. The Azimuthal quantum number for the valence electrons Ga^+ ion is.....
(Atomic number of Ga = 31) **(2021)**

57. The wavelenghts of electrons accelerated from rest through a potential difference of 40 kV is $x \times 10^{-12}$ m. the value of x is _____.
Given: Mass of electron = 9.1×10^{-31} kg
Charge on an electron = 1.6×10^{-19} kg
Planck's constant = 6.63×10^{-34} Js **(2021)**

58. Number of electrons that Vandium (Z = 23) has in p-orbitals is equal to_____. **(2021)**

59. A source of monochromatic radiation of wavelength 400 nm provides 1000 J of energy in 10 seconds. When this radiation falls on the surface of sodium, $x \times 10^{20}$ electrons are ejected per second. Assume that wavelength 400 nm is sufficient for ejection of electron from the surface of sodium metal. The value of x is .....
($h = 6.626 \times 10^{-34}$ Js) **(2021)**

60. An accelerted electron, has a speed of 5×10^6 ms^{-1} with an uncertainty of 0.02%. The uncertainty in finding its location while in motion is $x \times 10^{-9}$ m. The value of x is____.
[Use mass of electron = 9.1×10^{-31} Kg. $h = 6.63 \times 10^{-34}$ Js, $\pi = 3.14$] **(2021)**

61. Given below are two statements;
Statement-I : Rutherford's gold foil experiment cannot explain the line spectrum of hydrogen atom.
Statement-II: Bohr's model of hydrogen atom contradicts Heisenberg's uncertainty principle.
In the light of the above statements, choose the most appropriate answer from the options given below: **(2021)**
(*a*) Both statement I and statement II are true.
(*b*) Statement I is false but statement II is false.
(*c*) Statement I is true but statement II is false.
(*d*) Both statement I and statement II are false.

62. If the Thompson model of the atom was correct, the then result of Rutherford's gold foil experiment would have been: **(2021)**
(*a*) α-particles pass through the gold foil deflected by small angles and with reduced speed.
(*b*) All α-particles get bounced back by 180°.
(*c*) All of the α-particles pass through the gold foil without decrease in speed.
(*d*) α-particles are deflected over a wide range of angles.

63. Given below are two statements;
Statement-I: According to Bohr's model of an atom, qualitatively the magnitude of velocity of electron increases with decrease in positive charges on the nucleus as there is no strong hold on the electron by the nucleus.
Statement-II: According to Bohr's model of an atom, qualitatively the magnitude of velocity of electron increases with decrease in principal quantum number.
In the light of the above statements, choose the most appropriate answer from the options gien below: **(2021)**
(*a*) Statement-I is true but Statement-II is false.
(*b*) Statement-I is false but Statement-II is true.
(*c*) Both Statement-I and Statement-II are false.
(*d*) Both Statement-I and Statement-II are true.

64. A metal surface is exposed to 500 nm radiation. The threshold frequency of the metal for photoelectric current is 4.3×10^{14} Hz. The velocity of ejected electron is _____ $\times 10^5$ ms^{-1}.
[Use : $h = 6.63 \times 10^{-34}$ Js, $m_e = 9.0 \times 10^{-31}$ Kg] **(2021)**

65. The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is equal to $\frac{h^2}{xma_0^2}$. The value of 10x is ?____.
(a_0 is radius of Bohr's orbit)
[Given : $\pi = 3.14$] **(2021)**

66. The number of f electrons in the ground state electronic configuration of Np(Z = 93) is____. **(2021)**

67. The number of photons emitted by a monochromatic (single frequency) infrared range finder of power 1mW and wavelength of 1000nm, in 0.1 second is $x \times 10^{13}$. The value of x is____.
($h = 6.63 \times 10^{-34}$ Js, $c = 3.00 \times 10^8$ ms^{-1}) **(2021)**

68. The Eu^{2+} ion is a strong reducing agent in spite of its ground state electronic configuration (outermost) : [Atomic number of Eu = 63]. **(2021)**
(*a*) $4f^8\ 6s^2$ (*b*) $4f^7$ (*c*) $4f^7\ 6s^2$ (*d*) $4f^6$

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69. Identify the element for which electronic configuration in +3 oxidation state is $[Ar]3d^5$. **(2021)**

(*a*) Ru (*b*) Co (*c*) Mn (*d*) Fe

70. Consider the following pairs of electrons: **(2022)**

(A) (*a*) $n = 3, l = 1, m_l = 1, m_s = +\frac{1}{2}$

(*b*) $n = 3, l = 2, m_l = 1, m_s = +\frac{1}{2}$

(B) (*a*) $n = 3, l = 2, m_l = -2, m_s = -\frac{1}{2}$

(*b*) $n = 3, l = 2, m_l = -1, m_s = -\frac{1}{2}$

(C) (*a*) $n = 4, l = 2, m_l = 2, m_s = +\frac{1}{2}$

(*b*) $n = 3, l = 2, m_l = 2, m_s = +\frac{1}{2}$

The pairs of electron present in degenerate orbitals is/are:

(*a*) Only A (*b*) Only B

(*c*) Only C (*d*) Both (B) and (C)

71. If the radius of the 3rd Bohr's orbit of hydrogen atom is r_3 and the radius of 4th Bohr's orbit is r_4.

Then : **(2022)**

(*a*) $r_4 = \frac{9}{16}r_3$ (*b*) $r_4 = \frac{16}{9}r_3$ (*c*) $r_4 = \frac{3}{4}r_3$ (*d*) $r_4 = \frac{4}{3}r_3$

72. Which of the following sets of quantum numbers is not allowed? **(2022)**

(*a*) $n = 3, l = 2, m_l = 0, s = +\frac{1}{2}$

(*b*) $n = 3, l = 2, m_l = -2, s = +\frac{1}{2}$

(*c*) $n = 3, l = 3, m_l = -3, s = -\frac{1}{2}$

(*d*) $n = 3, l = 0, m_l = 0, s = -\frac{1}{2}$

73. The wavelength of an electron and a neutron will become equal when the velocity of the electron is x times the velocity of neutron. The value of x is_____. (Nearest Integer)

(Mass of electron is $9.1 \times 10–31$ kg and mass of neutron is $1.6 \times 10–27$ kg) **(2022)**

74. Consider an imaginary ion $^{48}_{22}X^{3-}$. The nucleus contains 'a' % more neutrons than the number of electrons in the ion. the value of 'a' is____. **(2022)**

75. Uncertainty in position of a moving particle is10^{-7} m and uncertainty velocity is 2.4×10^{-24} m/sec, then mass of particle is $[X] \times 10^{-5}$ Kg value of X is [Report your answer to nearest integer] **(2022)**

76. Which set of quantum number represent degenerate orbital? **(2022)**

(I) $n = 3, l = 2, m = 0, s = \frac{-1}{2}$ & $n = 3, l = 2$ $m = -1$ $s = \frac{+1}{2}$

(II) $n = 2, l = 1, m = 1, s = \frac{-1}{2}$ & $n = 3, l = 1$ $m = 1$ $s = \frac{+1}{2}$

(III) $n = 4, l = 2, m = -1, s = \frac{1}{2}$ & $n = 3, l = 2$ $m = -1$ $s = \frac{1}{2}$

(*a*) I (*b*) II

(*c*) III (*d*) None of these

JEE ADVANCED

77. The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is [a_0 is Bohr radius]: **(2012)**

(*a*) $\frac{h^2}{4\pi^2 m a_0^2}$ (*b*) $\frac{h^2}{16\pi^2 m a_0^2}$

(*c*) $\frac{h^2}{32\pi^2 m a_0^2}$ (*d*) $\frac{h^2}{64\pi^2 m a_0^2}$

78. The periodic table consists of 18 groups. An isotope of copper, on bombardment with protons, undergoes a nuclear reaction yielding element X as shown below. To which group, element X belongs in the periodic table?

$^{63}_{29}Cu + {}^1_1H \rightarrow 6{}^1_0n + {}^4_2\alpha + 2{}^1_1H + X$ **(2012)**

79. In the nuclear transmutation; **(2013)**

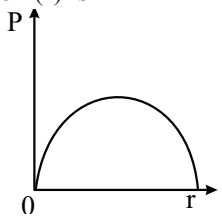
$^9_4Be + X \longrightarrow {}^8_4Be + Y$ (X, Y) is (are):

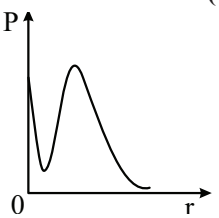
(*a*) (γ, n) (*b*) (p, D) (*c*) (n, D) (*d*) (γ, p)

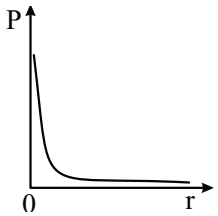
80. In an atom, the total number of electrons having quantum numbers n = 4, $|m_\ell| = 1$ and $m_s = -1/2$ is _____. **(2014)**

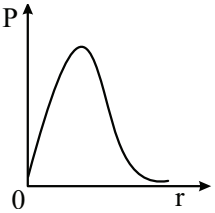
81. Not considering the electronic spin, the degeneracy of the second excited state (n = 3) of H atom is 9, while the degeneracy of the second excited state of H^- is _____. **(2015)**

82. P is the probability of finding the 1s electron of hydrogen atom in a spherical shell of infinitesimal thickness (dr) at a distance (r) from the nucleus. The volume of this shell is $4\pi r^2 dr$. The qualitative sketch of the dependence of (P) on (r) is **(2016)**

(*a*) 

(*b*) 

(*c*) 

(*d*) 

Comprehension (Q. 83 to 86): Answer 7, 8 and 9 by appropriately matching the information given in the three columns of the following table.

The wave function $\Psi_{n\ l,\ m_l}$ is a is a mathematical function whose value depends upon spherical polar coordinates (r, θ, ϕ) of the electron and characterized by the quantum numbers n, l and m_l. Here r is distance from nucleus, θ is colatitude and ϕ is azimuth. ln the mathematical functions given in the Table, Z is atomic number and a_0 is Bohr radius. **(2017)**

Column 1

(I) 1s orbital (II) 2s orbital

(III) $2p_z$ orbital (IV) $3d_z^2$ orbital

Column 2

(*i*) $\Psi_{n,l,m_l} \propto \left(\frac{Z}{a_0}\right)^{\frac{3}{2}} e^{-\left(\frac{Zr}{a_0}\right)}$

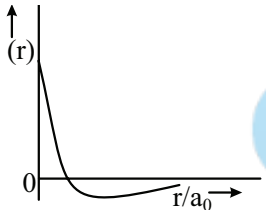
(*ii*) One radial node

(*iii*) $\Psi_{n,l,m_l} \propto \left(\frac{Z}{a_0}\right)^{\frac{5}{2}} re^{-\left(\frac{Zr}{2a_0}\right)} \cos\theta$

(*iv*) xy-plane is a nodal plane

Column 3

(P) Ψ_n, l, m_l,



(Q) Probability density at nucleus $\propto \frac{1}{a_o^3}$.

(R) Probability density is maximum at nucleus.

(S) Energy needed to excite electron from n = 2 state to n = 4 state is $\frac{27}{32}$ times the energy needed to excite electron from n = 2 state to n = 6 state.

83. For He^+ ion, the only **INCORRECT** combination is:

(*a*) (I) (i) (S) (*b*) (II) (ii) (Q)

(*c*) (I) (iii) (R) (*d*) (I) (i) (R)

84. For the given orbital in Column 1, the only **CORRECT** combination for any hydrogen-like species is:

(*a*) (II) (ii) (P) (*b*) (I) (ii) (S)

(*c*) (IV) (iv) (R) (*d*) (III) (iii) (P)

85. For hydrogen atom, the only **CORRECT** combination is:

(*a*) (I) (i) (P) (*b*) (I) (iv) (R)

(*c*) (II) (i) (Q) (*d*) (I) (i) (S)

86. The ground state energy of hydrogen atom is –13.6 eV. Consider an electronic state ψ of He^+ whose energy, azimuthal quantum number and magnetic quantum number are –3.4 eV, 2 and 0 respectively. Which of the following statement(s) is(are) true for the state ψ? **(2019)**

(*a*) It has 2 angular nodes.

(*b*) It has 3 radial nodes.

(*c*) It is a 4d state.

(*d*) The nuclear charge experienced by the electron in this state is less than 2e, where e is the magnitude of the electronic charge.

Comprehension (Q. 87 to 90): Answer the following by appropriately matching the lists based on the information given in the paragraph

Consider the Bohr's model of a one-electron atom where the electron moves around the nucleus. In the following List-I contains some quantities for the nth orbit of the atom and List-II contains options showing how they depend on n. **(2019)**

	List-I	**List-II**
(I)	Radius of the nth orbit.	(P) $\propto n^{-2}$
(II)	Angular momentum of the electron in the nth orbit.	(Q) $\propto n^{-1}$
(III)	Kinetic energy of the electron in the nth orbit.	(R) $\propto n^0$
(IV)	Potential energy of the electron in the nth orbit.	(S) $\propto n^1$
		(T) $\propto n^2$
		(U) $\propto n^{1/2}$

87. Which of the following options has the correct combination considering List-I and List-II?

(*a*) (II), (R) (*b*) (I), (P)

(*c*) (I), (T) (*d*) (II), (Q)

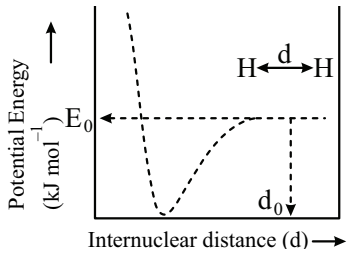
88. Which of the following options has the correct combination considering List-I and List-II?

(*a*) (III), (S) (*b*) (IV), (Q)

(*c*) (IV), (U) (*d*) (III), (P)

89. The figure below is the plot of potential energy versus internuclear distance (d) of H_2 molecule in the electronic ground state. What is the value of the net potential energy E_0 (as indicated in the figure) in kJ mol^{-1}, for $d=d_0$ at which the electron-electron repulsion and the nucleus-nucleus repulsion energies are absent? As reference, the potential energy of H atom is taken as zero when its electron and the nucleus are infinitely far apart.

Use Avogadro constant as 6.023×10^{-23} mol^{-1}. **(2020)**



90. Consider a Helium (He) atom that absorbs a photon of wavelength 330 nm. The change in the velocity (in cm s^{-1}) of He atom after the photon absorption is _____.

(Assume: Momentum is conserved when photon is absorbed).

Use: Planck constant = 6.6×10^{-34} Js, Avogadro number = 6×10^{23} mol^{-1}, molar mass of He = 4 g mol^{-1} **(2021)**

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ANSWER KEY

CONCEPT APPLICATION

1. (*a*)	**2.** (*d*)	**3.** (*b*)	**4.** (*b*)	**5.** (*c*)	**6.** (*a*)	**7.** (*b*)	**8.** (*c*)	**9.** (*b*)	**10.** (*c*)
11. (*a*)	**12.** (*b*)	**13.** (*a*)	**14.** (*a*)	**15.** (*c*)	**16.** (*b*)	**17.** (*b*)	**18.** (*a*)	**19.** (*d*)	**20.** (*a*)
21. (*c*)	**22.** (*d*)	**23.** (*c*)	**24.** (*b*)	**25.** (*b*)	**26.** (*a*)	**27.** (*d*)	**28.** (*a*)	**29.** (*b*)	**30.** (*b*)
31. (*a*)	**32.** (*c*)	**33.** (*d*)	**34.** (*a*)	**35.** (0.423Å)		**36.** (*c*)			

EXERCISE-1 (TOPICWISE)

1. (*a*)	**2.** (*c*)	**3.** (*b*)	**4.** (*d*)	**5.** (*c*)	**6.** (*d*)	**7.** (*c*)	**8.** (*d*)	**9.** (*a*)	**10.** (*d*)
11. (*d*)	**12.** (*c*)	**13.** (*a*)	**14.** (*a*)	**15.** (*d*)	**16.** (*c*)	**17.** (*c*)	**18.** (*c*)	**19.** (*d*)	**20.** (*c*)
21. (*a*)	**22.** (*b*)	**23.** (*a*)	**24.** (*a*)	**25.** (*b*)	**26.** (*a*)	**27.** (*b*)	**28.** (*b*)	**29.** (*a*)	**30.** (*b*)
31. (*b*)	**32.** (*a*)	**33.** (*d*)	**34.** (*c*)	**35.** (*c*)	**36.** (*c*)	**37.** (*b*)	**38.** (*c*)	**39.** (*d*)	**40.** (*d*)
41. (*c*)	**42.** (*c*)	**43.** (*c*)	**44.** (*a*)	**45.** (*b*)	**46.** (*c*)	**47.** (*d*)	**48.** (*d*)	**49.** (*b*)	**50.** (*b*)
51. (*c*)	**52.** (*d*)	**53.** (*a*)	**54.** (*d*)	**55.** (*b*)	**56.** (*b*)	**57.** (*a*)	**58.** (*b*)	**59.** (*b*)	**60.** (*b*)
61. (*d*)	**62.** (*c*)	**63.** (*a*)	**64.** (*c*)	**65.** (*c*)	**66.** (*c*)	**67.** (*c*)	**68.** (*d*)	**69.** (*d*)	**70.** (*b*)

EXERCISE-2 (LEARNING PLUS)

1. (*a*)	**2.** (*c*)	**3.** (*d*)	**4.** (*c*)	**5.** (*a*)	**6.** (*b*)	**7.** (*a*)	**8.** (*b*)	**9.** (*b*)	**10.** (*d*)
11. (*c*)	**12.** (*b*)	**13.** (*d*)	**14.** (*d*)	**15.** (*a*)	**16.** (*c*)	**17.** (*c*)	**18.** (*b*)	**19.** (*b*)	**20.** (*c*)
21. (*d*)	**22.** (*b*)	**23.** (*d*)	**24.** (*b*)	**25.** (*d*)	**26.** (*a*)	**27.** (*b*)	**28.** (*c*)	**29.** (*c*)	**30.** (*c*)
31. (*b*)	**32.** (*d*)	**33.** (*c*)	**34.** (*b*)	**35.** (*b*)	**36.** (*a*)	**37.** (*b*)	**38.** (*c*)	**39.** (*b*)	**40.** (*c*)

EXERCISE-3 (JEE ADVANCED LEVEL)

1. (*a,c*)	**2.** (*a,b,c*)	**3.** (*a,b,c,d*)	**4.** (*b,c*)	**5.** (*a,b,c,d*)	**6.** (*a,c,d*)	**7.** (*b,c,d*)	**8.** (*a,b,c*)	**9.** (*a,b,c*)	**10.** (*a*)
11. (*b*)	**12.** (*c*)	**13.** (*d*)	**14.** (*a*)	**15.** (*c*)	**16.** (*a*)	**17.** (*b*)	**18.** (*c*)	**19.** (*d*)	**20.** (*a,c*)
21. (*b,c,d*)	**22.** (*c,d*)	**23.** (*a,c*)	**24.** (*a,b,c*)	**25.** (*b,c*)	**26.** (*b*)	**27.** (*c,d*)	**28.** (*a,b,c*)	**29.** (*a,b,c*)	**30.** [14]
31. [6]	**32.** [6]	**33.** [5]	**34.** [26]	**35.** [4]	**36.** [5]	**37.** [20]	**38.** [3]	**39.** [4]	**40.** $\frac{5R}{9}$
41. 1240 kJ/mol.		**42.** [31]	**43.** [12]	**44.** [5]	**45.** [12]	**46.** –435.2 eV		**47.** [90 nm]	**48.** [91 nm]
49. 973.5 Å		**50.** [5×10^{-26} m]		**51.** [3.63×10^{6} m^{-1}]		**52.** [300303]			

EXERCISE-4 (PAST YEAR QUESTIONS)

JEE Main

1. (*d*)	**2.** (*d*)	**3.** (*d*)	**4.** (*a*)	**5.** (*d*)	**6.** (*a*)	**7.** (*d*)	**8.** (*c*)	**9.** (*c*)	**10.** (*c*)
11. (*c*)	**12.** (*d*)	**13.** (*a*)	**14.** (*c*)	**15.** (*b*)	**16.** (*b*)	**17.** (*b*)	**18.** (*d*)	**19.** (*d*)	**20.** (*d*)
21. (*b*)	**22.** (*d*)	**23.** (*d*)	**24.** (*b*)	**25.** (*d*)	**26.** (*b*)	**27.** (*d*)	**28.** (*c*)	**29.** (*c*)	**30.** (*d*)
31. [222]	**32.** (*d*)	**33.** (*a*)	**34.** (*b*)	**35.** (*d*)	**36.** [142]	**37.** (*d*)	**38.** (*b*)	**39.** (*c*)	**40.** (*b*)
41. (*a*)	**42.** [2]	**43.** [0]	**44.** [7]	**45.** [50]	**46.** [2]	**47.** (*b*)	**48.** [180]	**49.** (*d*)	**50.** (*d*)
51. [1.17]	**52.** [9]	**53.** [3]	**54.** [0]	**55.** (*c*)	**56.** [0]	**57.** [6]	**58.** [12]	**59.** [2]	**60.** [58]
61. (*a*)	**62.** (*a*)	**63.** (*b*)	**64.** [5]	**65.** [3155]	**66.** [4]	**67.** [50]	**68.** (*b*)	**69.** (*d*)	**70.** (*b*)
71. (*b*)	**72.** (*c*)	**73.** [1758]	**74.** [4]	**75.** [22]	**76.** [1]				

JEE Advanced

77. (*c*)	**78.** [8]	**79.** (*a,b*)	**80.** [6]	**81.** [3]	**82.** (*d*)	**83.** (*c*)	**84.** (*a*)	**85.** (*d*)	**86.** (*a,c*)
87. (*c*)	**88.** (*d*)	**89.** [–5246.49]		**90.** [30]					

CHAPTER

3 States of Matte

INTRODUCTION

Matter as we know broadly exists in three states.

There are always two opposite tendencies between particles of matter which determine the state of matter.

- Inter molecular attractive forces.
- The molecular motion / random motion.

Matter

Properties	Solid state	Liquid state	Gaseous state
Attractive force	• Large	• Smaller	• Almost zero
Thermal motion	• Almost zero	• Greater	• Random motion
Volume	• Fixed volume	• Fixed volume	• Varies with container
Geometry	• Definite	• Not definite	• Not definite

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In this chapter the properties and behaviour of the gases will be analysed and discussed in detail. These properties are measured with the help of the gas laws as proposed by Boyle,Charles,Gay Lussac etc.

GASEOUS LAW

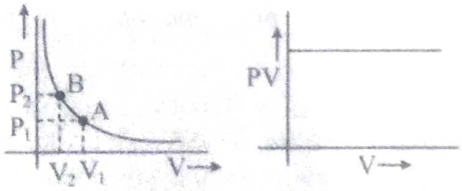
Boyle's Law and Measurement

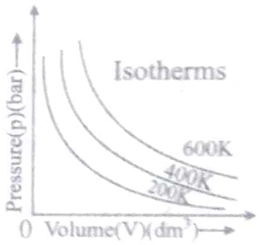
For a fixed amount of gas at constant temperature, the volume occupied by the gas is inversely proportional to the pressure applied on the gas or pressure of the gas.

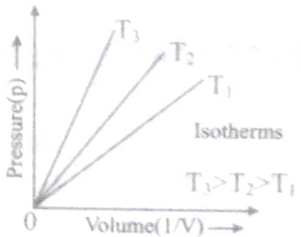
$$V \propto \frac{1}{P} \text{ (at constant T and n)}$$

Hence PV = constant

This constant will be dependent on the amount of the gas and temperature of the gas.







Application of Boyle's Law : For the two points 'A' and 'B'

$P_1V_1 = K$ & $P_2V_2 = K$

hence it follows that $P_1V_1 = P_2V_2$.

Atmospheric pressure: The pressure exerted by atmosphere on earth's surface at sea level is called 1 atm.

1 atm = 1.013×10^5 N/m^2 = 1.013 bar = 760 torr

Train Your Brain

Example 1: A gas is initially at 1 atm pressure. To compress it to 1/4 th of initial volume, what will be the pressure required?

Sol. $P_1 = 1$ atm $\quad V_1 = V$

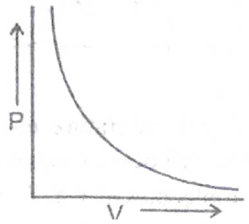
$P_2 = ?$ $\quad V_2 = \frac{V}{4}$

$P_1V_1 = P_2V_2$ at const. T & n

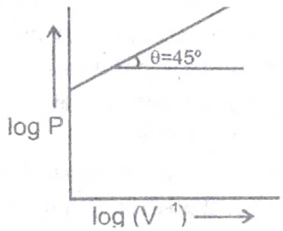
$$P_2 = \frac{P_1V_1}{V_2} = \frac{1 \text{ atm} \times V}{\frac{V}{4}} = 4 \text{ atm}$$

Example 2: Which of the following curve does not represent Boyle's law?

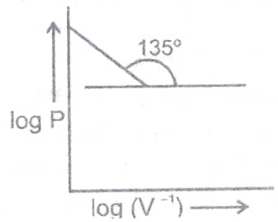
(*a*)



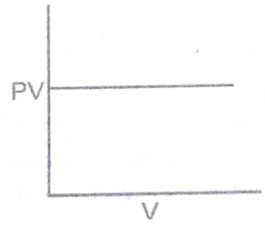
(*b*)



(*c*)



(*d*)



Sol. (*c*)

$PV = K$

$P = \frac{K}{V} = KV^{-1}$

$\log P = \log K + \log V^{-1}$

It follows equation of line

then option c is incorrect

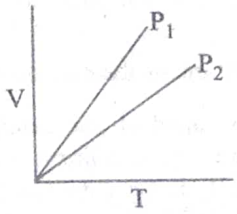
Concept Application

1. A 2.24 L cylinder of oxygen at 1 atm and 273 K is found to develop a leakage. When the leakage was plugged the pressure dropped to 570 mm of Hg. The number of moles of gas that escaped will be:

   (*a*) 0.025 (*b*) 0.050

   (*c*) 0.075 (*d*) 0.09

2. V versus T curves at constant pressure P_1 and P_2 for an ideal gas are shown in fig. Which is correct?



(*a*) $P_1 > P_2$ (*b*) $P_1 < P_2$

(*c*) $P_1 = P_2$ (*d*) All of these

Charle's Law

For a fixed amount of gas at constant pressure, volume occupied by the gas is directly proportional to temperature of the gas on absolute scale of temperature.

$V \propto T$ or $V = kT$

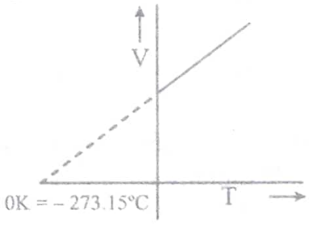
$\frac{V}{T}$ = constant where 'k' is a proportionality constant and is dependent on amount of gas and pressure.

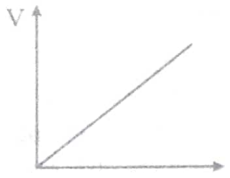
$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

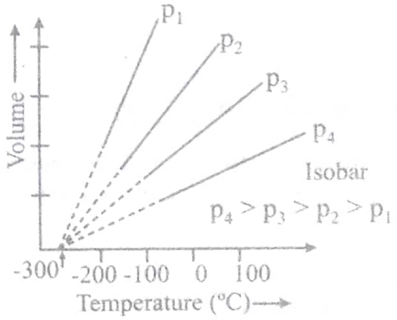
Temperature on absolute scale, kelvin scale or ideal gas scale.

$V = a + bt$

Temperature on centigrade scale.







Relation : $T = t + 273$

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- Since volume is proportional to absolute temperature. The volume of a gas should be theoretically zero at absolute zero temperature.
- Infact no substance exists as gas at a temperature near absolute zero, though the straight line plots can be extra plotted to zero volume. Absolute zero can never be attained practically though it can be approached only.
- By considering –273.15°C as the lowest approachable limit, Kelvin developed temperature scale which is known as absolute scale.

Gay-lussac's Law

For a fixed amount of gas at constant volume, pressure of the gas is directly proportional to temperature of the gas on absolute scale of temperature.

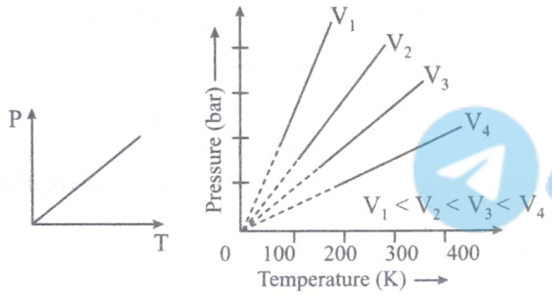
$P \alpha T$

$\frac{P}{T}$ = constant (dependent on amount and volume of gas)

$\frac{P_1}{T_1} = \frac{P_2}{T_2}$ (temperature on absolute scale originally)

The law was developed on the centigrade scale, where it was found that pressure is a linear function of temperature $P = P_0 + bt$ where 'b' is a constant and P_0 is pressure at zero degree centigrade.

Isochore:



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Example 3: If the temperature of a particular amount of gas is increased from 27°C to 57°C. Find final volume of the gas, if initial volume = 1L and assume pressure is constant.

Sol. $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

$\frac{1}{(273+27)} = \frac{V_2}{(273+57)}$ So $V_2 = 1.1L$

Example 4: The temperature of a certain mass of a gas in a rigid container is doubled. If initially the gas is at 1 atm pressure. Find the % increase in pressure?

Sol. $\frac{P_1}{T_1} = \frac{P_2}{T_2}; \frac{1}{T} = \frac{P_2}{2T}$

$\%\ \text{increase} = \frac{2-1}{1} \times 100 = 100\%$

Example 5: The temperature of a certain mass of a gas was increased from 27°C to 37°C at constant volume. What will be the pressure of the gas.

Sol. $\frac{P_1}{T_1} = \frac{P_2}{T_2}; \frac{P}{300} = \frac{P_2}{310}; P_2 = \frac{31}{30}P$

Concept Application

3. If the pressure of a gas contained in a closed vessel is increased by 0.4% when heated by 1°C it's initial temperature must be :
   (a) 250 K (b) 250°C
   (c) 25°C (d) 25 K
4. Equal weights of ethane & hydrogen are mixed in an empty container at 25°C, the fraction of the total pressure exerted by hydrogen is:
   (a) 1: 2 (b) 1: 1
   (c) 1: 16 (d) 15: 16
5. A mixture of hydrogen and oxygen at one bar pressure contains 20% by weight of hydrogen. Partial pressure of hydrogen will be
   (a) 0.2 bar (b) 0.4 bar
   (c) 0.6 bar (d) 0.8 bar

Avogadro's Hypothesis

For similar values of pressure & temperature equal number of molecules of different gases will occupy equal volume.

$N_1 \longrightarrow$ V (volume of N_1 molecules at P & T of one gas)

$N_2 \longrightarrow$ V (volume of N_2 molecules at P & T of second gas)

$\Rightarrow$ Molar volume & volume occupied by one mole of each and every gas under similar conditions will be equal.

One mole of any gas or a combination of gases occupies 22.413996 L of volume at STP.

The previous standard is still often used, and applies to all chemistry data more than decade old, in this definition **Standard Temperature and Pressure STP** denotes the same temperature of 0°C (273.15K), but a slightly higher pressure of 1 atm (101.325 kPa).

Standard Ambient Temperature and Pressure (SATP), Conditions are also used in some scientific works. SATP conditions means 298.15 K and 1 bar (i.e. exactly 10^5 Pa) At SATP (1 bar and 298.15 K), the molar volume of an ideal gas is 24.789 L mol^{-1}

EQUATION OF STATE

Combining all the gas relations in a single expression which describes relationship between pressure, volume and temperature, of a given mass of gas we get an expression known as equation of state.

$\frac{PV}{T}$ = constant (dependent on amount of the gas (n)).

$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

Ideal gas Equation:

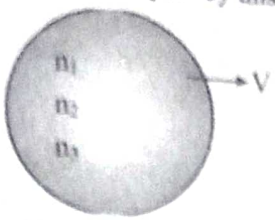
$\frac{PV}{nT}$ = R (universal gas constant)

R = 8.314 J/Kmole ≈ 25/3 = 1.987 cal/k·mole ≈ 2
= 0.0821 Latm/K/mole ≈ 1/12

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DALTON'S LAW OF PARTIAL PRESSURE

In a mixture of non reacting gases partial pressure of any component of gas is defined as pressure exerted by this component if whole of volume of mixture had been occupied by this component.



Partial pressure of first component gas

$$P_1 = \frac{n_1 RT}{V}; P_2 = \frac{n_2 RT}{V}; P_3 = \frac{n_3 RT}{V}$$

Total pressure = $P_1 + P_2 + P_3$.

Dalton's law: For a non reacting gaseous mixture total pressure of the mixture is the summation of partial pressure of the different component gases.

$$P_{Total} = P_1 + P_2 + P_3 = \frac{(n_1 + n_2 + n_3)RT}{V}$$

$$\frac{P_1}{P_T} = \frac{n_1}{n_T} = X_1 \text{ (mole fraction of first component of gas)}$$

$$\frac{P_2}{P_T} = \frac{n_2}{n_T} = X_2 \text{ (mole fraction of second component of gas)}$$

$$\frac{P_3}{P_T} = \frac{n_3}{n_T} = X_3 \text{ (mole fraction of third component of gas)}$$

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Example 6: The partial pressure of hydrogen in a flask containing two grams of hydrogen and 32 gm of sulphur dioxide is

(*a*) 1/16th of the total pressure
(*b*) 1/9th of the total pressure
(*c*) 2/3 of the total pressure
(*d*) 1/8th of the total pressure

Sol. $n_{H_2} = \frac{2g}{2\,g/mol} = 1mol.$ $n_{SO_2} = \frac{32g}{64\,g/mol} = 0.5$ mol

$$\therefore P_{H_2} = \frac{n_{H_2}}{(n_{H_2} + n_{SO_2})} \times P_T = \frac{1}{(1+0.5)} \times P_T = \frac{2}{3} P_T.$$

Example 7: Equal volume of two gases which do not react together are enclosed in separate vessels. Their pressures are 10 mm and 400 mm respectively. If the two vessels are joined together, then what will be the pressure of the resulting mixture (temperature remaining constant):

(*a*) 120 mm (*b*) 500 mm
(*c*) 1000 mm (*d*) 205 mm

Sol. Let, vol of containers be V & temperature be T

$P_1 = 10$ mm $P_2 = 400$ mm

$$\therefore n_1 = \frac{P_1 V}{RT} \quad \& \quad n_2 = \frac{P_2 V}{RT}$$

$$\therefore n_1 + n_2 = \frac{(P_1 + P_2) \times V}{RT} \quad \ldots (i)$$

After joining two containers final volume = (V + V) = 2V (for gases)

$$\therefore P_{final} = \frac{(n_1 + n_2)RT}{V_{final}} = \frac{(P_1 + P_2) \times V}{RT} \times \frac{RT}{2V}$$

$$= \frac{(P_1 + P_2)}{2} = \frac{(10 + 400)mm}{2} = 205 \text{ mm}.$$

Concept Application

6. Which of the following gas mixture is not applicable for Dalton's law of partial pressure?
   (*a*) SO_2 and Cl_2 (*b*) CO_2 and N_2
   (*c*) CO and CO_2 (*d*) CO and N_2
7. A 2.5L flask contains 0.25 mol each of SO_2 and CO_2 gas at 27°C. Calculate the partial pressure exerted by SO_2 CO_2 and total pressure respectively.
   (*a*) 2.46, 2.46, 4.92 (*b*) 1.23, 2.46, 2.46
   (*c*) 2.46, 1.23, 1.23 (*d*) None of these
8. 56 g of nitrogen and 96 g of oxygen are mixed isothermaly and at a total pressure of 10 atm. The partial pressures of oxygen and nitrogen (in atm) are respectively:
   (*a*) 4, 6 (*b*) 5, 5
   (*c*) 2, 8 (*d*) 6, 4
9. Equal masses of methane and oxygen are mixed in an empty container at 25°C. The fraction of the total pressure exerted by oxygen is:
   (*a*) $\frac{2}{3}$ (*b*) $\frac{1}{3} \times \frac{273}{298}$
   (*c*) $\frac{1}{3}$ (*d*) $\frac{1}{2}$

ANALYSIS OF GASEOUS MIXTURE

Vapour density: Vapour density of any gas is defined as the density of any gas with respect to density of the H_2 gas under identical conditions of temperature T and pressure P.

$$\text{vapour density} = \frac{\text{density of gas at T \& P}}{\text{density of } H_2 \text{ under same P \& T}}$$

$$P = \frac{m}{V}\cdot\frac{RT}{M} \Rightarrow P = \rho\frac{RT}{M} \Rightarrow \rho = \frac{PM}{RT}$$

$$\text{Vapour density} = \frac{PM_{gas}RT}{RT\,PM_{H_2}} = \frac{M_{gas}}{M_{H_2}} = \frac{M_{gas}}{2}$$

M_{gas} (molar mass of gas) = 2 × vapour density

Average molecular mass of gaseous mixture : Total mass of the mixture divided by total no. of moles in the mixture

$$M_{mix} = \frac{\text{Total mass of mixture}}{\text{Total no. of moles in mixture}}$$

If we have

'n_1', 'n_2' and 'n_3' are moles of three different gases having of molar mass 'M_1', 'M_2' and 'M_3' respectively.

$$M_{mix} = \frac{n_1M_1 + n_2M_2 + n_3M_3}{n_1 + n_2 + n_3}$$

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Example 8: Calculate the mean molar mass of a mixture of gases having 7 g of Nitrogen, 22 g of CO_2 and 5.6 litres of CO at STP.

Sol. Moles of N_2 = 7/28 = 1/4

Moles of CO_2 = 22/44 = 1/2

Moles of CO = 5.6/22.4 = 1/4

$$\text{Mean molar mass} = M_{mix} = \frac{n_1M_1 + n_2M_2 + n_3M_3}{n_1 + n_2 + n_3}$$

$$= \frac{\frac{1}{4}\times 28 + \frac{1}{2}\times 44 + \frac{1}{4}\times 28}{\frac{1}{4}+\frac{1}{2}+\frac{1}{4}}$$

$$= (7 + 7 + 22)/1 = 36$$

Example 9: A gaseous mixture of H_2 and CO_2 gas contains 66 mass % of CO_2. The vapour density of the mixture is:

(*a*) 6.1 (*b*) 5.4

(*c*) 2.7 (*d*) 10.8

Sol. (*c*) Average molar mass = $\frac{\text{Total mass}}{\text{Total moles}}$

$$= \frac{100}{\frac{66}{44}+\frac{34}{2}} = \frac{100}{18.5}$$

$$\text{Vapour density} = \frac{\text{molar mass}}{2} = 2.7$$

Concept Application

10. A and B are two identical vessels. A contains 15 g ethane at 1atm and 298 K. The vessel B contains 75 g of a gas X_2 at same temperature and pressure. The vapour density of X_2 is:

(*a*) 75 (*b*) 150

(*c*) 37.5 (*d*) 45

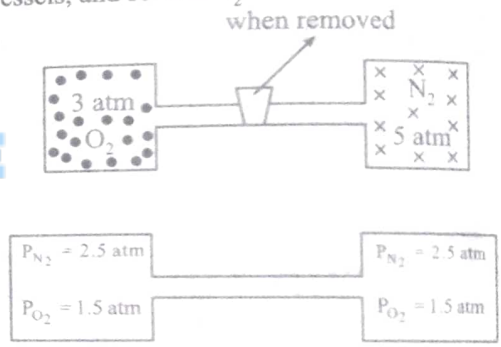
11. The vapour density of a mixture containing NO_2 and N_2O_4 is 27.6. The mole fraction of N_2O_4 in the mixture is:

(*a*) 0.1 (*b*) 0.2

(*c*) 0.5 (*d*) 0.8

GRAHAM'S LAW OF DIFFUSION/EFFUSION

Diffusion : Net spontaneous flow of gaseous molecules from region of high concentration (higher partial pressure) to the region of lower concentration or lower partial pressure. Flow will be from both sides, both gas will try to equalise its partial pressure in both the vessels, and so will O_2.



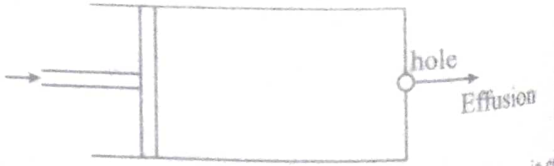
Graham's Law

"Under similar conditions of pressure (partial pressure) the rate of diffusion of different gases is inversely proportional to square root of the density of different gases."

❖ Rate of diffusion $r \propto \frac{1}{\sqrt{d}}$ {d = density of gas}

$$\frac{r_1}{r_2} = \frac{\sqrt{d_2}}{\sqrt{d_1}} = \frac{\sqrt{M_2}}{\sqrt{M_1}} = \sqrt{\frac{V.D_2}{V.D_1}}$$ V.D is vapour density

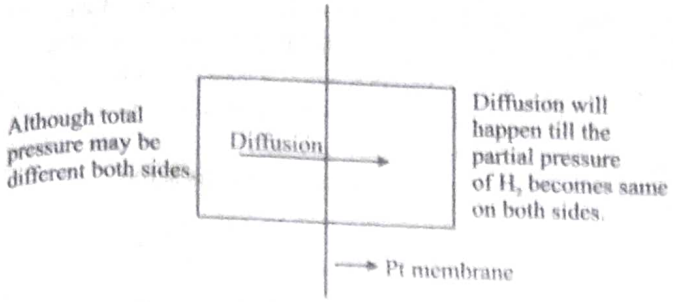
Effusion: (forced diffusion) a gas is made to diffuse through a hole by the application of external pressure.



Diffusion: Flow of gas to occupy available volume is called Diffusion.

Selective diffusion: If one or more than one components of a mixture are allowed to diffuse and others are not allowed then it is selective diffusion of those components.

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Platinum allows only H_2 gas to pass through it
Rate of diffusion and effusion can be termed as

r = change in pressure flow rate = $\frac{dP}{dt}$

r = volume flow rate = $\frac{dV_{out}}{dt}$

r = moles flow rate = $\frac{dn_{out}}{dt}$

r = distance travelled by gaseous molecules per unit time = $\frac{dx}{dt}$

- The general form of the graham's law of diffusion can be stated as follows, when one or all of the parameters are varied.

$$\text{rate} \propto \frac{PA}{\sqrt{TM}}$$

P – Pressure, A – area of hole, T – Temp., M – mol. wt.

- If partial pressure of gases are not equal.

Then rate of diffusion is found to be proportional to partial pressure & inversely proportional to square root of molecular mass.

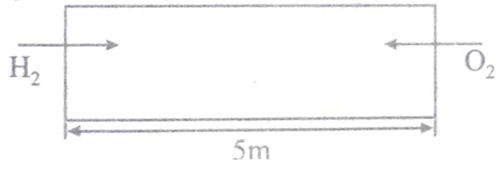
$r \propto P$

$$r \propto \frac{1}{\sqrt{M}}; \frac{r_1}{r_2} = \frac{P_1}{P_2}\sqrt{\frac{M_2}{M_1}}$$

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Example 10: In a tube of length 5 m having 2 identical holes at the opposite ends. H_2 & O_2 are made to effuse into the tube from opposite ends under identical conditions. Find the point where gases will meet for the first time.



Sol. $\frac{r_1}{r_2} = \frac{dx_1}{dt} \times \frac{dt}{dx_2} = \sqrt{\frac{M_2}{M_1}}$

$\frac{dx_1}{dx_2} = \sqrt{\frac{32}{2}} = 4$

$\Rightarrow \frac{\text{distance travelled by } H_2}{\text{distance travelled by } O_2} = 4$

$\frac{x}{(5-x)} = 4$

$x = (5 - x)\,4$; $x = 20 - 4x$; $5x = 20$; $x = 4$ from H_2 side

Example 11: Assume that you have a sample of hydrogen gas containing H_2, HD and D_2 that you want to separate into pure components (H = ^{1}H and D = ^{2}H). What are the relative rates of diffusion of the three molecules according to Graham's law?

Sol. Since D_2 is the heaviest of the three molecules, it will diffuse most slowly, and let we call its relative rate 1.00. We can then compare HD and H_2 with D_2.

Comparing HD with D_2, we have

$$\text{rate of diffusion} \propto \sqrt{\frac{1}{\text{molar mass}}}$$

$$\frac{\text{Rate of HD diffusion}}{\text{Rate of } D_2 \text{ diffusion}} = \sqrt{\frac{\text{Molecular mass of } D_2}{\text{Molecular mass of HD}}}$$

$$= \sqrt{\frac{4.0\text{ amu}}{3.0\text{ amu}}} = 1.15$$

Comparing H_2 with D_2 we have

$$\frac{\text{Rate of } H_2 \text{ diffusion}}{\text{Rate of } D_2 \text{ diffusion}} = \sqrt{\frac{\text{Mass of } D_2}{\text{Mass of } H_2}}$$

$$= \sqrt{\frac{4.0\text{ amu}}{2.0\text{ amu}}} = 1.41$$

Thus, the relative rates of diffusion are $H_2(1.41) > HD(1.15) > D_2(1.00)$.

Concept Application

12. X mL of H_2 gas effuses through a hole in a container in 5 seconds. The time taken for the effusion of the same volume of the gas specified below under identical condition is:

(*a*) 10sec:He (*b*) 20sec : O_2
(*c*) 25sec:CO (*d*) 55sec:CO_2

13. At identical temperature and pressure, the rate of diffusion of hydrogen gas is $3\sqrt{3}$ times that of a hydrocarbon having molecular formula C_nH_{2n-2}. What is the value of n?

(*a*) 1 (*b*) 4
(*c*) 3 (*d*) 8

14. A gaseous mixture containing He, CH_4 and SO_2 was allowed to effuse through a fine hole then find what molar ratio of gases coming out initially? If mixture contain He, CH_4 and SO_2 in 1 : 2 : 3 mole ratio.

(*a*) 2 : 2 : 3 (*b*) 6 : 6 : 1
(*c*) $\sqrt{2} : \sqrt{2} : 3$ (*d*) 4 : 4 : 3

KINETIC ENERGY AND MOLECULAR SPEEDS

Molecules of gases remain in continuous motion. While moving they collide with each other and with the walls of the container. This results in change of their speed and redistribution of energy. So the speed and energy of all the molecules of the gas at any instant are not the same. Thus, we can obtain only average value of speed of molecules. If there are n number of molecules in a sample and their individual speeds are u_1, u_2, ...u_n, then average speed of molecules u_{av} can be calculated as follows:

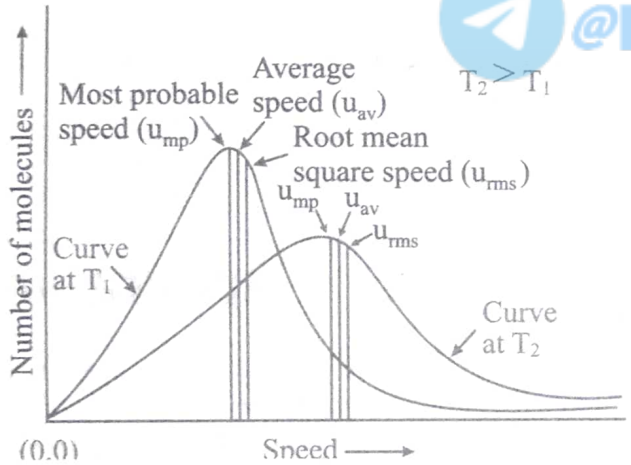
$$u_{av} = \frac{u_1 + u_2 + ...u_n}{n} = \sqrt{\frac{8RT}{\pi M}}$$

R = universal gas constant

T = absolute temperature

M = Molar mass of the gas

Maxwell and Boltzmann have shown that actual distribution of molecular speeds depends on temperature and molecular mass of a gas. Maxwell derived a formula for calculating the number of molecules possessing a particular speed. shows schematic plot of number of molecules v/s. molecular speed at two different temperatures T_1 and T_2 (T_2 is higher than T_1). The distribution of speeds shown in the plot is called Maxwell-Boltzmann distribution of speeds.

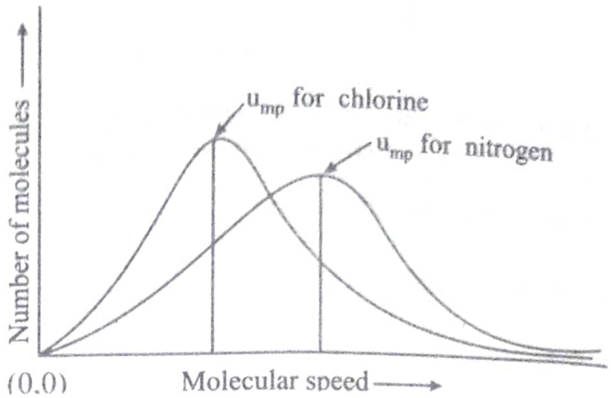


Maxwell-Boltzmann distribution of speeds

The graph shows that number of molecules possessing very high and very low speed is very small. The maximum in the curve represents speed possessed by maximum number of molecules. This speed is called **most probable speed,** u_{mp}.

$$u_{mp} = \sqrt{\frac{2RT}{M}}$$

This is very close to the average speed of the molecules. On increasing the temperature most probable speed increases. Also, speed distribution curve broadens at higher temperature. Broadening of the curve shows that number of molecules moving at higher speed increases. Speed distribution also depends upon mass of molecules. At the same temperature, gas molecules with heavier mass have slower speed than lighter gas molecules. For example, at the same temperature lighter nitrogen molecules move faster than heavier chlorine molecules. Hence, at any given temperature, nitrogen molecules have higher value of most probable speed than the chlorine molecules. Look at the molecular speed distribution curve of chlorine and nitrogen given in. Though at a particular temperature the individual speed of molecules keep changing, the distribution of speeds remains same.



Distribution of molecular speeds for chlorine and nitrogen 300 K

We know that kinetic energy of a particle is given by the expression:

$$\text{Kinetic Energy} = \frac{1}{2}mu^2$$

Therefore, if we want to know average translational kinetic energy, $\frac{1}{2}m\overline{u^2}$, for the movement of a gas particle in a straight line, we require the value of mean of square of speeds, $\overline{mu^2}$, of all molecules. This is represented as follows:

$$\overline{u^2} = \frac{u_1^2 + u_2^2 + ...u_n^2}{n}$$

The mean square speed is the direct measure of the average kinetic energy of gas molecules. If we take the square root of the mean of the square of speeds then we get a value of speed which is different from most probable speed and average speed. This speed is called **root mean square speed** and is given by the expression as follows:

$$u_{rms} = \sqrt{\overline{u^2}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3P}{d}} = \sqrt{\frac{3PV}{M}}$$

Root mean square speed, average speed and the most probable speed have following relationship:

$$u_{rms} > u_{av} > u_{mp}$$

The ratio between the three speeds is given below:

$$u_{mp} : u_{av} : u_{rms} = 1 : 1.128 : 1.224$$

KINETIC MOLECULAR THEORY OF GASES

So far we have learnt the laws (e.g., Boyle's law, Charles' law etc.) which are concise statements of experimental facts observed in the laboratory by the scientists. Conducting careful experiments is an important aspect of scientific method and it tells us how the particular system is behaving under different conditions. However

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ıce the experimental facts are established, a scientist is curious know why the system is behaving in that way. For example, gas ws help us to predict that pressure increases when we compress ases but we would like to know what happens at molecular level hen a gas is compressed ? A theory is constructed to answer such estions. A theory is a model (i.e., a mental picture) that enables ; to better understand our observations. The theory that attempts elucidate the behaviour of gases is known as kinetic molecular eory. Assumptions or postulates of the kinetic molecular theory f gases are given below. These postulates are related to atoms ıd molecule which cannot be seen, hence it is said to provide a icroscopic model of gases.

- Gases consist of large number of identical particles (atoms or molecules) that are so small and so far apart on the average that the actual volume of the molecules is negligible in comparison to the empty space between them. They are considered as point masses. This assumption explains the great compressibility of gases.
- There is no force of attraction between the particles of a gas at ordinary temperature and pressure. The support for this assumption comes from the fact that gases expand and occupy all the space available to them.
- Particles of a gas are always in constant and random motion. If the particles were at rest and occupied fixed positions, then a gas would have had a fixed shape which is not observed.
- Particles of a gas move in all possible directions in straight lines. During their random motion, they collide with each other and with the walls of the container. Pressure is exerted by the gas as a result of collision of the particles with the walls of the container.
- Collisions of gas molecules are perfectly elastic. This means that total energy of molecules before and after the collision remains same. There may be exchange of energy between colliding molecules, their individual energies may change, but the sum of their energies remains constant. If there were loss of kinetic energy, the motion of molecules will stop and gases will settle down. This is contrary to what is actually observed.
- At any particular time, different particles in the gas have different speeds and hence different kinetic energies. This assumption is reasonable because as the particles collide, we expect their speed to change. Even if initial speed of all the particles was same, the molecular collisions will disrupt this uniformity. Consequently, the particles must have different speeds, which go on changing constantly. It is possible to show that though the individual speeds are changing, the distribution of speeds remains constant at a particular temperature.
- If a molecule has variable speed, then it must have a variable kinetic energy. Under these circumstances, we can talk only about average kinetic energy. In kinetic theory, it is assumed that average kinetic energy of the gas molecules is directly proportional to the absolute temperature. It is seen that on heating a gas at constant volume, the pressure increases. On heating the gas, kinetic energy of the particles increases and these strike the walls of the container more frequently, thus, exerting more pressure.

Kinetic theory of gases allows us to derive theoretically, all the gas laws studied in the previous sections. Calculations and predictions based on kinetic theory of gases agree very well with the experimental observations and thus establish the correctness of this model.



Train Your Brain

Example 12: Find the temperature at which methane and ethane will have the same rms speed as carbon dioxide at 400°C.

Sol. $v_{rms} = \sqrt{\frac{3RT}{M}}$

$$\Rightarrow \frac{T_1}{M_1} = \frac{T_2}{M_2}$$

$$(a)\ T_{CH_4} = \frac{673}{44} \times 16 = 244.73K$$

$$(b)\ T_{C_2H_6} = \frac{673}{44} \times 30 = 458.86K$$

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Example 13: Two gases A and B have the same value of most probable speed at 298 K for A and 150 K for B. Calculate the ratio of their molor masses.

Sol. Most Probable $= \left(\sqrt{\frac{2RT}{M}}\right)$

$$\frac{T_A}{M_A} = \frac{T_B}{M_B}$$

$$\therefore \frac{M_A}{M_B} = \frac{T_A}{T_B} = \frac{298}{150} \approx 2$$

$$\therefore M_A : M_B = 2 : 1$$

Example 14: Calculate the ratio of kinetic energy per mole of nitrogen at 27°C and hydrogen at 127°C.

Sol. Kinetic energy (K.E) $= \frac{3}{2}RT$

$$KE_{N_2} = \frac{3}{2}RT_{N_2}$$

$$KE_{H_2} = \frac{3}{2}RT_{H_2}$$

$$\frac{KE_{N_2}}{KE_{H_2}} = \frac{T_{N_2}}{T_{H_2}} = \frac{300}{400} = \frac{3}{4}$$

Concept Application

15. The R.M.S. speed of the molecules of a gas of density 4 kg m^{-3} and pressure 1.2×10^5 N m^{-2} is:

(*a*) 120 ms^{-1} (*b*) 300 ms^{-1}

(*c*) 600 ms^{-1} (*d*) 900 ms^{-1}

16. The temperature of an ideal gas is increased from 120 K to 480 K. If at 120 K the root-mean-square velocity of the gas molecules is v, at 480 K it becomes :

(*a*) 4v (*b*) 2v

(*c*) v/2 (*d*) v/4

17. Helium atom is two times heavier than a hydrogen molecule. At 298 K, the average kinetic energy of a helium atom is

(*a*) Two times that of a hydrogen molecules

(*b*) Same as that of a hydrogen molecules

(*c*) Four times that of a hydrogen molecules

(*d*) Half that of a hydrogen molecules

18. Initially, the root mean square (rms) velocity of N_2 molecules at certain temperature is u. If this temperature is doubled and all the nitrogen molecules dissociate into nitrogen atoms, then the new rms velocity will be:

(*a*) 2 u (*b*) 14 u

(*c*) u / 2 (*d*) 4 u

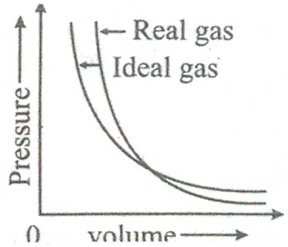
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REAL GASES

- Real gases do not obey the ideal gas laws exactly under all conditions of temperature and pressure.
- Real gases deviates from ideal behaviour because Real gas molecules have a finite volume {since on liquefaction real gases occupy a finite volume}.
- Inter molecular attractive forces between real gas molecules is not zero {Real gases can be converted into liquid where as ideal gases can't be}.
- Deviation of real gases from ideal behaviour can be measured by using compresibility factor : (Z)

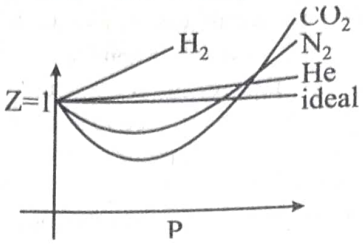
$$Z = \frac{(PV)_{real}}{(PV)_{ideal}} ; (PV)_{ideal} = nRT$$

$$Z = \frac{PV}{nRT} = \frac{PV_m}{RT}$$, V_M is volume of one mole of gas or molar volume.

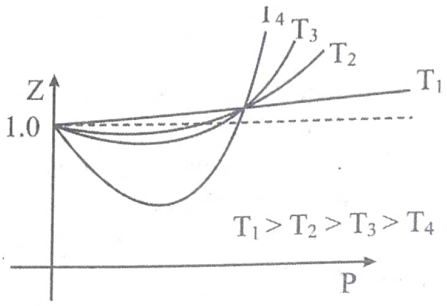


$$Z = \frac{V_m}{V_{m\ ideal}}$$

Variation of Z with pressure at constant temperature:



Variation of Z with pressure at different temperature (for a gas):



Conclusions:

Z = 1 for ideal gas ;

Z > 1 at all pressures for He/H_2

Z < 1 at low pressure (for all other gases) ; Z > 1 at high pressure (for all other gases)

Vander Waal's Equation of Real Gases

The ideal gas equation does not consider the effect of attractive forces and molecular volume. Vander Waal's corrected the ideal gas equation by taking the effect of

(*a*) Molecular volume (*b*) Molecular attraction

- Volume correction:

Ideal gas equation:

$P_i V_i = nRT$; In the equation 'V_i' stands for the volume which is available for free movement of the molecules.

V_{ideal} = volume available for free movement of gaseous molecule

hence, $V_i = V -$ {volume not available for free movement} For an ideal gas

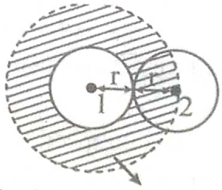
$V_i = V$ {V = volume of container}

but for a real gas

$V_i \neq V$, as all the volume is not available for free movement

Molecules have finite volume : The volume that is not available for free movement is called excluded volume.

let us see, how this excluded volume is calculated.



Excluded volume
(not available for free momement)

For above example, the entire shaded region is excluded, as it's centre of mass cannot enter this region.

If both molecules were ideal, then they would not have experienced any excluded volume but not in the case, of real gas as the centre of mass of '2' cannot go further.

Hence for this pair of real gas molecules,

$$\text{Excluded volume per molecule} = \frac{1}{2}\left\{\frac{4}{3}\pi (2r)^3\right\}$$

$$= 4\left\{\frac{4}{3}\pi r^3\right\}$$

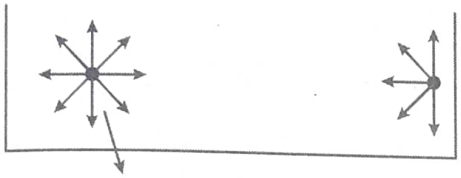
$$= \text{excluded volume per mole of gas (b)} = N_A\, 4\left\{\frac{4}{3}\pi r^3\right\}$$

$= 4 \times N_A \times$ Volume of individual molecule

for n moles, excluded volume $= nb$

$V_i = V - nb$ volume correction

❖ **Pressure correction or effect of molecular attraction forces:**



Molecule in the middle of container

Due to these attraction, speed during collisions will be reduced

Momentum will be less

Force applied will be less

Pressure will be less.

$P_{ideal} = P + \{\text{correction term}\}$

Correction term α no. of molecules attracting the colliding molecule $\propto (n/V)$.

Correction term α density of molecules $\propto (n/V)$.

no. of collision α density of molecules $\propto \left(\frac{n}{V}\right)$

$$\text{net correction term} \propto \left(\frac{n}{V}\right)\left(\frac{n}{V}\right) = \frac{an^2}{V^2}$$

'a' is constant of proportionality

and this is dependent on force of attraction

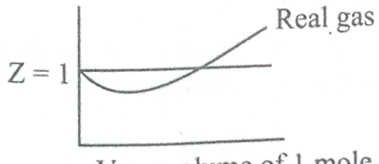
Stronger the force of attraction greater will be 'a' (Constant)

$$P_i = P + \frac{an^2}{V^2}$$

Vander waal's equation is $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$

VERIFICATION OF VANDER WAAL'S EQUATIONS

Variation of Z with P for vander waals' equation at any temp.



V_m = volume of 1 mole

Vander waal equation for 1mole:

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

❖ **AT LOW PRESSURE** (at separate temp.)

At low pressure V_m will be high.

Hence b can be neglected in comparision to V_m. but $\frac{a}{V_m^2}$ can't be neglected as pressure is low

Thus equation would be $\left(P + \frac{a}{V_m^2}\right) V_m = RT$

$$PV_m + \frac{a}{V_m} = RT$$

$$\frac{PV_m}{RT} + \frac{a}{V_m RT} = 1$$

$$Z = 1 - \frac{a}{V_m RT} \quad Z < 1$$

Real gas is easily compressible as compared to an ideal gas.

❖ **AT HIGH PRESSURE** (moderate temperature)

V_m will be low so b can't be neglected in comparision to V_m but $\frac{a}{V_m^2}$ can be neglected as compared to much higher values of P.

Then vander Waals' equation will be

$P(V_m - b) = RT$

$PV_m - Pb = RT$

$$\frac{PV_m}{RT} = \frac{Pb}{RT} + 1$$

$$Z = \frac{Pb}{RT} + 1 \quad (Z > 1)$$

If $Z > 1$, then gas is more difficult to compress as compared to an ideal gas.

❖ At low pressure and very high temperature.

V_m will be very large hence 'b' can be neglected and $\frac{a}{V_m^2}$ can also be neglected as V_m is very large

$PV_m = RT$ (ideal gas condition)

❖ For H_2 or He $a \cong 0$ because molecules are smaller in size or vander Waal's forces will be very weak, these are non polar so no dipole-dipole interactions are present in the actions.

$P(V_m - b) = RT$

$$\text{So } Z = 1 + \frac{Pb}{RT}$$

❖ 'a' factor depends on inter molecular attractive forces.

❖ 'a' factor for polar molecule > 'a' factor for non polar molecule.

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Train Your Brain

Example 15: Arrange following in decreasing 'a' factor (H_2O, CO_2, Ar)

Sol. $H_2O > CO_2 > Ar$

For non polar molecules: Greater the size or surface area, greater will be vander waal 's forces, so greater will be 'a' constant.

Example 16: Arrange following gases according to 'a' He, Ar, Ne, Kr.

Ans. $a_{Kr} > a_{Ar} > a_{Ne} > a_{He}$

- More 'a' factor means higher will be boiling point.
- **Liquification pressure:** Is the pressure required to convert gas into liquid.
  for easy liquefaction a↑ and P↓
  When $Z < 1, V_m < V_{m,\text{ideal}}$ ⇒ easily liquifiable
  $Z > 1, V_m > V_{m,\text{ideal}}$ ⇒ more difficult to compress.

Example 17: Arrange the following according to liquification pressure: n-pentane ; iso-pentane , neo pentane.

$a_{\text{n-pentene}} > a_{\text{iso-pentane}} > a_{\text{neo-pentane}}$

liquification pressure = LP

Sol. $L_{P_{n-\text{pentane}}} < L_{P_{\text{iso pentane}}} < L_{P_{\text{neo pentane}}}$

- b is roughly related with size of the molecule.

$$b = N_A\, 4\left\{\frac{4}{3}\pi r^3\right\}$$

Example 18: Two vander waals gases have same value of b but different a values. Which of these would occupy greater volume under identical conditions?

Sol. If two gases have same value of 'b' but different values of 'a', then the gas having a larger value of 'a' will occupy lesser volume. This is because the gas with a larger value of 'a' will have larger force of attraction and hence lesser distance between its molecules.

Ideal gases

- Zero volume
- Zero attractive force
- PV = nRT
- Z = 1

Real gases

- Corrected equation $\left(P+\frac{an^2}{V^2}\right)(V-nb) = nRT$
- non-zero volume
- some intermolecular force

(+)ve

- Z > 1
- Repulsive forces
- Difficult to compress
- Difficult to Liquify

(-)ve

- Z < 1
- Attractive forces
- Easy to compress
- Easy to Liquify

Correction for intermolecular attraction

Correction for molecular volume

$$\left(P+\frac{an^2}{V^2}\right)(V-nb) = nRT$$

Concept Application

19. A gas behaves like an ideal gas at:
(*a*) high pressure and low temperature
(*b*) low pressure and high temperature
(*c*) high pressure and high temperature
(*d*) low pressure and low temperature

20. Vander Waals constant b of helium is 24 mL mol^{-1}. Find molecular diameter of helium.
(*a*) 1.335×10^{-10} cm (*b*) 1.335×10^{-8} cm
(*c*) 2.67×10^{-8} cm (*d*) 4.34×10^{-8} cm

21. For a gas deviation from ideal behaviour is maximum at:
(*a*) 0°C and 1.0 atm (*b*) 100°C and 2.0 atm
(*c*) –13°C and 1.0 atm (*d*) –13°C and 2.0 atm

22. Consider the equation $Z = \frac{PV}{RT}$. Which of the following statements is correct?
(*a*) When Z >1, real gases are easier to compress than the ideal gas
(*b*) When Z = 1, real gases get compressed easily
(*c*) When Z > 1, real gases are difficult to compress
(*d*) When Z = 1, real gases are difficult to compress

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CRITICAL CONSTANTS

- **Critical temperature** of a gas is the temperature at or abov which vapor of the gas cannot be liquefied, no matter hov much pressure is applied.
- At critical temperature liquid phase passes into gaseous stat imperceptibly and continuously.
- A gas below the critical temperature can be liquefied b applying pressure and is called vapour of the substance.
- At critical temperature, density of vapour phase equal t density of liquid phase.
- Pressure required to liquefy a gas at its critical temperature is called **critical pressure** (P_C).
- The volume occupied by 1 mole of the gas at critical temperature and critical pressure is **critical volume** (V_C).
- **Relationship between critcal constants and Van der Waal constants**

$$P_C = \frac{a}{27b^2};\ V_C = 3b;\ T_C = \frac{8a}{27Rb}$$

Boyle's Temperature (T_B)

- It is that temperature at which a real gas behaves ideally over a large pressure range as shown:

$$T_B = \frac{a}{Rb}$$

The numerical value of $\frac{RT_C}{P_C V_{m,C}} = \frac{8}{3}$ (on substituting T_C, P_C, $V_{m,C}$). Thus one can write that the value of RT/PV at critical conditions is 8/3 of normal conditions.

i.e., $\frac{RT_C}{P_C V_{m,C}} = \frac{8}{3} = \frac{8}{3}\frac{RT}{PV}\left(\because \frac{RT}{PV} = 1\right)$

$\Rightarrow$ At critical conditions, $Z = \frac{P_C V_{m,C}}{RT_C} = \frac{3}{8}$

Note : Boyle's temperature of a gas is always higher than its critical temperature (T_C).

- Reduced pressure is P/P_C
- Reduced volume is V/V_C
- Reduced temperature T/T_C

COLLISION FREQUENCY AND MEAN FREE PATH

The collision rate (i.e., the number of collisions taking place in unit time per unit volume) is called collision frequency (z) and is given by

$$z = \frac{\pi n^2 \sigma^2 u_{avg}}{\sqrt{2}}$$

The average of distances travelled by a molecule between successive collisions is called mean free path (λ).

$$\lambda = \frac{1}{\sqrt{2}\pi n \sigma^2}$$

where n is the number of molecules per unit molar volume.

$$n = \frac{6.02 \times 10^{23}}{0.0224} m^{-3}$$

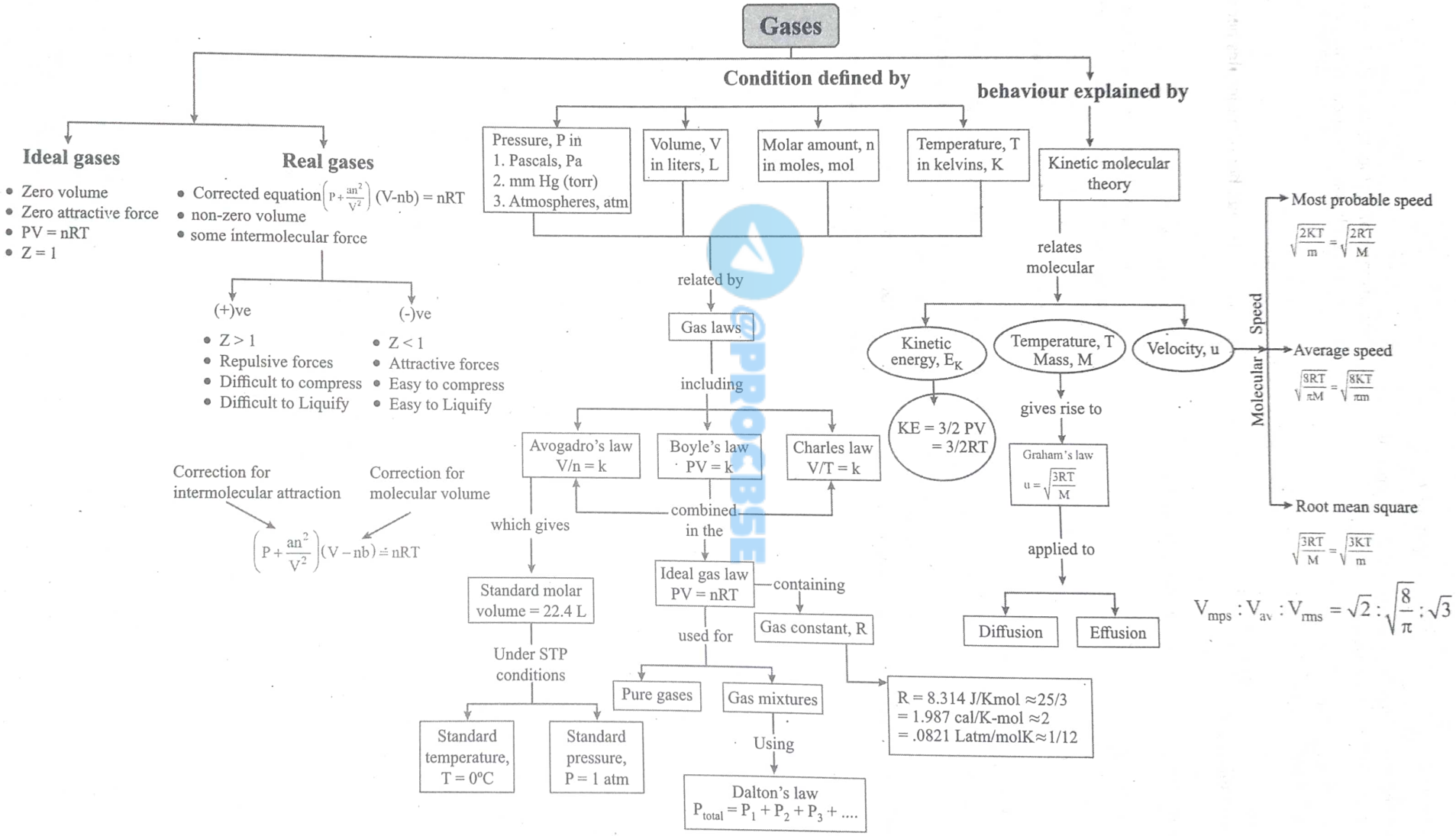
u_{avg} = average velocity, σ = collision diameter i.e., the minimum distance between the centres of two molecules when at the point of collision.

Also based on kinetic theory of gases, mean free path $\lambda \propto \frac{T}{P}$.

Thus,

- Larger the size of the molecules, smaller the mean free path.
- Greater the number of molecules per unit volume, smaller the mean free path.
- Larger the temperature, larger the mean free path.
- Larger the pressure, smaller the mean free path.

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Gases
Condition defined by
behaviour explained by
Ideal gases
Zero volume
Zero attractive force
PV = nRT
Z = 1
Real gases
Corrected equation $\left(P+\frac{an^2}{V^2}\right)(V-nb)=nRT$
non-zero volume
some intermolecular force
(+)ve
Z > 1
Repulsive forces
Difficult to compress
Difficult to Liquify
(-)ve
Z < 1
Attractive forces
Easy to compress
Easy to Liquify
Correction for intermolecular attraction
Correction for molecular volume
$\left(P+\frac{an^2}{V^2}\right)(V-nb)=nRT$
Pressure, P in
1. Pascals, Pa
2. mm Hg (torr)
3. Atmospheres, atm
Volume, V in liters, L
Molar amount, n in moles, mol
Temperature, T in kelvins, K
related by
Gas laws
including
Avogadro's law V/n = k
Boyle's law PV = k
Charles law V/T = k
combined in the
which gives
Ideal gas law PV = nRT
containing
Gas constant, R
Standard molar volume = 22.4 L
Under STP conditions
Standard temperature, T = 0°C
Standard pressure, P = 1 atm
used for
Pure gases
Gas mixtures
Using
Dalton's law $P_{total}=P_1+P_2+P_3+....$
R = 8.314 J/Kmol ≈25/3
= 1.987 cal/K-mol ≈2
= .0821 Latm/molK≈1/12
Kinetic molecular theory
relates molecular
Kinetic energy, E_K
Temperature, T Mass, M
Velocity, u
KE = 3/2 PV = 3/2RT
gives rise to
Graham's law $u=\sqrt{\frac{3RT}{M}}$
applied to
Diffusion
Effusion
Molecular Speed
Most probable speed $\sqrt{\frac{2KT}{m}}=\sqrt{\frac{2RT}{M}}$
Average speed $\sqrt{\frac{8RT}{\pi M}}=\sqrt{\frac{8KT}{\pi m}}$
Root mean square $\sqrt{\frac{3RT}{M}}=\sqrt{\frac{3KT}{m}}$
$V_{mps}:V_{av}:V_{rms}=\sqrt{2}:\sqrt{\frac{8}{\pi}}:\sqrt{3}$

Short Notes

Gas's Law

Boyle's law:

$$V \propto \frac{1}{P}(n, T = \text{const}) \quad \boxed{P_1V_1 = P_2V_2}$$

Charle's law:

$$V \propto T(n, P = \text{const}) \quad \frac{V_2}{V_1} = \frac{T_2}{T_1}$$

Gay lussac's law:

$$P \propto T(n, V = \text{const}) \quad \frac{P_2}{P_1} = \frac{T_2}{T_1}$$

Avogadro's law:

$V \propto$ moles $\propto$ number of molecules (P,T = const)

Ideal gas equation $PV = nRT$

$R = 0.0821 L\ atm\ mol^{-1}\ K^{-1}$

$R = 8.314\ J\ K^{-1}\ mol^{-1}$ or $8.314\ N \times m\ K^{-1}\ mol^{-1}$

$R = 2 cal\ K^{-1}\ mol^{-1}$, $R = 8.314 \times 10^7 erg\ K^{-1}\ mol^{-1}$

Graham's Diffusion Law

It is applicable for non reacting gases

$$r \propto \frac{1}{\sqrt{d}}; r \propto \frac{1}{\sqrt{VD}}; r \propto \frac{1}{\sqrt{M_w}} (P, T = \text{constant})$$

$$VD = \frac{d_{gas}}{d_{H_2}} = \frac{M_w}{2}$$

Rate of diffusion

$$r = \frac{l_{\text{diffused gas}}}{t_{\text{time taken}}}; r = \frac{V_{\text{diffused gas}}}{t_{\text{time taken}}}; r = \frac{n_{\text{diffused gas}}}{t_{\text{time taken}}}$$

(When, l = distance traveled by diffused gas)

Kinetic Gas Equation : $PV = 1/3\ mN\ V^2_{rms}$

Dalton's Law of Partial Pressure

$$P_{mixture} = \underbrace{P_1 + P_2 + P_3}_{\text{Partial pressure}} \ldots\ldots (T \& V \text{ const.})$$

$$P_{\text{moist gas}} = P_{\text{dry gas}} + P_{\text{water vapours}}$$

It is applicable for non reacting gases.

Methods of determination of partial pressure

(P_A & P_B are partial pressure)

❖From ideal gas equation

$P_AV = n_A RT$ & $P_BV = n_B RT$

❖In the form of mole fraction.

$$P_A = X_AP_T = \frac{n_A}{n_T} P_T;$$

$$P_B = X_BP_T = \frac{n_B}{n_T} P_T$$

$[X_A + X_B = 1]$

P_T = sum of partial pressure of all gases

❖In the form of volume fraction.

$$P_A = \frac{V_A}{V} P_T \text{ \& } P_B = \frac{V_B}{V} P_T$$

❖If individual pressure and individual volume are given

$$P_A = \frac{V_A}{V} P_1 \text{ and } P_B = \frac{V_B}{V} P_2$$

P_1, P_2 = pressure of gases before mixing

P_A, P_B = pressure of gases after mixing

Average Kinetic Energy (KE_{av})

$$K.E_{av} = \frac{3}{2} nRT \quad (n \text{ moles})$$

$$K.E_{av} = \frac{3}{2} RT \quad (1 \text{ mol or } N_A \text{ molecules})$$

$$K.E_{av} = \frac{3}{2} KT \quad (1 \text{ molecule})$$

$K_B = 1.38 \times 10^{-23}\ JK^{-1}\ molecule^{-1}$

K_B is called Boltzman's constant

$$u_{rms} = \sqrt{\frac{v_1^2 + v_2^2 + \ldots v_n^2}{N}}$$

$$u_{av} = \frac{v_1 + v_2 + v_3 \ldots v_n}{N}$$

$$u_{rms} = \sqrt{\frac{3RT}{Mw}}; u_{av} = \sqrt{\frac{8}{\pi}\frac{RT}{Mw}}; u_{mp} = \sqrt{\frac{2RT}{Mw}}$$

$$u_{rms} = \sqrt{\frac{3PV}{Mw}}; u_{av} = \sqrt{\frac{8}{\pi}\frac{PV}{Mw}}; u_{mp} = \sqrt{\frac{2PV}{Mw}}$$

$$u_{rms} = \sqrt{\frac{3P}{d}}; u_{av} = \sqrt{\frac{8}{\pi}\frac{P}{d}}; u_{mp} = \sqrt{\frac{2P}{d}}$$

$$u_{rms} : u_{av} : u_{mp} = \sqrt{3} : \sqrt{\frac{8}{\pi}} : \sqrt{2}$$

$$= 1 : 0.92 : 0.82$$

$$u_{mp} : u_{av} : u_{rms} = \sqrt{2} : \sqrt{\frac{8}{\pi}} : \sqrt{3}$$

$$= 1 : 1.128 : 1.224$$

Compressibility factor

$$(z) = \frac{(V_m)obs}{V_i} = \frac{P(V_m)obs}{RT}$$

IF $z = 1$, the gas show ideal gas behaviour.

IF $z > 1$, the gas show positive deviation.

IF $z < 1$, the gas show negative deviation.

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Vanderwaal's Equation

Vanderwaal's Equation	
$\left(P+\frac{an^2}{V^2}\right)(V-nb)=nRT$ $P_i=P_R+\frac{an^2}{V^2} \Rightarrow P_i > P_R$ a increases force of attraction increases liquification increases; b increases effective size of molecule increases, incompressible volume increases, compressible vol. decreases	❖At high pressure, Vanderwaal's eqⁿ is $\boxed{PV_m - Pb = RT}$ ❖At low pressure or Moderate pressure vanderwaal's eqⁿ is $\boxed{PV_m+\frac{a}{V_m}=RT}$ ❖At very low pressure, high temp. Vanderwaal's Equation is $\boxed{VP=nRT}$ Ideal gas behaviour. ❖Gases having higher value of a; will have higher T_C; higher rate of liquefaction.

Solved Examples

1. A rubber balloon contains some solid marbles each of volume 10 ml. A gas is filled in the balloon at a pressure of 2 atm and the total volume of the balloon is 1 litre in this condition. If the external pressure is increased to 4 atm the volume of Balloon becomes 625 ml. Find the number of marbles present in the balloon.

Sol. Let the no. of marbles be = n.

Volume of marble = 10 n ml.

Volume of balloon earlier = 1000 ml.

Later = 625 ml.

Now for the gas inside the balloon temperature and amount of the gas is constant, hence boyles law can be applied

$P_1V_1 = P_2V_2$

$4 \times (625 - 10n) = 2 \times (1000 - 10n)$

$625 \times 4 = 2000 - 20n + 40n$

$625 \times 4 - 2000 = 20n$

$\frac{625\times4-2000}{20} = n.\ \frac{125}{5} = n;\ n = 25$

2. If water is used in place of mercury then what should be minimum length of Barometer tube to measure normal atmospheric pressure.

Sol. $P_{Hg} = P_{H_2O} = P_{atm}$.

Pressure = ρgh

$0.76\ m \times 13.6 \times g = h_{H_2O} \times 1 \times g$

$h_{H_2O} = 0.76 \times 13.6 = 10.336\ m$

3. An open container of volume V contains air at temperature 27°C or 300 K. The container is heated to such a temperature so that amount of gas coming out is 2/3 of

(*a*) amount of gas initially present in the container.

(*b*) amount of gas finally remaining in the container.

Find the temperature to which the container should be heated.

Sol. (*a*) Here, P & V are constant, n & T are changing. Let, initially the amount of gas present be n & temp is 27°C or 300 K. Finally amount of gas present in container $= n - \frac{2}{3}n = \left(\frac{1}{3}\times n\right)$ & final temp. be T.

Then using $n_1T_1 = n_2T_2$, we have,

$n \times 300 = \frac{n_2}{3} \times T_2 \Rightarrow T_2 = 900\ K$

i.e., final temp = 900 K

(*b*) Let there be x moles of gas remaining in the container, $\frac{2}{3}$ of x come out

$\therefore\ (x + \frac{2}{3}x) = n \Rightarrow \frac{5x}{3} = n\ \therefore\ x = \frac{3n}{5}$

$\therefore$ Using $n_1T_1 = n_2T_2\ n \times 300\ K = \frac{3n}{5} \times T_2$

$\therefore\ T_2 = 500\ K$

Final temperature = 500 K

4. Find the lifting power of a 100 litre balloon filled with He at 730 mm and 25°C (Density of air = 1.25 g/L).

Sol. Since, $PV = nRT\ PV = \frac{W}{M}RT$

$\therefore\ W = \frac{PVM}{RT} = \frac{730}{760} \times \frac{100\times4}{0.082\times298}\ g$

i.e., Wt. of He = 15.72 g

Wt. of air displaced = 100 × 1.25 g/L = 125g

$\therefore$ Lifting power of the balloon = 125g – 15.72 g = 109.28 g

5. Some spherical balloons each of volume 2 litre are to be filled with hydrogen gas at one atm & 27°C from a cylinder of volume 4 litres. The pressure of the H_2 gas inside the cylinder is 20 atm at 127°C. Find number of balloons which can be filled using this cylinder. Assume that temperature of the cylinder is 27°C.

Sol. No. of moles of gas taken initially $n = \frac{PV}{RT} = \frac{20\times4}{R\times400} = 2.43$

No. of moles of gas left in cylinder $= \frac{1\times4}{R\times300} = 0.162$

No. of moles of gas to be filled in balloons

$= 2.43 - 0.162 = 2.268$

Let we have 'n' balloons that we can fill

No. of moles of gas that can be filled in 1 balloon

$= \frac{1\times2}{0.082\times300} = 0.081$

$\therefore\ 0.081 \times n = 2.268$

$n = 28$ balloons.

6. Four one litre flasks are separately filled with the gases O_2, F_2, CH_4 and CO_2 under the same conditions. The ratio of number of molecules in these gases:

(*a*) 2 : 2 : 4 : 3 (*b*) 1 : 1 : 1 : 1

(*c*) 1 : 2 : 3 : 4 (*d*) 2 : 2 : 3 : 4

Sol. (*b*) According to avogadro's hypothesis.

All the flasks contains same no. of molecules

$\therefore$ Ratio of no. of molecules of O_2, F_2, CH_4 & CO_2

= 1 : 1 : 1 : 1

7. A sample of water gas has a composition by volume of 50% H_2, 45% CO and 5% CO_2. Calculate the volume in litres at STP at which water gas which on treatment with excess of steam will produce 5 litre of H_2. The equation for the reaction is:

$CO + H_2O \rightarrow CO_2 + H_2$

Sol. If x L CO is needed then

Volume of H_2 in water gas $= \left(\frac{x}{0.45}\times50\%\right)L$

$= \left(\frac{x}{0.45}\times\frac{1}{2}\right)L = \frac{x}{0.9}L$

But, from equation : $CO + H_2O \rightarrow CO_2 + H_2$

& Gay-Lussac's law, we get, that the volume of H_2 Produced = volume of CO taken.

$\therefore$ Volume of H_2 due to reaction = x L

$\therefore$ Total volume of $H_2 = \left(\frac{x}{0.9} + x\right)L = 5\ L$

$\Rightarrow \frac{1.9x}{0.9} = 5\ L$

$\therefore\ x = \frac{0.9\times5}{1.9} \Rightarrow 2.37$

$\therefore$ Volume of water gas

$= \frac{x}{0.45}L = \frac{2.37}{0.45}L = 5.263$ L Ans.

8. The stop cock connecting the two bulbs of volume 5 litre and 10 litre containing as ideal gas at 9 atm and 6 atm respectively, is opened. What is the final pressure if the temperature remains same.

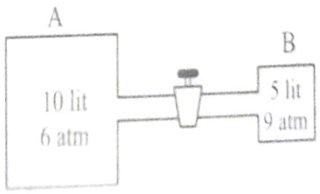
Sol. After the opening of the stop cock the pressure of the each bulb will remain same.

$n = \frac{PV}{RT}$

At the beginning, the no. of moles of gas in A $= \frac{10\times6}{RT}$

At the beginning, the no. of moles of gas in B $= \frac{5\times9}{RT}$

$\therefore$ Total no. of moles at the beginning $= \frac{105}{RT}$



Total no. of moles of gas before opening the stop cock = total no. of moles of gas after opening stopcock $= \frac{105}{RT}$

$\therefore$ Pressure after the opening of the stop cock

$P = \frac{105}{RT}\times\frac{RT}{V_{total}} = \frac{105}{10+5} = 7$ atm

9. A mixture of NO_2 & CO having total volume of 100 ml contains 70 ml of NO_2 at 1 atm, mixture is left for some time and same NO_2 get dimerised to N_2O_4 such that final volume of the mixture become 80 ml at 1 atm, calculate the mole fraction of NO_2 in final equilibrium mixture.

Sol. Initial volume of NO_2 = 70 ml

Initial volume of CO = 100 – 70 = 30 ml

Final volume of mixture = 80 ml

Let the volume of NO_2 in final mixture be x

Let 'v' ml NO_2 be converted to N_2O_4

$2NO_2 \rightarrow N_2O_4$

V V/2

Hence final volume

= volume of CO + volume of NO_2 left + volume of N_2O_4 formed

$30 + 70 - V + V/2 = 80$

$V = 40$ ml

Hence volume of NO_2 left = 70 – V = 30 ml

Now as volume $\propto$ moles

$\therefore$ mole fraction = volume fraction $= \frac{30}{80} = \frac{3}{8}$

10. 5 ml of H_2 gas diffuses out in 1 sec from a hole. Find the volume of O_2 that will diffuse out from the same hole under identical conditions in 2 sec.

@PROCBSE

Sol Rate of diffusion of $H_2 = \frac{5\,ml}{1\,sec} = 5\ ml/s = r_{H_2}$ (say)

rate of diffusion $\propto \sqrt{\frac{1}{\text{molar mass}}}$

$\therefore \quad r_{O_2} = r_{H_2} \times \frac{1}{4} = 5\ ml/s \times \frac{1}{4}$

$\therefore$ Volume of O_2 diffused in 2.0 seconds $= \frac{5}{4} \times 2$ ml $= 2.5$ ml.

11. A vessel contains H_2 & O_2 in the molar ratio of 8 : 1 respectively. This mixture of gases is allowed to diffuse through a hole, find composition of the mixture coming out of the hole.

Sol. Here, $n_{H_2} : n_{O_2} = 8 : 1$ & $\frac{r_{H_2}}{r_{O_2}} = \frac{n_{H_2}}{n_{O_2}} \sqrt{\frac{M_{O_2}}{M_{H_2}}}$

$\Rightarrow \quad \frac{r_{H_2}}{r_{O_2}} = \frac{8}{1} \times \sqrt{\frac{32}{2}} = \frac{32}{1}$

$\Rightarrow \quad \frac{(\text{no. of moles of } H_2 \text{ coming out})/\Delta t}{(\text{no. of moles of } O_2 \text{ coming out})/\Delta t} = \frac{32}{1}$

$\therefore$ Required composition of $H_2 : O_2$ coming out $= 32 : 1$.

12. If for two gases of molecular weights M_A and M_B at temperature T_A and T_B ; $T_A M_B = T_B M_A$, then which property has the same magnitude for both the gases.

(*a*) Density (*b*) Pressure

(*c*) KE per mol (*d*) RMS speed

Sol. (*d*) Given that $T_A M_B = T_B M_A \Rightarrow \frac{T_A}{M_A} = \frac{T_B}{M_B}$

But, r.m.s. $= \sqrt{\frac{3RT}{M}}$

$\therefore \quad r.m.s_A = \sqrt{\frac{3RT_A}{M_A}}$ & $r.m.s_B = \sqrt{\frac{3RT_B}{M_B}}$

$\therefore \quad r.m.s_A = r.m.s_B$

13. Under critical states of a gas for one mole of a gas, compressibility factor is

(*a*) $\frac{3}{8}$ (*b*) $\frac{8}{3}$

(*c*) 1 (*d*) $\frac{1}{4}$

Sol. (*a*) For 1 mole of gas $Z = \frac{P_C V_C}{RT_C}$ (Under critical condition)

But, $P_C = \frac{a}{27b^2}$, $V_C = 3b$, $T_C = \frac{8a}{27Rb}$

$\therefore \quad Z = \left(\frac{a}{27b^2}\right) \times \frac{3b}{R} \times \frac{27Rb}{8a} = \frac{3}{8}$

14. Two closed vessel A and B of equal volume containing air at pressure P_1 and temperature T_1 are connected to each other through a narrow open tube. If the temperature of one is now maintained at T_1 and other at T_2 (where $T_1 > T_2$) then that what will be the final pressure?

Sol.

P_1, T_1		P_1, T_1
Volume = V		Volume = V

Initial total moles $= \frac{P_1 V}{RT_1} + \frac{P_1 V}{RT_1} = \frac{2P_1 V}{RT_1}$

Final condition

P_f, T_1		P_f, T_2
V		V

Initial moles = final moles

$\frac{2P_1 V}{RT_1} = \frac{P_f V}{RT_1} + \frac{P_f V}{RT_2}$

$\frac{2P_1}{T_1} = P_f \left(\frac{T_1 + T_2}{T_1 T_2}\right)$

$\boxed{P_f = \frac{2P_1 T_2}{T_1 + T_2}}$

15. One litre of a gaseous mixture of two gases effuses in 311 seconds while 2 litres of oxygen takes 20 minutes. The vapour density of gaseous mixture containing CH_4 and H_2 is:

Sol. Rate $= \frac{\text{Volume diffused}}{\text{time taken}}$

$r_1 = \frac{1}{311} \Rightarrow$ rate of mixture

$r_2 = \frac{2}{20 \times 60} \Rightarrow$ rate of oxygen

$\frac{r_1}{r_2} = \sqrt{\frac{32}{\text{molar mass}}} = \frac{\frac{1}{311}}{\frac{2}{20 \times 60}}$

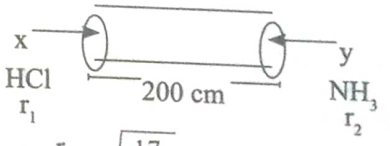
$\Rightarrow \sqrt{\frac{32}{\text{molar mass}}} = \frac{600}{311}$

Molar mass of mixture = 8.6

vapour density $= \frac{\text{molar mass}}{2} = \frac{8.6}{2} = 4.3$

16. A straight glass tube as shown, has 2 inlets X & Y at the two ends of 200 cm long tube. HCl gas through inlet X and NH_3 gas through inlet Y are allowed to enter in the tube at the same time and under the identical conditions. At a point P inside the tube both the gases meet first. The distance of point P from X is:

Sol.



$\frac{r_1}{r_2} = \sqrt{\frac{17}{36.5}}$

@PROBSE

Let both the gases meet in time t

$\sqrt{17}t + \sqrt{36.5}t = 200$

$t = \frac{200}{\sqrt{17} + \sqrt{36.5}}$

$= \frac{200}{4.12 + 6.04}$

$= \frac{200}{10.16}$

$= 19.68$ sec

Distance from x will be

$= 19.68 \times \sqrt{17}$

$= 81.14$ cm

17. A teacher enters a classroom from front door while a student enter from back door. There are 13 equidistant rows of benches in the classroom. The teacher releases N_2O, the laughing gas, from the first bench while the student releases the weeping gas ($C_6H_{11}OBr$) from the last bench. At which row will the students starts laughing and weeping simultaneously.

Sol. Let distance between two bench is x.

Total distance covered = 12x

$\frac{r_{N_2O}}{r_{tear\ gas}} = \sqrt{\frac{179}{44}}$

Let in time 't' both the gases meet

$\sqrt{179}t + \sqrt{144}t = 12x$

$t = \frac{12x}{\sqrt{179} + \sqrt{144}}$

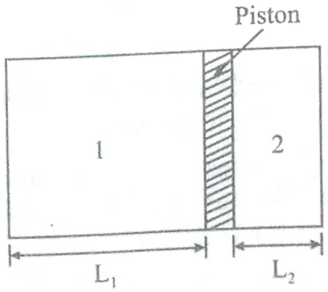
$= \frac{12x}{20}$

distance at which both gases will meet

$= \sqrt{179} \times \frac{12x}{20}$

$= 8x \Rightarrow 9^{th}$ bench

18. The closed cylinder shown in figure has a freely moving piston separating chambers 1 and 2. Chamber 1 contains 280 mg of N_2 gas, and chamber 2 contains 200 mg of helium gas. When equilibrium is established, what will be the ratio α_2/α_1? (Molecular weights of N_2 and He are 28 and 4).



Sol. $n_{N_2} = \frac{280 \times 10^{-3}}{28}$ $\qquad$ $n_{He} = \frac{200 \times 10^{-3}}{4}$

$= 10^{-2}$ $\qquad$ $= 5 \times 10^{-2}$

—— α_1 —— $\qquad$ —— α_2 ——

At equilibrium pressure in both the chamber will be same i.e.

$P_1 = P_2$

$\frac{10^{-2}RT}{\alpha_1 A} = \frac{5 \times 10^{-2}RT}{\alpha_2 A}$

$\boxed{\frac{\alpha_2}{\alpha_1} = 5}$

19. Consider the arrangment of bulbs shown below.



If the pressure of the system when all the stopcocks are opened is x (in atm) then find 100 x?

(760 mm Hg = 1 atm)

Sol. Initial mole = final mole

$\frac{300 \times 1}{RT} + \frac{240 \times 1}{RT} + \frac{440 \times 0.5}{RT} = \frac{P2.5}{RT}$

$760 = P \times 2.5$

$P = \frac{760}{2.5}$ mm

$P = \frac{2}{5}$ atm $= x$

$100x = 100 \times \frac{2}{5} = 40$ atm

20. A closed container of volume 30 litre contains a mixture of nitrogen and oxygen gases, at a temperature of 27°C and pressure of 4 atm. The total mass of the mixture is 148 gm. The moles of individual gases in the container are (Take R = 0.08 litre atm/moleK)

Sol. $V = 30L$

$T = 300K$

$P = 4atm$

$PV = nRT$

$4 \times 30 = n \times 0.08 \times 300$

$n = 5$ moles

Let moels of $N_2 = n_1$

moles of $O_2 = 5 - n_1 = n_2$

$28n_1 + 32(5 - n_1) = 148$

$160 - 4n_1 = 148$

$4n_1 = 12$

$n_1 = 3$ & $n_2 = 2$

@PROCBSE

Exercise-1 (Topicwise)

IDEAL GAS EQUATION OF GAS LAWS

1. If P, V, T represent pressure, volume and temperature of the gas, the correct representation of Boyle's law is

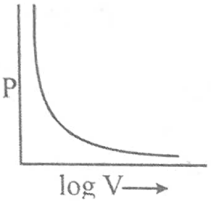
(*a*) $V \propto \frac{1}{T}$ (at constant P)

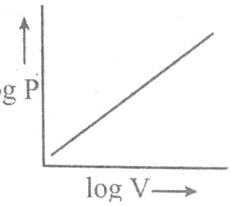
(*b*) $PV = RT$

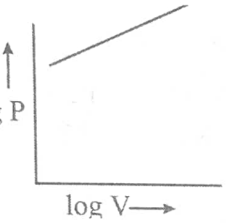
(*c*) $V \propto 1/P$ (at constant T)

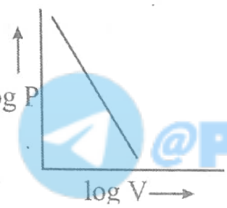
(*d*) $PV = nRT$

2. Which of the following represent log P vs log V variation as per Boyle's law ?

(*a*) 

(*b*) 

(*c*) 

(*d*) 

3. If $20cm^3$ gas at 1 atm is expanded to 50 cm^3 at constant T, then what is the final pressure of gas?

(*a*) $20 \times \frac{1}{50}$ (*b*) $50 \times \frac{1}{20}$

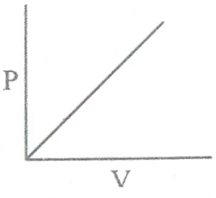
(*c*) $1 \times \frac{1}{20} \times 50$ (*d*) None of these

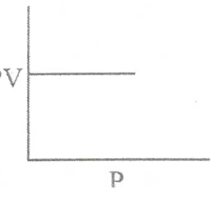
4. At the definite temperature, the volume of a definite mass of gas is 10 L at 5 atm pressure. At the same temperature, if the pressure of the gas is decreased to 1 atm, the volume of same gas becomes:

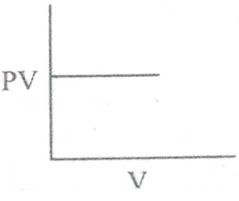
(*a*) 50 L (*b*) 2 L

(*c*) 5 L (*d*) 0.5 L

5. Which of the following graphs represent Boyle's law?

(*a*) 

(*b*) 

(*c*) 

(*d*) (*b*) & (*c*) both

6. Which of the following expression at constant pressure represents Charle's law?

(*a*) $V \propto \frac{1}{T}$ (*b*) $V \propto \frac{1}{T^2}$

(*c*) $V \propto T$ (*d*) $V \propto d$

7. If two moles of an ideal gas at 546K occupy a volume of 44.8 litres, the pressure must be

(*a*) 2 atm (*b*) 3 atm

(*c*) 4 atm (*d*) 1 atm

8. Correct gas equation is

(*a*) $\frac{V_1T_2}{P_1} = \frac{V_2T_1}{P_2}$ (*b*) $\frac{P_1V_1}{P_2V_2} = \frac{T_1}{T_2}$

(*c*) $\frac{P_1T_2}{V_1} = \frac{P_2V_2}{T_2}$ (*d*) $\frac{V_1V_2}{T_1T_2} = P_1P_2$

9. The vapour density of a gas is 11.2. The volume occupied by 11.2 g of this gas at N.T.P. is

(*a*) 1 L (*b*) 11.2 L

(*c*) 22.4 L (*d*) 20 L

10. A pre-weighed vessel was filled with oxygen at N.T.P. and weighed. It was then evacuated, filled with SO_2 at the same temperature and pressure, and again weighed. The weight of oxygen will be

(*a*) The same as that of SO_2

(*b*) Half of that of SO_2

(*c*) Twice that of SO_2

(*d*) One fourth that of SO_2

11. In the gas equation PV = nRT, the value of universal gas constant would depend only on -

(*a*) The nature of the gas

(*b*) The pressure of the gas

(*c*) The temperature of the gas

(*d*) The units of measurement

12. When the pressure of 5L of N_2 is doubled and its temperature is raised from 300K to 600K, the final volume of the gas would be-

(*a*) 10 L (*b*) 5 L

(*c*) 15 L (*d*) 20 L

13. One litre of a gas collected at NTP when kept at 2 atmospheric pressure and 27°C will occupy

(*a*) $\frac{300}{2 \times 273}$ litre (*b*) $\frac{2 \times 300}{273}$ litre

(*c*) $\frac{273}{2 \times 300}$ litre (*d*) $\frac{2 \times 273}{300}$ litre

@PROGRESS

14. 273 mL of a gas at STP was taken to 27°C and 600 mm pressure. The final volume of the gas would be
(a) 273 ml (b) 300 ml
(c) 380 ml (d) 586 ml

15. 10 g of hydrofluoric acid occupy 5.6 litre volume at NTP. The empirical formula of the gas is HF. The molecular formula of the gas will be (At. mass of fluorine = 19)
(a) H_4F_4 (b) HF
(c) H_3F_3 (d) H_2F_2

16. A fire extinguisher contains 4.4 kg of CO_2. The volume of CO_2 delivered by this fire extinguisher at room temperature is
(a) 24.5 litre (b) 100 × 24.5 litre
(c) 10 × 24.5 litre (d) 1000 × 24.5 litre

17. The density of a gas at 27°C and 1 atm is d. Pressure remaining constant, at which of the following temperatures will its density become 0.75 d?
(a) 20°C (b) 30°C
(c) 400 K (d) 300 K

18. Two sealed containers of same capacity at the same temperature are filled with 44 g of hydrogen gas in one and 44 g of CO_2 in the other. If the pressure of CO_2 is 1 atm, what is the pressure in the hydrogen container?
(a) 1 atm (b) 0 atm
(c) 22 atm (d) 44 atm

DALTON'S LAW OF PARTIAL PRESSURES

19. Dalton's law of partial pressures are applicable to
(a) Non reacting gases
(b) Ideal gases
(c) Temperature of the component gases in the mixture remain same
(d) All of these

20. 1000 ml of a gas A at 600 torr and 500 ml of gas B at 800 torr are placed in a 2L flask. The final pressure will be
(a) 2000 torr (b) 1000 torr
(c) 500 torr (d) 1400 torr

MIXING OF GASES

21. If a mixture containing 3 moles of hydrogen and 1 mole of nitrogen is converted completely into ammonia, the ratio of initial and final volume under the same temperature and pressure would be :
(a) 3 : 1 (b) 1 : 3
(c) 2 : 1 (d) 1 : 2

GRAHAM'S LAW OF DIFFUSION

22. A certain gas effuses out of two different vessels A and B. A has a circular orifice while B has a square orifice of length equal to the radius of the orifice of vessel A. The ratio of rate of diffusion of the gas from vessel A to that from vessel B is
(a) π : 1 (b) 1 : π
(c) 1 : 1 (d) 3 : 2

23. If three unreactive gases having partial pressures P_A, P_B and P_C and their moles are 1, 2 and 3 respectively, then their total pressure will be
(a) $P = P_A + P_B + P_C$ (b) $P = \frac{P_A + P_B + P_C}{6}$
(c) $P = \frac{\sqrt{P_A + P_B + P_C}}{3}$ (d) None of these

24. Rate of diffusion of a gas is
(a) Directly proportional to its density
(b) Directly proportional to its molecular mass
(c) Directly proportional to the square root of its molecular mass
(d) Inversely proportional to the square root of its molecular mass

25. The densities of two gases are in the ratio of 1 : 16. The ratio of their rates of diffusion is
(a) 16 : 1 (b) 4 : 1
(c) 1 : 4 (d) 1 : 16

26. A bottle of ammonia and a bottle of dry hydrogen chloride connected through a long tube are opened simultaneously at both ends, the white ammonium chloride ring first formed will be
(a) At the centre of the tube
(b) Near the hydrogen chloride bottle
(c) Near the ammonia bottle
(d) Throughout the length of the tube

27. In an effusion experiment, it required 40 s for a certain number of moles of a gas of unknown molar mass to pass through a small orifice into a vacuum. Under the same conditions, 16 s were required for the same number of moles of O_2 to effuse. What is the molar mass of the unknown gas?
(a) 5.1 g/mol (b) 12.8 g/mol
(c) 80 g/mol (d) 200 g/mol

28. 50 ml of a gas A diffuse through a membrane in the same time as for the diffusion of 40 ml of gas B under identical conditions of pressure and temperature. If the molecular mass of A is 64, molar mass of B would be
(a) 100 (b) 250
(c) 200 (d) 80

KINETIC THEORY OF GASES

29. Internal energy and pressure of a gas per unit volume are related as
(a) $P = \frac{2}{3}E$ (b) $P = \frac{3}{2}E$
(c) $P = \frac{1}{2}E$ (d) $P = 2E$

30. In deriving the kinetic gas equation, use is made of the root mean square velocity of the molecules because it is
(a) The average velocity of the molecules
(b) The most probable velocity of the molecules
(c) The square root of the average square velocity of the molecules
(d) The most accurate form in which velocity can be used in these calculations

@PROCBSE

31. The density of a gas A is three times that of a gas B at same temperature. If the molecular mass of A is M, the molecular mass of B is

(a) 3M (b) $\sqrt{3}$M

(c) M/3 (d) M$\sqrt{3}$

32. The ratio of root mean square velocity to average velocity of gas molecules at a particular temperature is

(a) 1.086 : 1 (b) 1 : 1.086

(c) 2 : 1.086 (d) 1.086 : 2

33. The ratio among most probable velocity, mean velocity and root mean square velocity is given by

(a) 1 : 2 : 3

(b) $1 : \sqrt{2} : \sqrt{3}$

(c) $\sqrt{2} : \sqrt{3} : \sqrt{8/\pi}$

(d) $\sqrt{2} : \sqrt{8/\pi} : \sqrt{3}$

34. At constant volume, for a fixed number of moles of a gas, the pressure of the gas increases with increase in temperature due to

(a) Increase in the average molecular speed

(b) Increased rate of collision amongst molecules

(c) Increase in molecular attraction

(d) Decrease in mean free path

35. Molecular velocities of the two gases at the same temperature are u_1 and u_2. Their masses are m_1 and m_2 respectively. Which of the following expressions is correct?

(a) $\frac{m_1}{u_1^2} = \frac{m_2}{u_2^2}$ (b) $m_1u_1 = m_2u_2$

(c) $\frac{m_1}{u_1} = \frac{m_2}{u_2}$ (d) $m_1u_1^2 = m_2u_2^2$

36. If a gas is allowed to expand at constant temperature, then

(a) number of molecules of the gas decreases

(b) the kinetic energy of the gas molecules decreases

(c) the kinetic energy of the gas molecules increases

(d) the kinetic energy of the gas molecules remains the same

37. At constant volume, for a fixed number of mole of a gas, the pressure of the gas increases with rise of temperature due to

(a) increase in average molecular speed

(b) increased rate of collisions amongst molecules

(c) increase in molecular attraction

(d) decrease in mean free path

38. In a closed flask of 5 litre, 1.0 g of H_2 is heated from 300-600 K. Which statement is not correct?

(a) The rate of collision increases.

(b) The energy of gaseous molecules increases.

(c) The number of mole of the gas increases.

(d) Pressure of the gas increases.

39. The molecules of which of the following gas have highest speed?

(a) Hydrogen at – 50°C (b) Methane at 298K

(c) Nitrogen at 1000°C (d) Oxygen at 0°C

REAL GAS

Vander Waal Equation and Virial Equation of State

40. Which of the following satisfies the greater compressibility of real gas?

(a) $Z < 1$

(b) At the higher pressure

(c) Above the Boyle's temperature

(d) Lesser the value of "a" but higher value of "b"

41. Gases deviate from the ideal gas behaviour because their molecules

(a) Possess negligible volume

(b) Have forces of attraction between them

(c) Are polyatomic

(d) Are not attracted to one another

42. The compressibility factor of a gas is defined as $Z = PV/RT$. The compressibility factor of an ideal gas is

(a) 0 (b) Infinity

(c) 1 (d) –1

43. Vander Waal's equation of state is obeyed by real gases. For n moles of a real gas, the expression will be

(a) $\left(\frac{P}{n} + \frac{na}{V^2}\right)\left(\frac{V}{n-b}\right) = RT$

(b) $\left(P + \frac{a}{V^2}\right)(V - b) = nRT$

(c) $\left(P + \frac{na}{V^2}\right)(nV - b) = nRT$

(d) $\left(P + \frac{n^2a}{V^2}\right)(V - nb) = nRT$

44. When is deviation more in the behaviour of a gas from the ideal gas equation $PV = nRT$?

(a) At high temperature and low pressure

(b) At low temperature and high pressure

(c) At high temperature and high pressure

(d) At low temperature and low high pressure

45. The Vander Waal's constant 'a' for the gases O_2, N_2, NH_3 and CH_4 are 1.3, 1.390, 4.170 and 2.253 L^2 atm mol^{-2} respectively. The gas which can be most easily liquefied is

(a) O_2 (b) N_2

(c) NH_3 (d) CH_4

46. The compressibility factor of a gas is less than 1 at STP. Its molar volume V_m will be

(a) $V_m > 22.42$ (b) $V_m < 22.42$

(c) $V_m = 22.42$ (d) None of these

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47. The critical temperature of a gas is that temperature
(a) above which it can no longer remain in the gaseous state.
(b) above which it cannot be liquefied by pressure.
(c) at which it solidifies.
(d) at which the volume of the gas becomes zero.

48. In van der Waals' equation of state for a non-ideal gas, the term that accounts for intermolecular force is
(a) $\left(P+\frac{a}{V^2}\right)$ (b) $(V-b)$
(c) RT (d) $\frac{1}{RT}$

Exercise-2 (Learning Plus)

1. A thin balloon filled with air at 47°C has a volume of 3 litre. If on placing it in a cooled room, its volume becomes 2.7 litre, then the temperature of room is:
(a) 42°C (b) 100°C
(c) 15°C (d) 200°C

2. At constant temperature, in a given mass of an ideal gas
(a) The ratio of pressure and volume always remain constant
(b) Volume always remain constant
(c) Pressure always remain constant
(d) The product of pressure and volume always remain constant

3. Three flasks of equal volumes contain CH_4, CO_2 and Cl_2 gases respectively. They will contain equal number of molecules if
(a) The mass of all the gases is same
(b) The moles of all the gas is same but temperature is different
(c) Pressure of all the flasks are same
(d) Temperature, pressure are same in all the flasks

4. A certain mass of a gas occupies a volume of 2 litres at STP. Keeping the pressure constant, at what temperature would the gas occupy a volume of 4 litres?
(a) 546°C (b) 273°C
(c) 100°C (d) 50°C

5. Two flasks A and B of 500 ml each are respectively filled with O_2 and SO_2 at 300 K and 1 atm pressure. The flasks will contain
(a) The same number of atoms
(b) The same number of molecules
(c) More number of moles in flask A as compared to flask B
(d) The same amount of gases

6. A gas is heated from 0°C to 100°C at 1.0 atm pressure. If the initial volume of the gas is 10.0L, its final volume would be:
(a) 7.32L (b) 10.00L
(c) 13.66L (d) 20.00L

7. Under what conditions will a pure sample of an ideal gas not only exhibit a pressure of 1 atm but also a concentration of 1 mol litre^{-1}. [R = 0.082 litre atm mol^{-1} K^{-1}]
(a) at S.T.P.
(b) when V = 22.42 L
(c) when T = 12 K
(d) impossible under any condition

8. A 0.5 dm^3 flask contains gas 'A' and 1 dm^3 flask contains gas 'B' at the same temperature. If density of A = 3.0 gm dm^{-3} and that of B = 1.5 gm dm^{-3} and the molar mass of A = 1/2 of B, then the ratio of pressure exerted by gases is
(a) $P_A/P_B = 2$ (b) $P_A/P_B = 1$
(c) $P_A/P_B = 4$ (d) $P_A/P_B = 3$.

9. The density of neon will be highest at
(a) STP (b) 0°C, 2 atm
(c) 273°C, 1 atm (d) 273°C, 2 atm

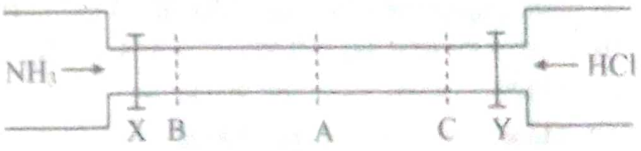
10. Equal weights of ethane & hydrogen are mixed in an empty container at 25°C, the fraction of the total pressure exerted by hydrogen is :
(a) 1: 2 (b) 1: 1
(c) 1: 16 (d) 15: 16

11. A mixture of hydrogen and oxygen at one bar pressure contains 20% by weight of hydrogen. Partial pressure of hydrogen will be
(a) 0.2 bar (b) 0.4 bar
(c) 0.6 bar (d) 0.8 bar

12. The rates of diffusion of SO_3, CO_2, PCl_3 and SO_2 are in the following order -
(a) $PCl_3 > SO_3 > SO_2 > CO_2$
(b) $CO_2 > SO_2 > PCl_3 > SO_3$
(c) $SO_2 > SO_3 > PCl_3 > CO_2$
(d) $CO_2 > SO_2 > SO_3 > PCl_3$

13. 20L of SO_2 diffuses through a porous partition in 60 seconds. Volume of O_2 diffuse under similar conditions in 30 seconds, will be
(a) 12.14L (b) 14.14L
(c) 18.14L (d) 28.14L

14. See the figure-1:



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The valves X and Y are opened simultaneously. The white fumes of NH_4Cl will first form at:

(*a*) A

(*b*) B

(*c*) C

(*d*) A,B and C simultaneously

15. X ml of H_2 gas effuses through a hole in a container in 5 sec. The time taken for the effusion of the same volume of the gas specified below under identical conditions is:

(*a*) 10 sec, He (*b*) 20 sec, O_2

(*c*) 25 sec, CO_2 (*d*) 55 sec, CO_2

16. The rate of diffusion of 2 gases 'A' and 'B' are in the ratio 16: 3. If the ratio of their masses present in the mixture is 2 : 3, then

(*a*) The ratio of their molar masses is 16 : 1.

(*b*) The ratio of their molar masses is 1 : 4.

(*c*) The ratio of their moles present inside the container is 8 : 3.

(*d*) (*b*) & (*c*) both

17. Three footballs are respectively filled with nitrogen, hydrogen and helium. If the leaking of the gas occurs with time from the same area of hole, then the ratio of the rate of leaking of gases ($r_{N_2} : r_{H_2} : r_{He}$) from three footballs (in equal time interval) is:

(*a*) $\left(1:\sqrt{14}:\sqrt{7}\right)$ (*b*) $\left(\sqrt{14}:\sqrt{7}:1\right)$

(*c*) $\left(\sqrt{7}:1:\sqrt{14}\right)$ (*d*) $\left(1:\sqrt{7}:\sqrt{14}\right)$

18. The volume of CO_2 produced by the combustion of 40 ml of gaseous acetone in excess of oxygen, is

(*a*) 40 ml (*b*) 80 ml

(*c*) 60 ml (*d*) 120 ml

19. Temperature at which r.m.s. speed of O_2 is equal to that of neon at 300K, is :

(*a*) 280 K (*b*) 480 K

(*c*) 680 K (*d*) 180 K

20. The temperature of an ideal gas is increased from 120 K to 480K. If at 120K the root-mean-square velocity of the gas molecules is v, then at 480K, it becomes :

(*a*) 4v (*b*) 2v

(*c*) v/2 (*d*) v/4

21. The ratio between the r.m.s. velocity of H_2 at 50 K and that of O_2 at 800K is:

(*a*) 4 (*b*) 2

(*c*) 1 (*d*) 1/4

22. At what temperature will the total KE of 0.3 mol of He be the same as the total KE of 0.40 mol of Ar at 400 K?

(*a*) 533 K (*b*) 400 K

(*c*) 346 K (*d*) 300 K

23. By how many folds the temp of a gas would increase when the r.m.s. velocity of gas molecules in a closed container of fixed volume is increased from 5×10^4 cm s^{-1} to 10×10^4 cm s^{-1}?

(*a*) 0.5 times (*b*) 2 times

(*c*) 4 times (*d*) 16 times.

24. The correct relationship between T_C, T_B and T_i is

(*a*) $T_C < T_B < T_i$ (*b*) $T_C > T_B > T_i$

(*c*) $T_C < T_i < T_B$ (*d*) $T_i < T_B < T_C$

25. In vander Waal's equation of state for a non ideal gas, the term that accounts for intermolecular forces is:

(*a*) nb (*b*) nRT

(*c*) n^2a/V^2 (*d*) $(nRT)^{-1}$

26. The correct order of normal boiling points of O_2, N_2, NH_3 and CH_4, for whom the values of vander Waal's constant 'a' are 1.360, 1.390, 4.170 and 2.253 L^2 atm mol^{-2} respectively, is:

(*a*) $O_2 < N_2 < NH_3 < CH_4$ (*b*) $O_2 < N_2 < CH_4 < NH_3$

(*c*) $NH_3 < CH_4 < N_2 < O_2$ (*d*) $NH_3 < CH_4 < O_2 < N_2$

27. A real gas obeying Vander Waal's equation will resemble ideal gas, if the:

(*a*) Constants a & b are small

(*b*) A is large & b is small

(*c*) A is small & b is large

(*d*) Constant a & b are large

28. For the non-zero values of force of attraction between gas molecules, gas equation will be:

(*a*) $PV = nRT - \frac{n^2a}{V}$ (*b*) $PV = nRT + nbP$

(*c*) $PV = nRT$ (*d*) $P = \frac{nRT}{V-b}$

29. Compressibility factor for H_2 behaving as real gas is:

(*a*) 1 (*b*) $\left(1-\frac{a}{RTV}\right)$

(*c*) $\left(1+\frac{Pb}{RT}\right)$ (*d*) $\frac{RTV}{(1-a)}$

30. At low pressures (For 1 mole), the Vander Waal's equation is written as

$$\left[p+\frac{a}{V^2}\right]V = RT$$

The compressibility factor is then equal to:

(*a*) $\left(1-\frac{a}{RTV}\right)$ (*b*) $\left(1-\frac{RTV}{a}\right)$

(*c*) $\left(1+\frac{a}{RTV}\right)$ (*d*) $\left(1+\frac{RTV}{a}\right)$

31. Calculate the radius of He atoms if its Vander Waal's constant 'b' is 24 ml mol^{-1}.
(Note 1ml = 1cubic centimeter)

(*a*) 1.355 Å (*b*) 1.314 Å

(*c*) 1.255 Å (*d*) 0.355 Å

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32. Which of the following gas has higher value of the van der waals constant 'b'?
(a) N_2 (b) O_2 (c) O_3 (d) F_2

33. A certain saturated hydrocarbon effuses 0.525 times fast as methane. The molecular formula of the hydrocarbon (a compound made up of only carbon and hydrogen) is
(a) C_4H_8 (b) C_4H_{10}
(c) C_3H_6 (d) C_3H_8

34. Consider the gaseous reaction, $N_2 + 3H_2 \rightarrow 2NH_3$
1 mol N_2 and 4 mole H_2 are taken in 15 L flask at 27°C. After the completion of the reaction, 5L of H_2O is added to the reaction vessel, pressure setup in the flask would be (assume conversion to be 100%)
(a) $\frac{3\times0.0821\times300}{15}$ atm (b) $\frac{2\times0.0821\times300}{10}$ atm
(c) $\frac{1\times0.0821\times300}{15}$ atm (d) $\frac{1\times0.0821\times300}{10}$ atm

35. 20 ml of O_2 contracts to 17 ml when subjected to silent electric discharge in an ozoniser. What further contraction would be observed if the partially ozonised oxygen is treated with oil of cinnamon (which absorbs ozone). All volumes are reported at the same temperature and pressure.
(a) 2 ml (b) 8 ml (c) 6 ml (d) 4 ml

36. NH_3 is liquefied more easily than N_2, hence
(a) $a_{NH_3} > a_{N_2}$ and $b_{NH_3} > b_{N_2}$
(b) $a_{NH_3} > a_{N_2}$ but $b_{NH_3} < b_{N_2}$
(c) $a_{NH_3} < a_{N_2}$ but $b_{NH_3} > b_{N_2}$
(d) None

37. The temperature of an ideal gas is reduced from 927°C to 27°C. The r.m.s. velocity of the molecules becomes
(a) Double the initial value
(b) Half of the initial value
(c) Four times the initial value
(d) Ten times the initial value

38. An open vessel containing air is heated from 300 K to 400 K. The fraction of air, which goes out with respect to originally present is
(a) 3/4 (b) 1/4 (c) 2/3 (d) 1/8

39. 4.0 g of argon has pressure 'P' at temperature 'T' K in a vessel. On keeping the sample at 50° higher temperature, 0.8 g gas was given out to maintain the pressure 'P'. The original temperature was
(a) 73 K (b) 100 K
(c) 200 K (d) 510 K

40. The pressure of the atmosphere 100 miles above the surface of earth is about 7.6×10^{-6} mm of Hg and its temperature is −23°C. How many molecules are there in 1 ml of a gas at this altitude?
(a) 2.93×10^{11} (b) 7.58×10^{10}
(c) 0.05×10^{11} (d) 2.07×10^{10}

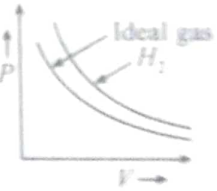
41. The density of phosphorus vapour at 310°C and 775 torr is 2.64 g/l. What is the molecular formula of phosphorus?
(a) P (b) P_2 (c) P_4 (d) P_8

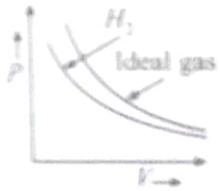
42. A gas cylinder containing cooking gas can withstand a pressure of 14.9 atm. The pressure gauge of cylinder indicates 12 atm at 27°C. Due to sudden fire in building the temperature starts rising. The temperature at which the cylinder explodes is
(a) 42.5°C (b) 67.8°C
(c) 99.5°C (d) 25.7°C

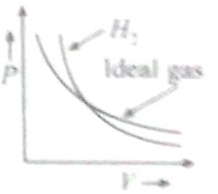
43. At constant pressure what would be the percentage decrease in the density of an ideal gas for a 10% increase in the temperature.
(a) 10% (b) 9.1%
(c) 11% (d) 12.09%

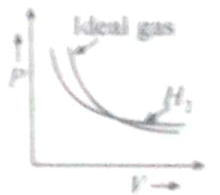
44. The diameter of a balloon filled by 8 moles of He gas is 10 cm. If 7 moles of He gas e uses out in the night, then what should be the diameter of balloon in the next morning? Assume constant pressure and temperature for the gas.
(a) 5 cm (b) $\left(\frac{7}{8}\right)^{1/3} \times 5$ cm
(c) $\left(\frac{8}{7}\right)^{1/3} \times 5$ cm (d) 20 cm

45. A volume of 50 L of dry N_2 is passed through 36 g of water at 300 K. After the passage of gas, there is a loss of 1.20 g in water. The vapour pressure of water at 300 K is
(a) 0.033 mm Hg (b) 24.96 mm Hg
(c) 16.05 mm Hg (d) 38.73 mm Hg

46. Which of the following graph is correct for hydrogen gas at 0°C?
(a) 

(b) 

(c) 

(d) 

47. The equation of state for a gas is P(V − nb) = nRT, where b and R are constants. If the pressure and temperature are such that Vm = 10 b, then what is the compressibility factor of gas in this condition?
(a) 10/9 (b) 9/11
(c) 11/10 (d) 10

48. Under what pressure will carbon dioxide have the density of 2.2 g/litre at 300 K? For CO_2, a = 3.6 atm $L^2 mol^{-2}$ and b = 0.05 mol^{-1} L.
(a) 1.23 atm (b) 1.28 atm
(c) 2.46 atm (d) 0.64 atm

49. At moderate pressure, the compressibility factor for a gas is given as , where P is in bar and T is in Kelvin. What is the Boyle's temperature of the gas?
(a) 168 K (b) 480 K
(c) 58.8 K (d) 575 K

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Exercise-3 (JEE Advanced Level)

MULTIPLE CORRECT TYPE QUESTIONS

1. A gas cylinder containing cooking gas can withstand a pressure of 14.9 atmosphere. The pressure guaze of cylinder indicates 12 atmosphere at 27 °C. Due to sudden fire in the building temperature starts rising. The temperature at which cylinder will explode is :

(a) 372.5 K (b) 99.5 °C

(c) 199 °C (d) 472.5 k

2. Select the correct observation for a 8.21L container, filled with 2 moles of He at 300 K.,

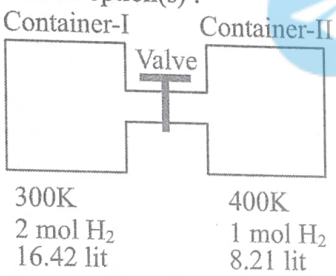
(a) It has pressure 6 atm.

(b) If it is an open rigid container, its pressure increases to 8 atm on heating to 400 K.

(c) If it is closed non-rigid (like thin skin balloon), its volume increases to 16.42 lit. on heating to 600 K.

(d) When connected with another similar empty container maintained at 150 K while maintaining original container at 300 K, pressure reduces to $\frac{2}{3}$ atm.

3. Select the **correct** option(s) :



(a) Pressure in container-I is 3 atm before opening the valve.

(b) Pressure after opening the valve is 3.57 atm.

(c) Moles in each compartment are same after opening the valve.

(d) Pressure in each compartment are same after opening the valve.

4. Which of the following statements are correct ?

(a) Helium diffuses at a rate 8.65 times as much as CO does.

(b) Helium escapes at a rate 2.65 times as fast as CO does.

(c) Helium escapes at a rate 4 times as fast as CO_2 does.

(d) Helium escapes at a rate 4 times as fast as SO_2 does.

5. The rate of diffusion of 2 gases 'A' and 'B' are in the ratio of 16: 3. If the ratio of their masses present in the mixture is 2 : 3, then

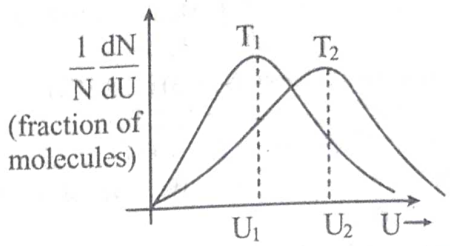
(a) The ratio of their molar masses is 16 : 1.

(b) The ratio of their molar masses is 1 : 4.

(c) The ratio of their moles present inside the container is 1 : 24.

(d) The ratio of their moles present inside the container is 8 : 3.

6.



In the above maxwellian plot at two different temperatures, which of the following statements may be true?

(a) Area under the two plots is the same.

(b) Fraction of molecules with speed u_1 at T_1 > fraction of molecules having speed u_2 at T_2.

(c) $U_2 > U_1$ and $T_2 > T_1$

(d) U_{MPS} at $T_1 < U_{MPS}$ at T_2 ; U_{RMS} at $T_1 < U_{RMS}$ at T_2

7. If a closed container of volume 200L of O_2 gas (ideal gas) at 1 atm & 200 K is taken to another planet. Find the pressure of oxygen gas at the planet at 821 K in same container

(a) $\frac{10}{e^{100}}$ (b) $\frac{20}{e^{50}}$

(c) 1 atm (d) 4.1 atm

8. Which of the following are correct statements?

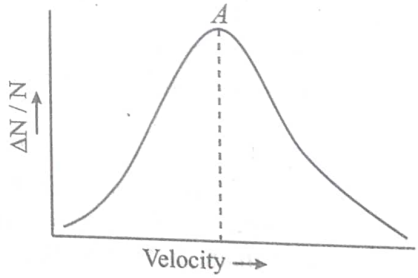
(a) van der Waals' constant 'a' is a measure of attractive force.

(b) van der Waals' constant 'b' is also called co-volume or excluded volume.

(c) b is expressed in L mol^{-1}.

(d) b is one-third of critical volume.

9. Point A in the given curve shifts to higher value of velocity if



(a) T is increased

(b) P is decreased

(c) V is decreased

(d) molecular weightM is decreased

10. According to the kinetic theory of gases

(a) pressure of a gas is due to collisions of molecules with each other.

(b) kinetic energy is proportional to square root of the temperature.

(c) pressure of a gas is due to collisions of molecules against the sides of the container.

(d) there is no force of attraction between gas molecules.

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11. For two gases, A and B with molecular weights M_A and M_B, it is observed that at a certain temperature T, the mean velocity of A is equal to the V_{rms} of B. Thus the mean velocity of A can be made equal to the mean velocity of B, if
(a) A is at temperature T and B at T′ such that T > T′.
(b) Temperature of A is lowered to T_2 while B is at T such that $T_2 < T$.
(c) Both A and B are raised to a higher temperature.
(d) Heat energy is supplied to A.

12. If for two gases of molecular weights M_A and M_B at temperature T_A and T_B, it is observed that $T_A M_B = T_B M_A$, then which property has the same magnitude for both the gases.
(a) PV if mass of gases taken are same
(b) Pressure
(c) KE per mol
(d) V_{rms}

13. Molecular attraction and size of the molecules in a gas are not negligible at
(a) Critical point
(b) High pressure
(c) High temperature and low pressure
(d) low temperature and high pressure

14. Select correct statements
(a) Vapour may be condensed to liquid by the application of pressure.
(b) To liquefy a gas one must lower the temperature below T_c and apply pressure.
(c) At T_c there is no distinction between liquid and vapour state.
(d) At T_c, density of liquid is very high as compared to its gaseous state.

15. Which of the following statement is/are correct?
(a) All real gases are less compressible than ideal gas at high pressure.
(b) Hydrogen and helium are more compressible than ideal gas for all values of pressure.
(c) Except H_2 and He, the compressibility factor $Z = \left(\frac{PV}{nRT}\right) < 1$ for all gases at low pressure.
(d) The compressibility factor of real gases is independent of temperature.

16. Precisely 1 mole of helium and 1 mole of neon are placed in a container. Indicate the correct statements about the system
(a) Molecules of the two gases strike the wall of the container with same frequency.
(b) Molecules of helium strike the wall more frequently.
(c) Molecules of helium has greater average molecular speed.
(d) Helium exerts larger pressure.

17. Which of the following graphs are correct for ideal gases?
(a) $\frac{1}{dC}\cdot\left(\frac{dN}{N}\right)$ vs C → $T_1 < T_2$; T_2 K; T_1 K
(b) $\frac{1}{dC}\cdot\left(\frac{dN}{N}\right)$ vs C → $T_1 < T_2$; T_1 K; T_2 K
(c) $\frac{1}{dC}\cdot\left(\frac{dN}{N}\right)$ vs C → $M_1 < M_2$; M_2 K; M_1 K
(d) $\frac{1}{dC}\cdot\left(\frac{dN}{N}\right)$ vs C → $M_2 < M_1$; M_2; M_1

18. The critical temperature and pressure of a gas are 340 K and 42 atm, respectively. The gas can be liquefied at
(a) 340 K and 45 atm (b) 340 K and 40 atm
(c) 350 K and 100 atm (d) 320 K and 42 atm

19. Which of the following statement(s) is/are correct?
(a) The physical state of the system at critical condition cannot be defined.
(b) All the physical properties of liquid and gaseous state of a substance become identical at critical condition.
(c) For a gas, there is no distinction between liquid and vapour state at critical condition.
(d) Easily liquefiable gases have higher Boyle's temperature while the gases di cult to liquefy has lower Boyle's temperature.

20. For a real gas (van der Waals gas), behaving ideally, the pressure may be
(a) $a.b.V_m$ (b) $\frac{V_m}{a.b}$
(c) $\frac{a}{V_m.b}$ (d) $\frac{b}{V_m.a}$

21. Van der Waals constant 'a' can be expressed as
(a) $\frac{9RT_CV_C}{8}$ (b) $3P_CV_C^2$
(c) $\frac{3}{8}.\frac{P_CV_C}{T_C}$ (d) $\frac{27}{64}.\frac{R^2T_C^2}{P_C}$

COMPREHENSION BASED QUESTIONS

Comprehension (Q. 22 to 24): Two containers X & Y are present with container X consisting of some mass of He at some temperature while container Y having double the volume as that

of container X & kept at same temperature containing same mass of H_2 gas as the mass of Helium gas. Based on this data & the following conditions, answer the question that follows:

Assume sizes of H_2 molecule & He atom to be same & size of H–atom to be half to that of He–atom & only bimolecular collisions to be occuring.

Condition I: All except one atom of He are stationary in container X & all molecules of H_2 are moving in container Y.

Condition II: Both containers contain all moving molecules.

22. Assuming **condition I** to be applicable & if no. of *total collisions* occuring per unit time is 'A' in container X, then no. of total collisions made by any one molecule in container Y will be:

(*a*) A (*b*) $\sqrt{3}$A

(*c*) $\frac{A}{\sqrt{2}}$ (*d*) None of these

23. Assuming **condition II** then ratio of 'total no. of collisions per unit volume per unit time' in container X & container Y is (container X : container Y)

(*a*) 1:1 (*b*) $\sqrt{2}:1$

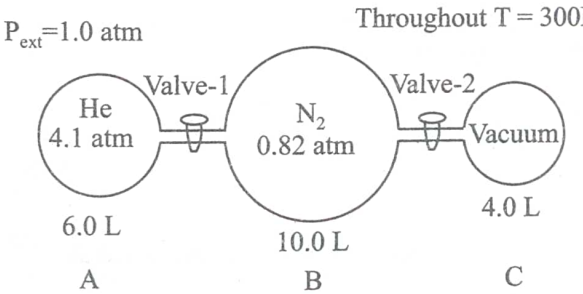
(*c*) $1:\sqrt{2}$ (*d*) 4:1

24. Assuming **condition II** to be applicable, if temperature only of container Y is doubled to that of original (causing dissociation of all H_2 gas into H gaseous atoms) then, if no. of total collisions per unit volume per unit time in container X is A then, no. of 'total collisions made by all molecules per unit volume in container Y would be

(*a*) $2\sqrt{2}$A (*b*) $\sqrt{2}$A

(*c*) $8\sqrt{2}$A (*d*) None of these

Comprehension (Q. 25 to 28): The figure given below shows three glass chambers that are connected by valves of negligible volume. At the outset of an experiment, the valves are closed and the chambers contain the gases as detailed in the diagram. All the chambers are at the temperature of 300K and external pressure of 1.0 atm.



25. What will be the work done by N_2 gas when the valve–2 is opened and valve–1 remaining closed?

(*a*) 8.2 L atm (*b*) –8.2 atm

(*c*) 0 (*d*) –3.28 L atm

26. Which of the following represents the total kinetic energy of all the gas molecules after both valves are opened? (R = 0.082 atm L K^{-1} mol^{-1} = 8.314 L K^{-1} mol^{-1})

(*a*) 2836.2 J (*b*) 3280.0 J

(*c*) 4520.6 (*d*) 4982.4 J

27. Suppose that after both valves are opened, the entire apparatus is cooled to lower the final internal pressure to 1.0 atm. What is the contribution of N_2 gas to this final pressure?

(*a*) 0.40 atm (*b*) 0.35 atm

(*c*) 0.30 atm (*d*) 0.25 atm

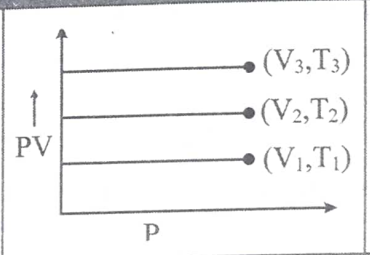
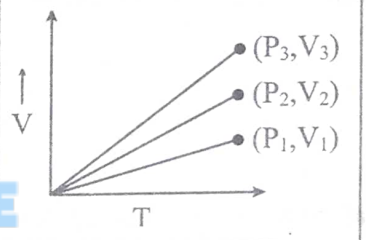
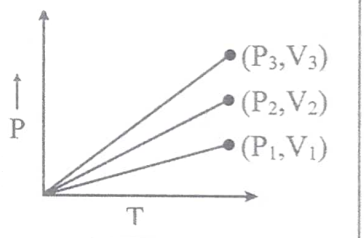
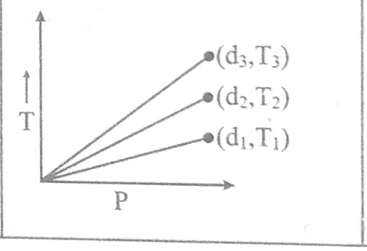
28. What is the total pressure in chamber B if only valve–1 is opened?

(*a*) 0.31 atm (*b*) 2.05 atm

(*c*) 2.46 atm (*d*) 3.10 atm

MATCH THE COLUMN TYPE QUESTIONS

29. For a fixed amount of the gas, match the two columns:

Column-I		Column-II	
A.		p.	$T_3 > T_2 > T_1$
B.		q.	$P_1 > P_2 > P_3$
C.		r.	$V_1 > V_2 > V_3$
D.		s.	$d_1 > d_2 > d_3$

(*a*) A → s; B → q,s; C → r; D → p,r

(*b*) A → q,s; B → s; C → r; D → p,r

(*c*) A → s; B → r; C → q,s; D → p,r

(*d*) A → s; B → q,s; C → p,r; D → r

30. The vander waal gas constant 'a' is given by

(*a*) $\frac{1}{3}V_c$ (*b*) $3P_C V_C^2$

(*c*) $\frac{1}{8}\frac{RT_C}{P_C}$ (*d*) $\frac{27}{64}\frac{R^2 T_C^2}{P_C}$

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31. For gaseous state, which of the following is incorrect?
(*a*) Thermal energy = Molecular attraction
(*b*) Thermal energy >> Molecular attraction
(*c*) Thermal energy << Molecular attraction
(*d*) Molecular force >> attraction in liquid

32. Select the incorrect statement(s):
(*a*) At Boyle's temperature, a real gas behaves like an ideal gas irrespective of pressure.
(*b*) At critical condition, a real gas behaves like an ideal gas.
(*c*) On increasing the temperature four times,collision frequency (Z_{11}) becomes double at constant volume.
(*d*) At high pressure, Vander Waal's constant 'b' dominates over 'a'.

33. Match the column & choose the correct option among the options given.
(**Note:** Each statement in column-I has one or more matches in column-II.)

	Column-I		Column-II
A.	Critical temperature	p.	Gas can be liquefied.
B.	Boyle's temperature	q.	Deviate from ideal gas equation.
C.	Compressibility factor (Z) < 1	r.	Gas follows the ideal gas equation.
D.	High temperature and low pressure	s.	Assumption of no intermolecular force of attraction is valid.

(*a*) A → p, q; B → r, s; C → p, q; D → r, s
(*b*) A → p, r; B → r, s; C → p, r; D → r, s
(*c*) A → r, s; B → p, q; C → r, s; D → p, q
(*d*) A → p, q; B → p, q; C → r, s; D → r, s

34. Match the column & choose the correct option among the options given.
(**Note:** Each statement in column-I has one or more matches in column-II.)

	Column-I		Column-II
A.	Diffusion of gas	p.	High when molecular mass is low
B.	Compressibility of gas (Z) < 1	q.	$V_{real} < V_{ideal}$
C.	V_{rms}	r.	Increases with increase in temperature
D.	Liquefaction of gas	s.	Attractive force dominates

(*a*) A → p, r; B → p, r, s; C → p, r ; D → p, q, s
(*b*) A → p, r; B → q, r, s; C → p, r ; D → q, s
(*c*) A → p, r; B → q, r, s; C → p, q, r; D → p, q, s
(*d*) A → p, r; B → p, q; C → p, q, s; D → p, q, s

35. Match the column & choose the correct option among the options given.
(**Note:** Each statement in column-I has only one match in column-II.)

	Column-I		Column-II
A.	Compressibility factor (Z) for ideal gas	p.	$\frac{3}{8}$
B.	Z for real gas at low P	q.	(1 + Pb/RT)
C.	Z for real gas at high P	r.	1
D.	Z for critical state	s.	(1 – a/RTV)

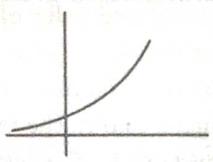
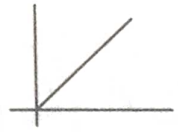
(*a*) A → p; B → s;C → q; D → r
(*b*) A → r; B → q; C → s; D → p
(*c*) A → r; B → s; C → q; D → p
(*d*) A → p; B → q; C → s; D → r

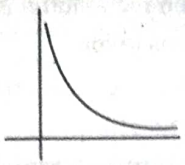
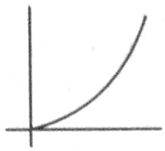
36. Match the columns & choose the correct option among the options given.
(**Note:** Each statement in column-I has one or more matches in column-II.)

	Column-I		Column-II
A.	H_2 gas at 300 K	p.	Maximum translational K.E. per mol.
B.	CH_4 gas at 400 K	q.	Maximum translational K.E. per gram.
C.	O_2 gas at 900 K	r.	Minimum translational K.E. per mol.
D.	He gas at 600 K	s.	Minimum translational K.E. per gram.

(*a*) A → q, r; B → p; C → s; D → q
(*b*) A → q, r; B → q; C → p; D → s
(*c*) A → q, r; B → p; C → q; D → s
(*d*) A → q, r; B → s; C → p; D → q

37. Match the description in Column-I with graph in Column-II for 'n' moles of an ideal gas at constant temperature & choose the correct option among the options given.
(**Note:** Each statement in column-I has only one match in column-II.)

	Column-I		Column-II
A.	P/V vs. P	p.	
B.	P/V vs. V	q.	

C.	(V/P) Vs $\frac{1}{P^2}$	r.	
D.	P/V vs. log P	s.	

(*a*) A → r; B → r; C → q; D → p
(*b*) A → s; B → r; C → p; D → q
(*c*) A → s; B → r; C → q; D → p
(*d*) A → r; B → q; C → p; D → s

NUMERICAL TYPE QUESTIONS

38. A vertical cylinder containing some ideal gas is closed at both ends. A frictionless piston is fitted in the cylinder such that it divides the cylinder into two equal parts at 300K. Moles of gas in one part is double than that in other part. To what temperature (**in Kelvin**) the cylinder should be heated or cooled such that the volume of upper part becomes three times that of lower part?

39. 500 ml of a vessel contains an ideal gas saturated with water vapours at 3.2 atm and 47°C. Now the entire mass is transferred into vessel of 1800 ml capacity maintained at 87°C. The final pressure (**in atm**) becomes (vapour pressure of water at 47°C and 87°C are 0.2 atm and 0.6 atm, respectively) _____.

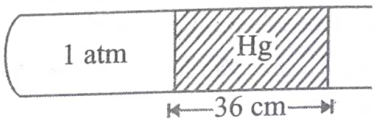
40. At 400 K, the root mean square (rms) speed of gas X (molecular mass = 40) is equal to the most probable speed of gas Y at 60K. The molecular mass of the gas Y is _____.

41. The molecular weight of a gas which diffuse through a porous plug at 1/6th of the speed of hydrogen under identical conditions, is

42. A gaseous mixture containing equal moles of H_2, O_2 and He is subjected to series of effusion steps. The composition (by moles) of effused mixture after 4 effusion steps is x : 1 : y respectively. Then, find the value of $\left(\frac{x}{y}\right)$.

43. Air is trapped in a horizontal glass tube by 40cm mercury column as shown below:

If the tube is held vertical keeping the open end up. length of air coumn shrink to 19cm. Length in cm by which the mercury column shifts down is _____.



44. An iron meteorite was analysed for its isotopic argon content. The amount of Ar^{36} was 1.911 mm^3 (STP) per kg of meteorite. If each Ar^{36} atom had been formed by a single cosmic event, then how many such events must there have been per kg of meteorite? If the answer is X, then the value of $X/10^{16}$ is (R = 8.4 J/K-mol, $N_A = 6.0 \times 10^{23}$)

45. A vessel of volume 3.0 L contains ideal gas at the temperature 0°C. After a portion of the gas has been let out, the pressure in the vessel decreases by 0.8 atm (the temperature remaining constant). The gas density under the normal condition (0°C and 1 atm) is 1.25 g/L. The mass of released gas in (g) is

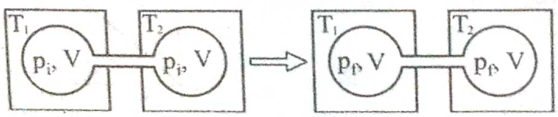
46. Two identical vessels are connected by a tube with a valve letting the gas to pass from one vessel into the other if the pressure difference $\Delta P \geq 2.0$ atm. Initially, there was a vacuum in one vessel while the other contained ideal gas at a temperature 300 K and pressure 4.0 atm. Then both vessels were heated to a temperature 600 K. Up to what value will be the pressure (in atm) in the first vessel (which had a vacuum initially) increase?

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Exercise-4 (Past Years Questions)

JEE MAIN

1. Two closed bulbs of equal volume (V) containing an ideal gas initially at pressure p_i and temperature T_1 are connected through a narrow tube of negligible volume as shown in the figure below. The temperature of one of the bulbs is then raised to T_2. The final pressure p_f is: **(2016)**



(*a*) $p_i\left(\frac{T_1T_2}{T_1+T_2}\right)$ (*b*) $2p_i\left(\frac{T_1}{T_1+T_2}\right)$

(*c*) $2p_i\left(\frac{T_2}{T_1+T_2}\right)$ (*d*) $2p_i\left(\frac{T_1T_2}{T_1+T_2}\right)$

2. The volume of gas A is twice that of gas B. The compressibility factor of gas A is thrice that of gas B at same temperature. The pressure of gases for equal moles are **(2019)**

(*a*) $3P_A = 2P_B$ (*b*) $2P_A = 3P_B$

(*c*) $P_A = 3P_B$ (*d*) $P_A = 2P_B$

3. 0.5 moles of gas A and x moles of gas B exert a pressure of 200 Pa in a container of volume 10 m^3 at 1000K. Given: R is the gas constant in JK^{-1} mol^{-1}, x is **(2019)**

(a) $\frac{2R}{4+R}$ (b) $\frac{2R}{4-R}$

(c) $\frac{4+R}{2R}$ (d) $\frac{4-R}{2R}$

4. Consider the van der Waals constants, a and b, for the following gases: **(2019)**

Gas	Ar	Ne	Kr	Xe
a/(atm dm^6 mol^{-2})	1.3	0.2	5.1	4.1
b/(10^{-2} dm^3 mol^{-1})	3.2	1.7	1.0	5.0

Which gas is expected to have the highest critical temperature?

(a) Kr (b) Ne

(c) Ar (d) Xe

5. At a given temperature T, gases Ne, Ar, Xe and Kr are found to deviate from ideal gas behaviour. Their equation of state is given as **(2019)**

$$P = \frac{RT}{V-b} \text{ at T.}$$

Here, b is the van der Waals constant. which gas will exhibit steepest increase in the plot of Z(compression factor) vs p?

(a) Ne (b) Ar

(c) Xe (d) Kr

6. Consider the following table: (2019)

Gas	a/(k pa dm^6 mol^{-10})	b/(dm^3 mol^{-1})
A	642.32	0.05192
B	155.21	0.04136
C	431.91	0.05196
D	155.21	0.4382

a and b are vander waals constant. The correct statement about the gases is :

(a) Gas C will occupy lesser volume than gas A; gas B will be lesser compressible than gas D.

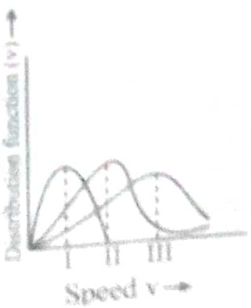
(b) Gas C will occupy more volume than gas A; gas B will be lesser compressible than gas D.

(c) Gas C will occupy more volume than gas A; gas B will be more compressible than gas D.

(d) Gas C will occupy lesser volume than gas A; gas B will be more compressible than gas D.

7. Points I, II and III in the following plot respectively correspond to **(2019)**

(V_{mp}: most probable velocity)



(a) V_{mp} of N_2 (300K); V_{mp} of H_2(300K); V_{mp} of O_2 (400K)

(b) V_{mp} of H_2 (300K); V_{mp} of N_2(300K); V_{mp} of O_2 (400K)

(c) V_{mp} of O_2 (400K); V_{mp} of N_2(300K); V_{mp} of H_2 (300K)

(d) V_{mp} of N_2 (300K); V_{mp} of O_2(400K); V_{mp} of H_2 (300K)

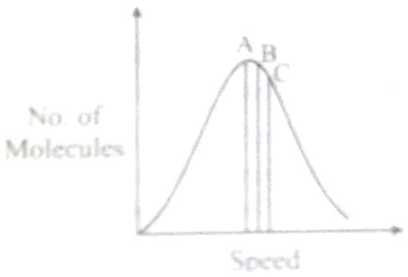
8. The noble gas that does NOT occur in the atmosphere is: **(2019)**

(a) He (b) Rn

(c) Ne (d) Kr

9. Identify the correct of A, B and C in the following graph from the options below: **(2020)**

Root mean square speed (V_{rms}); most probable speed (V_{mp}); Average speed (V_{av})



(a) A – V_{mp}; B–V_{rms}; C – V_{av}

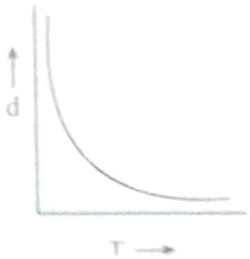
(b) A – V_{rms}; B–V_{mp}; C – V_{av}

(c) A – V_{av}; B–V_{rms}; C – V_{mp}

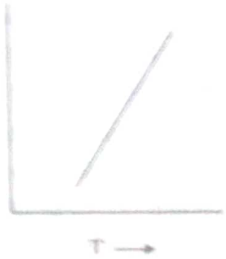
(d) A – V_{av}; B–V_{mp}; C–V_{rms}

10. Which one of the following graphs is not correct for ideal gas? **(2020)**

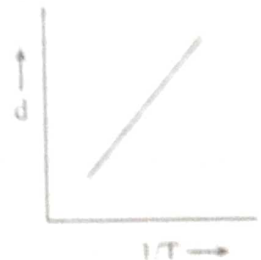
(a)



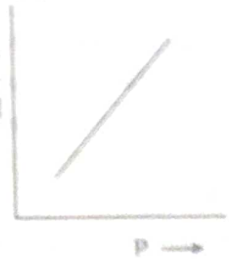
(b)



(c)



(d)



d = Density, P = Pressure, T = Temperature

11. A mixture of one mole each of H_2, He and O_2 each are enclosed in a cylinder of volume V at temperature T. If the partial pressure of H_2 is 2 atm, the total pressure of the gases in the cylinder is **(2020)**

(a) 14 atm (b) 38 atm

(c) 22 atm (d) 6 atm

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12. A spherical balloon of radius 3 cm containing helium gas has a pressure of 48×10^{-3} bar. At the same temperature, the pressure, of a spherical balloon of radius 12 cm containing the same amount of gas will be _____ $\times 10^{-6}$ bar. **(2020)**

13. An empty LPG cylinder weighs 14.8kg. When full, it weighs 29.0kg and shows a pressure of 3.47 atm. In the course of use at ambient temperature, the mass of the cylinder is reduced to 23.0kg. The final pressure inside the cylinder is _____ atm (Nearest integer)
(Assume LPG to be an ideal gas) **(2021)**

14. When 5.1g of solid NH_4HS is introduced into a two litre evacuated flask at 27°C,20% of the solid decomposes into gaseous ammonia and hydrogen sulphide. The K_p for the reaction at 27°C is $x \times 10^{-2}$.
The value of x is _____(Integer answer)
[Given: R = 0.082 L atm K^{-1} mol^{-1}] **(2021)**

15. An LPG cylinder contains gas at a pressure of 300kPa at 27°C. The cylinder can withstand the pressure of 1.2×10^6 Pa. The room in which the cylinder is kept catches fire. The minimum temperature at which the bursting of cylinder will take place is _____ °C. (Nearest integer) **(2021)**

16. A home owner uses 4.00×10^3 m^3 of methane (CH_4) gas, (assume CH_4 is an ideal gas) in a year to heat his home. Under the pressure of 1.0 atm and 300 K, mass of gas used is $x \times 10^5$g. The value of x is _____(Nearest integer)
(Given: R = 0.083 L atm K^{-1} mol^{-1}) **(2021)**

17. At 300 K, a sample of 3.0 g of gas A occupies the same volume as 0.2 g of hydrogen at 200 K at the same pressure. The molar mass of gas A is_____g mol^{-1} (nearest integer). Assume that the behaviour of gases as ideal. (Given: The molar mass of hydrogen (H_2) gas is 2.0 g mol^{-1}) **(2021)**

18. A rigid nitrogen tank stored inside a laboratory has a pressure of 30 atm at 06:00 am when the temperature is 27°C. At 03:00 pm, when the temperature is 45°C, the pressure in the tank will be_____atm. [nearest integer] **(2022)**

19. A sealed flask with a capacity of 2 dm^3 contains 11 g of propane gas. The flask is so weak that it will burst if the pressure becomes 2 MPa. The minimum temperature at which the flask will burst is _____°C. [Nearest integer]
(Given: R = 8.3 J K^{-1} mol^{-1}. Atomic masses of C and H are 12u and lu respectively.) (Assume that propane behaves as an ideal gas.) **(2022)**

20. A mixture of hydrogen and oxygen contains 40% hydrogen by mass when the pressure is 2.2 bar. The partial pressure of hydrogen is _____ bar. (Nearest Integer) **(2022)**

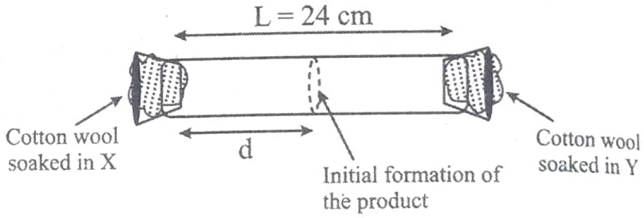
21. For a real gas at 25°C temperature and high pressure (99 bar), the value of compressibility factor is 2, so the value of Vander Waal's constant 'b' should be _____ $\times 10^{-2}$ L mol^{-1} (Nearest integer)
(Given: R = 0.083 L bar K^{-1} mol^{-1}) **(2022)**

JEE ADVANCED

22. The atomic masses of He and Ne are 4 and 20 a.m.u., respectively. The value of the de Broglie wavelength of He gas at –73°C is "M" times that of the de Broglie wavelength of Ne at 727°C. M is___. **(2013)**

Paragraph for questions 23 and 24

X and Y are two volatile liquids with molar weights of 10 g mol^{-1} and 40 g mol^{-1} respectively. Two cotton plugs, one soaked in X and the other soaked in Y, are simultaneously placed at the ends of a tube of length L = 24 cm, as shown in the figure. The tube is filled with an inert gas at 1 atmosphere pressure and a temperature of 300 K. Vapours of X and Y react to form a product which is first observed at a distance d cm from the plug soaked in X. Take X and Y to have equal molecular diameters and assume ideal behaviour for the inert gas and the two vapours.

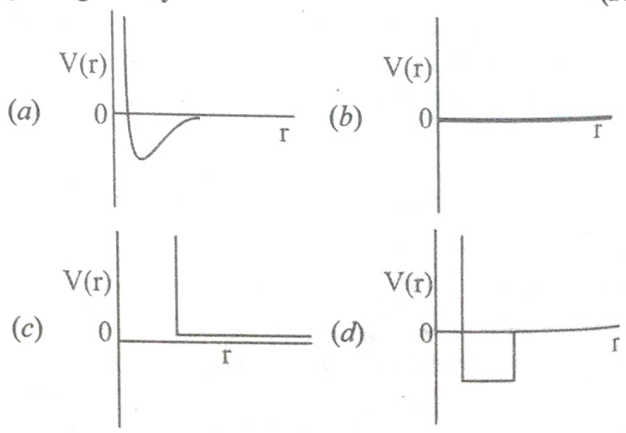


23. The value of **d** in cm (shown in the figure), as estimated from Graham's law, is: **(2014)**
(*a*) 8 (*b*) 12
(*c*) 16 (*d*) 20

24. The experimental value of **d** is found to be smaller than the estimate obtained using Graham's law. This is due to **(2014)**
(*a*) larger mean free path for **X** as compared to that of **Y**
(*b*) larger mean free path for **Y** as compared to that of **X**
(*c*) increased collision frequency of **Y** with the inert gas as compared to that of **X** with the inert gas
(*d*) increased collision frequency of **X** with the inert gas as compared to that of **Y** with the inert gas

25. A closed vessel with rigid walls contains 1 mol of $^{238}_{92}U$ and 1 mol of air at 298K. Considering complete decay of $^{238}_{92}U$ to $^{206}_{82}Pb$, the ratio of the final pressure to the initial pressure of the system at 298K, is _____ **(2015)**

26. One mole of a monoatomic real gas satisfies the equation $p(V - b) = RT$ where b is a constant. The relationship of interatomic potential V(r) and interatomic distance r for the gas is given by **(2015)**



27. The diffusion coefficient of an ideal gas is proportional to its mean free path and mean speed. The absolute temperature of an ideal gas is increased 4 times and its pressure is increased 2 times. As a result, the diffusion coefficient of this gas increases x times. The value of x is **(2016)**

28. A closed tank has two compartments A and B, both filled with oxygen (assumed to be ideal gas). The partition separating the two compartments is fixed and is a perfect heat insulator (Figure 1). If the old partition is replaced by a new partition which can slide and conduct heat but does NOT allow the gas to leak across (Figure 2), the volume (in m^3) of the compartment A after the system attains equilibrium is____. **(2018)**

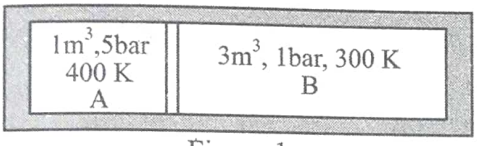


Figure 1

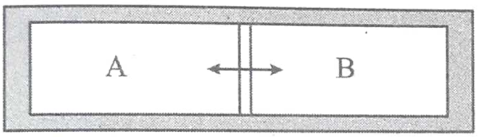


Figure 2

29. Which of the following statement(s) is (are) correct regarding the root mean square speed (U_{rms}) and average translational kinetic energy (ε_{av}) of a molecule in a gas at equilibrium? **(2019)**

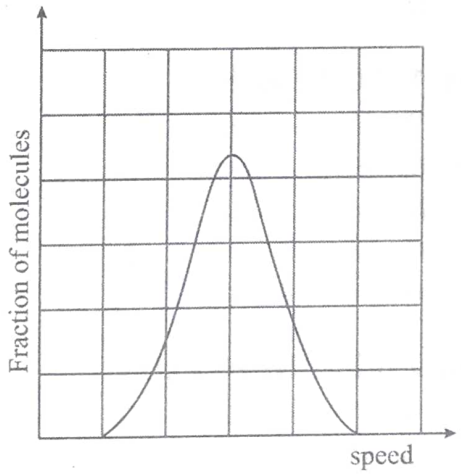
(*a*) U_{rms} is doubled when its temperature is increased four times.

(*b*) ε_{av} at a given temperature does not depend on its molecular mass.

(*c*) U_{rms} is inversely proportional to the square root of its molecular mass.

(*d*) ε_{av} is doubled when its temperature is increased four times.

30. If the distribution of molecular speeds of a gas is as per the figure shown below, then the ratio of the most probable, the average and the roots mean square speeds, respectively, is **(2020)**



(*a*) 1:1:1 (*b*) 1:1:1.224

(*c*) 1:1.128:1.224 (*d*) 1:1.128:1

ANSWER KEY

CONCEPT APPLICATION

1. (a)	2. (b)	3. (a)	4. (d)	5. (d)	6. (a)	7. (a)	8. (d)	9. (c)	10. (a)
11. (a)	12. (b)	13. (b)	14. (d)	15. (b)	16. (b)	17. (b)	18. (a)	19. (b)	20. (c)
21. (d)	22. (c)								

EXERCISE-1 (TOPICWISE)

1. (c)	2. (d)	3. (a)	4. (a)	5. (d)	6. (c)	7. (a)	8. (b)	9. (b)	10. (b)
11. (d)	12. (b)	13. (a)	14. (c)	15. (d)	16. (b)	17. (c)	18. (c)	19. (d)	20. (c)
21. (c)	22. (a)	23. (a)	24. (d)	25. (b)	26. (b)	27. (d)	28. (a)	29. (a)	30. (d)
31. (c)	32. (a)	33. (d)	34. (a)	35. (d)	36. (d)	37. (a)	38. (c)	39. (a)	40. (a)
41. (b)	42. (c)	43. (d)	44. (b)	45. (c)	46. (b)	47. (b)	48. (a)		

EXERCISE-2 (LEARNING PLUS)

1. (c)	2. (d)	3. (d)	4. (b)	5. (b)	6. (c)	7. (c)	8. (c)	9. (b)	10. (d)
11. (d)	12. (d)	13. (b)	14. (c)	15. (b)	16. (d)	17. (a)	18. (d)	19. (b)	20. (b)
21. (c)	22. (a)	23. (c)	24. (a)	25. (c)	26. (b)	27. (a)	28. (a)	29. (c)	30. (a)
31. (a)	32. (c)	33. (b)	34. (d)	35. (c)	36. (b)	37. (b)	38. (b)	39. (c)	40. (a)
41. (c)	42. (c)	43. (b)	44. (a)	45. (b)	46. (a)	47. (a)	48. (a)	49. (b)	

EXERCISE-3 (JEE ADVANCED LEVEL)

1. (a,b)	2. (a,c)	3. (a,d)	4. (b,d)	5. (b,d)	6. (a,b,c,d)	7. (d)	8. (a,b,c,d)	9. (a,d)	10. (c,d)
11. (a,d)	12. (a,d)	13. (a,b,d)	14. (a,b,c)	15. (a,c)	16. (b,c)	17. (a,d)	18. (a,d)	19. (a,b,c,d)	20. (c)
21. (a,b,d)	22. (d)	23. (c)	24. (b)	25. (c)	26. (d)	27. (d)	28. (b)	29. (a)	30. (b,d)
31. (a,c,d)	32. (a,b)	33. (a)	34. (b)	35. (c)	36. (d)	37. (c)	38. [90]	39. [0001]	40. [4]
41. [72]	42. [4]	43. [10]	44. [5]	45. [3]	46. [3]				

EXERCISE-4 (PAST YEAR QUESTIONS)

JEE Main

1. (c)	2. (b)	3. (d)	4. (a)	5. (c)	6. (c)	7. (d)	8. (b)	9. (d)	10. (b)
11. (d)	12. [750]	13. [2]	14. (6)	15. [927]	16. [26]	17. [45]	18. [32]	19. [1655]	20. [2]
21. [25]									

JEE Advanced

22. [5]	23. (c)	24. (d)	25. [9]	26. (c)	27. [4]	28. [2.22]	29. (a,b,c)	30. (b)